

Topological Energy Condensate Theory (TECT): 12

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A. Mass protection from residual valley symmetry

We now close the mass-protection question at the theorem level.

a. Setup. Assume that opposite-valley pairing has already been established, so that the isolated shell-relevant doubled block is

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}, \quad h_-(p) = -h_+(-p), \quad (1)$$

and that, after a linear change of momentum coordinates and a shell-adapted basis choice, the linearized crossing takes the standard Weyl-paired form

$$H_D(p) = \sum_{i=1}^3 v_i p_i \alpha_i, \quad \alpha_i := \tau_z \otimes \sigma_i. \quad (2)$$

Here σ_i acts on the two-component Weyl doublet at each valley, while τ_i acts on the valley pair $(+G_*, -G_*)$.

We also assume that at $p = 0$ each valley carries the same irreducible little-group representation ρ , as required by opposite-valley pairing.

b. Goal. We want to determine all *constant Hermitian operators* on the doubled block that are compatible with the residual symmetries, and then identify which of them are genuine Dirac masses.

1. 1. Classification of constant little-group-scalar operators

Let K be a constant Hermitian operator on the doubled block. Write it in 2×2 valley-block form:

$$K = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A, D, B \in M_2(\mathbb{C}). \quad (3)$$

Assume that K preserves the little-group action at the crossing point. Since the two valley blocks carry the same irreducible representation ρ , Schur's lemma gives

$$A = a_+ \mathbf{1}_2, \quad D = a_- \mathbf{1}_2, \quad B = b \mathbf{1}_2, \quad (4)$$

with $a_\pm \in \mathbb{R}$ and $b \in \mathbb{C}$. Therefore every little-group-scalar constant Hermitian operator is of the form

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2, \quad (5)$$

for real coefficients a_0, a_3, m_1, m_2 .

Thus the space of symmetry-preserving constant operators is four-dimensional, but only two of them are genuine Dirac-mass candidates.

[Classification of constant Dirac masses] For the kinetic matrices $\alpha_i = \tau_z \otimes \sigma_i$, the constant Hermitian matrices that anticommute with all α_i are exactly

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}. \quad (6)$$

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Take a general constant Hermitian matrix in block form

$$M = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}.$$

The condition $\{M, \alpha_i\} = 0$ for all $i = 1, 2, 3$, with

$$\alpha_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix},$$

gives

$$\{A, \sigma_i\} = 0, \quad \{D, \sigma_i\} = 0, \quad B\sigma_i - \sigma_i B = 0, \quad B^\dagger \sigma_i - \sigma_i B^\dagger = 0, \quad (7)$$

for all i . Since the Pauli matrices generate $M_2(\mathbb{C})$, the first two conditions imply

$$A = 0, \quad D = 0,$$

while the commutation condition implies

$$B = b \mathbf{1}_2$$

for some $b \in \mathbb{C}$. Hence

$$M = \begin{pmatrix} 0 & b \mathbf{1}_2 \\ \bar{b} \mathbf{1}_2 & 0 \end{pmatrix} = (\Re b) \tau_x \otimes \mathbf{1}_2 + (\Im b) \tau_y \otimes \mathbf{1}_2. \quad (8)$$

This proves (6).

a. Interpretation. The terms

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2$$

are the *only* constant Hermitian terms that open a genuine Dirac gap on the doubled block. Indeed, for

$$H_D^{(m)}(p) = \sum_i v_i p_i \alpha_i + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2, \quad (9)$$

one has

$$\left(H_D^{(m)}(p)\right)^2 = \sum_i v_i^2 p_i^2 + (m_1^2 + m_2^2), \quad (10)$$

so the spectrum is

$$E(p) = \pm \sqrt{\sum_i v_i^2 p_i^2 + m_1^2 + m_2^2}. \quad (11)$$

By contrast,

$$\mathbf{1}_4$$

is a trivial overall energy shift, while

$$\tau_z \otimes \mathbf{1}_2$$

is a valley-energy splitting term, not a genuine Dirac mass. Such a term must be eliminated separately by the already-isolated opposite-valley matching / energy-locking condition if one wants the crossing exactly at zero energy.

2. Residual valley symmetry

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2. \quad (12)$$

Then

$$[Q_V, H_D(p)] = 0 \quad (13)$$

for the massless doubled kinetic operator (2). Hence the doubled block has a natural residual valley symmetry

$$\Psi \mapsto U_\theta \Psi, \quad U_\theta := e^{i\theta Q_V/2}. \quad (14)$$

Now compute the transformation of the two mass generators:

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos \theta (\tau_x \otimes \mathbf{1}_2) + \sin \theta (\tau_y \otimes \mathbf{1}_2), \quad (15)$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin \theta (\tau_x \otimes \mathbf{1}_2) + \cos \theta (\tau_y \otimes \mathbf{1}_2). \quad (16)$$

Equivalently,

$$U_\theta((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2)U_\theta^{-1} = e^{\pm i\theta}((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2). \quad (17)$$

Thus the Dirac-mass operators carry nonzero valley charge.

Therefore, if the residual valley $U(1)_V$ symmetry is exact, any constant Dirac mass term is forbidden. [Mass protection from residual valley symmetry] Assume:

1. the low-energy doubled shell block has the form (2);
2. the constant operator under consideration preserves the little-group action at $p = 0$;
3. the residual valley symmetry generated by $Q_V = \tau_z \otimes \mathbf{1}_2$ is exact on the isolated doubled block.

Then every symmetry-allowed constant Hermitian operator on the doubled block is of the form

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2, \quad (18)$$

and in particular the Dirac-mass coefficients vanish:

$$m_1 = m_2 = 0. \quad (19)$$

Hence no constant Dirac mass term is allowed, i.e. mass protection holds.

By (5), every constant little-group-scalar Hermitian operator is

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2.$$

Exact $U(1)_V$ invariance requires

$$U_\theta K U_\theta^{-1} = K \quad \text{for all } \theta.$$

The terms proportional to $\mathbf{1}_4$ and $\tau_z \otimes \mathbf{1}_2$ are invariant, while (15)–(17) show that the pair

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2$$

rotates nontrivially and is therefore not invariant unless

$$m_1 = m_2 = 0.$$

By Lemma (6), these are precisely all genuine constant Dirac masses. Hence they are all forbidden.

a. Weaker version. The full continuous $U(1)_V$ is stronger than necessary. The discrete remnant

$$S_V := \tau_z \otimes \mathbf{1}_2, \quad S_V^2 = \mathbf{1}_4, \quad (20)$$

already suffices, because

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \quad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \quad (21)$$

So an exact residual $\mathbb{Z}_2^V$ also forbids all constant Dirac masses.

3. 3. Stability under remote-sector elimination

We now show that the same symmetry protects against radiatively induced constant masses from the remote sector. Let the full Hamiltonian be decomposed as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p)^\dagger \\ V(p) & H_R(p) \end{pmatrix}, \quad (22)$$

where H_R is the remote sector and V is the coupling between the doubled shell block and the remote sector.

Assume that the full microscopic theory preserves the residual valley symmetry, meaning that there exists a unitary action $U_\theta \oplus U_{R,\theta}$ such that

$$(U_\theta \oplus U_{R,\theta})H_{\text{full}}(p)(U_\theta \oplus U_{R,\theta})^{-1} = H_{\text{full}}(p). \quad (23)$$

Then the Schur-complement effective shell Hamiltonian

$$H_{\text{eff}}(p, \omega) = H_D(p) + V(p)^\dagger (\omega - H_R(p))^{-1} V(p) \quad (24)$$

inherits the same symmetry:

$$U_\theta H_{\text{eff}}(p, \omega) U_\theta^{-1} = H_{\text{eff}}(p, \omega). \quad (25)$$

In particular, the zero-momentum self-energy

$$\Sigma(0, 0) := V(0)^\dagger (-H_R(0))^{-1} V(0) \quad (26)$$

must commute with Q_V :

$$[Q_V, \Sigma(0, 0)] = 0. \quad (27)$$

If we also impose little-group scalarity at the crossing point, then by the same Schur/representation argument

$$\Sigma(0, 0) = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2. \quad (28)$$

Hence no radiatively generated term proportional to

$$\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2$$

can appear.

[Radiative stability of mass protection] If the residual valley symmetry is exact in the full microscopic theory and remote-sector elimination preserves that symmetry, then no symmetry-preserving self-energy correction can generate a constant Dirac mass on the isolated doubled shell block.

a. Exact logical status. This closes the mass-protection problem at the theorem level, *provided* the residual valley symmetry is exact. What still remains open is the microscopic identification of that residual symmetry in the full TECT/Class II realization.

So the honest conclusion is:

Residual valley symmetry ($U(1)_V$ or $\mathbb{Z}_2^V$) $\implies$ all constant Dirac masses are forbidden.

(29)

$$\mathcal{H}_D = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad H_D(p) = \sum_i v_i p_i (\tau_z \otimes \sigma_i).$$

$$\text{End}_{G_*}(\mathcal{H}_D)_{\text{const}} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_z \otimes \mathbf{1}_2, \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

$$\mathcal{M}_D = \{M : \{M, \tau_z \otimes \sigma_i\} = 0 \ \forall i\} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

If the residual symmetry generated by

$$Q_V = \tau_z \otimes \mathbf{1}_2$$

is exact, then

$$[Q_V, H_{\text{eff}}(p, \omega)] = 0,$$

hence the symmetry-allowed constant sector reduces to

$$\text{End}_{G_* \cap U(1)_V}(\mathcal{H}_D)_{\text{const}} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_z \otimes \mathbf{1}_2\}.$$

Therefore

$$\mathcal{M}_D \cap \text{End}_{G_* \cap U(1)_V}(\mathcal{H}_D)_{\text{const}} = \{0\}.$$

Hence all genuine constant Dirac masses are forbidden:

$$m_1 = m_2 = 0.$$

So:

exact residual valley symmetry $\implies$ mass protection

with radiative stability under symmetry-preserving remote-sector elimination.

B. Summary

$$\mathcal{H}_D = \mathcal{H}_+ \oplus \mathcal{H}_-, \quad H_D(p) = \sum_i v_i p_i (\tau_z \otimes \sigma_i).$$

$$\text{End}_{G_*}(\mathcal{H}_D)_{\text{const}} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_z \otimes \mathbf{1}_2, \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

$$\mathcal{M}_D = \{M : \{M, \tau_z \otimes \sigma_i\} = 0 \ \forall i\} = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

If the residual symmetry generated by

$$Q_V = \tau_z \otimes \mathbf{1}_2$$

is exact, then

$$[Q_V, H_{\text{eff}}(p, \omega)] = 0,$$

hence the symmetry-allowed constant sector reduces to

$$\text{End}_{G_* \cap U(1)_V}(\mathcal{H}_D)_{\text{const}} = \text{span}_{\mathbb{R}}\{\mathbf{1}_4, \tau_z \otimes \mathbf{1}_2\}.$$

Therefore

$$\mathcal{M}_D \cap \text{End}_{G_* \cap U(1)_V}(\mathcal{H}_D)_{\text{const}} = \{0\}.$$

Hence all genuine constant Dirac masses are forbidden:

$$m_1 = m_2 = 0.$$

So:

exact residual valley symmetry $\implies$ mass protection

with radiative stability under symmetry-preserving remote-sector elimination.

C. Little-group mismatch implies $V(0) = 0$

a. Important correction of notation. In the spectral-isolation decomposition,

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad (30)$$

the symbol $V(p)$ denotes the *off-diagonal inter-sector mixing block*, not a vacuum-energy scalar. Thus $V(0) = 0$ means *absence of constant hybridization at the crossing*, not vanishing vacuum energy.

b. Setup. Let H_* be the little group at the selected shell crossing $p = 0$. Assume that the full Hilbert space decomposes as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \quad (31)$$

where

- $\mathcal{H}_D$ is the isolated doubled shell block carrying the local Dirac sector,
- $\mathcal{H}_R$ is the orthogonal remote sector.

Assume further that H_* acts unitarily and preserves this decomposition:

$$U(h)\mathcal{H}_D \subset \mathcal{H}_D, \quad U(h)\mathcal{H}_R \subset \mathcal{H}_R, \quad h \in H_*. \quad (32)$$

Thus we obtain restricted representations

$$U_D(h) := U(h)|_{\mathcal{H}_D}, \quad U_R(h) := U(h)|_{\mathcal{H}_R}. \quad (33)$$

At $p = 0$, write the Hamiltonian in block form:

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}. \quad (34)$$

c. Equivariance hypothesis. Assume that the microscopic Hamiltonian is H_* -equivariant at the crossing:

$$U(h) H_{\text{full}}(0) U(h)^{-1} = H_{\text{full}}(0) \quad \forall h \in H_*. \quad (35)$$

[The constant mixing block is an intertwiner] Under (32) and (35), the off-diagonal block satisfies

$$U_D(h) V(0) U_R(h)^{-1} = V(0) \quad \forall h \in H_*, \quad (36)$$

hence

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \quad (37)$$

Since the decomposition $\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$ is H_* -invariant, the symmetry operator has block form

$$U(h) = \begin{pmatrix} U_D(h) & 0 \\ 0 & U_R(h) \end{pmatrix}.$$

Substituting this and (34) into (35), we obtain

$$\begin{pmatrix} U_D(h) & 0 \\ 0 & U_R(h) \end{pmatrix} \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix} \begin{pmatrix} U_D(h)^{-1} & 0 \\ 0 & U_R(h)^{-1} \end{pmatrix} = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}.$$

Comparing the off-diagonal block gives

$$U_D(h) V(0) U_R(h)^{-1} = V(0)$$

for all $h \in H_*$, which is exactly the condition that $V(0)$ be an H_* -intertwiner.

d. Irreducible decomposition. Let $\{W_\lambda\}_\lambda$ be a complete set of inequivalent irreducible H_* -modules. Write the isotypic decompositions

$$\mathcal{H}_D \cong \bigoplus_\lambda m_\lambda^{(D)} W_\lambda, \quad \mathcal{H}_R \cong \bigoplus_\lambda m_\lambda^{(R)} W_\lambda, \quad (38)$$

where $m_\lambda^{(D)}, m_\lambda^{(R)} \in \mathbb{N}_0$ are multiplicities.

Then standard representation theory yields

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_\lambda \text{Mat}_{m_\lambda^{(D)} \times m_\lambda^{(R)}}(\mathbb{C}). \quad (39)$$

Therefore

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \iff m_\lambda^{(D)} m_\lambda^{(R)} = 0 \text{ for all } \lambda. \quad (40)$$

This is the precise algebraic meaning of *little-group mismatch*.

[Little-group mismatch implies $V(0) = 0$] Assume:

1. $\mathcal{H}_{\text{full}}(0)$ is H_* -equivariant;
2. $\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$ is an H_* -invariant decomposition;
3. $\mathcal{H}_D$ and $\mathcal{H}_R$ have no common irreducible H_* -component, equivalently

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0. \quad (41)$$

Then

$$V(0) = 0. \quad (42)$$

By the intertwiner lemma, $V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D)$. By assumption (41), this Hom-space is zero. Hence $V(0) = 0$.
[Irreducible special case] If $\mathcal{H}_D$ and $\mathcal{H}_R$ are irreducible H_* -modules and

$$\mathcal{H}_D \not\cong \mathcal{H}_R, \quad (43)$$

then

$$V(0) = 0. \quad (44)$$

By Schur's lemma,

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

for inequivalent irreducible modules. Apply the theorem.

e. First nontrivial order in momentum. Once $V(0) = 0$, the leading mixing begins at higher order in p . If the momentum vector transforms in the little-group representation R_p , then the linear coefficient must lie in

$$V_i \in \text{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D). \quad (45)$$

Therefore:

- if $\text{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D) = 0$, then

$$V(p) = O(|p|^2); \quad (46)$$

- if this Hom-space is nonzero, then generically

$$V(p) = O(|p|). \quad (47)$$

In either case, together with a remote spectral gap $\Delta > 0$, one obtains the Schur-complement estimate

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right) \quad (48)$$

whenever $V(p) = O(|p|)$.

f. Important honesty clause. The statement

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = \text{Hom}(H_{R,\text{shell}}, H_{D,\text{shell}}) \otimes \text{Hom}(H_{R,\text{int}}, H_{D,\text{int}}) \quad (49)$$

is *not* a general identity. It is valid only under an additional product-group/product-representation assumption, e.g. if

$$H_* = G_1 \times G_2, \quad \mathcal{H}_D \cong D_1 \boxtimes D_2, \quad \mathcal{H}_R \cong R_1 \boxtimes R_2.$$

Without such an assumption, the correct and fully general theorem is the intertwiner theorem above.

g. Conclusion. The exact theorem-level closure is therefore

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \implies V(0) = 0.} \quad (50)$$

This is the precise content of the little-group mismatch condition.

D. Summary

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}, \quad \mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R.$$

If the little group H_* acts unitarily and

$$U(h) H_{\text{full}}(0) U(h)^{-1} = H_{\text{full}}(0), \quad U(h) \mathcal{H}_D \subset \mathcal{H}_D, \quad U(h) \mathcal{H}_R \subset \mathcal{H}_R,$$

then

$$U_D(h) V(0) U_R(h)^{-1} = V(0),$$

so

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Writing

$$\mathcal{H}_D \cong \bigoplus_{\lambda} m_{\lambda}^{(D)} W_{\lambda}, \quad \mathcal{H}_R \cong \bigoplus_{\lambda} m_{\lambda}^{(R)} W_{\lambda},$$

one has

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_{\lambda} \text{Mat}_{m_{\lambda}^{(D)} \times m_{\lambda}^{(R)}}(\mathbb{C}).$$

Hence

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \iff m_{\lambda}^{(D)} m_{\lambda}^{(R)} = 0 \text{ for all } \lambda.$$

Therefore the little-group mismatch condition

$$\boxed{\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0}$$

implies

$$\boxed{V(0) = 0.}$$

Once $V(0) = 0$, the linear mixing coefficient belongs to

$$\text{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D).$$

Thus generically

$$V(p) = O(|p|),$$

and with a remote gap $\Delta > 0$,

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1} V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right).$$

So the theorem-level closure is:

$$\boxed{\text{little-group mismatch} \implies V(0) = 0 \implies \text{no constant hybridization at the crossing.}}$$

E. Remote-gap positivity as an independent spectral input

a. Logical goal. We now close the statement

$$\Delta > 0$$

in the precise theorem-level form that is actually justified by the current TECT framework.

The correct statement is not that the Class II correction *creates* a remote gap automatically. Rather, the remote gap is an *independent spectral property* of the remote sector, and the Class II correction is compatible with it and preserves it under a small bounded perturbation.

b. Remote-sector definition. Let the full microscopic Hamiltonian be decomposed as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad (51)$$

where $H_D(p)$ is the isolated doubled shell Dirac block and $H_R(p)$ is the remote sector.

The remote-gap condition is

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad \text{for all sufficiently small } p. \quad (52)$$

This is the correct spectral statement already isolated in the TECT Dirac program.

1. A. Why a positive shell determinant does not by itself prove $\Delta > 0$

Let

$$\det M = A(B^2 + C^2) \quad (53)$$

be the shell-level Weyl/Dirac determinant of the enriched local block. This quantity controls the nondegeneracy of the projected shell Dirac symbol, but it does *not* determine the full spectrum of the complementary operator H_R .

Indeed, two distinct operators may have the same shell-level determinant $\det M$ while having different remote spectra. Therefore any positivity statement of the form

$$\delta(\det M) > 0$$

can at most be regarded as a compatibility or heuristic indication, not as a theorem proving

$$\text{spec}(H_R) \cap (-\Delta, \Delta) = \emptyset.$$

Hence the theorem-level route to $\Delta > 0$ must be spectral, not determinant-based.

2. B. Unperturbed remote gap

Assume there exists an unperturbed microscopic realization with block decomposition

$$H_{\text{full}}^{(0)}(p) = \begin{pmatrix} H_D^{(0)}(p) & V^{(0)}(p) \\ V^{(0)}(p)^\dagger & H_R^{(0)}(p) \end{pmatrix}, \quad (54)$$

and suppose the remote sector has a uniform positive gap:

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \quad \text{for all sufficiently small } p, \quad (55)$$

for some $\Delta_0 > 0$.

This is the genuine microscopic input.

3. C. Class II correction as a bounded perturbation

Now perturb the theory by a self-adjoint Class II correction:

$$H_{\text{full}}(p) = H_{\text{full}}^{(0)}(p) + \delta H(p), \quad (56)$$

and write the remote block as

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p). \quad (57)$$

Assume that the remote correction is uniformly bounded in operator norm:

$$\sup_{|p| \leq p_*} \|\delta H_R(p)\| \leq \varepsilon. \quad (58)$$

[Remote-gap stability under bounded perturbation] Assume (55) and (58). If

$$\varepsilon < \Delta_0, \quad (59)$$

then the perturbed remote sector still satisfies a positive uniform gap

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset, \quad \Delta := \Delta_0 - \varepsilon > 0, \quad (60)$$

for all sufficiently small p .

Fix p in the allowed small-momentum region and let $\lambda \in \text{spec}(H_R(p))$. By standard spectral stability for self-adjoint bounded perturbations,

$$\text{dist}(\lambda, \text{spec}(H_R^{(0)}(p))) \leq \|\delta H_R(p)\|. \quad (61)$$

Since $\text{spec}(H_R^{(0)}(p))$ avoids $(-\Delta_0, \Delta_0)$, every point of the unperturbed spectrum has absolute value at least Δ_0 . Therefore any point of the perturbed spectrum must satisfy

$$|\lambda| \geq \Delta_0 - \|\delta H_R(p)\| \geq \Delta_0 - \varepsilon = \Delta. \quad (62)$$

Since $\Delta > 0$, we conclude

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.$$

a. Immediate corollary (the P6 compatibility form). If one imposes the stronger smallness condition

$$\|\delta H_R\| < \frac{\Delta_0}{2}, \quad (63)$$

then

$$\Delta \geq \Delta_0 - \|\delta H_R\| > \frac{\Delta_0}{2} > 0. \quad (64)$$

This is the exact theorem-level meaning of the earlier ‘‘Class II is gap-compatible’’ statement.

4. D. Resolvent bound in the low-energy window

Let E satisfy

$$|E| < \frac{\Delta}{2}. \quad (65)$$

Then by (60),

$$\text{dist}(E, \text{spec}(H_R(p))) \geq \frac{\Delta}{2}, \quad (66)$$

hence the resolvent is well-defined and obeys

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}. \quad (67)$$

This is the exact place where the remote gap enters the Schur-complement control of the low-energy Dirac block.

5. E. Independence of the leading Dirac cone

Assume in addition that

$$V(0) = 0, \quad \|V(p)\| \leq c|p| \quad (68)$$

for small p . Then the Schur complement gives the effective low-energy operator

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p), \quad \Sigma(E, p) := V(p)(H_R(p) - E)^{-1}V(p)^\dagger. \quad (69)$$

Using (67) and (68),

$$\|\Sigma(E, p)\| \leq \|V(p)\|^2 \|(H_R(p) - E)^{-1}\| \leq \frac{2c^2}{\Delta} |p|^2. \quad (70)$$

Therefore

$$H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right). \quad (71)$$

So the remote sector affects the low-energy Dirac cone only through subleading curvature corrections. In this precise sense, the positive remote gap is *independent* of the shell-level linear Dirac structure.

6. F. Honest TECT conclusion

The theorem-level status is therefore:

1. The existence of a remote spectral gap $\Delta > 0$ is an independent microscopic spectral check.
2. The Class II correction does not by itself prove the existence of that gap.
3. If an already-gapped realization exists and the Class II correction is sufficiently small in operator norm, then the positive remote gap persists.
4. Once this gap persists, the remote sector contributes only $O(|p|^2/\Delta)$ corrections to the isolated Dirac block.

Hence the mathematically honest closure is

$$\boxed{\Delta_0 > 0 \text{ for the unperturbed remote sector} \quad + \quad \|\delta H_R\| < \Delta_0 \implies \Delta \geq \Delta_0 - \|\delta H_R\| > 0.} \quad (72)$$

F. Summary

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.$$

The remote-gap condition is a spectral property of H_R , not a consequence of the shell-level determinant

$$\det M = A(B^2 + C^2).$$

Assume an unperturbed realization

$$H_R^{(0)}(p)$$

with

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset, \quad \Delta_0 > 0,$$

and a bounded Class II correction

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \quad \sup_{|p| \leq p_*} \|\delta H_R(p)\| \leq \varepsilon.$$

Then, if

$$\varepsilon < \Delta_0,$$

the perturbed remote sector still has a positive gap

$$\Delta = \Delta_0 - \varepsilon > 0, \quad \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.$$

For $|E| < \Delta/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}.$$

If also

$$V(0) = 0, \quad \|V(p)\| \leq c|p|,$$

then the Schur-complement self-energy satisfies

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger, \quad \|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta}|p|^2,$$

hence

$$H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).$$

Therefore:

the remote gap is an independent microscopic spectral input,

Class II preserves it under sufficiently small bounded perturbations,

and the gapped remote sector modifies the Dirac block only at $O(|p|^2/\Delta)$.

G. Final Spectral Isolation Theorem

a. Goal. We now combine the three independently closed ingredients

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection},$$

into a single theorem proving spectral isolation of the local doubled Dirac block and stability of the massless Dirac cone.

1. A. Local doubled Dirac block

Assume that an opposite shell pair $\pm G_*$ exists and that the shell-relevant doubled block has already been reduced to the 4×4 Dirac form

$$H_D(p) = \Gamma_i(Mp)_i, \tag{73}$$

where

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}, \quad M \in M_{3 \times 3}(\mathbb{R}), \quad \det M \neq 0. \tag{74}$$

Equivalently, in the coefficient language used in the shell-level reduction,

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \quad \det M = A(B^2 + C^2) \neq 0. \tag{75}$$

Hence the isolated shell block is a nondegenerate massless Dirac block.

2. B. Full microscopic block decomposition

Let the full microscopic Hamiltonian near the selected crossing be written as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad (76)$$

where H_D acts on the local four-dimensional doubled shell sector $\mathcal{H}_D$, and H_R acts on the remote complement $\mathcal{H}_R$.

3. C. Three structural inputs, now theorem-level

We now insert the three conditions in their theorem-level form.

a. (C1) Zero constant mixing from little-group mismatch. Assume the little group H_* at $p = 0$ preserves the decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

and that

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0. \quad (77)$$

Then the constant mixing block must vanish:

$$V(0) = 0. \quad (78)$$

Moreover, the first nontrivial term is at least linear:

$$\|V(p)\| \leq c|p| \quad \text{for small } p. \quad (79)$$

b. (C2) Positive remote gap from independent spectral input. Assume the unperturbed remote sector $H_R^{(0)}(p)$ has a positive gap $\Delta_0 > 0$:

$$\text{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \emptyset \quad \text{for } |p| \leq p_*, \quad (80)$$

and the correction satisfies

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \quad \sup_{|p| \leq p_*} \|\delta H_R(p)\| \leq \varepsilon < \Delta_0. \quad (81)$$

Then the corrected remote sector still has a positive gap

$$\Delta := \Delta_0 - \varepsilon > 0, \quad \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset. \quad (82)$$

c. (C3) Mass protection from residual valley symmetry. Assume the local doubled shell block carries an exact residual symmetry generated by

$$Q_V = \tau_z \otimes \mathbf{1}_2 \quad (83)$$

or an equivalent $\mathbb{Z}_2$ remnant, such that all constant Dirac masses are forbidden. Equivalently, for the constant Hermitian mass space

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}, \quad (84)$$

one has

$$\mathcal{M}_D \cap \mathcal{A}_{\text{res}} = \{0\}, \quad (85)$$

where $\mathcal{A}_{\text{res}}$ is the algebra of symmetry-allowed constant operators. Hence no constant Dirac mass is generated.

4. D. Schur-complement reduction

Fix an energy window

$$|E| < \frac{\Delta}{2}. \quad (86)$$

Then the remote resolvent exists and satisfies

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}. \quad (87)$$

Therefore the Schur-complement effective Hamiltonian on $\mathcal{H}_D$ is

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p), \quad \Sigma(E, p) := V(p)(H_R(p) - E)^{-1}V(p)^\dagger. \quad (88)$$

Using (79) and (87),

$$\|\Sigma(E, p)\| \leq \|V(p)\|^2 \|(H_R(p) - E)^{-1}\| \leq \frac{2c^2}{\Delta} |p|^2. \quad (89)$$

Hence

$$H_{\text{eff}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right). \quad (90)$$

5. E. Exclusion of a generated mass term

The estimate (90) alone still allows, in principle, a constant $O(1)$ contribution if symmetry permits it. However, by (85), all symmetry-allowed constant Dirac masses vanish. Therefore the effective Hamiltonian has no constant mass deformation:

$$H_{\text{eff}}(0, 0) = 0 \quad \text{in the Dirac-mass channel.} \quad (91)$$

6. F. Final theorem

[Final Spectral Isolation Theorem] Assume:

1. the local opposite-valley doubled shell block exists and has the nondegenerate Dirac form (73) with $\det M \neq 0$;
2. the little-group mismatch condition (77) holds, hence $V(0) = 0$ and $\|V(p)\| \leq c|p|$;
3. the remote sector has a positive uniform gap $\Delta > 0$ in the sense of (82);
4. a residual symmetry excludes all constant Dirac masses as in (85).

Then, for sufficiently small p and $|E| < \Delta/2$, the full microscopic Hamiltonian contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{eff}}(E, p) = \Gamma_i(Mp)_i + O\left(\frac{|p|^2}{\Delta}\right). \quad (92)$$

In particular, the full theory contains a stable massless Dirac cone.

Item (1) gives a nondegenerate local massless Dirac block. Item (2) removes constant inter-sector hybridization and makes the first nontrivial mixing at least linear in p . Item (3) gives the resolvent bound (87). Therefore the Schur-complement correction is quadratic in p , as in (89), and the effective Hamiltonian differs from the local Dirac block only by $O(|p|^2/\Delta)$. Item (4) excludes all symmetry-allowed constant Dirac mass terms, so no $O(1)$ gap can be generated. Hence the leading linear Dirac cone survives and remains spectrally isolated from all remote states.

H. Dirac Emergence Closure Theorem

We now combine the spectral-isolation theorem with the shell-level nonvanishing conditions. [Dirac Emergence Closure Theorem] Assume:

1. the shell-level witness conditions hold:

$$\widehat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad PVT^{(1)} \neq 0, \quad PVT^{(2)} \neq 0; \quad (93)$$

2. the hypotheses of the Final Spectral Isolation Theorem hold.

Then the full TECT linearized theory contains a stable low-energy massless Dirac fermion with effective Hamiltonian

$$H_{\text{Dirac}}(p) = \Gamma_i(Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad \{\Gamma_i, \Gamma_j\} = 2\delta_{ij}, \quad \det M \neq 0. \quad (94)$$

The shell-level witness conditions imply the existence of the nondegenerate local massless Dirac block. The Final Spectral Isolation Theorem shows that this block remains spectrally isolated and is corrected only at quadratic order by remote states. Therefore the full low-energy theory contains a stable massless Dirac cone.

a. Exact logical status. What is now genuinely closed is the theorem-level implication

(shell Dirac block)+(little-group mismatch)+(remote-gap positivity)+(mass protection) $\implies$ stable massless Dirac cone.

What may still remain open in an actual microscopic realization is only the direct verification of the input hypotheses from the final TECT linearization.

I. Summary

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad H_D(p) = \Gamma_i(Mp)_i, \quad \det M \neq 0.$$

If

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then

$$V(0) = 0, \quad \|V(p)\| \leq c|p|.$$

If the remote sector satisfies

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset,$$

then for $|E| < \Delta/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}.$$

Hence the Schur-complement self-energy obeys

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger, \quad \|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta}|p|^2.$$

If, in addition, a residual symmetry forbids all constant Dirac masses, then

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p) = \Gamma_i(Mp)_i + O\left(\frac{|p|^2}{\Delta}\right),$$

with no generated constant mass term.

Therefore:

$$\boxed{\text{local nondegenerate Dirac block} + \text{little-group mismatch} + \text{positive remote gap} + \text{mass protection}}$$

$$\boxed{\implies \text{spectrally isolated stable massless Dirac cone.}}$$

Combining this with the shell-level witness conditions

$$\hat{U}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad PVT^{(1)} \neq 0, \quad PVT^{(2)} \neq 0,$$

one obtains the final closure theorem:

$$\boxed{H_{\text{Dirac}}(p) = \Gamma_i(Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad \det M \neq 0.}$$

J. Witness-to-Weyl theorem for the BCC shell pair

a. Goal. We now turn the remaining local valley-existence problem into a finite-dimensional witness problem. The objective is to prove that once one constructs

1. one admissible longitudinal witness, and
2. one admissible transverse witness,

inside the actual symmetry-allowed microscopic tensor space, the local BCC shell crossing is generically Weyl.

1. A. Allowed microscopic tensor space and witness functionals

Fix a selected opposite BCC shell pair $\pm G_*$, and let

$$\hat{n} := \frac{G_*}{|G_*|}$$

be the distinguished axis, together with an adapted orthonormal transverse frame

$$\{e_1, e_2, \hat{n}\}.$$

Let V denote the finite-dimensional real vector space of symmetry-allowed microscopic shell tensors contributing to the first-order enriched shell Hamiltonian near G_* . The shell-projected first-order operator has the form

$$\delta L_{\text{enr}}^{(1)} = g_{\parallel} O_{\parallel} + g_I O_I + g_J O_J, \tag{95}$$

where $O_{\parallel}, O_I, O_J$ are the shell-adapted basis operators.

Define the dual witness functionals

$$W_{\parallel}(X) := \langle O_{\parallel}, X \rangle, \quad W_I(X) := \frac{1}{2} \langle O_I, X \rangle, \quad W_J(X) := \frac{1}{2} \langle O_J, X \rangle. \tag{96}$$

Then, by construction,

$$W_{\parallel}(\delta L_{\text{enr}}^{(1)}) = g_{\parallel}, \quad W_I(\delta L_{\text{enr}}^{(1)}) = g_I, \quad W_J(\delta L_{\text{enr}}^{(1)}) = g_J. \tag{97}$$

2. B. Weyl determinant in witness form

The local shell Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel}(X) (W_I(X)^2 + W_J(X)^2). \quad (98)$$

Therefore the shell crossing is Weyl exactly when

$$W_{\parallel}(X) \neq 0, \quad (W_I(X), W_J(X)) \neq (0, 0). \quad (99)$$

Define the transverse polynomial

$$D_{\perp}(X) := W_I(X)^2 + W_J(X)^2. \quad (100)$$

Then the Weyl condition is equivalently

$$W_{\parallel}(X) \neq 0, \quad D_{\perp}(X) \neq 0. \quad (101)$$

3. C. Two witness existence theorem

We now prove the main reduction theorem.

[Two-witness generic Weyl theorem] Let V be the finite-dimensional real vector space of symmetry-allowed microscopic shell tensors near the selected BCC shell pair. Assume:

1. there exists $N^{(\parallel)} \in V$ such that

$$W_{\parallel}(N^{(\parallel)}) \neq 0; \quad (102)$$

2. there exists $N^{(\perp)} \in V$ such that

$$D_{\perp}(N^{(\perp)}) = W_I(N^{(\perp)})^2 + W_J(N^{(\perp)})^2 \neq 0. \quad (103)$$

Then the set

$$V_{\text{Weyl}} := \{X \in V : W_{\parallel}(X) \neq 0, D_{\perp}(X) \neq 0\} \quad (104)$$

is a nonempty Zariski-open subset of V . In particular, it is open, dense, and of full Lebesgue measure in V . Hence the local shell crossing is generically Weyl on V .

Since $W_{\parallel}$ is a linear functional on V , the assumption

$$W_{\parallel}(N^{(\parallel)}) \neq 0$$

implies that $W_{\parallel}$ is not the zero functional. Therefore its zero locus

$$Z_{\parallel} := \{X \in V : W_{\parallel}(X) = 0\} \quad (105)$$

is a proper affine hyperplane in V , hence closed, nowhere dense, and measure zero.

Likewise, $D_{\perp}(X) = W_I(X)^2 + W_J(X)^2$ is a polynomial on V . The assumption

$$D_{\perp}(N^{(\perp)}) \neq 0$$

implies that $D_{\perp}$ is not the zero polynomial. Therefore its zero locus

$$Z_{\perp} := \{X \in V : D_{\perp}(X) = 0\} \quad (106)$$

is a proper algebraic subset of V , hence closed, nowhere dense, and measure zero.

Thus

$$V_{\text{Weyl}} = V \setminus (Z_{\parallel} \cup Z_{\perp})$$

is nonempty and Zariski-open. Since $Z_{\parallel} \cup Z_{\perp}$ is a proper algebraic subset, V_{Weyl} is also open, dense, and full measure in the ordinary Euclidean topology on V . Finally, by (101), every $X \in V_{\text{Weyl}}$ gives a Weyl crossing.

4. D. Simultaneous witness criterion

A useful variant is the one-shot version.

[Simultaneous witness criterion] If there exists $N \in V$ such that

$$W_{\parallel}(N) \neq 0, \quad D_{\perp}(N) \neq 0, \quad (107)$$

then the selected shell crossing is already Weyl at N , and generic Weyl nondegeneracy follows on an open dense full-measure subset of V .

Equation (107) implies both hypotheses of the theorem, with $N^{(\parallel)} = N^{(\perp)} = N$. Thus the theorem applies.

5. E. Canonical ambient witnesses and symmetry projection

The abstract theorem reduces the problem to a finite witness search. We now explain why the obstruction is not algebraic.

Before imposing symmetry projection, the ambient tensor space contains canonical trial tensors of the forms

$$X_{ija}^{(\alpha)} = e_{\alpha i} G_{*j} v_a, \quad Y_{ija}^{(\alpha)} = (e_{\alpha i} G_{*j} + G_{*i} e_{\alpha j}) v_a, \quad Z_{ija}^{(\alpha)} = (e_{\alpha i} G_{*j} - G_{*i} e_{\alpha j}) v_a, \quad (108)$$

for $\alpha \in \{1, 2\}$, where v_a is any internal vector satisfying $u^{(\alpha)} \cdot v \neq 0$.

These satisfy strictly nonzero shell overlaps:

$$L_{\alpha}(X^{(\alpha)}) \neq 0, \quad L_{\alpha}(Y^{(\alpha)}) \neq 0, \quad L_{\alpha}(Z^{(\alpha)}) \neq 0. \quad (109)$$

Therefore the obstruction to Weyl nondegeneracy is not the absence of candidate tensors in the ambient space, but only the possibility that the symmetry projection annihilates all such candidates.

This yields the practical finite microscopic task:

Construct one admissible symmetry-projected longitudinal witness and one admissible symmetry-projected transverse witness in the actual TECT microscopic tensor space V .

6. F. Practical finite reduction theorem

We summarize the exact status.

[Finite reduction of the local valley problem] For the selected BCC shell pair, the local Weyl-valley existence problem is reduced to the following finite-dimensional task:

construct

$$N^{(\parallel)} \in V \quad \text{with} \quad W_{\parallel}(N^{(\parallel)}) \neq 0,$$

and

$$N^{(\perp)} \in V \quad \text{with} \quad D_{\perp}(N^{(\perp)}) \neq 0.$$

Once this is achieved, Weyl nondegeneracy holds generically on an open dense full-measure subset of V .

Immediate from the Two-witness generic Weyl theorem.

a. Honest scope. This theorem does *not* yet prove that the final microscopic TECT linearization automatically supplies those witnesses. What it proves is sharper and cleaner:

the remaining local Weyl problem is no longer an abstract existence problem, but an explicit finite tensor-construction problem inside the actual microscopic symmetry-allowed space.

K. Summary

$$\delta L_{\text{enr}}^{(1)} = g_{\parallel} O_{\parallel} + g_I O_I + g_J O_J, \quad W_{\parallel}(X) = \langle O_{\parallel}, X \rangle, \quad W_I(X) = \frac{1}{2} \langle O_I, X \rangle, \quad W_J(X) = \frac{1}{2} \langle O_J, X \rangle.$$

The shell Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel}(X) (W_I(X)^2 + W_J(X)^2).$$

Define

$$D_{\perp}(X) := W_I(X)^2 + W_J(X)^2.$$

Then the crossing is Weyl exactly when

$$W_{\parallel}(X) \neq 0, \quad D_{\perp}(X) \neq 0.$$

Let V be the finite-dimensional symmetry-allowed microscopic tensor space. If there exist

$$N^{(\parallel)} \in V \quad \text{with} \quad W_{\parallel}(N^{(\parallel)}) \neq 0,$$

and

$$N^{(\perp)} \in V \quad \text{with} \quad D_{\perp}(N^{(\perp)}) \neq 0,$$

then

$$V_{\text{Weyl}} = \{X \in V : W_{\parallel}(X) \neq 0, D_{\perp}(X) \neq 0\}$$

is a nonempty Zariski-open subset of V .

Hence V_{Weyl} is open, dense, and full measure in V , and the selected shell crossing is generically Weyl.

Therefore the local BCC shell-valley problem is reduced exactly to the finite microscopic construction of two admissible witnesses:

one longitudinal witness + one transverse witness.

Once those exist, Weyl nondegeneracy is generic.

L. Explicit witness verification for the canonical minimal Class II representative

a. Goal. We now verify that the canonical minimal Class II microscopic representative produces explicit nonzero longitudinal and transverse witnesses inside the actual TECT microscopic tensor space. This closes the witness-existence step for the shell-level Weyl problem.

1. A. Canonical minimal Class II representative

Consider the canonical minimal Class II choice

$$m^a(P_0) = m \delta^{a3}, \tag{110}$$

and

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}, \tag{111}$$

with all other components zero.

Define

$$\mu := \frac{\lambda g_X}{M_X^2} m. \tag{112}$$

The induced Class II microscopic tensor is

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \tag{113}$$

Since only $m^3 \neq 0$, only the $a = 3$ channel contributes, and therefore

$$\delta N_{ij}^b = -\mu C_{ij}^{3b}. \quad (114)$$

Using (111), we obtain explicitly

$$\delta N_{ij}^3 = -\mu c_{\parallel} \hat{n}_i \hat{n}_j, \quad (115)$$

$$\delta N_{ij}^1 = -\mu (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}), \quad (116)$$

$$\delta N_{ij}^2 = -\mu (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}). \quad (117)$$

2. B. Axial-family identification

Recall that the shell-relevant axial family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (118)$$

Comparing (115)–(117) with (118), we identify the induced coefficients

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J. \quad (119)$$

Thus the canonical minimal Class II representative belongs exactly to the shell-relevant axial tensor family.

3. C. Witness extraction formulas

The previously established witness functionals are

$$W_{\parallel}[N] := \hat{n}_i \hat{n}_j N_{ij}^3, \quad (120)$$

$$W_I[N] := \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad (121)$$

$$W_J[N] := \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1). \quad (122)$$

For any tensor in the axial family (118), these extract exactly the coefficients:

$$W_{\parallel}[N] = A, \quad W_I[N] = B, \quad W_J[N] = C. \quad (123)$$

Applying (123) to δN , and using (119), we obtain

$$W_{\parallel}[\delta N] = -\mu c_{\parallel}, \quad (124)$$

$$W_I[\delta N] = -\mu c_I, \quad (125)$$

$$W_J[\delta N] = -\mu c_J. \quad (126)$$

4. D. Explicit longitudinal and transverse witnesses

Define the transverse polynomial

$$D_{\perp}[N] := W_I[N]^2 + W_J[N]^2. \quad (127)$$

Then for the canonical minimal representative,

$$D_{\perp}[\delta N] = \mu^2(c_I^2 + c_J^2). \quad (128)$$

Hence we immediately obtain:

[Explicit longitudinal witness] If

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (129)$$

then

$$W_{\parallel}[\delta N] \neq 0. \quad (130)$$

So the canonical minimal Class II representative is an admissible longitudinal witness.

Immediate from (124).

[Explicit transverse witness] If

$$\mu \neq 0, \quad (c_I, c_J) \neq (0, 0), \quad (131)$$

then

$$D_{\perp}[\delta N] \neq 0. \quad (132)$$

So the canonical minimal Class II representative is an admissible transverse witness.

Immediate from (128).

Combining both gives the stronger statement:

[Canonical Class II simultaneous witness theorem] Assume

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (133)$$

Then the canonical minimal Class II representative δN satisfies

$$W_{\parallel}[\delta N] \neq 0, \quad D_{\perp}[\delta N] \neq 0. \quad (134)$$

Therefore δN is a simultaneous admissible witness in the actual TECT/Class II microscopic tensor space.

By (124), (125), (126), and (128), the conditions (133) imply both (130) and (132).

5. E. Consequence for generic Weyl nondegeneracy

The previously proved two-witness generic Weyl theorem states that if the actual symmetry-allowed tensor space V contains

1. one admissible longitudinal witness with $W_{\parallel} \neq 0$, and
2. one admissible transverse witness with $D_{\perp} \neq 0$,

then the Weyl-good set

$$V_{\text{Weyl}} = \{N \in V : W_{\parallel}[N] \neq 0, D_{\perp}[N] \neq 0\} \quad (135)$$

is nonempty, Zariski-open, open dense, and full measure.

Since the canonical minimal Class II representative δN is itself such a simultaneous witness under (133), we conclude:

[Witness verification theorem for the TECT/Class II space] Let V_{TECT} be the actual symmetry-allowed microscopic tensor space containing the canonical minimal Class II representative. Assume (133). Then

$$V_{\text{Weyl}} = \{N \in V_{\text{TECT}} : W_{\parallel}[N] \neq 0, D_{\perp}[N] \neq 0\} \quad (136)$$

is a nonempty open dense full-measure subset of V_{TECT} . Hence local Weyl nondegeneracy is generic in the actual TECT/Class II microscopic tensor space.

Apply the two-witness generic Weyl theorem using δN itself as the required admissible witness.

6. *F. Purely induced Weyl regime*

A particularly important special case is the purely induced regime

$$A_0 = B_0 = C_0 = 0. \quad (137)$$

Then the effective coefficients satisfy

$$A_{\text{eff}} = \delta A = -\mu c_{\parallel}, \quad B_{\text{eff}} = \delta B = -\mu c_I, \quad C_{\text{eff}} = \delta C = -\mu c_J. \quad (138)$$

Therefore, under (133),

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0), \quad (139)$$

and the shell determinant becomes

$$\det M = (\kappa_* |G_*|)^3 (-\mu c_{\parallel}) \mu^2 (c_I^2 + c_J^2) \neq 0. \quad (140)$$

Thus the canonical minimal Class II representative does not merely preserve a pre-existing witness structure: it can generate a local Weyl witness structure by itself.

7. *G. Honest logical status*

What is now closed is:

1. an explicit admissible microscopic tensor δN has been constructed inside the actual TECT/Class II family;
2. its longitudinal witness equals $-\mu c_{\parallel}$;
3. its transverse witness polynomial equals $\mu^2(c_I^2 + c_J^2)$;
4. hence the witness-existence step is closed for this explicit microscopic realization;
5. therefore generic local Weyl nondegeneracy follows in the actual TECT/Class II tensor space, under the stated nonvanishing conditions.

What is not claimed here is UV uniqueness. The theorem uses only existence of one concrete admissible microscopic representative, which is exactly the level already justified in the current TECT program.

M. Summary

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

with all other components zero, and

$$\mu = \frac{\lambda g_X}{M_X^2} m.$$

Then

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}$$

gives

$$\delta N_{ij}^3 = -\mu c_{\parallel} \hat{n}_i \hat{n}_j, \quad \delta N_{ij}^1 = -\mu (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}), \quad \delta N_{ij}^2 = -\mu (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}).$$

Hence the induced axial coefficients are

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J.$$

Using

$$W_{\parallel}[N] = \hat{n}_i \hat{n}_j N_{ij}^3, \quad W_I[N] = \frac{1}{2}(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad W_J[N] = \frac{1}{2}(e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1),$$

we obtain

$$W_{\parallel}[\delta N] = -\mu c_{\parallel}, \quad W_I[\delta N] = -\mu c_I, \quad W_J[\delta N] = -\mu c_J.$$

Therefore

$$D_{\perp}[\delta N] = W_I[\delta N]^2 + W_J[\delta N]^2 = \mu^2(c_I^2 + c_J^2).$$

So if

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

then

$$W_{\parallel}[\delta N] \neq 0, \quad D_{\perp}[\delta N] \neq 0.$$

Hence the canonical minimal Class II representative is an explicit simultaneous admissible witness in the actual TECT/Class II microscopic tensor space.

By the previously proved two-witness generic Weyl theorem, it follows that

$$V_{\text{Weyl}} = \{N \in V_{\text{TECT}} : W_{\parallel}[N] \neq 0, D_{\perp}[N] \neq 0\}$$

is nonempty, open dense, and full measure.

Therefore local Weyl nondegeneracy is generic in the actual TECT/Class II microscopic tensor space.

In the purely induced case

$$A_0 = B_0 = C_0 = 0,$$

one has

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J,$$

hence

$$\det M = (\kappa_* |G_*|)^3 (-\mu c_{\parallel}) \mu^2 (c_I^2 + c_J^2) \neq 0.$$

Thus the canonical minimal Class II representative can itself generate the local Weyl witness structure.

N. Opposite-valley witness matching for the canonical minimal Class II representative

a. Goal. We prove that the canonical minimal Class II microscopic representative produces identical witness triples at the matched opposite shell pair $\pm G_*$. By the previously established witness criterion, this implies the standard opposite-valley pairing relation and hence Dirac doubling.

1. A. Matched opposite-valley frames

Let

$$\hat{n}_+ := \frac{G_*}{|G_*|}, \quad \hat{n}_- := -\hat{n}_+. \quad (141)$$

Choose a positively oriented orthonormal frame at $+G_*$,

$$\{e_{1+}, e_{2+}, \hat{n}_+\}, \quad e_{1+} \times e_{2+} = \hat{n}_+. \quad (142)$$

Define the matched opposite-valley frame at $-G_*$ by

$$e_{1-} := e_{1+}, \quad e_{2-} := -e_{2+}, \quad \hat{n}_- = -\hat{n}_+. \quad (143)$$

Then

$$e_{1-} \times e_{2-} = e_{1+} \times (-e_{2+}) = -\hat{n}_+ = \hat{n}_-, \quad (144)$$

so $\{e_{1-}, e_{2-}, \hat{n}_-\}$ is again a positively oriented orthonormal frame.

2. B. Canonical minimal Class II data at the two valleys

Assume the canonical minimal Class II data are *even under the matched opposite-valley transformation*, in the following precise sense:

$$m^a(+) = m \delta^{a3}, \quad m^a(-) = m \delta^{a3}, \quad (145)$$

and the shell-adapted tensors are

$$C_{ij}^{33}(+) = c_{\parallel} \hat{n}_{+i} \hat{n}_{+j}, \quad C_{ij}^{31}(+) = c_I \hat{n}_{+i} e_{1+j} + c_J \hat{n}_{+i} e_{2+j}, \quad C_{ij}^{32}(+) = -c_J \hat{n}_{+i} e_{1+j} + c_I \hat{n}_{+i} e_{2+j}, \quad (146)$$

$$C_{ij}^{33}(-) = c_{\parallel} \hat{n}_{-i} \hat{n}_{-j}, \quad C_{ij}^{31}(-) = c_I \hat{n}_{-i} e_{1-j} + c_J \hat{n}_{-i} e_{2-j}, \quad C_{ij}^{32}(-) = -c_J \hat{n}_{-i} e_{1-j} + c_I \hat{n}_{-i} e_{2-j}, \quad (147)$$

with all other components zero.

Thus the scalar coefficients

$$m, \quad c_{\parallel}, \quad c_I, \quad c_J \quad (148)$$

are identical at the two valleys; only the local frame data differ.

Define

$$\mu := \frac{\lambda g_X}{M_X^2} m. \quad (149)$$

3. C. Induced tensors at $\pm G_*$

The induced microscopic tensor is

$$\delta N_{ij}^b(\pm) = -\frac{\lambda g_X}{M_X^2} m^a(\pm) C_{ij}^{ab}(\pm). \quad (150)$$

Using $m^a(\pm) = m \delta^{a3}$, we obtain

$$\delta N_{ij}^b(\pm) = -\mu C_{ij}^{3b}(\pm). \quad (151)$$

Therefore, at $+G_*$,

$$\delta N_{ij}^3(+) = -\mu c_{\parallel} \hat{n}_{+i} \hat{n}_{+j}, \quad (152)$$

$$\delta N_{ij}^1(+) = -\mu (c_I \hat{n}_{+i} e_{1+j} + c_J \hat{n}_{+i} e_{2+j}), \quad (153)$$

$$\delta N_{ij}^2(+) = -\mu (-c_J \hat{n}_{+i} e_{1+j} + c_I \hat{n}_{+i} e_{2+j}), \quad (154)$$

while at $-G_*$,

$$\delta N_{ij}^3(-) = -\mu c_{\parallel} \hat{n}_{-i} \hat{n}_{-j}, \quad (155)$$

$$\delta N_{ij}^1(-) = -\mu (c_I \hat{n}_{-i} e_{1-j} + c_J \hat{n}_{-i} e_{2-j}), \quad (156)$$

$$\delta N_{ij}^2(-) = -\mu (-c_J \hat{n}_{-i} e_{1-j} + c_I \hat{n}_{-i} e_{2-j}). \quad (157)$$

Thus the induced tensor at each valley has the same axial-family coefficients in its own matched local frame.

4. D. Witness triples at the two valleys

Define the positive-valley witness functionals by

$$W_{\parallel}^+[N] := \hat{n}_{+i}\hat{n}_{+j}N_{ij}^3(+), \quad (158)$$

$$W_I^+[N] := \frac{1}{2} \left(e_{1+j}\hat{n}_{+i}N_{ij}^1(+) + e_{2+j}\hat{n}_{+i}N_{ij}^2(+) \right), \quad (159)$$

$$W_J^+[N] := \frac{1}{2} \left(e_{1+j}\hat{n}_{+i}N_{ij}^2(+) - e_{2+j}\hat{n}_{+i}N_{ij}^1(+) \right). \quad (160)$$

Similarly, define the negative-valley witness functionals in the matched opposite-valley frame:

$$W_{\parallel}^-[N] := \hat{n}_{-i}\hat{n}_{-j}N_{ij}^3(-), \quad (161)$$

$$W_I^-[N] := \frac{1}{2} \left(e_{1-j}\hat{n}_{-i}N_{ij}^1(-) + e_{2-j}\hat{n}_{-i}N_{ij}^2(-) \right), \quad (162)$$

$$W_J^-[N] := \frac{1}{2} \left(e_{1-j}\hat{n}_{-i}N_{ij}^2(-) - e_{2-j}\hat{n}_{-i}N_{ij}^1(-) \right). \quad (163)$$

Applying these to the canonical minimal representative δN , we obtain at the positive valley

$$W_{\parallel}^+[\delta N] = -\mu c_{\parallel}, \quad W_I^+[\delta N] = -\mu c_I, \quad W_J^+[\delta N] = -\mu c_J. \quad (164)$$

Exactly the same computation at the negative valley gives

$$W_{\parallel}^-[\delta N] = -\mu c_{\parallel}, \quad W_I^-[\delta N] = -\mu c_I, \quad W_J^-[\delta N] = -\mu c_J. \quad (165)$$

Therefore

$$W_{\parallel}^-[\delta N] = W_{\parallel}^+[\delta N], \quad W_I^-[\delta N] = W_I^+[\delta N], \quad W_J^-[\delta N] = W_J^+[\delta N]. \quad (166)$$

[Explicit opposite-valley witness matching theorem] Assume the canonical minimal Class II representative is even under the matched opposite-valley transformation in the sense of (145)–(148). Then its witness triples at $\pm G_*$ coincide:

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+). \quad (167)$$

Immediate from (164) and (165).

5. E. Consequence for opposite-valley pairing

The previously established witness-doubling theorem states that equality of the witness triples in matched opposite-valley frames is equivalent to the standard reduced pairing relation

$$h_-(p) = -h_+(-p) \quad (168)$$

after subtraction of the common scalar part.

Hence we obtain:

[Canonical Class II opposite-valley pairing] Under the hypotheses of the explicit opposite-valley witness matching theorem, the reduced valley Hamiltonians satisfy

$$h_-(p) = -h_+(-p) \quad (169)$$

in the canonical matched reduced basis.

6. F. Dirac doubling consequence

Assume in addition the nondegeneracy conditions

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (170)$$

Then at both valleys,

$$W_{\parallel}^{\pm} \neq 0, \quad (W_I^{\pm}, W_J^{\pm}) \neq (0, 0), \quad (171)$$

so each valley is locally Weyl.

Combining (171) with (169), we conclude that the doubled block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \quad (172)$$

is unitarily equivalent to a massless Dirac Hamiltonian.

[Canonical Class II local Dirac doubling theorem] Assume:

1. an opposite shell pair $\pm G_*$ exists;
2. the canonical minimal Class II representative is even under the matched opposite-valley transformation;
3. $\mu \neq 0$, $c_{\parallel} \neq 0$, and $(c_I, c_J) \neq (0, 0)$.

Then the canonical minimal Class II representative yields:

1. a Weyl valley at $+G_*$;
2. a Weyl valley at $-G_*$;
3. equality of the witness triples at the two valleys;
4. the standard opposite-valley relation $h_-(p) = -h_+(-p)$;
5. a doubled four-component massless Dirac block.

Items (1) and (2) follow from (171). Item (3) is (167). Item (4) follows from the witness-pairing equivalence. Item (5) is the standard opposite-valley Dirac doubling theorem.

a. Exact logical status. What is closed here is not the existence of the opposite valley from the full microscopic TECT PDE. What is closed is the following sharper statement:

once an opposite shell pair $\pm G_*$ is present and the canonical minimal Class II data are even under the matched valley transformation, the same explicit microscopic representative automatically produces matched witness triples and hence local Dirac doubling.

O. Summary

$$\hat{n}_- = -\hat{n}_+, \quad e_{1-} = e_{1+}, \quad e_{2-} = -e_{2+},$$

so that

$$e_{1-} \times e_{2-} = \hat{n}_-.$$

For the canonical minimal Class II representative, assume the scalar data are even under the matched opposite-valley transformation:

$$m^a(\pm) = m\delta^{a3}, \quad (c_{\parallel}, c_I, c_J)_- = (c_{\parallel}, c_I, c_J)_+ = (c_{\parallel}, c_I, c_J).$$

Then

$$\delta N_{ij}^b(\pm) = -\mu C_{ij}^{3b}(\pm), \quad \mu = \frac{\lambda g_X}{M_X^2} m,$$

gives

$$\begin{aligned}\delta N_{ij}^3(\pm) &= -\mu c_{\parallel} \hat{n}_{\pm i} \hat{n}_{\pm j}, \\ \delta N_{ij}^1(\pm) &= -\mu(c_I \hat{n}_{\pm i} e_{1\pm j} + c_J \hat{n}_{\pm i} e_{2\pm j}), \\ \delta N_{ij}^2(\pm) &= -\mu(-c_J \hat{n}_{\pm i} e_{1\pm j} + c_I \hat{n}_{\pm i} e_{2\pm j}).\end{aligned}$$

Therefore the witness triples at the two valleys are identical:

$$W_{\parallel}^{\pm} = -\mu c_{\parallel}, \quad W_I^{\pm} = -\mu c_I, \quad W_J^{\pm} = -\mu c_J.$$

Hence

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+).$$

By the previously established witness criterion, this is equivalent to

$$h_-(p) = -h_+(-p)$$

in the matched reduced basis.

If, in addition,

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

then both valleys are Weyl:

$$W_{\parallel}^{\pm} \neq 0, \quad (W_I^{\pm}, W_J^{\pm}) \neq (0, 0),$$

and the doubled block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

Thus:

canonical minimal Class II representative + matched opposite-valley evenness $\implies$ explicit witness matching $\implies$ local Dirac dou

P. Local Dirac Block Realization Theorem

a. Goal. We now assemble the previously established ingredients into a single theorem showing that the canonical minimal Class II microscopic representative realizes a stable massless Dirac fermion in the full low-energy sector, provided the remaining independent spectral/symmetry inputs are satisfied.

1. A. Canonical minimal Class II representative

Fix an opposite shell pair $\pm G_*$ and define

$$\mu := \frac{\lambda g_X}{M_X^2} m. \tag{173}$$

Take the canonical minimal Class II representative

$$m^a(P_0) = m \delta^{a3}, \tag{174}$$

and, in the matched local shell frames at $\pm G_*$,

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}, \tag{175}$$

with all other components zero.

The induced shell-relevant tensor is

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}, \tag{176}$$

hence

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J. \tag{177}$$

2. B. Effective shell coefficients and explicit witness nonvanishing

Let the baseline shell coefficients be (A_0, B_0, C_0) , and define the corrected coefficients

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J. \quad (178)$$

In particular, in the purely induced case

$$A_0 = B_0 = C_0 = 0, \quad (179)$$

one has

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J. \quad (180)$$

Assume the nonvanishing conditions

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0). \quad (181)$$

A sufficient explicit condition in the purely induced case is

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (182)$$

Then the shell determinant is

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2) \neq 0. \quad (183)$$

Equivalently, the local shell Hamiltonian at each valley is Weyl.

3. C. Opposite-valley witness matching

Assume the canonical Class II data are even under the matched opposite-valley transformation. Then the induced coefficients satisfy

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+. \quad (184)$$

Equivalently, the witness triples satisfy

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+). \quad (185)$$

Therefore the reduced valley Hamiltonians obey the standard opposite-valley pairing relation

$$h_-(p) = -h_+(-p). \quad (186)$$

Since both valleys are Weyl by (183), the doubled local shell block is a massless Dirac block:

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \sim \Gamma_i (M_{\text{eff}} p)_i. \quad (187)$$

4. D. Full microscopic block decomposition

Now decompose the full microscopic Hamiltonian near the selected crossing as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \quad (188)$$

where H_D acts on the four-dimensional local Dirac sector and H_R acts on the remote complement.

Assume the little-group mismatch condition

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0. \quad (189)$$

Then

$$V(0) = 0, \quad \|V(p)\| \leq c|p| \quad (190)$$

for small p .

Assume further that the remote sector is uniformly gapped:

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset \quad \text{for small } p, \quad (191)$$

with $\Delta > 0$. A sufficient theorem-level route is the existence of an already-gapped unperturbed realization plus a sufficiently small bounded Class II perturbation.

Finally assume a residual symmetry excludes all constant Dirac masses in the effective theory.

5. E. Schur-complement reduction

For $|E| < \Delta/2$, the remote resolvent exists and obeys

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}. \quad (192)$$

Hence the effective Hamiltonian on the four-dimensional low-energy sector is

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p), \quad \Sigma(E, p) := V(p)(H_R(p) - E)^{-1}V(p)^\dagger. \quad (193)$$

Using (190) and (192),

$$\|\Sigma(E, p)\| \leq \frac{2c^2}{\Delta}|p|^2. \quad (194)$$

Therefore

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right). \quad (195)$$

Because the residual symmetry forbids a constant Dirac mass term, no $O(1)$ mass deformation can appear. Thus the low-energy sector remains a massless Dirac cone.

6. F. Main theorem

[Local Dirac Block Realization Theorem] Assume:

1. an opposite shell pair $\pm G_*$ exists;
2. the canonical minimal Class II representative (174)–(175) is present in the actual microscopic TECT tensor space;
3. the effective shell coefficients satisfy

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0),$$

equivalently $\det M_{\text{eff}} \neq 0$;

4. the Class II data are even under the matched opposite-valley transformation, so that

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+;$$

5. the little-group mismatch condition holds:

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0;$$

6. the remote sector has a positive gap $\Delta > 0$;
7. a residual symmetry forbids all constant Dirac masses on the effective four-dimensional sector.

Then, for sufficiently small p , the full microscopic TECT linearization contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad \det M_{\text{eff}} \neq 0. \quad (196)$$

Consequently, the full microscopic low-energy theory contains a stable massless Dirac fermion.

By items (2)–(3), the canonical Class II representative produces nonzero shell witnesses and hence a Weyl block at each valley. By item (4), the witness triples match across the opposite valleys, so the doubled shell block is local Dirac. By item (5), the constant inter-sector mixing vanishes and the first nontrivial mixing is at least linear in momentum. By items (6)–(7), the Schur-complement correction is well-defined, quadratic in p , and cannot generate a constant Dirac mass. Therefore the four-dimensional local Dirac block survives in the full microscopic low-energy theory as a stable spectrally isolated massless Dirac sector.

7. *G. Strongest explicit sufficient form*

A concrete sufficient version is the following.
[Explicit sufficient realization criterion] Assume:

1. the opposite shell pair $\pm G_*$ survives in the actual microscopic realization;
2. the canonical minimal Class II representative is present and even under the matched opposite-valley transformation;
- 3.

$$\mu \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0);$$

4. little-group mismatch, remote-gap positivity, and mass-forbidding residual symmetry all hold.

Then the full microscopic low-energy theory contains a stable massless Dirac fermion with effective Hamiltonian

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right), \quad (197)$$

where in the purely induced case

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J. \quad (198)$$

a. Exact logical status. What is now closed is the following theorem chain:

explicit canonical Class II witness $\implies$ Weyl at each valley $\implies$ opposite-valley Dirac doubling $\implies$ spectrally isolated stable mass

What still remains outside the theorem itself is only the final microscopic verification that the actual TECT linearization satisfies the input hypotheses: survival of the opposite shell pair, little-group mismatch, remote-gap positivity, and the residual mass-forbidding symmetry.

Q. Summary

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j},$$

with

$$\mu = \frac{\lambda g_X}{M_X^2} m.$$

The induced shell coefficients are

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

If

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0),$$

then

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2) \neq 0,$$

so each valley is Weyl.

If the Class II data are even under the matched opposite-valley transformation, then

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+,$$

hence

$$h_-(p) = -h_+(-p),$$

so the doubled shell block is a local Dirac block:

$$H_D(p) \sim \Gamma_i(M_{\text{eff}}p)_i.$$

If, in addition,

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \quad \text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset,$$

and a residual symmetry forbids constant Dirac masses, then

$$V(0) = 0, \quad \|V(p)\| \leq c|p|,$$

and the Schur-complement effective Hamiltonian satisfies

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta}\right).$$

Therefore:

canonical minimal Class II representative + opposite-valley evenness + little-group mismatch + remote gap + mass-forbidding sy

$\implies$ full microscopic low-energy TECT sector contains a stable massless Dirac fermion.

R. Actual little-group mismatch audit for the BCC local Dirac block

a. Goal. We now reformulate the condition

$$V(0) = 0$$

at the level of the actual BCC representation content of the local Dirac block. The purpose is to replace the abstract sufficient condition

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

by a concrete sectorwise audit in the actual TECT shell construction.

1. A. Exact discrete little group versus IR axial remnant

Fix the selected BCC shell direction

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \quad G_* = |G_*|\hat{n}. \quad (199)$$

Then the exact ordinary cubic stabilizers are

$$\text{Stab}(+\hat{n}) \cong C_{2v}, \quad \text{Stab}(\{\pm\hat{n}\}) \cong D_{2h}. \quad (200)$$

Both groups are abelian, hence all ordinary complex irreducible representations are one-dimensional. Therefore:

$$\text{ordinary cubic little-group irreducibility does not force a 2D local sector.} \quad (201)$$

What remains true kinematically is the canonical splitting

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus \hat{n}^\perp, \quad (202)$$

i.e. one longitudinal direction plus a two-dimensional transverse plane. The actual two-dimensional shell structure arises only at the infrared axial-remnant level acting on $\hat{n}^\perp$, not from ordinary bare cubic irreps.

2. B. Actual representation content of the local Dirac block

Let

$$W_{\text{mix}} = \text{span}\{W^{(1)}, W^{(2)}\} = \hat{n}^\perp \otimes \hat{n} \quad (203)$$

denote the shell-transverse doublet, where

$$W_{ij}^{(\alpha)} = e_{\alpha i} \hat{n}_j, \quad \alpha = 1, 2. \quad (204)$$

Assume the internal channel contains a matching two-dimensional subspace

$$U_D = \text{span}\{u^{(1)}, u^{(2)}\} \quad (205)$$

transforming in the same transverse way. Then the shell-relevant local Dirac carrier is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2, \quad (206)$$

where

- $\mathbb{C}_{\text{valley}}^2$ is the opposite-shell pair space $\text{span}\{\phi_+, \phi_-\}$,
- $\mathbb{C}_{\text{int}}^2$ is the shell-relevant transverse/internal doublet.

Thus the actual local Dirac block is not an ordinary 2D irrep of C_{2v} or D_{2h} , but a four-dimensional shell-relevant carrier built from:

$$(\text{opposite shell pair}) \otimes (\text{matching transverse/internal doublet}). \quad (207)$$

3. C. Sectorwise decomposition of the remote space

Let the full Hilbert space decompose at the crossing point as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \quad \mathcal{H}_R = \bigoplus_a K_a, \quad (208)$$

where each K_a is H_* -invariant.

The practical shell-relevant labels carried by $\mathcal{H}_D$ are:

1. the selected shell-pair label $\pm G_*$,
2. transverse/non-volumetric character,
3. enriched internal doublet label,
4. residual grading/parity/chirality labels, if present,
5. any additional conserved microscopic labels carried by $\mathcal{H}_D$.

A remote sector K_a can mix at $p = 0$ with $\mathcal{H}_D$ only if it matches $\mathcal{H}_D$ in *all* of these shell-relevant labels.

4. D. Three canonical mismatch sectors

We now audit the three most important remote sectors explicitly.

a. Sector 1: higher-shell sector. Suppose $K_a \subset \mathcal{H}_{\text{high}}$, i.e. K_a belongs to a shell orbit different from the selected pair $\pm G_*$. Then K_a carries a different shell label from $\mathcal{H}_D$, hence

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0. \quad (209)$$

Therefore:

$$\text{different shell orbit} \implies V_{K_a \rightarrow D}(0) = 0. \quad (210)$$

b. Sector 2: same shell but non-enriched sector. Suppose K_a lies on the same shell orbit but does not carry the enriched internal doublet $\mathbb{C}_{\text{int}}^2$ required by $\mathcal{H}_D$. Then the internal matching condition fails, so

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0. \quad (211)$$

Hence:

$$\text{same shell but no enriched internal doublet} \implies V_{K_a \rightarrow D}(0) = 0. \quad (212)$$

c. Sector 3: wrong internal sector. Suppose K_a has a nontrivial internal structure, but not the one matching the shell-relevant internal doublet. Examples include:

- an internal singlet instead of a doublet,
- a different doublet not identified with the shell-relevant one,
- a larger representation whose shell-relevant multiplicity does not match.

Then the internal representation still mismatches, so

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0. \quad (213)$$

Therefore:

$$\text{wrong internal representation} \implies V_{K_a \rightarrow D}(0) = 0. \quad (214)$$

5. E. Sectorwise zero-mixing theorem

We now combine the above into the actual representation-audit theorem.
[Actual little-group mismatch audit] Assume:

1. the full Hamiltonian is H_* -equivariant at $p = 0$;
2. the local shell-relevant Dirac block has the actual representation content

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2;$$

3. the remote sector admits an H_* -invariant decomposition

$$\mathcal{H}_R = \bigoplus_a K_a;$$

4. for every sector K_a , at least one shell-relevant label mismatches that of $\mathcal{H}_D$.

Then

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a, \quad (215)$$

hence

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \quad (216)$$

and therefore

$$V(0) = 0. \quad (217)$$

Because the decomposition $\mathcal{H}_R = \bigoplus_a K_a$ is H_* -invariant, one has

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \text{Hom}_{H_*}(K_a, \mathcal{H}_D). \quad (218)$$

By assumption, each K_a fails to match $\mathcal{H}_D$ in at least one shell-relevant label. Hence no nonzero H_* -intertwiner can map K_a into $\mathcal{H}_D$, so

$$\text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0$$

for all a . Therefore the direct sum (218) vanishes, giving (216).

Since the Hamiltonian is H_* -equivariant at $p = 0$, the constant mixing block must satisfy

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Because the latter Hom-space is zero, we conclude

$$V(0) = 0.$$

6. F. Practical audit checklist

The theorem reduces the microscopic $V(0) = 0$ problem to the following finite checklist. A remote sector is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one of:

1. the selected shell-pair label $\pm G_*$,
2. transverse versus longitudinal/volumetric character,
3. presence of the enriched shell-relevant internal doublet,
4. residual grading / parity / chirality,
5. any extra conserved microscopic label.

Thus the actual microscopic task is no longer abstract: it is a sectorwise label audit of the remote spectrum.

a. Exact logical status. What is now proved is not that the BCC ordinary little group itself produces a 2D irrep. Rather, the exact ordinary little-group route is known to fail. The correct theorem is instead:

$$\text{actual local carrier } \mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2 \quad + \quad \text{sectorwise remote-label mismatch} \implies V(0) = 0.$$

This is the sharpest honest microscopic form of the little-group mismatch argument presently justified in TECT.

S. Summary

$$\text{Stab}(+\hat{n}) \cong C_{2v}, \quad \text{Stab}(\{\pm\hat{n}\}) \cong D_{2h},$$

and both groups are abelian, so all ordinary complex irreducible representations are one-dimensional.
Hence:

ordinary cubic little-group irreducibility does not force the local 2D/4D Dirac carrier.

The actual shell-relevant local Dirac block is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

where $\mathbb{C}_{\text{valley}}^2$ is the opposite-shell pair $\pm G_*$, and $\mathbb{C}_{\text{int}}^2$ is the matching transverse/internal doublet furnished at the IR axial-remnant level.

Let

$$\mathcal{H}_R = \bigoplus_a K_a$$

be an H_* -invariant decomposition of the remote sector. A sector K_a can mix at $p = 0$ with $\mathcal{H}_D$ only if it matches $\mathcal{H}_D$ in all shell-relevant labels:

$\pm G_*$, transverse/non-volumetric character, enriched internal doublet, grading/parity/chirality, extra conserved labels.

In particular:

$$\text{different shell orbit} \implies \text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0,$$

$$\text{same shell but no enriched internal doublet} \implies \text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0,$$

$$\text{wrong internal representation} \implies \text{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

Therefore, if every remote sector K_a fails to match $\mathcal{H}_D$ in at least one shell-relevant label, then

$$\text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

Since equivariance implies

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D),$$

we obtain

$V(0) = 0.$

Thus the actual microscopic little-group mismatch statement is:

$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2 \quad + \quad \text{sectorwise remote-label mismatch} \implies V(0) = 0.$

T. Remote-gap sectorwise lower-bound theorem in the minimal truncation

a. Goal. We now close the remote-gap problem at the level of the first nontrivial finite truncation. The point is to replace the abstract condition

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset$$

by an explicit finite-dimensional lower-bound criterion.

1. A. Minimal truncated Hilbert space

Consider the shell-relevant minimal truncation

$$\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}. \quad (219)$$

Here:

- $\mathcal{H}_D \cong \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$ is the local doubled Dirac carrier;
- $\mathcal{H}_{\text{vol}}$ is the volumetric/longitudinal sector;
- $\mathcal{H}_{\text{high}}$ is the higher-shell sector not belonging to the selected shell pair $\pm G_*$;
- $\mathcal{H}_{\text{non-enr}}$ is the same-shell but non-enriched sector;
- $\mathcal{H}_{\text{wrong-int}}$ contains modes with internal content not matching the selected shell doublet.

This is the smallest truncation in which all previously identified dangerous remote channels are explicitly present.

2. B. Block decomposition and zero constant Dirac-remote mixing

Write the truncated Hamiltonian as

$$H_{\text{full}}^{\text{tr}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V_{DR}(p)^\dagger & H_R(p) \end{pmatrix}, \quad (220)$$

with

$$H_R(p) = \begin{pmatrix} H_{\text{vol}}(p) & R_{\text{vol,high}}(p) & \cdots \\ R_{\text{high,vol}}(p) & H_{\text{high}}(p) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \quad (221)$$

Assume the shell-relevant label mismatch audit has already been verified for the minimal truncation. Then

$$V_{DR}(0) = 0. \quad (222)$$

Hence the only remaining spectral question is whether the remote block $H_R(p)$ can approach zero.

3. C. Diagonal sector-center energies and remote mixing norm

For the four remote sectors define the diagonal or sector-center energies

$$\mu_{\text{vol}}(p), \quad \mu_{\text{high}}(p), \quad \mu_{\text{non-enr}}(p), \quad \mu_{\text{wrong-int}}(p). \quad (223)$$

Set the pointwise remote diagonal distance from zero

$$\delta_0(p) := \min \left\{ |\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)| \right\}. \quad (224)$$

Now split the remote block as

$$H_R(p) = D_R(p) + R_R(p), \quad (225)$$

where $D_R(p)$ is the diagonal (or sector-center) part carrying the energies $\mu_a(p)$, and $R_R(p)$ is the residual remote-remote mixing. Define

$$\eta_R(p) := \|R_R(p)\|. \quad (226)$$

4. D. Pointwise lower bound

[Pointwise operator lower bound] For every $\psi \in \mathcal{H}_R$,

$$\|H_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p)) \|\psi\|. \quad (227)$$

Using $H_R = D_R + R_R$, the reverse triangle inequality gives

$$\|H_R\psi\| = \|(D_R + R_R)\psi\| \geq \|D_R\psi\| - \|R_R\psi\|.$$

By construction of $\delta_0(p)$, every vector in the remote sector satisfies

$$\|D_R(p)\psi\| \geq \delta_0(p) \|\psi\|.$$

Also,

$$\|R_R(p)\psi\| \leq \|R_R(p)\| \|\psi\| = \eta_R(p) \|\psi\|.$$

Combining the two inequalities yields (227).

[Pointwise remote-gap bound] If

$$\delta_0(p) > \eta_R(p), \quad (228)$$

then the remote spectrum has a positive pointwise gap:

$$\text{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \emptyset, \quad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0. \quad (229)$$

Since $H_R(p)$ is Hermitian, (227) implies that every singular value, hence every absolute eigenvalue, is bounded below by $\delta_0(p) - \eta_R(p)$. Therefore no eigenvalue can enter the interval

$$(-(\delta_0(p) - \eta_R(p)), \delta_0(p) - \eta_R(p)).$$

This proves (229).

5. E. Uniform local remote gap

Fix a neighborhood

$$N_\rho := \{p : |p| \leq \rho\}. \quad (230)$$

Define

$$\delta_{0,\rho} := \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} := \sup_{|p| \leq \rho} \eta_R(p). \quad (231)$$

[Uniform remote-gap sectorwise lower-bound theorem] If

$$\delta_{0,\rho} > \eta_{R,\rho}, \quad (232)$$

then for all $|p| \leq \rho$,

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset, \quad (233)$$

with

$$\Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0. \quad (234)$$

Apply the pointwise bound (229) for each $p \in N_\rho$:

$$\Delta(p) \geq \delta_0(p) - \eta_R(p).$$

Taking the infimum over $|p| \leq \rho$ gives

$$\inf_{|p| \leq \rho} \Delta(p) \geq \delta_{0,\rho} - \eta_{R,\rho} = \Delta_\rho > 0.$$

Hence (233) holds uniformly on N_ρ .

6. F. Gershgorin alternative

Sometimes a basis-dependent estimate is easier. Let the remote matrix entries in a fixed basis be $(H_R)_{ab}(p)$, and define the row sums

$$R_a(p) := \sum_{b \neq a} |(H_R)_{ab}(p)|. \quad (235)$$

Then Gershgorin's theorem gives the following sufficient condition:

$$\min_a (|(H_R)_{aa}(p)| - R_a(p)) > 0 \implies \text{spec}(H_R(p)) \cap (-\Delta_G(p), \Delta_G(p)) = \emptyset, \quad (236)$$

where

$$\Delta_G(p) := \min_a (|(H_R)_{aa}(p)| - R_a(p)). \quad (237)$$

Uniformly on N_ρ , define

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_a (|(H_R)_{aa}(p)| - R_a(p)). \quad (238)$$

If

$$\Delta_{G,\rho} > 0, \quad (239)$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \emptyset \quad (|p| \leq \rho). \quad (240)$$

This is often convenient in sparse truncations.

7. G. Sectorwise sufficient conditions

The theorem becomes especially transparent when each remote sector admits an individual lower bound:

$$|\mu_{\text{vol}}(p)| \geq \Delta_{\text{vol}}, \quad |\mu_{\text{high}}(p)| \geq \Delta_{\text{high}}, \quad |\mu_{\text{non-enr}}(p)| \geq \Delta_{\text{non-enr}}, \quad |\mu_{\text{wrong-int}}(p)| \geq \Delta_{\text{wrong-int}}, \quad (241)$$

for all $|p| \leq \rho$.

Then

$$\delta_{0,\rho} \geq \Delta_{\min} := \min\{\Delta_{\text{vol}}, \Delta_{\text{high}}, \Delta_{\text{non-enr}}, \Delta_{\text{wrong-int}}\}. \quad (242)$$

Therefore, if

$$\eta_{R,\rho} < \Delta_{\min}, \quad (243)$$

the remote block has a uniform gap

$$\Delta_\rho \geq \Delta_{\min} - \eta_{R,\rho} > 0. \quad (244)$$

8. H. Heavy volumetric sector as a natural remote contributor

In the EFT reduction of the heavy volumetric mode σ , the effective longitudinal stiffness is shifted by

$$(\lambda + 2\mu) \longmapsto (\lambda + 2\mu) + \frac{g^2}{M_\sigma^2}, \quad (245)$$

while heavy-sector effects are suppressed by powers of M_σ^{-2} . Therefore the volumetric sector is naturally pushed upward relative to the light transverse shell sector and contributes positively to the remote diagonal distance from zero.

This does not by itself prove the full remote gap, but it provides a natural microscopic mechanism making

$$\Delta_{\text{vol}} > 0$$

plausible and often large compared to the shell-relevant low-energy scale.

9. I. Consequence for spectral isolation

Assume also the already-established zero constant mixing and linear bound

$$V_{DR}(0) = 0, \quad \|V_{DR}(p)\| \leq c|p|. \quad (246)$$

Then, for $|E| < \Delta_\rho/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta_\rho}. \quad (247)$$

Hence the Schur-complement self-energy satisfies

$$\Sigma(E, p) = V_{DR}(p)(H_R(p) - E)^{-1}V_{DR}(p)^\dagger = O\left(\frac{|p|^2}{\Delta_\rho}\right). \quad (248)$$

Thus the remote gap theorem closes the final spectral input needed for the local Dirac block to remain dynamically isolated in the minimal truncation.

a. Honest logical status. What is proved here is the full remote-gap theorem in the minimal finite truncation. What is not yet proved is that the exact final microscopic TECT linearization contains no additional remote channels beyond this truncation, or that their inclusion preserves the same lower bound automatically. So the exact theorem-level statement is:

$$\text{minimal truncation} + \delta_{0,\rho} > \eta_{R,\rho} \implies \text{uniform remote gap } \Delta_\rho > 0.$$

This is already sufficient to convert the remote-gap problem from an abstract dynamical question into a finite matrix lower-bound problem.

U. Summary

$$\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

Write

$$H_{\text{full}}^{\text{tr}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V_{DR}(p)^\dagger & H_R(p) \end{pmatrix}, \quad H_R(p) = D_R(p) + R_R(p),$$

with

$$\delta_0(p) = \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\}, \quad \eta_R(p) = \|R_R(p)\|.$$

Then the pointwise operator bound is

$$\|H_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p))\|\psi\|.$$

Hence, if

$$\delta_0(p) > \eta_R(p),$$

the remote sector has a pointwise gap

$$\text{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \emptyset, \quad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0.$$

On a neighborhood

$$N_\rho = \{p : |p| \leq \rho\},$$

define

$$\delta_{0,\rho} = \inf_{|p| \leq \rho} \delta_0(p), \quad \eta_{R,\rho} = \sup_{|p| \leq \rho} \eta_R(p).$$

If

$$\delta_{0,\rho} > \eta_{R,\rho},$$

then

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho),$$

with

$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.$$

A Gershgorin alternative is:

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_a \left(|(H_R)_{aa}(p)| - \sum_{b \neq a} |(H_R)_{ab}(p)| \right) > 0$$

which also implies a uniform remote gap.

Therefore the remote-gap problem in the minimal truncation is reduced to an explicit finite-dimensional inequality:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \quad \text{or equivalently a Gershgorin lower bound.}}$$

If, in addition,

$$V_{DR}(0) = 0, \quad \|V_{DR}(p)\| \leq c|p|,$$

then for $|E| < \Delta_\rho/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta_\rho}, \quad \Sigma(E, p) = O\left(\frac{|p|^2}{\Delta_\rho}\right).$$

Thus the minimal truncation remote-gap theorem supplies the final spectral lower bound needed for local Dirac spectral isolation.

V. Full remote-tail stability theorem

a. Goal. We now push the remote-gap theorem one step beyond the minimal truncation. The purpose is to prove that once a positive remote gap is established on the minimal shell-relevant truncation, it survives in the full microscopic operator provided the additional remote tail is itself gapped and only weakly coupled to the truncated remote block.

1. A. Minimal remote gap as input

Assume that on the minimal truncation we already proved a uniform remote gap on the neighborhood

$$N_\rho := \{p : |p| \leq \rho\}, \quad (249)$$

namely

$$\text{spec}(H_R^{\text{tr}}(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset, \quad \Delta_\rho > 0, \quad |p| \leq \rho. \quad (250)$$

This is exactly what the sectorwise lower-bound theorem on the minimal truncation delivers.

2. B. Full remote decomposition

Now let the full remote space decompose as

$$\mathcal{H}_R^{\text{full}} = \mathcal{H}_R^{\text{tr}} \oplus \mathcal{H}_{\text{tail}}, \quad (251)$$

where $\mathcal{H}_R^{\text{tr}}$ is the remote sector already included in the minimal truncation, and $\mathcal{H}_{\text{tail}}$ contains all additional remote channels not retained there.

Write the full remote Hamiltonian in block form:

$$H_R^{\text{full}}(p) = \begin{pmatrix} H_R^{\text{tr}}(p) & C(p) \\ C(p)^\dagger & H_{\text{tail}}(p) \end{pmatrix}. \quad (252)$$

3. C. Tail-sector spectral hypothesis

Assume that the additional tail block is itself uniformly gapped away from zero:

$$\text{spec}(H_{\text{tail}}(p)) \cap (-\Delta_{\text{tail},\rho}, \Delta_{\text{tail},\rho}) = \emptyset, \quad \Delta_{\text{tail},\rho} > 0, \quad |p| \leq \rho. \quad (253)$$

Set

$$\Delta_{\min,\rho} := \min\{\Delta_\rho, \Delta_{\text{tail},\rho}\}. \quad (254)$$

4. D. Tail-coupling norm

Define the off-diagonal tail-coupling norm

$$\eta_{\text{tail},\rho} := \sup_{|p| \leq \rho} \|C(p)\|. \quad (255)$$

More generally, one may allow small additional bounded corrections on the diagonal blocks as well. In that case write

$$H_R^{\text{full}}(p) = D_{\text{full}}(p) + R_{\text{full}}(p), \quad (256)$$

where

$$D_{\text{full}}(p) = \begin{pmatrix} H_R^{\text{tr}}(p) & 0 \\ 0 & H_{\text{tail}}(p) \end{pmatrix}, \quad (257)$$

and $R_{\text{full}}(p)$ collects the tail coupling and any further bounded tail correction. Define

$$\eta_{\text{full},\rho} := \sup_{|p| \leq \rho} \|R_{\text{full}}(p)\|. \quad (258)$$

The off-diagonal-only case is recovered by $R_{\text{full}} = \begin{pmatrix} 0 & C \\ C^\dagger & 0 \end{pmatrix}$, in which case $\eta_{\text{full},\rho} = \eta_{\text{tail},\rho}$.

5. E. Operator lower bound

[Full remote lower bound] For every $\Psi \in \mathcal{H}_R^{\text{full}}$ and every $|p| \leq \rho$,

$$\|H_R^{\text{full}}(p)\Psi\| \geq (\Delta_{\min,\rho} - \eta_{\text{full},\rho}) \|\Psi\|. \quad (259)$$

By construction,

$$H_R^{\text{full}}(p) = D_{\text{full}}(p) + R_{\text{full}}(p).$$

Therefore

$$\|H_R^{\text{full}}(p)\Psi\| \geq \|D_{\text{full}}(p)\Psi\| - \|R_{\text{full}}(p)\Psi\|.$$

Since $D_{\text{full}}(p)$ is block diagonal with blocks $H_R^{\text{tr}}(p)$ and $H_{\text{tail}}(p)$, and each block is spectrally separated from zero by at least $\Delta_{\min,\rho}$, we have

$$\|D_{\text{full}}(p)\Psi\| \geq \Delta_{\min,\rho} \|\Psi\|.$$

Also,

$$\|R_{\text{full}}(p)\Psi\| \leq \|R_{\text{full}}(p)\| \|\Psi\| \leq \eta_{\text{full},\rho} \|\Psi\|.$$

Combining the two inequalities gives (259).

6. F. Full remote-tail stability theorem

[Full remote-tail stability theorem] Assume:

1. the minimal remote block satisfies the uniform gap condition (250);
2. the additional remote tail satisfies the uniform gap condition (253);
3. the full tail correction obeys

$$\eta_{\text{full},\rho} < \Delta_{\text{min},\rho}. \quad (260)$$

Then the full remote block has a uniform positive gap on N_ρ :

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_\rho^{\text{full}}, \Delta_\rho^{\text{full}}) = \emptyset, \quad |p| \leq \rho, \quad (261)$$

with

$$\Delta_\rho^{\text{full}} \geq \Delta_{\text{min},\rho} - \eta_{\text{full},\rho} > 0. \quad (262)$$

By Lemma (259), every vector in $\mathcal{H}_R^{\text{full}}$ satisfies

$$\|H_R^{\text{full}}(p)\Psi\| \geq (\Delta_{\text{min},\rho} - \eta_{\text{full},\rho})\|\Psi\|.$$

Since $H_R^{\text{full}}(p)$ is self-adjoint, this implies that every absolute eigenvalue is bounded below by $\Delta_{\text{min},\rho} - \eta_{\text{full},\rho}$. Thus the spectrum cannot enter

$$(-(\Delta_{\text{min},\rho} - \eta_{\text{full},\rho}), \Delta_{\text{min},\rho} - \eta_{\text{full},\rho}).$$

This proves (261) and (262).

7. G. Consequence for the Dirac Schur complement

Assume also the already-established microscopic facts

$$V_{DR}(0) = 0, \quad \|V_{DR}(p)\| \leq c|p|. \quad (263)$$

Then, for

$$|E| < \frac{\Delta_\rho^{\text{full}}}{2}, \quad (264)$$

the resolvent exists and satisfies

$$\|(H_R^{\text{full}}(p) - E)^{-1}\| \leq \frac{2}{\Delta_\rho^{\text{full}}}. \quad (265)$$

Hence the full Schur-complement self-energy obeys

$$\Sigma_{\text{full}}(E, p) = V_{DR}(p)(H_R^{\text{full}}(p) - E)^{-1}V_{DR}(p)^\dagger = O\left(\frac{|p|^2}{\Delta_\rho^{\text{full}}}\right). \quad (266)$$

Therefore the full low-energy effective Hamiltonian still has the same principal Dirac part:

$$H_{\text{eff}}^{\text{full}}(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta_\rho^{\text{full}}}\right). \quad (267)$$

8. H. Stronger interpretation

The theorem proves that once the minimal truncation is under control, the remaining extension to the full microscopic theory is reduced to two analytic estimates on the tail:

1. a uniform tail gap $\Delta_{\text{tail},\rho} > 0$,
2. a small tail-coupling norm $\eta_{\text{full},\rho} < \Delta_{\text{min},\rho}$.

Thus the full remote-gap problem is no longer a qualitative mystery; it is reduced to a quantitative operator-norm control problem.

a. Honest logical status. What is proved here is conditional but exact:

$$\text{minimal remote gap} + \text{tail remote gap} + \text{small tail coupling} \implies \text{full remote gap}.$$

What remains open is the actual derivation of those tail estimates from the complete microscopic TECT operator. That is now the final analytic step, not a representation-theoretic one.

W. Summary

$$\mathcal{H}_R^{\text{full}} = \mathcal{H}_R^{\text{tr}} \oplus \mathcal{H}_{\text{tail}}, \quad H_R^{\text{full}}(p) = \begin{pmatrix} H_R^{\text{tr}}(p) & C(p) \\ C(p)^\dagger & H_{\text{tail}}(p) \end{pmatrix}.$$

Assume the minimal remote block has a uniform gap

$$\text{spec}(H_R^{\text{tr}}(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset,$$

and the additional tail block has a uniform gap

$$\text{spec}(H_{\text{tail}}(p)) \cap (-\Delta_{\text{tail},\rho}, \Delta_{\text{tail},\rho}) = \emptyset.$$

Let

$$\Delta_{\text{min},\rho} := \min\{\Delta_\rho, \Delta_{\text{tail},\rho}\}, \quad \eta_{\text{full},\rho} := \sup_{|p| \leq \rho} \|R_{\text{full}}(p)\|,$$

where R_{full} contains the tail coupling and any further bounded tail correction.

Then for every $\Psi \in \mathcal{H}_R^{\text{full}}$,

$$\|H_R^{\text{full}}(p)\Psi\| \geq (\Delta_{\text{min},\rho} - \eta_{\text{full},\rho})\|\Psi\|.$$

Therefore, if

$$\eta_{\text{full},\rho} < \Delta_{\text{min},\rho},$$

the full remote block has a uniform positive gap

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_\rho^{\text{full}}, \Delta_\rho^{\text{full}}) = \emptyset, \quad \Delta_\rho^{\text{full}} \geq \Delta_{\text{min},\rho} - \eta_{\text{full},\rho} > 0.$$

If also

$$V_{DR}(0) = 0, \quad \|V_{DR}(p)\| \leq c|p|,$$

then for $|E| < \Delta_\rho^{\text{full}}/2$,

$$\|(H_R^{\text{full}}(p) - E)^{-1}\| \leq \frac{2}{\Delta_\rho^{\text{full}}},$$

and the Schur-complement self-energy satisfies

$$\Sigma_{\text{full}}(E, p) = O\left(\frac{|p|^2}{\Delta_{\rho}^{\text{full}}}\right).$$

Hence:

$$\boxed{\text{minimal remote gap} + \text{tail remote gap} + \text{small tail coupling} \implies \text{full remote gap}.}$$

So the extension from the minimal truncation to the full microscopic theory is reduced to two quantitative tail estimates:

$$\boxed{\Delta_{\text{tail}, \rho} > 0 \quad \text{and} \quad \eta_{\text{full}, \rho} < \Delta_{\text{min}, \rho}.$$

X. Tail coupling norm estimate theorem

a. Goal. We now derive an explicit quantitative upper bound for the full tail-coupling norm

$$\eta_{\text{full}, \rho} = \sup_{|p| \leq \rho} \|R_{\text{full}}(p)\|,$$

thereby reducing the remaining full remote-gap condition to a finite analytic inequality.

1. A. Setup

Recall the full remote decomposition

$$\mathcal{H}_R^{\text{full}} = \mathcal{H}_R^{\text{tr}} \oplus \mathcal{H}_{\text{tail}}, \quad (268)$$

with

$$H_R^{\text{full}}(p) = \begin{pmatrix} H_R^{\text{tr}}(p) & C(p) \\ C(p)^{\dagger} & H_{\text{tail}}(p) \end{pmatrix}. \quad (269)$$

More generally, we allow an additional bounded correction on the two diagonal blocks and write

$$H_R^{\text{full}}(p) = D_{\text{full}}(p) + R_{\text{full}}(p), \quad (270)$$

where

$$D_{\text{full}}(p) = \begin{pmatrix} H_R^{\text{tr}}(p) & 0 \\ 0 & H_{\text{tail}}(p) \end{pmatrix}. \quad (271)$$

Our aim is to bound

$$\eta_{\text{full}, \rho} := \sup_{|p| \leq \rho} \|R_{\text{full}}(p)\|. \quad (272)$$

2. B. Exact vanishing from shell-label/length-ratio obstruction

We first isolate the sectors whose coupling vanishes identically.

Let $K_a \subset \mathcal{H}_{\text{tail}}$ be an H_* -invariant tail sector. Suppose K_a differs from the local shell-relevant sector in its shell label, or more sharply, that the necessary mixed-shell length-ratio condition for nontrivial resonance fails. Then the corresponding direct shell-to-tail resonant coupling vanishes:

$$C_a^{\text{res}}(p) \equiv 0. \quad (273)$$

Therefore only the tail sectors not excluded by exact shell-label/metric obstruction contribute to the norm bound.

a. Consequence. The full tail-coupling norm decomposes as

$$\eta_{\text{full},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} \sup_{|p| \leq \rho} \|R_a(p)\|, \quad (274)$$

where $\mathcal{A}_{\text{act}}$ is the finite set of *active* tail sectors not killed by exact mismatch.

3. C. Derivative-order suppression

We now estimate each active sector.

Assume that locality and symmetry imply that the coupling to sector a has Taylor expansion

$$R_a(p) = \sum_{|\alpha|=r_a} p^\alpha T_{a,\alpha} + O(|p|^{r_a+1}), \quad r_a \geq 1, \quad (275)$$

where α is a multi-index and $T_{a,\alpha}$ are bounded operators. The condition $r_a \geq 1$ is the analytic expression of the fact that the constant term is forbidden:

$$R_a(0) = 0. \quad (276)$$

Define

$$K_{a,r_a} := \sum_{|\alpha|=r_a} \|T_{a,\alpha}\|. \quad (277)$$

Then for sufficiently small $|p|$,

$$\|R_a(p)\| \leq K_{a,r_a} |p|^{r_a} + O(|p|^{r_a+1}). \quad (278)$$

Hence there exists a constant $K_a > 0$ such that

$$\|R_a(p)\| \leq K_a |p|^{r_a} \quad (|p| \leq \rho). \quad (279)$$

[Derivative-order tail estimate] If $R_a(p)$ vanishes to order $r_a \geq 1$ at $p = 0$, then on a sufficiently small neighborhood N_ρ ,

$$\sup_{|p| \leq \rho} \|R_a(p)\| \leq K_a \rho^{r_a}. \quad (280)$$

Immediate from (279) by taking the supremum over $|p| \leq \rho$.

4. D. Heavy-mode suppression

Suppose now that the coupling to sector a is generated only after integrating out a heavy auxiliary block $H_a^{(H)}(p)$ with a uniform gap

$$\text{spec}(H_a^{(H)}(p)) \cap (-M_a, M_a) = \emptyset, \quad M_a > 0, \quad (281)$$

and that the induced effective coupling has Schur-complement form

$$R_a^{\text{ind}}(p, E) = X_a(p) (H_a^{(H)}(p) - E)^{-1} Y_a(p), \quad (282)$$

for $|E| < M_a/2$.

Then

$$\|(H_a^{(H)}(p) - E)^{-1}\| \leq \frac{2}{M_a}. \quad (283)$$

If furthermore

$$\|X_a(p)\| \leq A_a |p|^{m_a}, \quad \|Y_a(p)\| \leq B_a |p|^{n_a}, \quad (284)$$

then

$$\|R_a^{\text{ind}}(p, E)\| \leq \frac{2A_a B_a}{M_a} |p|^{m_a+n_a}. \quad (285)$$

[Heavy-mode suppressed tail estimate] Under (281)–(284),

$$\sup_{|p| \leq \rho} \|R_a^{\text{ind}}(p, E)\| \leq \frac{2A_a B_a}{M_a} \rho^{m_a+n_a} \quad (|E| < M_a/2). \quad (286)$$

Combine (282), (283), and (284), then take the supremum over $|p| \leq \rho$.

a. More general form. If the heavy integration is iterated or involves higher-order local operators, the same argument yields bounds of the generic form

$$\|R_a(p)\| \leq \frac{\tilde{K}_a}{M_a^{\sigma_a}} |p|^{r_a}, \quad \sigma_a \geq 1. \quad (287)$$

5. E. Global tail-coupling bound

Combining the exact-zero sectors, derivative suppression, and heavy-mode suppression, we obtain the main estimate.

[Tail coupling norm estimate theorem] Let $\mathcal{A}_{\text{act}}$ be the active tail sectors not excluded by exact shell-label/metric mismatch. Assume that for each $a \in \mathcal{A}_{\text{act}}$, one of the following holds:

1. *pure derivative suppression:*

$$\|R_a(p)\| \leq K_a |p|^{r_a}, \quad r_a \geq 1; \quad (288)$$

2. *heavy-mode suppressed derivative coupling:*

$$\|R_a(p)\| \leq \frac{\tilde{K}_a}{M_a^{\sigma_a}} |p|^{r_a}, \quad r_a \geq 1, \sigma_a \geq 1. \quad (289)$$

Then on $N_\rho = \{|p| \leq \rho\}$,

$$\eta_{\text{full}, \rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad (290)$$

where the first sum runs over purely derivative-suppressed sectors and the second over heavy-mode-induced sectors. In particular,

$$\eta_{\text{full}, \rho} \xrightarrow{\rho \rightarrow 0} 0. \quad (291)$$

By (274),

$$\eta_{\text{full}, \rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} \sup_{|p| \leq \rho} \|R_a(p)\|.$$

For each sector, apply either the derivative-order estimate (280) or the heavy-mode estimate (286) / (287). Summing the sectorwise bounds yields (290). Since every exponent $r_a \geq 1$, each term vanishes as $\rho \rightarrow 0$, proving (291).

6. F. Consequence for the full remote gap

Recall the previously proved full remote-tail stability theorem: if

$$\eta_{\text{full}, \rho} < \Delta_{\text{min}, \rho}, \quad (292)$$

then the full remote block has a uniform gap

$$\Delta_\rho^{\text{full}} \geq \Delta_{\text{min},\rho} - \eta_{\text{full},\rho} > 0. \quad (293)$$

Combining this with (290), we obtain the explicit sufficient inequality

$$\sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} < \Delta_{\text{min},\rho}. \quad (294)$$

Therefore the full remote gap is guaranteed whenever the momentum window ρ is sufficiently small relative to the active tail scales and heavy masses.

7. G. Honest logical status

What is now proved is the following:

1. exact shell-label/metric mismatch kills many tail sectors identically;
2. every remaining local tail coupling is suppressed by at least one power of $|p|$;
3. if the coupling is induced through heavy auxiliary blocks, it is additionally suppressed by inverse powers of the heavy gap;
4. therefore the full tail norm satisfies an explicit bound of the form

$$\eta_{\text{full},\rho} \lesssim \sum_a \rho^{r_a} + \sum_a \frac{\rho^{r_a}}{M_a^{\sigma_a}}.$$

What is *not* yet proved automatically is the actual numerical value of the coefficients $K_a, \tilde{K}_a$ and the tail gaps M_a in the final microscopic TECT operator. So the remaining work is no longer structural; it is a finite coefficient-estimation problem.

Y. Summary

$$\mathcal{H}_R^{\text{full}} = \mathcal{H}_R^{\text{tr}} \oplus \mathcal{H}_{\text{tail}}, \quad H_R^{\text{full}}(p) = D_{\text{full}}(p) + R_{\text{full}}(p).$$

If a tail sector is excluded by shell-label or mixed-shell length-ratio obstruction, then its resonant coupling vanishes identically:

$$R_a(p) \equiv 0.$$

For every active tail sector a , if locality and symmetry imply

$$R_a(0) = 0$$

and the first nonzero Taylor term is order $r_a \geq 1$, then

$$\|R_a(p)\| \leq K_a |p|^{r_a}, \quad \sup_{|p| \leq \rho} \|R_a(p)\| \leq K_a \rho^{r_a}.$$

If, more specifically, the coupling is generated by integrating out a heavy auxiliary block of gap M_a , with

$$R_a^{\text{ind}}(p, E) = X_a(p)(H_a^{(H)}(p) - E)^{-1} Y_a(p),$$

then for $|E| < M_a/2$,

$$\|R_a^{\text{ind}}(p, E)\| \leq \frac{2A_a B_a}{M_a} |p|^{m_a + n_a}.$$

Therefore the full tail-coupling norm obeys

$$\eta_{\text{full},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

In particular,

$$\eta_{\text{full},\rho} \rightarrow 0 \quad (\rho \rightarrow 0).$$

Combining this with the previously proved full remote-tail stability theorem, the full remote gap is guaranteed whenever

$$\sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} < \Delta_{\text{min},\rho}.$$

Hence the full remote-gap problem is reduced to the explicit analytic inequality

$$\eta_{\text{full},\rho} < \Delta_{\text{min},\rho}, \quad \eta_{\text{full},\rho} \lesssim \sum_a \rho^{r_a} + \sum_a \frac{\rho^{r_a}}{M_a^{\sigma_a}}.$$

Z. Tail-sector gap decomposition theorem

a. Goal. We now reduce the tail-gap problem to a sectorwise lower-bound problem. The purpose is to decompose the full tail block into structurally distinct sectors and prove that each sector carries a positive diagonal spectral distance from zero under explicit geometric or heavy-mode conditions.

1. A. Tail decomposition

Let the full remote space already be decomposed as

$$\mathcal{H}_R^{\text{full}} = \mathcal{H}_R^{\text{tr}} \oplus \mathcal{H}_{\text{tail}}, \quad (295)$$

where $\mathcal{H}_R^{\text{tr}}$ is the minimal remote block previously treated.

We now decompose the remaining tail into

$$\mathcal{H}_{\text{tail}} = \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}. \quad (296)$$

The meaning is:

- $\mathcal{H}_{\text{high}}$: states carried by shell orbits whose radius differs from the selected shell pair;
- $\mathcal{H}_{\text{vol}}$: heavy longitudinal / volumetric sector;
- $\mathcal{H}_{\text{wrong-int}}$: states with internal content spectrally separated from the selected enriched internal doublet;
- $\mathcal{H}_{\text{extra}}$: any remaining modes not belonging to the previous classes.

Write the tail Hamiltonian as

$$H_{\text{tail}}(p) = D_{\text{tail}}(p) + R_{\text{tail}}(p), \quad (297)$$

where $D_{\text{tail}}(p)$ is block diagonal with respect to (296), and $R_{\text{tail}}(p)$ contains all tail–tail off-diagonal mixing.

Define

$$\eta_{\text{tail},\rho} := \sup_{|p| \leq \rho} \|R_{\text{tail}}(p)\|. \quad (298)$$

2. B. Higher-shell sector lower bound

Assume the underlying microscopic quadratic kernel is of Brazovskii type near the selected shell radius q_* :

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \quad K > 0. \quad (299)$$

Let $\mathcal{Q}_{\text{high}}$ be the discrete set of shell momenta carried by $\mathcal{H}_{\text{high}}$, and assume a strict radial separation

$$\delta_{\text{shell}} := \inf_{Q \in \mathcal{Q}_{\text{high}}} ||Q|^2 - q_*^2| > 0. \quad (300)$$

Then every higher-shell diagonal energy satisfies

$$|\Lambda(Q)| \geq \Delta_{\text{high}}^{(0)}, \quad \Delta_{\text{high}}^{(0)} := \inf_{Q \in \mathcal{Q}_{\text{high}}} \left| r_* + K(|Q|^2 - q_*^2)^2 \right|. \quad (301)$$

In the standard soft-shell regime where the selected shell is the unique soft shell and all other shell radii remain noncritical, one has

$$\Delta_{\text{high}}^{(0)} > 0. \quad (302)$$

A sufficient explicit criterion is

$$K\delta_{\text{shell}}^2 > |r_*|, \quad (303)$$

in which case

$$\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| > 0. \quad (304)$$

3. C. Heavy volumetric sector lower bound

Assume the volumetric sector is generated by a heavy scalar/longitudinal block with mass gap $M_\sigma > 0$, so that its quadratic symbol has the form

$$H_{\text{vol}}^{(0)}(\omega, k) = M_\sigma^2 + c_\sigma^2 k^2 - \omega^2 \quad (\text{Euclidean/static reduction: } M_\sigma^2 + c_\sigma^2 k^2). \quad (305)$$

Then, in the static or low-energy regime $|\omega|, |k| \ll M_\sigma$,

$$\text{dist}(\text{spec}(H_{\text{vol}}^{(0)}), 0) \geq M_\sigma^2/2 \quad (306)$$

after restricting to a sufficiently small neighborhood. Equivalently, at the Hamiltonian level one may write

$$\Delta_{\text{vol}}^{(0)} \geq c_{\text{vol}} M_\sigma \quad (307)$$

for some positive model-dependent constant c_{vol} .

Thus the heavy volumetric block contributes a positive tail diagonal gap.

4. D. Wrong-internal sector lower bound

Let the selected shell-relevant internal doublet be $U_D \subset U_{\text{int}}$, and suppose the internal operator I admits an orthogonal decomposition

$$U_{\text{int}} = U_D \oplus U_{\text{wrong}}, \quad (308)$$

with a positive spectral separation

$$m_{\text{int}} := \text{dist}(\text{spec}(I|_{U_{\text{wrong}}}), \text{spec}(I|_{U_D})) > 0. \quad (309)$$

Then every state in $\mathcal{H}_{\text{wrong-int}}$ has a diagonal internal energy shift at least m_{int} away from the selected low-energy block. Hence

$$\Delta_{\text{wrong-int}}^{(0)} \geq m_{\text{int}} > 0. \quad (310)$$

5. E. Extra sectors

For any additional tail sector $K \subset \mathcal{H}_{\text{extra}}$, assume there is an explicit diagonal lower bound

$$\text{dist}(\text{spec}(H_K^{(0)}(p)), 0) \geq \Delta_K^{(0)} > 0 \quad (|p| \leq \rho). \quad (311)$$

This is the only input needed for the abstract theorem below.

6. F. Tail diagonal minimum

Define the sectorwise diagonal tail gap candidates

$$\Delta_{\text{high}}^{(0)}, \quad \Delta_{\text{vol}}^{(0)}, \quad \Delta_{\text{wrong-int}}^{(0)}, \quad \Delta_{\text{extra}}^{(0)} := \inf_{K \subset \mathcal{H}_{\text{extra}}} \Delta_K^{(0)}. \quad (312)$$

Set

$$\Delta_{\text{diag}, \rho}^{\text{tail}} := \min \left\{ \Delta_{\text{high}}^{(0)}, \Delta_{\text{vol}}^{(0)}, \Delta_{\text{wrong-int}}^{(0)}, \Delta_{\text{extra}}^{(0)} \right\}. \quad (313)$$

Then every vector $\Psi \in \mathcal{H}_{\text{tail}}$ satisfies

$$\|D_{\text{tail}}(p)\Psi\| \geq \Delta_{\text{diag}, \rho}^{\text{tail}} \|\Psi\| \quad (|p| \leq \rho). \quad (314)$$

7. G. Tail-sector gap theorem

[Tail-sector gap decomposition theorem] Assume:

1. the higher-shell sector satisfies the radial-separation condition (300) and hence $\Delta_{\text{high}}^{(0)} > 0$;
2. the volumetric sector is heavy, with $M_\sigma > 0$, hence $\Delta_{\text{vol}}^{(0)} > 0$;
3. the wrong-internal sector has a positive internal spectral separation $m_{\text{int}} > 0$, hence $\Delta_{\text{wrong-int}}^{(0)} > 0$;
4. every extra tail sector $K \subset \mathcal{H}_{\text{extra}}$ has a positive diagonal lower bound $\Delta_K^{(0)} > 0$;
5. the tail-tail mixing norm satisfies

$$\eta_{\text{tail}, \rho} < \Delta_{\text{diag}, \rho}^{\text{tail}}. \quad (315)$$

Then the full tail block has a uniform positive gap:

$$\text{spec}(H_{\text{tail}}(p)) \cap (-\Delta_{\text{tail}, \rho}, \Delta_{\text{tail}, \rho}) = \emptyset \quad (|p| \leq \rho), \quad (316)$$

with

$$\Delta_{\text{tail}, \rho} \geq \Delta_{\text{diag}, \rho}^{\text{tail}} - \eta_{\text{tail}, \rho} > 0. \quad (317)$$

By (297),

$$H_{\text{tail}}(p) = D_{\text{tail}}(p) + R_{\text{tail}}(p).$$

For any $\Psi \in \mathcal{H}_{\text{tail}}$,

$$\|H_{\text{tail}}(p)\Psi\| \geq \|D_{\text{tail}}(p)\Psi\| - \|R_{\text{tail}}(p)\Psi\|.$$

Using (314) and the definition of $\eta_{\text{tail}, \rho}$,

$$\|H_{\text{tail}}(p)\Psi\| \geq (\Delta_{\text{diag}, \rho}^{\text{tail}} - \eta_{\text{tail}, \rho}) \|\Psi\|.$$

Since $H_{\text{tail}}(p)$ is Hermitian, every absolute eigenvalue is bounded below by $\Delta_{\text{diag}, \rho}^{\text{tail}} - \eta_{\text{tail}, \rho}$, which proves (316) and (317).

8. *H. Combination with the full remote-tail theorem*

Recall the previous full remote-tail stability theorem: if the truncated remote block has gap $\Delta_\rho > 0$, the tail block has gap $\Delta_{\text{tail},\rho} > 0$, and the tail coupling to the truncated remote block is small, then the full remote block remains gapped.

Therefore, combining the present theorem with that theorem yields:

[Structured full remote-gap criterion] If

1. the minimal truncated remote block satisfies $\Delta_\rho > 0$;
2. the tail sectors satisfy the hypotheses of the Tail-sector gap decomposition theorem;
3. the coupling between $\mathcal{H}_R^{\text{tr}}$ and $\mathcal{H}_{\text{tail}}$ obeys the smallness condition of the full remote-tail stability theorem;

then the full remote block $H_R^{\text{full}}(p)$ has a uniform positive gap on N_ρ .

a. Exact logical status. What is now closed is the following structural reduction:

$$\text{full tail gap} \Leftarrow \begin{cases} \text{higher-shell radial separation,} \\ \text{heavy volumetric mass gap,} \\ \text{wrong-internal spectral separation,} \\ \text{small tail-tail mixing.} \end{cases}$$

Thus the tail-gap problem is reduced to explicit sectorwise lower bounds plus one norm estimate. What is still model-specific is the actual numerical evaluation of these quantities in the final microscopic TECT linearization.

. **Summary**

$$\mathcal{H}_{\text{tail}} = \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}, \quad H_{\text{tail}}(p) = D_{\text{tail}}(p) + R_{\text{tail}}(p).$$

Define

$$\eta_{\text{tail},\rho} = \sup_{|p| \leq \rho} \|R_{\text{tail}}(p)\|.$$

For the higher-shell sector, assume radial shell separation

$$\delta_{\text{shell}} = \inf_{Q \in \mathcal{Q}_{\text{high}}} \left| |Q|^2 - q_*^2 \right| > 0.$$

Then for a Brazovskii-type kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2,$$

a sufficient explicit bound is

$$K\delta_{\text{shell}}^2 > |r_*| \implies \Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| > 0.$$

For the volumetric sector, if the locking mode has mass gap $M_\sigma > 0$, then

$$\Delta_{\text{vol}}^{(0)} \geq c_{\text{vol}} M_\sigma > 0.$$

For the wrong-internal sector, if the internal operator has spectral separation

$$m_{\text{int}} = \text{dist}(\text{spec}(I|_{U_{\text{wrong}}}), \text{spec}(I|_{U_D})) > 0,$$

then

$$\Delta_{\text{wrong-int}}^{(0)} \geq m_{\text{int}} > 0.$$

Let

$$\Delta_{\text{diag},\rho}^{\text{tail}} = \min\{\Delta_{\text{high}}^{(0)}, \Delta_{\text{vol}}^{(0)}, \Delta_{\text{wrong-int}}^{(0)}, \Delta_{\text{extra}}^{(0)}\}.$$

If

$$\eta_{\text{tail},\rho} < \Delta_{\text{diag},\rho}^{\text{tail}},$$

then the full tail block has a uniform positive gap

$$\text{spec}(H_{\text{tail}}(p)) \cap (-\Delta_{\text{tail},\rho}, \Delta_{\text{tail},\rho}) = \emptyset, \quad \Delta_{\text{tail},\rho} \geq \Delta_{\text{diag},\rho}^{\text{tail}} - \eta_{\text{tail},\rho} > 0.$$

Therefore the tail-gap problem is reduced to four explicit checks:

higher-shell radial separation

heavy volumetric gap $M_\sigma > 0$

wrong-internal spectral separation $m_{\text{int}} > 0$

small tail-tail mixing $\eta_{\text{tail},\rho} < \Delta_{\text{diag},\rho}^{\text{tail}}$.

Combining this with the previous full remote-tail stability theorem yields a structured sufficient criterion for the full remote spectral gap.

. BCC shell-radius arithmetic theorem

a. Goal. We now compute explicitly the first radial gap above the selected BCC first shell. This closes the higher-shell part of the tail-gap theorem by replacing the abstract shell-separation parameter by an exact arithmetic value.

1. A. Reciprocal-lattice arithmetic for the selected BCC first shell

The selected first shell used throughout the BCC shell analysis is the cuboctahedral 12-star

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (318)$$

Equivalently, before normalization one works with the integer representatives

$$\tilde{S}_{\text{BCC}} = \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (319)$$

The natural reciprocal-lattice arithmetic associated with this first shell is the parity-even cubic lattice

$$L_{\text{BCC}}^* := \{(h, k, \ell) \in \mathbb{Z}^3 : h + k + \ell \equiv 0 \pmod{2}\}. \quad (320)$$

Its nonzero lattice vectors are precisely the reciprocal harmonics compatible with the cuboctahedral first shell.

2. B. Parity lemma for squared norms

[Parity-even norm lemma] For every $Q = (h, k, \ell) \in L_{\text{BCC}}^*$,

$$|Q|^2 = h^2 + k^2 + \ell^2 \in 2\mathbb{Z}. \quad (321)$$

Modulo 2, one has

$$n^2 \equiv n \pmod{2}$$

for every integer n . Therefore

$$h^2 + k^2 + \ell^2 \equiv h + k + \ell \equiv 0 \pmod{2}$$

because $Q \in L_{\text{BCC}}^*$ implies $h + k + \ell$ is even. Hence $|Q|^2$ is always an even integer.

a. Immediate consequence. The possible nonzero squared radii in L_{BCC}^* begin at the positive even integers

$$2, 4, 6, 8, \dots$$

subject to representability as sums of three squares.

3. C. First and second shell radii

[First shell] The smallest nonzero squared radius in L_{BCC}^* is

$$q_*^2 = 2. \quad (322)$$

It is realized exactly by the twelve vectors of type

$$(\pm 1, \pm 1, 0), \quad (\pm 1, 0, \pm 1), \quad (0, \pm 1, \pm 1). \quad (323)$$

By the parity-even norm lemma, every nonzero squared norm is an even positive integer, so the smallest possible value is at least 2. The vector $(1, 1, 0) \in L_{\text{BCC}}^*$ has squared norm 2, hence the minimum is exactly 2. Its orbit under sign changes and coordinate permutations is precisely (323), containing 12 vectors.

[Second shell] The next squared radius above $q_*^2 = 2$ is

$$q_2^2 = 4. \quad (324)$$

It is realized by the six vectors of type

$$(\pm 2, 0, 0), \quad (0, \pm 2, 0), \quad (0, 0, \pm 2). \quad (325)$$

Again by the parity-even norm lemma, the next possible squared radius above 2 is at least 4. The vector $(2, 0, 0) \in L_{\text{BCC}}^*$ has squared norm 4, so $q_2^2 = 4$. The corresponding orbit is (325).

[Exact first radial shell gap] In the integer reciprocal convention,

$$\delta_{\text{shell}} := \min_{Q \in L_{\text{BCC}}^* \setminus \tilde{S}_{\text{BCC}}} ||Q|^2 - q_*^2| = q_2^2 - q_*^2 = 4 - 2 = 2. \quad (326)$$

4. D. Unit-shell normalization

The shell-selection literature often normalizes the first shell to unit radius by dividing all vectors by $\sqrt{2}$, cf. (318). In this normalized convention,

$$\hat{q}_*^2 = 1, \quad \hat{q}_2^2 = \frac{4}{2} = 2. \quad (327)$$

Therefore the normalized shell gap is

$$\hat{\delta}_{\text{shell}} = \hat{q}_2^2 - \hat{q}_*^2 = 2 - 1 = 1. \quad (328)$$

So:

$\delta_{\text{shell}} = 2 \quad (\text{integer reciprocal convention}), \quad \hat{\delta}_{\text{shell}} = 1 \quad (\text{unit-shell convention}).$

5. E. First few higher shells

For later bookkeeping, the first few shells of L_{BCC}^* are:

$$q_1^2 = 2 \quad \text{from} \quad (\pm 1, \pm 1, 0)\text{-type}, \quad (329)$$

$$q_2^2 = 4 \quad \text{from} \quad (\pm 2, 0, 0)\text{-type}, \quad (330)$$

$$q_3^2 = 6 \quad \text{from} \quad (\pm 2, \pm 1, \pm 1)\text{-type}, \quad (331)$$

$$q_4^2 = 8 \quad \text{from} \quad (\pm 2, \pm 2, 0)\text{-type}. \quad (332)$$

Thus there is no shell radius between $q_*^2 = 2$ and $q_2^2 = 4$.

6. F. Explicit higher-shell diagonal gap

Now take the Brazovskii-type quadratic kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \quad K > 0. \quad (333)$$

For every higher-shell vector $Q \in L_{\text{BCC}}^* \setminus \tilde{S}_{\text{BCC}}$, one has

$$||Q|^2 - q_*^2| \geq \delta_{\text{shell}} = 2 \quad (334)$$

in the integer convention. Therefore

$$|\Lambda(Q)| \geq K \delta_{\text{shell}}^2 - |r_*| = 4K - |r_*|. \quad (335)$$

Hence the higher-shell diagonal spectral lower bound satisfies

$$\Delta_{\text{high}}^{(0)} \geq 4K - |r_*|. \quad (336)$$

In particular, if

$$4K > |r_*|, \quad (337)$$

then

$$\Delta_{\text{high}}^{(0)} > 0. \quad (338)$$

In the normalized unit-shell convention, $\hat{\delta}_{\text{shell}} = 1$, so the same argument gives

$$\hat{\Delta}_{\text{high}}^{(0)} \geq K - |r_*|. \quad (339)$$

Thus, if

$$K > |r_*|, \quad (340)$$

the normalized higher-shell diagonal gap is positive.

7. G. The theorem

[BCC shell-radius arithmetic theorem] Let

$$L_{\text{BCC}}^* = \{(h, k, \ell) \in \mathbb{Z}^3 : h + k + \ell \equiv 0 \pmod{2}\}$$

be the reciprocal lattice corresponding to the selected BCC first shell. Then:

1. the first shell has squared radius

$$q_*^2 = 2$$

and consists of the 12 vectors of type $(\pm 1, \pm 1, 0)$;

2. the second shell has squared radius

$$q_2^2 = 4$$

and consists of the 6 vectors of type $(\pm 2, 0, 0)$;

3. the exact first shell-separation parameter is

$$\delta_{\text{shell}} = 2$$

in the integer reciprocal convention, equivalently

$$\hat{\delta}_{\text{shell}} = 1$$

in the unit-shell normalization;

4. for the Brazovskii-type diagonal kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \quad K > 0,$$

the higher-shell diagonal gap obeys

$$\Delta_{\text{high}}^{(0)} \geq 4K - |r_*|$$

in integer convention, and

$$\widehat{\Delta}_{\text{high}}^{(0)} \geq K - |r_*|$$

in unit-shell convention.

a. Interpretation. This closes the higher-shell contribution to the tail-gap problem numerically. The only remaining parts of the tail-gap theorem are now the volumetric gap, wrong-internal spectral separation, and tail-tail mixing norms.

. Summary

$$L_{\text{BCC}}^* = \{(h, k, \ell) \in \mathbb{Z}^3 : h + k + \ell \equiv 0 \pmod{2}\}.$$

For every $Q = (h, k, \ell) \in L_{\text{BCC}}^*$,

$$|Q|^2 = h^2 + k^2 + \ell^2 \equiv h + k + \ell \equiv 0 \pmod{2},$$

so every nonzero squared radius is an even positive integer.

The first shell is

$$q_*^2 = 2, \quad \widetilde{S}_{\text{BCC}} = \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\},$$

and the second shell is

$$q_2^2 = 4, \quad \{(\pm 2, 0, 0), (0, \pm 2, 0), (0, 0, \pm 2)\}.$$

Therefore the exact first shell-separation parameter is

$$\delta_{\text{shell}} = q_2^2 - q_*^2 = 4 - 2 = 2.$$

After unit-shell normalization by $1/\sqrt{2}$,

$$\widehat{q}_*^2 = 1, \quad \widehat{q}_2^2 = 2, \quad \widehat{\delta}_{\text{shell}} = 1.$$

For a Brazovskii-type diagonal kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \quad K > 0,$$

one obtains the explicit higher-shell lower bound

$$\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| = 4K - |r_*|$$

in the integer convention, and

$$\widehat{\Delta}_{\text{high}}^{(0)} \geq K - |r_*|$$

in the normalized convention.

Hence the higher-shell contribution to the tail-gap theorem is numerically closed.

. **Wrong-internal sector spectral separation theorem**

a. Goal. We now close the wrong-internal part of the remote-gap problem at the finite-dimensional internal-operator level. The objective is to replace the abstract condition

$$m_{\text{int}} > 0$$

by an explicit spectral-separation criterion on the internal shell-point Hamiltonian.

1. *A. Selected internal doublet and orthogonal complement*

Let U_{int} be the finite-dimensional internal carrier at the selected shell point G_* . Assume that the shell-point Hamiltonian on this internal space is a Hermitian operator

$$I_0 : U_{\text{int}} \rightarrow U_{\text{int}}. \quad (341)$$

Assume there exists a selected two-dimensional internal doublet

$$U_D \subset U_{\text{int}}, \quad \dim U_D = 2, \quad (342)$$

with orthogonal projector P , and define the orthogonal complement

$$U_{\text{wrong}} := U_D^\perp, \quad Q := 1 - P. \quad (343)$$

The shell-point protected-doublet condition is

$$PI_0P = E_*P. \quad (344)$$

This is exactly the form forced by an effective transverse $SO(2)$ symmetry on the selected shell-transverse plane.

2. *B. General complement-gap criterion*

Define the shifted complement operator

$$K_{\text{wrong}} := Q(I_0 - E_*)Q \quad \text{on } U_{\text{wrong}}. \quad (345)$$

[General wrong-internal spectral separation theorem] Assume (344). If K_{wrong} is invertible, then the wrong-internal sector is spectrally separated from the selected doublet by

$$m_{\text{int}}^{(0)} := \text{dist}(E_*, \text{spec}(QI_0Q)) = \|K_{\text{wrong}}^{-1}\|^{-1} > 0. \quad (346)$$

Equivalently,

$$\text{spec}(QI_0Q) \cap (E_* - m_{\text{int}}^{(0)}, E_* + m_{\text{int}}^{(0)}) = \emptyset. \quad (347)$$

Since QI_0Q is Hermitian on the finite-dimensional space U_{wrong} , its spectrum is real and finite. The operator $K_{\text{wrong}} = Q(I_0 - E_*)Q$ is Hermitian and invertible by assumption, hence

$$0 \notin \text{spec}(K_{\text{wrong}}).$$

Therefore

$$\text{dist}(0, \text{spec}(K_{\text{wrong}})) > 0.$$

Because

$$\text{spec}(K_{\text{wrong}}) = \text{spec}(QI_0Q) - E_*,$$

we obtain

$$m_{\text{int}}^{(0)} = \text{dist}(E_*, \text{spec}(QI_0Q)) = \text{dist}(0, \text{spec}(K_{\text{wrong}})).$$

For an invertible finite-dimensional Hermitian operator, the distance from zero to the spectrum equals the inverse norm of the resolvent at zero:

$$\text{dist}(0, \text{spec}(K_{\text{wrong}})) = \|K_{\text{wrong}}^{-1}\|^{-1}.$$

This proves (346).

3. C. Min-max form

The previous theorem is exact but abstract. A more concrete form uses sign-definite complement blocks. Assume U_{wrong} splits into I_0 -invariant subspaces

$$U_{\text{wrong}} = U_+ \oplus U_-, \quad (348)$$

such that

$$QI_0Q|_{U_+} \geq (E_* + m_+)\mathbf{1}, \quad QI_0Q|_{U_-} \leq (E_* - m_-)\mathbf{1}, \quad (349)$$

with

$$m_+ > 0, \quad m_- > 0. \quad (350)$$

[Min-max wrong-internal gap criterion] Under (344) and (349),

$$m_{\text{int}}^{(0)} \geq \min\{m_+, m_-\} > 0. \quad (351)$$

Every eigenvalue of $QI_0Q|_{U_+}$ is at least $E_* + m_+$, and every eigenvalue of $QI_0Q|_{U_-}$ is at most $E_* - m_-$. Hence the distance from E_* to the spectrum of the full complement is at least $\min\{m_+, m_-\}$. Equivalently, this is the Rayleigh–Ritz/min-max statement on the two invariant subspaces.

4. D. Minimal $\mathbb{C}^3$ shell-point carrier

We now specialize to the minimal shell-point internal carrier already isolated in the BCC shell analysis. At a selected shell point $G_* \neq 0$, one has the canonical orthogonal splitting

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_{\mathbb{C}}^{\perp}, \quad \hat{n} = \frac{G_*}{|G_*|}. \quad (352)$$

Assume the shell-point Hamiltonian possesses the effective transverse $SO(2)$ symmetry, so that

$$I_0|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2. \quad (353)$$

Then we take

$$U_D = \hat{n}_{\mathbb{C}}^{\perp}, \quad U_{\text{wrong}} = \mathbb{C}\hat{n}. \quad (354)$$

Let the longitudinal shell-point energy be

$$I_0\hat{n} = E_L\hat{n}. \quad (355)$$

[Minimal-carrier wrong-internal gap] Under (353) and (355),

$$m_{\text{int}}^{(0)} = |E_L - E_*|. \quad (356)$$

In particular, if

$$E_L \neq E_*, \quad (357)$$

then the wrong-internal sector is spectrally separated from the selected doublet.

On the one-dimensional complement $U_{\text{wrong}} = \mathbb{C}\hat{n}$, the operator QI_0Q has the single eigenvalue E_L . Hence

$$\text{spec}(QI_0Q) = \{E_L\},$$

so

$$m_{\text{int}}^{(0)} = \text{dist}(E_*, \{E_L\}) = |E_L - E_*|.$$

5. E. Stability under bounded internal perturbations

We now allow a bounded perturbation of the internal shell-point operator:

$$I = I_0 + \delta I. \quad (358)$$

Assume

$$\|Q \delta I Q\| \leq \varepsilon_{\text{int}}. \quad (359)$$

[Wrong-internal gap stability] Assume the unperturbed wrong-internal gap $m_{\text{int}}^{(0)} > 0$. Under (359), the perturbed complement spectrum satisfies

$$\text{dist}(E_*, \text{spec}(QIQ)) \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}. \quad (360)$$

In particular, if

$$\varepsilon_{\text{int}} < m_{\text{int}}^{(0)}, \quad (361)$$

then

$$m_{\text{int}} := \text{dist}(E_*, \text{spec}(QIQ)) \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}} > 0. \quad (362)$$

Since

$$QIQ = QI_0Q + Q\delta IQ,$$

the standard spectral variation bound for Hermitian finite-dimensional operators gives

$$\text{dist}(\text{spec}(QIQ), \text{spec}(QI_0Q)) \leq \|Q\delta IQ\| \leq \varepsilon_{\text{int}}.$$

Hence every eigenvalue of QIQ stays within ε_{int} of the unperturbed complement spectrum. Since the unperturbed spectrum is at least $m_{\text{int}}^{(0)}$ away from E_* , the perturbed spectrum remains at least $m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}$ away.

6. F. Interpretation for the tail-gap theorem

The wrong-internal contribution to the tail diagonal gap was previously denoted

$$\Delta_{\text{wrong-int}}^{(0)}. \quad (363)$$

The present theorem identifies it concretely as

$$\Delta_{\text{wrong-int}}^{(0)} = m_{\text{int}}^{(0)} = \text{dist}(E_*, \text{spec}(QI_0Q)), \quad (364)$$

and in the minimal $\mathbb{C}^3$ shell-point carrier,

$$\Delta_{\text{wrong-int}}^{(0)} = |E_L - E_*|. \quad (365)$$

Thus the wrong-internal tail-gap condition is no longer abstract. It is reduced to the finite-dimensional spectral separation between the selected protected doublet and its orthogonal complement at the shell point.

a. Exact logical status. What is now closed is:

$$\text{selected shell-point internal doublet } U_D \quad + \quad Q(I_0 - E_*)Q \text{ invertible} \implies m_{\text{int}}^{(0)} > 0.$$

In the minimal $\mathbb{C}^3$ carrier this reduces further to the explicit scalar condition

$$E_L \neq E_*.$$

What is not yet derived automatically from the final microscopic TECT PDE is the actual numerical value of $E_L - E_*$ (or, in the general case, of $\text{dist}(E_*, \text{spec}(QI_0Q))$). But the wrong-internal gap problem itself has now been reduced completely to a finite-dimensional Hermitian block calculation.

. Summary

$$U_{\text{int}} = U_D \oplus U_{\text{wrong}}, \quad P = \Pi_{U_D}, \quad Q = \Pi_{U_{\text{wrong}}}, \quad PI_0P = E_*P.$$

Define the shifted wrong-internal block

$$K_{\text{wrong}} := Q(I_0 - E_*)Q.$$

If K_{wrong} is invertible, then the wrong-internal spectral separation is

$$m_{\text{int}}^{(0)} = \text{dist}(E_*, \text{spec}(QI_0Q)) = \|K_{\text{wrong}}^{-1}\|^{-1} > 0.$$

A min-max sufficient form is: if

$$QI_0Q|_{U_+} \geq (E_* + m_+)\mathbf{1}, \quad QI_0Q|_{U_-} \leq (E_* - m_-)\mathbf{1},$$

then

$$m_{\text{int}}^{(0)} \geq \min\{m_+, m_-\} > 0.$$

In the minimal shell-point carrier

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_{\mathbb{C}}^{\perp},$$

if effective transverse $SO(2)$ gives

$$I_0|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_*\mathbf{1}_2,$$

and

$$I_0\hat{n} = E_L\hat{n},$$

then

$$U_D = \hat{n}_{\mathbb{C}}^{\perp}, \quad U_{\text{wrong}} = \mathbb{C}\hat{n}, \quad m_{\text{int}}^{(0)} = |E_L - E_*|.$$

Under a bounded perturbation $I = I_0 + \delta I$ with

$$\|Q\delta IQ\| \leq \varepsilon_{\text{int}},$$

the gap is stable:

$$m_{\text{int}} = \text{dist}(E_*, \text{spec}(QIQ)) \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}.$$

Hence, if $\varepsilon_{\text{int}} < m_{\text{int}}^{(0)}$, then

$$m_{\text{int}} > 0.$$

Therefore the wrong-internal sector gap problem is reduced completely to a finite-dimensional Hermitian block calculation:

$$\boxed{\Delta_{\text{wrong-int}}^{(0)} = m_{\text{int}}^{(0)} = \text{dist}(E_*, \text{spec}(QI_0Q))},$$

and in the minimal $\mathbb{C}^3$ carrier simply

$$\boxed{\Delta_{\text{wrong-int}}^{(0)} = |E_L - E_*|}.$$

. Shell-point longitudinal–transverse splitting theorem

a. Goal. We now sharpen the wrong-internal gap criterion by extracting the longitudinal and transverse shell-point eigenvalues from the same effective symbol. This reduces the condition

$$m_{\text{int}}^{(0)} > 0$$

to a single explicit scalar inequality.

1. A. Effective shell-point assumptions

Let $G_* \neq 0$ be the selected shell point and define

$$\hat{n} := \frac{G_*}{|G_*|}, \quad q_* := |G_*|. \quad (366)$$

Assume the shell-point internal carrier is the minimal complex three-dimensional space

$$U_{\text{int}} = \mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_{\mathbb{C}}^{\perp}. \quad (367)$$

We impose two effective shell-point hypotheses:

1. **Protected transverse doublet.** There is an effective transverse $SO(2)$ symmetry acting on $\hat{n}_{\mathbb{C}}^{\perp}$, hence the shell-point Hamiltonian H_0 satisfies

$$H_0|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2. \quad (368)$$

2. **Longitudinal-only volumetric coupling.** The heavy volumetric mode σ couples only to the longitudinal displacement and, after integration out, modifies only the longitudinal block by the standard positive stiffening term

$$\frac{g^2 k^2}{M_{\sigma}^2 + c_{\sigma}^2 k^2}. \quad (369)$$

These are exactly the two structures already isolated in the TECT infrared analysis.

2. B. Effective shell-point symbol

Under effective isotropy, the quadratic elastic symbol has the Lamé form

$$D_{ij}(k) = \mu k^2 \delta_{ij} + (\lambda + \mu) k_i k_j = (\lambda + 2\mu) k^2 P_{ij}^L + \mu k^2 P_{ij}^T, \quad (370)$$

where

$$P_{ij}^L = \hat{k}_i \hat{k}_j, \quad P_{ij}^T = \delta_{ij} - \hat{k}_i \hat{k}_j. \quad (371)$$

After integrating out the heavy volumetric mode, the longitudinal block is shifted while the transverse block remains unchanged:

$$D_{ij}^{\text{eff}}(k) = \left[(\lambda + 2\mu) k^2 + \frac{g^2 k^2}{M_{\sigma}^2 + c_{\sigma}^2 k^2} \right] P_{ij}^L + \mu k^2 P_{ij}^T. \quad (372)$$

Evaluating at the selected shell point $k = G_*$, we obtain the shell-point operator

$$H_0 = E_L P_L + E_* P_T, \quad (373)$$

with

$$E_*(G_*) = \mu q_*^2, \quad (374)$$

$$E_L(G_*) = (\lambda + 2\mu) q_*^2 + \frac{g^2 q_*^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2}. \quad (375)$$

Here

$$P_L = \hat{n} \hat{n}^{\dagger}, \quad P_T = \mathbf{1}_3 - P_L. \quad (376)$$

3. C. Exact longitudinal–transverse splitting

Subtracting (374) from (375), we obtain

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right]. \quad (377)$$

Thus the shell-point symbol takes the explicit spectral form

$$H_0 = E_* \mathbf{1}_3 + (E_L - E_*) P_L. \quad (378)$$

Hence the transverse plane $\hat{n}_\mathbb{C}^\perp$ is a protected exact doublet of energy E_* , while the longitudinal line $\mathbb{C}\hat{n}$ has energy E_L .

[Shell-point longitudinal–transverse splitting theorem] Under the effective shell-point assumptions above, the shell-point operator on $\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_\mathbb{C}^\perp$ has the exact spectral decomposition

$$H_0 = E_L P_L + E_* P_T, \quad (379)$$

with

$$E_*(G_*) = \mu q_*^2, \quad E_L(G_*) = (\lambda + 2\mu)q_*^2 + \frac{g^2 q_*^2}{M_\sigma^2 + c_\sigma^2 q_*^2}, \quad (380)$$

and splitting

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right]. \quad (381)$$

Equation (370) gives the isotropic elastic decomposition into one longitudinal and two transverse sectors. Equation (372) shows that the heavy volumetric mode contributes only to the longitudinal block. Evaluating at $k = G_*$ yields (373), with eigenvalues (374) and (375). Subtracting them gives (381).

4. D. Wrong-internal gap in the minimal shell-point carrier

Take the selected protected doublet to be

$$U_D = \hat{n}_\mathbb{C}^\perp, \quad U_{\text{wrong}} = \mathbb{C}\hat{n}. \quad (382)$$

Then the wrong-internal spectral gap is precisely

$$m_{\text{int}}^{(0)} = |E_L - E_*|. \quad (383)$$

Therefore:

[Explicit wrong-internal gap formula] In the minimal shell-point carrier,

$$m_{\text{int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|. \quad (384)$$

Since $U_{\text{wrong}} = \mathbb{C}\hat{n}$ is one-dimensional and $H_0\hat{n} = E_L\hat{n}$, the complement spectrum is $\{E_L\}$, while the selected doublet energy is E_* . Hence

$$m_{\text{int}}^{(0)} = |E_L - E_*|.$$

Substituting (381) gives (384).

5. E. Sufficient conditions for nonvanishing

The exact condition for wrong-internal separation is simply

$$(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \neq 0. \quad (385)$$

This is already enough to guarantee

$$m_{\text{int}}^{(0)} > 0.$$

A stronger sufficient condition is positivity:

$$(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} > 0. \quad (386)$$

Under (386),

$$E_L > E_*, \quad m_{\text{int}}^{(0)} = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right] > 0. \quad (387)$$

a. Genericity remark. If $g \neq 0$, then the cancellation condition

$$(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} = 0$$

defines a codimension-one locus in the parameter space. Hence nonzero splitting is generic.

6. F. Stability under bounded shell-point perturbations

Let the shell-point operator be perturbed:

$$H = H_0 + \delta H. \quad (388)$$

Assume the perturbation respects the selected decomposition to leading order and satisfies

$$\|Q \delta H Q\| \leq \varepsilon_{\text{int}}, \quad Q = P_L. \quad (389)$$

Then by the finite-dimensional spectral stability theorem already proved for the wrong-internal block,

$$m_{\text{int}} \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}. \quad (390)$$

In particular, if

$$\varepsilon_{\text{int}} < m_{\text{int}}^{(0)}, \quad (391)$$

then

$$m_{\text{int}} > 0. \quad (392)$$

7. G. Honest logical status

What is now genuinely closed is the following conditional theorem chain:

$$\text{effective transverse } SO(2) \quad + \quad \text{longitudinal-only volumetric locking} \implies H_0 = E_L P_L + E_* P_T \implies m_{\text{int}}^{(0)} = |E_L - E_*|.$$

In the minimal shell-point carrier this becomes the explicit scalar formula

$$m_{\text{int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.$$

What remains open is not the structure of the gap, but only the actual microscopic verification that the shell-point problem of the full TECT linearization indeed falls into this effective shell-point class.

. **Summary**

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_{\mathbb{C}}^{\perp}, \quad P_L = \hat{n}\hat{n}^{\dagger}, \quad P_T = \mathbf{1}_3 - P_L, \quad q_* = |G_*|.$$

If the shell-point Hamiltonian has an effective transverse $SO(2)$, then

$$H_0|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2.$$

If, moreover, the heavy volumetric mode couples only to the longitudinal sector and has gap $M_{\sigma} > 0$, then the effective shell-point symbol is

$$H_0 = E_L P_L + E_* P_T,$$

with

$$E_*(G_*) = \mu q_*^2, \quad E_L(G_*) = (\lambda + 2\mu)q_*^2 + \frac{g^2 q_*^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2}.$$

Therefore the longitudinal–transverse splitting is

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right].$$

In the minimal shell-point carrier

$$U_D = \hat{n}_{\mathbb{C}}^{\perp}, \quad U_{\text{wrong}} = \mathbb{C}\hat{n},$$

the wrong-internal gap is exactly

$$m_{\text{int}}^{(0)} = |E_L - E_*| = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right|.$$

Hence the exact sufficient condition for wrong-internal separation is

$$(\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \neq 0.$$

A stronger sufficient condition is

$$(\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} > 0,$$

which implies

$$E_L > E_*, \quad m_{\text{int}}^{(0)} > 0.$$

Under a bounded perturbation $H = H_0 + \delta H$ with

$$\|Q \delta H Q\| \leq \varepsilon_{\text{int}}, \quad Q = P_L,$$

the gap is stable:

$$m_{\text{int}} \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}.$$

Thus the wrong-internal gap is reduced to an explicit shell-point scalar splitting formula.

. **Master explicit spectral inequality theorem**

a. Goal. We now combine all previously established explicit lower bounds and norm estimates into one master inequality guaranteeing full microscopic Dirac isolation.

1. A. Structural inputs already reduced to explicit quantities

Fix the selected shell pair $\pm G_*$ and write

$$q_* := |G_*|. \quad (393)$$

Assume the local shell-level Dirac block has already been realized, i.e.

$$H_D(p) = \Gamma_i(M_{\text{eff}}p)_i, \quad \det M_{\text{eff}} \neq 0, \quad (394)$$

with opposite-valley matching and no constant Dirac mass allowed by the residual low-energy symmetry.

We now collect the explicit remote-sector lower bounds.

a. (1) *Higher-shell diagonal bound.* For the BCC reciprocal arithmetic, the first shell-separation parameter is

$$\delta_{\text{shell}} = 2 \quad (395)$$

in the integer reciprocal convention, so the Brazovskii-type diagonal kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2 \quad (396)$$

gives

$$\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| = 4K - |r_*|. \quad (397)$$

(Equivalently, in the unit-shell normalization one has $\hat{\Delta}_{\text{high}}^{(0)} \geq K - |r_*|$.)

b. (2) *Heavy volumetric diagonal bound.* From longitudinal-only volumetric locking, the heavy scalar σ yields a positive longitudinal heavy scale. We write its induced remote lower bound as

$$\Delta_{\text{vol}}^{(0)} \geq c_{\text{vol}}M_\sigma, \quad c_{\text{vol}} > 0, \quad M_\sigma > 0. \quad (398)$$

c. (3) *Wrong-internal diagonal bound.* At the selected shell point, effective transverse $SO(2)$ gives an exact protected transverse doublet

$$H_0|_{\hat{n}_C^\perp} = E_* \mathbf{1}_2,$$

while longitudinal-only volumetric locking gives

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right].$$

Hence the wrong-internal gap is

$$\Delta_{\text{wrong-int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|. \quad (399)$$

d. (4) *Extra remote sectors.* Let

$$\Delta_{\text{extra},\rho}^{(0)} > 0 \quad (400)$$

be any explicit lower bound for all additional remote channels not covered by the three classes above.

2. B. Explicit diagonal benchmark

Define the global explicit diagonal remote benchmark

$$\Delta_{\text{bench},\rho} := \min \left\{ 4K - |r_*|, c_{\text{vol}}M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{extra},\rho}^{(0)} \right\}. \quad (401)$$

(If one works in the normalized unit-shell convention, replace $4K - |r_*|$ by $K - |r_*|$.)

By construction, every remote diagonal channel lies at least $\Delta_{\text{bench},\rho}$ away from zero.

3. C. Total remote mixing budget

Let $D_{\text{rem}}(p)$ be the corresponding block-diagonal benchmark operator on the full remote sector, and let

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p). \quad (402)$$

Define the total remote mixing budget

$$\eta_{\text{rem},\rho} := \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|. \quad (403)$$

This includes:

- all remote–remote off-diagonal mixing,
- all tail couplings,
- all bounded residual corrections to the diagonal benchmark.

By the previously proved tail-coupling estimate theorem, one may bound this norm explicitly by

$$\eta_{\text{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \quad r_a \geq 1. \quad (404)$$

4. D. Explicit remote-gap theorem

[Master explicit spectral inequality theorem] Assume:

1. the local shell-level Dirac block (394) exists and is nondegenerate;
2. little-group mismatch gives

$$V(0) = 0, \quad \|V(p)\| \leq c|p|; \quad (405)$$

3. a residual symmetry forbids all constant Dirac mass terms;
4. the explicit diagonal benchmark $\Delta_{\text{bench},\rho}$ is positive:

$$\Delta_{\text{bench},\rho} > 0; \quad (406)$$

5. the total remote mixing budget satisfies the master inequality

$$\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho}. \quad (407)$$

Then the full remote block has a uniform positive spectral gap on $N_\rho = \{|p| \leq \rho\}$:

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset, \quad |p| \leq \rho, \quad (408)$$

with

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0. \quad (409)$$

Consequently, for $|E| < \Delta_{\text{full},\rho}/2$,

$$\|(H_R^{\text{full}}(p) - E)^{-1}\| \leq \frac{2}{\Delta_{\text{full},\rho}}, \quad (410)$$

and the Schur-complement effective Hamiltonian satisfies

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}} p)_i + O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right). \quad (411)$$

Hence the full microscopic low-energy sector contains a stable massless Dirac cone.

By construction of the diagonal benchmark, every eigenvalue of $D_{\text{rem}}(p)$ has absolute value at least $\Delta_{\text{bench},\rho}$ for $|p| \leq \rho$. Hence for every Ψ in the full remote sector,

$$\|D_{\text{rem}}(p)\Psi\| \geq \Delta_{\text{bench},\rho}\|\Psi\|.$$

Using $H_R^{\text{full}} = D_{\text{rem}} + R_{\text{rem}}$, the reverse triangle inequality gives

$$\|H_R^{\text{full}}(p)\Psi\| \geq \|D_{\text{rem}}(p)\Psi\| - \|R_{\text{rem}}(p)\Psi\| \geq (\Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho})\|\Psi\|.$$

By (407), the right-hand side is positive, so

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset$$

with $\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho}$.

For $|E| < \Delta_{\text{full},\rho}/2$, the resolvent bound (410) follows immediately. Then, together with $V(0) = 0$ and $\|V(p)\| \leq c|p|$,

$$\Sigma(E, p) = V(p)(H_R^{\text{full}}(p) - E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right).$$

Therefore the effective Hamiltonian equals the local Dirac block plus a quadratic correction. Finally, the residual symmetry excludes any constant Dirac mass term, so the leading linear massless Dirac cone persists.

5. E. Strong explicit sufficient form

Using (404), a concrete sufficient inequality is

$$\sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} < \min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{extra},\rho}^{(0)} \right\}. \quad (412)$$

Under this inequality, the full microscopic Dirac isolation follows immediately.

a. Normalized-shell version. If all shell radii are normalized so that $q_*^2 = 1$, replace the first benchmark entry $4K - |r_*|$ by $K - |r_*|$.

6. F. Exact logical status

What is now mathematically closed is:

higher-shell arithmetic+heavy volumetric gap+wrong-internal splitting+tail-coupling norm estimate $\implies$ one explicit master ine

The remaining open work is no longer conceptual: it is the actual coefficient estimation of

$$\Delta_{\text{extra},\rho}^{(0)}, \quad K_a, \tilde{K}_a, M_a,$$

inside the final microscopic TECT operator.

Thus the full low-energy Dirac-isolation problem has been reduced to a finite explicit spectral inequality problem.

. Summary

$$\Delta_{\text{high}}^{(0)} \geq 4K - |r_*|, \quad \Delta_{\text{vol}}^{(0)} \geq c_{\text{vol}} M_\sigma, \quad \Delta_{\text{wrong-int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,$$

together with

$$\Delta_{\text{extra},\rho}^{(0)} > 0.$$

Define the explicit remote diagonal benchmark

$$\Delta_{\text{bench},\rho} := \min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{extra},\rho}^{(0)} \right\}.$$

Let

$$H_R^{\text{full}}(p) = D_{\text{rem}}(p) + R_{\text{rem}}(p), \quad \eta_{\text{rem},\rho} = \sup_{|p| \leq \rho} \|R_{\text{rem}}(p)\|.$$

By the tail-coupling estimate theorem,

$$\eta_{\text{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.$$

If

$$\Delta_{\text{bench},\rho} > \eta_{\text{rem},\rho},$$

then the full remote block has a uniform positive gap

$$\text{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \emptyset, \quad \Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0.$$

If also

$$V(0) = 0, \quad \|V(p)\| \leq c|p|,$$

and a residual symmetry forbids constant Dirac masses, then

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}} p)_i + O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right),$$

so the full microscopic low-energy sector contains a stable massless Dirac cone.

Therefore the entire spectral-isolation problem is reduced to the single explicit inequality

$$\min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{extra},\rho}^{(0)} \right\} > \eta_{\text{rem},\rho}.$$

. Extra-sector coercivity theorem

a. Goal. We now eliminate the last abstract slot in the master spectral inequality, namely the quantity

$$\Delta_{\text{extra},\rho}^{(0)},$$

by deriving it from coercivity of the quadratic form on the remaining extra remote sector.

1. A. Extra remote sector and quadratic form

Let the full tail sector be decomposed as

$$\mathcal{H}_{\text{tail}} = \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}, \quad (413)$$

and let

$$H_{\text{extra}}^{(0)}(p) \quad (414)$$

denote the unperturbed block acting on $\mathcal{H}_{\text{extra}}$.

Assume $H_{\text{extra}}^{(0)}(p)$ is self-adjoint for all $|p| \leq \rho$, and define the corresponding quadratic form

$$\mathfrak{q}_{\text{extra},p}[\psi] := \langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle, \quad \psi \in \mathcal{D}_{\text{extra}}, \quad (415)$$

on a common dense form domain $\mathcal{D}_{\text{extra}}$.

2. B. Uniform coercivity hypothesis

We assume the extra sector is uniformly coercive on the low-momentum neighborhood

$$N_\rho := \{p : |p| \leq \rho\}, \quad (416)$$

in the sense that there exists $c_{\text{extra},\rho} > 0$ such that

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq c_{\text{extra},\rho} \|\psi\|^2 \quad \forall \psi \in \mathcal{D}_{\text{extra}}, \quad \forall |p| \leq \rho. \quad (417)$$

This is the natural positivity statement for all additional remote channels not covered by the higher-shell, volumetric, or wrong-internal classifications.

3. C. Coercivity implies spectral gap

[Extra-sector coercivity theorem] Assume (417). Then the unperturbed extra block has a uniform positive spectral lower bound:

$$\text{spec}(H_{\text{extra}}^{(0)}(p)) \subset [c_{\text{extra},\rho}, \infty) \quad (|p| \leq \rho). \quad (418)$$

In particular,

$$\Delta_{\text{extra},\rho}^{(0)} := \inf_{|p| \leq \rho} \text{dist}(0, \text{spec}(H_{\text{extra}}^{(0)}(p))) \geq c_{\text{extra},\rho} > 0. \quad (419)$$

Fix $|p| \leq \rho$. By the spectral theorem (or min-max principle), a self-adjoint operator satisfies

$$\inf \text{spec}(H_{\text{extra}}^{(0)}(p)) = \inf_{\psi \neq 0} \frac{\langle \psi, H_{\text{extra}}^{(0)}(p) \psi \rangle}{\|\psi\|^2}.$$

By (417), the Rayleigh quotient is bounded below by $c_{\text{extra},\rho}$. Hence

$$\inf \text{spec}(H_{\text{extra}}^{(0)}(p)) \geq c_{\text{extra},\rho}.$$

This proves (418). Taking the infimum over $|p| \leq \rho$ yields (419).

4. D. Two-sided coercivity variant

In some models the extra block may not be strictly positive but only bounded away from zero on both sides. Then one assumes the stronger absolute coercivity condition

$$\|H_{\text{extra}}^{(0)}(p)\psi\| \geq c_{\text{extra},\rho} \|\psi\| \quad \forall \psi \in \mathcal{D}(H_{\text{extra}}^{(0)}(p)), \quad |p| \leq \rho. \quad (420)$$

This implies

$$\text{spec}(H_{\text{extra}}^{(0)}(p)) \cap (-c_{\text{extra},\rho}, c_{\text{extra},\rho}) = \emptyset. \quad (421)$$

In the present TECT application, however, the extra sector is naturally treated as part of the remote positive-energy complement, so the one-sided coercivity form (417) is the relevant one.

5. E. Stability under bounded extra-sector corrections

Let the actual extra block be

$$H_{\text{extra}}(p) = H_{\text{extra}}^{(0)}(p) + \delta H_{\text{extra}}(p), \quad (422)$$

with bounded correction satisfying

$$\varepsilon_{\text{extra},\rho} := \sup_{|p| \leq \rho} \|\delta H_{\text{extra}}(p)\| < c_{\text{extra},\rho}. \quad (423)$$

[Extra-sector gap stability] Under (417) and (423),

$$\text{spec}(H_{\text{extra}}(p)) \subset [c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho}, \infty) \quad (|p| \leq \rho), \quad (424)$$

hence

$$\Delta_{\text{extra},\rho} \geq c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho} > 0. \quad (425)$$

For every ψ ,

$$\langle \psi, H_{\text{extra}}(p)\psi \rangle = \langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle + \langle \psi, \delta H_{\text{extra}}(p)\psi \rangle.$$

By coercivity,

$$\langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle \geq c_{\text{extra},\rho} \|\psi\|^2.$$

By boundedness of the perturbation,

$$|\langle \psi, \delta H_{\text{extra}}(p)\psi \rangle| \leq \|\delta H_{\text{extra}}(p)\| \|\psi\|^2 \leq \varepsilon_{\text{extra},\rho} \|\psi\|^2.$$

Therefore

$$\langle \psi, H_{\text{extra}}(p)\psi \rangle \geq (c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho}) \|\psi\|^2.$$

Applying the min-max principle again proves (424) and (425).

6. F. Replacement of the last unknown slot in the master inequality

The previous master explicit spectral inequality theorem contained the benchmark quantity

$$\Delta_{\text{extra},\rho}^{(0)}. \quad (426)$$

The present theorem identifies a concrete sufficient realization:

$$\Delta_{\text{extra},\rho}^{(0)} \geq c_{\text{extra},\rho}, \quad (427)$$

and, after bounded extra-sector perturbation,

$$\Delta_{\text{extra},\rho} \geq c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho}. \quad (428)$$

Therefore the master benchmark may be replaced by

$$\Delta_{\text{bench},\rho}^{\text{coer}} := \min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho} \right\}. \quad (429)$$

7. G. Coercive master inequality

[Coercive master explicit spectral inequality] Assume:

1. the local shell Dirac block exists and is nondegenerate;
2. little-group mismatch gives $V(0) = 0$ and $\|V(p)\| \leq c|p|$;
3. residual symmetry forbids constant Dirac masses;

4. the extra remote sector is coercive with constant $c_{\text{extra},\rho} > 0$;
5. the bounded extra-sector correction satisfies $\varepsilon_{\text{extra},\rho} < c_{\text{extra},\rho}$;
6. the total remote mixing budget satisfies

$$\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{coer}}. \quad (430)$$

Then the full remote block has a uniform positive gap

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho}^{\text{coer}} - \eta_{\text{rem},\rho} > 0, \quad (431)$$

and consequently

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right). \quad (432)$$

Hence the full microscopic low-energy sector contains a stable massless Dirac cone.

The previous theorem replaces the extra-sector slot by a positive explicit lower bound $c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho}$. Substituting this into the master benchmark and applying the previously proved master explicit spectral inequality theorem yields the result.

a. Exact logical status. What is now closed is:

$$\text{extra-sector gap} \Leftarrow \text{coercivity of the extra quadratic form}$$

and therefore the master inequality no longer needs a separate abstract $\Delta_{\text{extra},\rho}^{(0)}$ input. The remaining open work is the actual extraction of the coercivity constant $c_{\text{extra},\rho}$ from the final microscopic TECT quadratic form.

. Summary

$$\mathcal{H}_{\text{tail}} = \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{wrong-int}} \oplus \mathcal{H}_{\text{extra}}, \quad \mathfrak{q}_{\text{extra},p}[\psi] = \langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle.$$

If the extra-sector quadratic form is uniformly coercive on $N_\rho = \{|p| \leq \rho\}$,

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq c_{\text{extra},\rho} \|\psi\|^2, \quad c_{\text{extra},\rho} > 0,$$

then

$$\text{spec}(H_{\text{extra}}^{(0)}(p)) \subset [c_{\text{extra},\rho}, \infty), \quad \Delta_{\text{extra},\rho}^{(0)} \geq c_{\text{extra},\rho} > 0.$$

If

$$H_{\text{extra}}(p) = H_{\text{extra}}^{(0)}(p) + \delta H_{\text{extra}}(p), \quad \varepsilon_{\text{extra},\rho} = \sup_{|p| \leq \rho} \|\delta H_{\text{extra}}(p)\| < c_{\text{extra},\rho},$$

then the extra-sector gap is stable:

$$\Delta_{\text{extra},\rho} \geq c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho} > 0.$$

Hence the abstract benchmark slot $\Delta_{\text{extra},\rho}^{(0)}$ in the master spectral inequality can be replaced by

$$c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho}.$$

The fully explicit benchmark becomes

$$\Delta_{\text{bench},\rho}^{\text{coer}} := \min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho} \right\}.$$

If

$$\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{coer}},$$

then the full remote block has a uniform positive gap

$$\Delta_{\text{full},\rho} \geq \Delta_{\text{bench},\rho}^{\text{coer}} - \eta_{\text{rem},\rho} > 0,$$

and therefore

$$H_{\text{eff}}(E, p) = \Gamma_i(M_{\text{eff}}p)_i + O\left(\frac{|p|^2}{\Delta_{\text{full},\rho}}\right).$$

Thus the extra-sector slot in the master inequality is reduced to coercivity of the extra quadratic form.

. Principal-symbol coercivity theorem

a. Goal. We now derive the extra-sector coercivity constant $c_{\text{extra},\rho}$ directly from the principal symbol of the extra-sector operator, together with a quantitative lower-order perturbation bound.

1. A. Extra-sector operator and quadratic form

Let

$$H_{\text{extra}}^{(0)}(p) = P_{\text{pr}}(p, D) + B_{\text{lo}}(p) \quad (433)$$

be the unperturbed extra-sector operator on the low-momentum neighborhood

$$N_\rho := \{p : |p| \leq \rho\}. \quad (434)$$

Assume $P_{\text{pr}}(p, D)$ is a formally self-adjoint principal differential/pseudodifferential operator of order $m \geq 1$, and $B_{\text{lo}}(p)$ is a lower-order symmetric remainder.

Let the associated quadratic form be

$$\mathfrak{q}_{\text{extra},p}[\psi] = \langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle = \mathfrak{q}_{\text{pr},p}[\psi] + \mathfrak{b}_{\text{lo},p}[\psi]. \quad (435)$$

2. B. Uniform principal-symbol positivity

Let $s_{\text{pr}}(p, \xi)$ denote the Hermitian principal symbol of $P_{\text{pr}}(p, D)$ on the extra-sector fiber. Assume there exists $\kappa_{\text{extra}} > 0$ such that

$$\langle u, s_{\text{pr}}(p, \xi)u \rangle \geq \kappa_{\text{extra}} \langle \xi \rangle^m \|u\|^2 \quad \forall u, \forall |p| \leq \rho, \forall \xi. \quad (436)$$

This is the uniform elliptic/coercive positivity of the principal part on the extra sector.

3. C. Spectral exclusion of the low shell inside the extra sector

Because the extra sector is, by definition, orthogonal to the selected low-energy shell carrier, one has a sectorwise Poincaré/Fourier lower bound. Assume there exists $\Lambda_{\text{extra},\rho} > 0$ such that

$$\|\psi\|_{H^{m/2}}^2 \geq \Lambda_{\text{extra},\rho} \|\psi\|^2 \quad \forall \psi \in \mathcal{D}_{\text{extra}}, \quad |p| \leq \rho. \quad (437)$$

This constant encodes the fact that the extra sector does not contain the selected soft shell modes.

a. Interpretation. In concrete TECT truncations, $\Lambda_{\text{extra},\rho}$ can arise from:

- shell-radius separation (higher-shell exclusion),
- heavy longitudinal/volumetric frequency exclusion,
- internal-sector orthogonality constraints,
- or a direct finite-dimensional spectral exclusion in the Bloch basis.

4. D. Lower-order relative bound

Assume the lower-order remainder is form-bounded relative to the $H^{m/2}$ -norm:

$$|\mathfrak{b}_{\text{lo},p}[\psi]| \leq \alpha_{\text{extra}} \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 + \beta_{\text{extra},\rho} \|\psi\|^2, \quad (438)$$

with

$$0 \leq \alpha_{\text{extra}} < 1, \quad \beta_{\text{extra},\rho} \geq 0, \quad |p| \leq \rho. \quad (439)$$

This is the standard small relative-boundedness condition needed to preserve coercivity.

5. E. Principal-form lower bound

[Principal coercive form bound] Under (436), there exists a constant $c_{\text{pr}} > 0$ (which may be taken as κ_{extra} after normalization) such that

$$\mathfrak{q}_{\text{pr},p}[\psi] \geq \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 \quad \forall \psi \in \mathcal{D}_{\text{extra}}, \quad |p| \leq \rho. \quad (440)$$

This is the standard Gårding/elliptic lower-bound consequence of the uniform positivity of the principal symbol (436). Because the symbol is Hermitian and uniformly positive, the quadratic form of the principal operator dominates the $H^{m/2}$ -norm up to normalization.

6. F. Coercivity constant from principal positivity

[Principal-symbol coercivity theorem] Assume (436), (437), and (438). Then the extra-sector quadratic form satisfies

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq \left((1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} \right) \|\psi\|^2 \quad \forall \psi \in \mathcal{D}_{\text{extra}}, \quad |p| \leq \rho. \quad (441)$$

In particular, if

$$(1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} > \beta_{\text{extra},\rho}, \quad (442)$$

then the extra sector is uniformly coercive with explicit constant

$$c_{\text{extra},\rho} = (1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} > 0. \quad (443)$$

From (435),

$$\mathfrak{q}_{\text{extra},p}[\psi] = \mathfrak{q}_{\text{pr},p}[\psi] + \mathfrak{b}_{\text{lo},p}[\psi].$$

Using (440) and (438),

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 - \alpha_{\text{extra}} \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 - \beta_{\text{extra},\rho} \|\psi\|^2.$$

Hence

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq (1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 - \beta_{\text{extra},\rho} \|\psi\|^2.$$

Now apply the sectorwise Poincaré/Fourier exclusion bound (437):

$$\|\psi\|_{H^{m/2}}^2 \geq \Lambda_{\text{extra},\rho} \|\psi\|^2.$$

Therefore

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq \left((1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} \right) \|\psi\|^2.$$

This proves (441), and under (442) the explicit coercivity constant is (443).

7. G. Spectral consequence

By the extra-sector coercivity theorem already proved, (443) implies

$$\Delta_{\text{extra},\rho}^{(0)} \geq c_{\text{extra},\rho} = (1 - \alpha_{\text{extra}})\kappa_{\text{extra}}\Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} > 0. \quad (444)$$

Thus the abstract extra-sector slot in the master inequality is replaced by a completely explicit principal-symbol expression.

8. H. TECT specialization

In the TECT setting, the motivation for the decomposition

$$H_{\text{extra}}^{(0)} = P_{\text{pr}} + B_{\text{lo}}$$

is as follows.

1. The exact BCC linearized Hessian $D_{ij}(k)$ exists and decomposes exactly into an isotropic Lamé-type part plus a cubic anisotropy remainder $\Delta D_{ij}(k)$. Thus the natural principal part is the isotropic positive sector, while $\Delta D_{ij}(k)$ is a lower-order anisotropic remainder in the infrared analysis.
2. Integrating out the heavy volumetric mode yields a controlled local $1/M_\sigma^2$ expansion with explicit remainder bounds. Therefore the heavy-induced corrections naturally enter the lower-order perturbation budget $\beta_{\text{extra},\rho}$.
3. The extra sector is by definition orthogonal to the selected soft shell carrier, so its Fourier support or Bloch labels exclude the selected low shell, which is exactly the source of the constant $\Lambda_{\text{extra},\rho} > 0$.

Hence the theorem fits the actual TECT analytic architecture.

9. I. Master replacement

The master benchmark from the previous step was

$$\Delta_{\text{bench},\rho}^{\text{coer}} := \min \left\{ 4K - |r_*|, c_{\text{vol}}M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho} \right\}. \quad (445)$$

Using (443), this becomes the fully analytic benchmark

$$\Delta_{\text{bench},\rho}^{\text{pr}} := \min \left\{ 4K - |r_*|, c_{\text{vol}}M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, (1 - \alpha_{\text{extra}})\kappa_{\text{extra}}\Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} \right\}. \quad (446)$$

Hence the entire full remote-gap problem reduces to the explicit inequality

$$\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{pr}}. \quad (447)$$

a. Exact logical status. What is now proved is:

principal-symbol positivity + sectorwise Fourier/Bloch exclusion + small lower-order remainder $\implies c_{\text{extra},\rho} > 0$.

So the extra-sector slot is no longer abstract. The only remaining open work is the actual extraction of the numerical constants

$$\kappa_{\text{extra}}, \quad \Lambda_{\text{extra},\rho}, \quad \alpha_{\text{extra}}, \quad \beta_{\text{extra},\rho}$$

from the final microscopic TECT operator.

. **Summary**

$$H_{\text{extra}}^{(0)}(p) = P_{\text{pr}}(p, D) + B_{\text{lo}}(p), \quad \mathfrak{q}_{\text{extra},p}[\psi] = \langle \psi, H_{\text{extra}}^{(0)}(p)\psi \rangle.$$

Assume:

$$\langle u, s_{\text{pr}}(p, \xi)u \rangle \geq \kappa_{\text{extra}} \langle \xi \rangle^m \|u\|^2,$$

$$\|\psi\|_{H^{m/2}}^2 \geq \Lambda_{\text{extra},\rho} \|\psi\|^2,$$

$$|\mathfrak{b}_{\text{lo},p}[\psi]| \leq \alpha_{\text{extra}} \kappa_{\text{extra}} \|\psi\|_{H^{m/2}}^2 + \beta_{\text{extra},\rho} \|\psi\|^2, \quad 0 \leq \alpha_{\text{extra}} < 1.$$

Then

$$\mathfrak{q}_{\text{extra},p}[\psi] \geq \left((1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} \right) \|\psi\|^2.$$

Therefore, if

$$(1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} > \beta_{\text{extra},\rho},$$

the extra sector is uniformly coercive with explicit constant

$$c_{\text{extra},\rho} = (1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} > 0,$$

hence

$$\Delta_{\text{extra},\rho}^{(0)} \geq c_{\text{extra},\rho} > 0.$$

Substituting into the master benchmark gives

$$\Delta_{\text{bench},\rho}^{\text{pr}} := \min \left\{ 4K - |r_*|, c_{\text{vol}} M_{\sigma}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_{\sigma}^2 + c_{\sigma}^2 q_*^2} \right|, (1 - \alpha_{\text{extra}}) \kappa_{\text{extra}} \Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho} \right\}.$$

Hence the full remote-gap condition is reduced to the explicit analytic inequality

$$\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{pr}}.$$

. **Bloch-exclusion constant theorem**

a. Goal. We now identify the abstract exclusion constant $\Lambda_{\text{extra},\rho}$ appearing in the principal-symbol coercivity theorem with an explicit Bloch/Fourier shell-exclusion constant.

1. *A. Bloch harmonic representation near the selected shell point*

Let the periodic BCC background define a reciprocal lattice $\mathcal{G}$, and let the Bloch fiber at quasi-momentum k act on periodic functions

$$u_k(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x}, \quad \delta \phi_k(x) = \sum_{G \in \mathcal{G}} c_G e^{i(k+G) \cdot x}. \quad (448)$$

Fix a selected shell point $G_* \in S_{\text{BCC}}$ and set

$$k = G_* + p, \quad |p| \leq \rho. \quad (449)$$

The exact Bloch operator is diagonal in the bare part:

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G) \delta_{G,G'} + \widehat{V}(G-G'), \quad (450)$$

with bare dispersion

$$h_0(q) = r + \kappa |q|^2 + \lambda |q|^4, \quad q_*^2 = -\frac{\kappa}{2\lambda} \quad (\kappa < 0, \lambda > 0). \quad (451)$$

2. B. Selected shell carrier and extra sector

Let $S_* \subset \mathcal{G}$ denote the finite set of reciprocal harmonics retained in the local shell carrier, chosen so that

$$h_0(G_* + p + G)$$

is near the shell minimum for $G \in S_*$. This is exactly the shell-projected Bloch truncation used in the local valley theorem.

Define the extra Bloch sector as the orthogonal complement of the selected shell carrier:

$$\mathcal{H}_{\text{extra}}(p) := \left\{ u = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x} : c_G = 0 \text{ for all } G \in S_* \right\}. \quad (452)$$

Thus every extra-sector state is supported only on excluded harmonics $G \notin S_*$.

3. C. Shell-distance exclusion constant

Define the shell-distance exclusion constant

$$\delta_{\text{Bloch}, \rho} := \inf_{\substack{|p| \leq \rho \\ G \notin S_*}} \left| |G_* + p + G|^2 - q_*^2 \right|. \quad (453)$$

[Positivity of the Bloch exclusion constant] Assume S_* contains all reciprocal harmonics whose bare radius lies on the selected soft shell

$$|G_* + G|^2 = q_*^2$$

inside the chosen local truncation. Then, for sufficiently small ρ ,

$$\delta_{\text{Bloch}, \rho} > 0. \quad (454)$$

At $p = 0$, the excluded set $G \notin S_*$ contains no harmonic with shell radius q_*^2 by construction, so

$$\delta_{\text{Bloch}, 0} = \inf_{G \notin S_*} \left| |G_* + G|^2 - q_*^2 \right| > 0$$

because the reciprocal lattice is discrete and S_* is finite. The map

$$(p, G) \mapsto \left| |G_* + p + G|^2 - q_*^2 \right|$$

is continuous in p and the excluded harmonic set is discrete, so the positive minimum persists for sufficiently small $|p| \leq \rho$.

4. D. Explicit Bloch lower bound

For any $u \in \mathcal{H}_{\text{extra}}(p)$,

$$u(x) = \sum_{G \notin S_*} c_G e^{iG \cdot x}. \quad (455)$$

Then

$$\sum_{G \notin S_*} (|G_* + p + G|^2 - q_*^2)^2 |c_G|^2 \geq \delta_{\text{Bloch}, \rho}^2 \sum_{G \notin S_*} |c_G|^2. \quad (456)$$

Equivalently,

$$\left\| (|D + G_* + p|^2 - q_*^2) u \right\|_{L^2}^2 \geq \delta_{\text{Bloch}, \rho}^2 \|u\|_{L^2}^2. \quad (457)$$

Thus the natural extra-sector exclusion constant in the Brazovskii shell geometry is

$$\Lambda_{\text{extra}, \rho}^{\text{Bloch}} = \delta_{\text{Bloch}, \rho}^2. \quad (458)$$

5. *E. Principal-symbol coercivity from shell exclusion*

Consider the extra-sector principal part

$$P_{\text{pr}}(p) = K(|D + G_* + p|^2 - q_*^2)^2, \quad K > 0. \quad (459)$$

Then for every $u \in \mathcal{H}_{\text{extra}}(p)$,

$$\langle u, P_{\text{pr}}(p)u \rangle = K \sum_{G \notin S_*} (|G_* + p + G|^2 - q_*^2)^2 |c_G|^2 \geq K \delta_{\text{Bloch}, \rho}^2 \|u\|_{L^2}^2. \quad (460)$$

Hence the principal-symbol positivity constant and exclusion constant may be identified as

$$\kappa_{\text{extra}} = K, \quad \Lambda_{\text{extra}, \rho} = \delta_{\text{Bloch}, \rho}^2. \quad (461)$$

Substituting into the previously proved principal-symbol coercivity theorem yields

$$c_{\text{extra}, \rho} = (1 - \alpha_{\text{extra}}) K \delta_{\text{Bloch}, \rho}^2 - \beta_{\text{extra}, \rho}, \quad (462)$$

provided

$$(1 - \alpha_{\text{extra}}) K \delta_{\text{Bloch}, \rho}^2 > \beta_{\text{extra}, \rho}. \quad (463)$$

6. *F. Explicit BCC first-shell estimate*

We now specialize to the normalized BCC first-shell convention

$$q_*^2 = 1, \quad (464)$$

and assume S_* contains all first-shell harmonics. Then every excluded harmonic $Q := G_* + G$ satisfies either

$$|Q|^2 = 0 \quad (465)$$

or

$$|Q|^2 \geq 2, \quad (466)$$

because the next BCC shell has normalized squared radius 2.

a. Case 1: $Q = 0$. Then

$$||Q + p|^2 - 1| = ||p|^2 - 1| \geq 1 - \rho^2. \quad (467)$$

b. Case 2: $|Q|^2 \geq 2$. Then $|Q| \geq \sqrt{2}$, so

$$\begin{aligned} ||Q + p|^2 - 1| &= ||Q|^2 - 1 + 2Q \cdot p + |p|^2| \\ &\geq (|Q|^2 - 1) - 2|Q|\rho - \rho^2 \\ &\geq 1 - 2\sqrt{2}\rho - \rho^2. \end{aligned} \quad (468)$$

Therefore, for all excluded harmonics,

$$\delta_{\text{Bloch}, \rho} \geq \min\{1 - \rho^2, 1 - 2\sqrt{2}\rho - \rho^2\} = 1 - 2\sqrt{2}\rho - \rho^2. \quad (469)$$

In particular, if

$$0 < \rho < \sqrt{3} - \sqrt{2}, \quad (470)$$

then

$$\delta_{\text{Bloch}, \rho} > 0. \quad (471)$$

[Explicit BCC Bloch-exclusion coercivity] Under (470),

$$c_{\text{extra}, \rho} \geq (1 - \alpha_{\text{extra}}) K (1 - 2\sqrt{2}\rho - \rho^2)^2 - \beta_{\text{extra}, \rho}. \quad (472)$$

Hence a sufficient explicit positivity condition is

$$(1 - \alpha_{\text{extra}}) K (1 - 2\sqrt{2}\rho - \rho^2)^2 > \beta_{\text{extra}, \rho}. \quad (473)$$

7. *G. Master inequality replacement*

The master explicit benchmark from the previous step contained the extra-sector slot

$$(1 - \alpha_{\text{extra}})\kappa_{\text{extra}}\Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho}.$$

Using (461), this becomes

$$(1 - \alpha_{\text{extra}})K\delta_{\text{Bloch},\rho}^2 - \beta_{\text{extra},\rho}. \quad (474)$$

Thus the fully analytic benchmark is

$$\Delta_{\text{bench},\rho}^{\text{Bloch}} := \min \left\{ 4K - |r_*|, c_{\text{vol}}M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, (1 - \alpha_{\text{extra}})K\delta_{\text{Bloch},\rho}^2 - \beta_{\text{extra},\rho} \right\}. \quad (475)$$

Hence the full remote-gap condition is reduced to the explicit inequality

$$\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{Bloch}}. \quad (476)$$

a. Exact logical status. What is now proved is:

$$\text{Bloch harmonic exclusion from the selected shell carrier} \implies \Lambda_{\text{extra},\rho} = \delta_{\text{Bloch},\rho}^2.$$

So the extra-sector coercivity constant is no longer abstract. In the normalized BCC first-shell setting, it admits the explicit lower bound

$$\delta_{\text{Bloch},\rho} \geq 1 - 2\sqrt{2}\rho - \rho^2.$$

The remaining work is purely coefficient-level: estimate α_{extra} , $\beta_{\text{extra},\rho}$, and $\eta_{\text{rem},\rho}$ in the final microscopic TECT operator.

. Summary

$$u_k(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x}, \quad \delta\phi_k(x) = \sum_{G \in \mathcal{G}} c_G e^{i(k+G) \cdot x}, \quad k = G_* + p.$$

Let $S_* \subset \mathcal{G}$ be the selected finite shell carrier, and define the extra Bloch sector by

$$\mathcal{H}_{\text{extra}}(p) = \left\{ u = \sum c_G e^{iG \cdot x} : c_G = 0 \text{ for } G \in S_* \right\}.$$

Define the shell-distance exclusion constant

$$\delta_{\text{Bloch},\rho} := \inf_{\substack{|p| \leq \rho \\ G \notin S_*}} \left| |G_* + p + G|^2 - q_*^2 \right|.$$

If S_* contains all selected soft-shell harmonics, then

$$\delta_{\text{Bloch},\rho} > 0$$

for sufficiently small ρ .

For every $u \in \mathcal{H}_{\text{extra}}(p)$,

$$\sum_{G \notin S_*} (|G_* + p + G|^2 - q_*^2)^2 |c_G|^2 \geq \delta_{\text{Bloch},\rho}^2 \sum_{G \notin S_*} |c_G|^2.$$

Hence for the Brazovskii principal part

$$P_{\text{pr}}(p) = K \left(|D + G_* + p|^2 - q_*^2 \right)^2,$$

one has

$$\langle u, P_{\text{pr}}(p)u \rangle \geq K \delta_{\text{Bloch},\rho}^2 \|u\|_{L^2}^2.$$

Therefore

$$\kappa_{\text{extra}} = K, \quad \Lambda_{\text{extra},\rho} = \delta_{\text{Bloch},\rho}^2,$$

and the extra-sector coercivity constant becomes

$$c_{\text{extra},\rho} = (1 - \alpha_{\text{extra}}) K \delta_{\text{Bloch},\rho}^2 - \beta_{\text{extra},\rho}.$$

In the normalized BCC first-shell convention $q_*^2 = 1$, if S_* contains all first-shell harmonics, then every excluded harmonic satisfies either $|Q|^2 = 0$ or $|Q|^2 \geq 2$, so

$$\delta_{\text{Bloch},\rho} \geq 1 - 2\sqrt{2}\rho - \rho^2.$$

Thus for

$$0 < \rho < \sqrt{3} - \sqrt{2},$$

one obtains the explicit bound

$$c_{\text{extra},\rho} \geq (1 - \alpha_{\text{extra}}) K (1 - 2\sqrt{2}\rho - \rho^2)^2 - \beta_{\text{extra},\rho}.$$

So the master explicit benchmark becomes

$$\Delta_{\text{bench},\rho}^{\text{Bloch}} := \min \left\{ 4K - |r_*|, c_{\text{vol}} M_\sigma, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, (1 - \alpha_{\text{extra}}) K \delta_{\text{Bloch},\rho}^2 - \beta_{\text{extra},\rho} \right\},$$

and the full remote-gap condition is

$$\boxed{\eta_{\text{rem},\rho} < \Delta_{\text{bench},\rho}^{\text{Bloch}}}.$$

. Class II microscopic PDE: coupled UV system and effective reduced equation

a. Goal. We now write the actual PDE system generated by the Class II mediated completion. The logic is:

minimal triplet Brazovskii PDE + Class II auxiliary mediator $\implies$ coupled UV PDE $\implies$ effective shell-enriched PD

1. A. Minimal microscopic condensate PDE already fixed

The minimal microscopic TECT field is a complex triplet

$$\Psi(x, t) \in \mathbb{C}^3. \quad (477)$$

Within the minimal local, isotropic, $U(3)$ -invariant, second-order-in-time, fourth-order-in-space truncation, the microscopic action is

$$S_0[\Psi] = \int dt d^3x \left[\frac{1}{2} \partial_t \Psi^\dagger \partial_t \Psi - \frac{\mu^2}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi - \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - \frac{g}{4} (\Psi^\dagger \Psi)^2 \right]. \quad (478)$$

Its Euler–Lagrange equation is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi. \quad (479)$$

This is the already-established minimal microscopic TECT PDE.

2. B. Why Class II must be added

The shell-level Dirac reduction requires enriched microscopic tensors N_{ij}^a entering a first-order shell operator of the form

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a. \quad (480)$$

The minimal triplet Brazovskii PDE (479) fixes the shell instability and the internal CP^2 geometry, but does not by itself yet produce a fully explicit microscopic source for (480). This is precisely why the Class II mediated completion was introduced.

3. C. Class II UV completion: mediator field

Introduce a heavy auxiliary internal vector mediator

$$X_i^a(x), \quad i = 1, 2, 3, \quad a = 1, 2, 3, \quad (481)$$

and consider the UV-completed action

$$S_{\text{UV}}^{(II)}[\Psi, P, \psi, X] = S_0[\Psi] + \int dt d^3x \left[\frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi] \right]. \quad (482)$$

Here:

$$J_i^a[\Psi] := (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi), \quad (483)$$

and

$$K_i^a[P, \psi] \sim (T^a \partial_i P) \mathcal{O}[\psi]. \quad (484)$$

At the linearized shell-relevant level, one may take

$$K_i^a[P, \psi] \sim \partial_i \delta P^a \psi. \quad (485)$$

The exact choice of $\mathcal{O}[\psi]$ depends on the reduced test-mode sector, but this does not affect the basic Euler–Lagrange structure below.

4. D. Euler–Lagrange equations of the coupled Class II system

We now vary (482).

a. (1) *Equation for the heavy mediator X_i^a .* Since X_i^a is auxiliary (no derivatives on X), its equation is algebraic:

$$\frac{\delta S_{\text{UV}}^{(II)}}{\delta X_i^a} = M_X^2 X_i^a + \alpha J_i^a[\Psi] + \beta K_i^a[P, \psi] = 0. \quad (486)$$

Hence

$$X_i^a = -\frac{\alpha}{M_X^2} J_i^a[\Psi] - \frac{\beta}{M_X^2} K_i^a[P, \psi]. \quad (487)$$

b. (2) *Equation for Ψ .* Varying with respect to $\Psi^\dagger$ gives

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{S}_\Psi^{(II)}[X, \Psi, P, \psi], \quad (488)$$

where the Class II source is

$$\mathcal{S}_\Psi^{(II)} = -\alpha \partial_i (X_i^a T^a \Psi) + \alpha X_i^a T^a \partial_i \Psi + \dots. \quad (489)$$

The omitted terms come from the precise symmetrization convention of $J_i^a[\Psi]$; they are all bilinear in X and Ψ , with one spatial derivative.

c. (3) *Equation for the projector/tangent fluctuation.* If P (or equivalently its tangent fluctuation $\delta P^a \sim \xi^a$) is treated as an explicit dynamical variable, then varying with respect to P gives

$$\mathcal{L}_P[\delta P] + \beta \mathcal{J}_P[X, \psi] = 0, \quad (490)$$

where $\mathcal{L}_P$ is the intrinsic projector/tangent-sector operator and $\mathcal{J}_P$ is the Class II source induced by $X_i^a K_i^a[P, \psi]$.

Thus the actual microscopic Class II PDE is a coupled system:

$$\begin{aligned} \partial_t^2 \Psi &= -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{S}_\Psi^{(II)}[X, \Psi, P, \psi], \\ M_X^2 X_i^a &= -\alpha J_i^a[\Psi] - \beta K_i^a[P, \psi], \\ \mathcal{L}_P[\delta P] &= -\beta \mathcal{J}_P[X, \psi]. \end{aligned} \quad (491)$$

5. E. Effective PDE after eliminating the heavy mediator

Substitute (487) back into the action. Completing the square in X_i^a gives

$$S_{\text{UV}}^{(II)} = S_0[\Psi] - \frac{1}{2M_X^2} \int dt d^3x (\alpha J_i^a[\Psi] + \beta K_i^a[P, \psi])^2. \quad (492)$$

Expanding the square,

$$S_{\text{eff}}^{(II)} = S_0[\Psi] - \frac{\alpha^2}{2M_X^2} \int J_i^a J_i^a - \frac{\alpha\beta}{M_X^2} \int J_i^a K_i^a - \frac{\beta^2}{2M_X^2} \int K_i^a K_i^a. \quad (493)$$

The load-bearing mixed term is

$$S_{\text{mix}}^{(II)} = -\frac{\alpha\beta}{M_X^2} \int dt d^3x J_i^a[\Psi] K_i^a[P, \psi]. \quad (494)$$

Therefore the effective Ψ -equation becomes

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{S}_{\Psi, \text{eff}}^{(II)}[\Psi, P, \psi], \quad (495)$$

with

$$\mathcal{S}_{\Psi, \text{eff}}^{(II)} = -\frac{\alpha\beta}{M_X^2} \frac{\delta}{\delta \Psi^\dagger} \left(J_i^a[\Psi] K_i^a[P, \psi] \right) + \dots. \quad (496)$$

6. F. Locked-background reduction to the shell-enriched tensor PDE

Now expand around the locked BCC background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger, \quad (497)$$

and write

$$\Psi = \Psi_0 + \sigma z_0 + \psi, \quad \delta P \sim \xi^a. \quad (498)$$

Then the shell current contains the key piece

$$J_i^a[\Psi] = (\partial_i \phi_0) C^a[\psi] + \dots, \quad (499)$$

and

$$K_i^a[P, \psi] \sim \partial_i \xi^a \psi. \quad (500)$$

Therefore the mixed term gives

$$L_{\text{mix}}^{(2)} \sim -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) C^a[\psi] \partial_i \xi^a. \quad (501)$$

After integration by parts / shell reduction / projection onto the shell-relevant internal doublet, this becomes precisely

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a, \quad (502)$$

with

$$N_{ij}^a \sim \frac{\alpha \beta}{M_X^2} \times (\text{projected shell/tangent tensor coefficient}). \quad (503)$$

Thus the honest effective PDE produced by Class II is:

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a \Psi + \dots \quad (504)$$

where the dots denote higher-order shell-irrelevant, heavy-suppressed, or nonleading fluctuation terms.

7. G. Canonical minimal Class II representative

For the canonical minimal Class II representative, one fixes

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}, \quad (505)$$

with all other components zero. Then the induced shell tensor lies in the axial family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (506)$$

and produces the explicit corrected coefficients

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad \mu = \frac{\lambda g_X}{M_X^2} m. \quad (507)$$

So the PDE (504) is not vague anymore: in the canonical Class II realization, its shell-relevant enriched tensor is exactly the one already used in the Weyl test.

8. H. Exact logical status

What is now genuinely closed is:

minimal microscopic PDE + Class II auxiliary mediator $\implies$ coupled UV PDE $\implies$ effective shell-enriched PDE with shell-relevant term

$$N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a.$$

What is *not* yet closed is:

1. that this Class II completion is the unique UV realization;
2. that the resulting full corrected PDE automatically satisfies

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection;}$$

3. that the full microscopic TECT chooses exactly this representative rather than a shell-equivalent alternative.

But the PDE itself is now explicit.

. Summary

$$\Psi(x, t) \in \mathbb{C}^3, \quad S_0[\Psi] = \int dt d^3x \left[\frac{1}{2} \partial_t \Psi^\dagger \partial_t \Psi - \frac{\mu^2}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi - \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - \frac{g}{4} (\Psi^\dagger \Psi)^2 \right].$$

Its Euler–Lagrange equation is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi.$$

Class II adds a heavy auxiliary internal vector mediator

$$X_i^a(x)$$

through

$$S_{\text{UV}}^{(II)} = S_0[\Psi] + \int dt d^3x \left[\frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi] \right].$$

The Euler–Lagrange equations are

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + \mathcal{S}_\Psi^{(II)}[X, \Psi, P, \psi],$$

$$M_X^2 X_i^a + \alpha J_i^a[\Psi] + \beta K_i^a[P, \psi] = 0,$$

$$\mathcal{L}_P[\delta P] = -\beta \mathcal{J}_P[X, \psi].$$

Eliminating the auxiliary field gives

$$X_i^a = -\frac{\alpha}{M_X^2} J_i^a[\Psi] - \frac{\beta}{M_X^2} K_i^a[P, \psi],$$

hence the effective action contains the mixed term

$$S_{\text{mix}}^{(II)} = -\frac{\alpha\beta}{M_X^2} \int J_i^a[\Psi] K_i^a[P, \psi].$$

Around the locked BCC background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger,$$

this reduces after shell projection to

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a, \quad N_{ij}^a \sim \frac{\alpha\beta}{M_X^2} \times (\text{projected tensor coefficient}).$$

Therefore the effective Class II-enriched microscopic PDE is

$$\boxed{\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a \Psi + \dots}$$

For the canonical minimal Class II representative,

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

so the shell-relevant corrected coefficients are

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

Thus the PDE is now explicit at the Class II level:

$$\text{minimal triplet Brazovskii wave equation} + \text{Class II mediator} \implies \text{shell-enriched effective PDE}.$$

. **Linearization of the Class II-enriched PDE and shell-projected coefficient extraction**

a. Honest scope. The full infinite-dimensional Bloch Hessian of the periodic BCC condensate is still an open microscopic calculation. What we now close is the shell-projected linearization of the Class II-enriched effective PDE, i.e. the exact level at which the already-established Weyl/Dirac matching theorem operates.

1. *A. Class II-enriched effective PDE*

Start from the effective Class II-enriched PDE

$$\partial_t^2 \Psi = -\mu_0^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a \Psi + \dots, \quad (508)$$

where $\phi_0(x)$ is the locked BCC amplitude background and N_{ij}^a is the shell-relevant enriched tensor produced by the Class II mediator. The dots denote shell-irrelevant, higher-order, or heavy-suppressed terms.

Let the locked background be

$$\Psi_0(x) = \phi_0(x) z_0, \quad \phi_0(x) = 2A \sum_{m=1}^6 \cos(Q_m \cdot x), \quad (509)$$

with $Q_m \in S_{\text{BCC}}$.

Linearize around Ψ_0 :

$$\Psi = \Psi_0 + \delta \Psi. \quad (510)$$

At the shell-effective level used in the existing matching theorem, all background-induced linear terms are collected into a periodic linear operator $\mathcal{U}_{\text{per}}(x)$, so the linearized equation takes the form

$$\partial_t^2 \delta \Psi = \mathcal{L}_0 \delta \Psi + \mathcal{U}_{\text{per}}(x) \delta \Psi + N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a \delta \Psi + \dots, \quad (511)$$

with

$$\mathcal{L}_0 := -\mu_0^2 + Z \nabla^2 - Y \nabla^4. \quad (512)$$

2. *B. Bloch ansatz near a selected shell vector*

Fix one selected shell vector $G_* \in S_{\text{BCC}}$, and write

$$k = G_* + p, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2, \quad \hat{n} = \frac{G_*}{|G_*|}. \quad (513)$$

Use the Bloch ansatz

$$\delta \Psi_k(x, t) = e^{-i\omega t} e^{ik \cdot x} u_p(x), \quad u_p(x + R) = u_p(x), \quad (514)$$

and expand the periodic part in reciprocal harmonics:

$$u_p(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x}. \quad (515)$$

Then the linearized Bloch problem becomes

$$\omega^2 u_p = H_{\text{Bloch}}^{(II)}(G_* + p) u_p, \quad (516)$$

where

$$H_{\text{Bloch}}^{(II)}(k) = H_{\text{Bloch}}^{(0)}(k) + \delta H_{\text{enr}}(k), \quad (517)$$

and

$$\left(H_{\text{Bloch}}^{(0)}(k) \right)_{G, G'} = h_0(k + G) \delta_{G, G'} + \widehat{\mathcal{U}}_{G-G'}, \quad (518)$$

while the enriched term comes from the shell-current operator

$$\delta H_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a. \quad (519)$$

3. C. Shell projection

Let $S_* \subset \mathcal{G}$ be the selected finite shell carrier, and denote by Π_{S_*} the corresponding projector. The shell-projected Bloch operator is

$$H_{\text{shell}}^{(II)}(p) = \Pi_{S_*} H_{\text{Bloch}}^{(II)}(G_* + p) \Pi_{S_*}. \quad (520)$$

Assume, as in the already-established matching theorem, that inside $H_{\text{shell}}^{(II)}(p)$ there is a spectrally isolated complex two-dimensional valley subspace H_* . Let P_* be the orthogonal projector onto H_* .

Then the linearized reduced operator is

$$h_+^{\text{lin}}(p) = P_* \left(p_i K_i^{(II)} \right) P_* \Big|_{H_*}, \quad K_i^{(II)} := \partial_{k_i} H_{\text{Bloch}}^{(II)}(k) \Big|_{k=G_*}. \quad (521)$$

4. D. Pauli decomposition and coefficient extraction

Since $h_+^{\text{lin}}(p)$ is Hermitian on $H_* \cong \mathbb{C}^2$, it decomposes as

$$h_+^{\text{lin}}(p) = a_0(p) \mathbf{1} + a_\mu(p) \sigma^\mu, \quad \mu = 1, 2, 3. \quad (522)$$

The coefficient matrix is

$$M_{\mu i} = \frac{1}{2} \text{Tr}_{H_*} \left(\sigma^\mu P_* K_i^{(II)} P_* \right), \quad (523)$$

so that

$$a_\mu(p) = M_{\mu i} p_i. \quad (524)$$

This is exactly the already-established projector extraction theorem for the local Weyl data.

5. E. Shell-adapted axial family and contraction formulas

Now impose the shell-adapted axial tensor family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (525)$$

Then the shell-level matching formulas are

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (526)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad (527)$$

$$\beta = \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2). \quad (528)$$

For the axial family (525), these reduce to

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C. \quad (529)$$

Therefore the projected positive-valley Hamiltonian becomes

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2). \quad (530)$$

6. F. Weyl coefficient matrix and determinant

The corresponding Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad (531)$$

hence

$$\det M = \lambda_{\parallel}(\alpha^2 + \beta^2). \quad (532)$$

Equivalently, in the axial tensor coefficients,

$$\det M = (\kappa_* |G_*|)^3 A(B^2 + C^2). \quad (533)$$

7. G. Canonical Class II representative

For the canonical minimal Class II representative, the induced tensor lies in the same axial family and simply renormalizes the coefficients:

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad \mu = \frac{\lambda g_X}{M_X^2} m. \quad (534)$$

Therefore

$$\lambda_{\parallel}^{\text{eff}} = \kappa_* |G_*| A_{\text{eff}}, \quad \alpha_{\text{eff}} = \kappa_* |G_*| B_{\text{eff}}, \quad \beta_{\text{eff}} = \kappa_* |G_*| C_{\text{eff}}, \quad (535)$$

and the corrected determinant is

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2). \quad (536)$$

8. H. Exact logical status

What is closed here is:

$$\text{Class II-enriched effective PDE} \implies \text{shell-projected first-order Bloch block} \implies (\lambda_{\parallel}, \alpha, \beta) \implies \det M.$$

What is still not closed is the full infinite-dimensional Bloch-Hessian derivation proving that the actual microscopic TECT operator admits exactly this isolated shell block. But at the shell-projected level used by the existing Weyl/Dirac matching theorem, the coefficient extraction is now explicit and complete.

. Summary

$$\partial_t^2 \Psi = -\mu_0^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi + N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s^a \Psi + \dots$$

Linearizing around the locked BCC background

$$\Psi_0 = \phi_0 z_0, \quad \phi_0(x) = 2A \sum_{m=1}^6 \cos(Q_m \cdot x),$$

and using the Bloch ansatz

$$\delta \Psi_k(x, t) = e^{-i\omega t} e^{i(G_* + p) \cdot x} u_p(x), \quad u_p(x) = \sum_{G \in \mathcal{G}} c_G e^{iG \cdot x},$$

one obtains the shell-projected Bloch operator

$$H_{\text{shell}}^{(II)}(p) = \Pi_{S_*} H_{\text{Bloch}}^{(II)}(G_* + p) \Pi_{S_*}.$$

Projecting to the isolated two-dimensional valley space H_* , the reduced linear operator is

$$h_+^{\text{lin}}(p) = P_*(p_i K_i^{(II)}) P_*|_{H_*}.$$

For the shell-adapted axial tensor family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

the matching formulas are

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C.$$

Hence the positive-valley Hamiltonian is

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

The Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det M = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Equivalently,

$$\det M = (\kappa_* |G_*|)^3 A (B^2 + C^2).$$

For the canonical Class II representative,

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J,$$

so

$$\lambda_{\parallel}^{\text{eff}} = \kappa_* |G_*| A_{\text{eff}}, \quad \alpha_{\text{eff}} = \kappa_* |G_*| B_{\text{eff}}, \quad \beta_{\text{eff}} = \kappa_* |G_*| C_{\text{eff}},$$

and

$$\boxed{\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2)}.$$

. Negative-valley shell projection of the Class II-enriched PDE

a. Honest scope. What we close here is the shell-projected opposite-valley pairing implied by the Class II-enriched effective PDE. We do *not* claim that the full infinite-dimensional periodic Bloch Hessian has been explicitly diagonalized in the final microscopic TECT model. Rather, we show that once the shell-projected valley reduction exists, the Class II-enriched PDE reproduces the standard opposite-valley pairing formula at that level.

1. A. Positive-valley reduced operator

From the previous step, the shell-projected positive-valley linearized operator takes the form

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (537)$$

with

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C, \quad (538)$$

where the shell-adapted axial tensor family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (539)$$

2. B. Matched opposite-valley frame

At the opposite shell point $-G_*$, define the matched frame

$$\hat{n}_- = -\hat{n}_+ = -\hat{n}, \quad e_{1-} = e_1, \quad e_{2-} = -e_2. \quad (540)$$

Then

$$e_{1-} \times e_{2-} = e_1 \times (-e_2) = -\hat{n} = \hat{n}_-, \quad (541)$$

so the frame remains right-handed at the negative valley.

The local momentum decomposition near $-G_*$ is

$$k = -G_* + p, \quad p = p_{\parallel} \hat{n}_- + p_1 e_{1-} + p_2 e_{2-}. \quad (542)$$

Equivalently, expressed in the original $+$ -frame,

$$p = -p_{\parallel} \hat{n} + p_1 e_1 - p_2 e_2. \quad (543)$$

3. C. Negative-valley tensor coefficients

Let the shell-relevant microscopic tensor near $-G_*$ be denoted $N_{ij}^a(-)$. Define the reduced coefficients in the matched opposite-valley frame by

$$A^- = \kappa_* |G_*| \hat{n}_{-i} \hat{n}_{-j} N_{ij}^3(-), \quad (544)$$

$$B^- = \frac{\kappa_* |G_*|}{2} \left(e_{1-j} \hat{n}_{-i} N_{ij}^1(-) + e_{2-j} \hat{n}_{-i} N_{ij}^2(-) \right), \quad (545)$$

$$C^- = \frac{\kappa_* |G_*|}{2} \left(e_{2-j} \hat{n}_{-i} N_{ij}^1(-) - e_{1-j} \hat{n}_{-i} N_{ij}^2(-) \right). \quad (546)$$

The standard microscopic sufficient condition for opposite-valley pairing is exactly

$$A^- = A^+, \quad B^- = B^+, \quad C^- = C^+, \quad (547)$$

in the matched opposite-valley frame. This is already the known finite tensor criterion for

$$h_-(p) = -h_+(-p).$$

4. D. Class II evenness condition

For the canonical Class II completion, the induced tensor is linear in the background vector $m^a(P_0)$ and the mediator kernel C_{ij}^{ab} . If these are even under the matched opposite-valley transformation, then the induced shell-relevant corrections satisfy

$$\delta A^- = \delta A^+, \quad \delta B^- = \delta B^+, \quad \delta C^- = \delta C^+. \quad (548)$$

Therefore, if the baseline shell construction already has

$$A_0^- = A_0^+, \quad B_0^- = B_0^+, \quad C_0^- = C_0^+,$$

then the Class II corrected coefficients still satisfy

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+. \quad (549)$$

In particular, for the canonical minimal representative,

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad (550)$$

and the same formula holds at both valleys under the evenness hypothesis.

5. E. Negative-valley reduced Hamiltonian

Define

$$\lambda_{\parallel}^{-} = \kappa_* |G_*| A^{-}, \quad \alpha^{-} = \kappa_* |G_*| B^{-}, \quad \beta^{-} = \kappa_* |G_*| C^{-}. \quad (551)$$

Then the negative-valley reduced operator in its own matched frame is

$$h_{\parallel}^{\text{lin}}(p) = \lambda_{\parallel}^{-} p_{\parallel} \sigma_3 + \alpha^{-} (p_1 \sigma_1 + p_2 \sigma_2) + \beta^{-} (-p_2 \sigma_1 + p_1 \sigma_2). \quad (552)$$

Now use the matching condition

$$\lambda_{\parallel}^{-} = \lambda_{\parallel}^{+}, \quad \alpha^{-} = \alpha^{+}, \quad \beta^{-} = \beta^{+}. \quad (553)$$

Because the valley label is reversed, the standard canonical reduced basis uses the same coefficient triple but opposite shell point. Evaluating $h_{+}^{\text{lin}}(-p)$ gives

$$h_{+}^{\text{lin}}(-p) = -\lambda_{\parallel} p_{\parallel} \sigma_3 - \alpha (p_1 \sigma_1 + p_2 \sigma_2) - \beta (-p_2 \sigma_1 + p_1 \sigma_2). \quad (554)$$

Therefore

$$-h_{+}^{\text{lin}}(-p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad (555)$$

which coincides with (552) under (553). Hence

$$h_{-}^{\text{lin}}(p) = -h_{+}^{\text{lin}}(-p). \quad (556)$$

6. F. Dirac doubling

Once both valleys are Weyl and satisfy (556), the doubled Hamiltonian

$$H_D^{\text{lin}}(p) = \begin{pmatrix} h_{+}^{\text{lin}}(p) & 0 \\ 0 & h_{-}^{\text{lin}}(p) \end{pmatrix} \quad (557)$$

takes the canonical opposite-valley form

$$H_D^{\text{lin}}(p) = \begin{pmatrix} d_i(p) \sigma_i & 0 \\ 0 & -d_i(-p) \sigma_i \end{pmatrix}, \quad (558)$$

and therefore is unitarily equivalent to a massless Dirac Hamiltonian.

7. G. Exact logical status

What is closed here is:

$$\text{Class II-enriched shell-projected PDE} \implies A_{\text{eff}}^{-} = A_{\text{eff}}^{+}, \quad B_{\text{eff}}^{-} = B_{\text{eff}}^{+}, \quad C_{\text{eff}}^{-} = C_{\text{eff}}^{+} \implies h_{-}^{\text{lin}}(p) = -h_{+}^{\text{lin}}(-p).$$

What is still not closed is the full infinite-dimensional proof that the exact microscopic periodic Bloch Hessian admits no additional symmetry-breaking corrections beyond this shell-projected sector.

. Summary

$$h_{+}^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad \lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C.$$

At the opposite valley use the matched frame

$$\hat{n}_{-} = -\hat{n}, \quad e_{1-} = e_1, \quad e_{2-} = -e_2.$$

Define

$$A^- = \kappa_* |G_*| \hat{n}_{-i} \hat{n}_{-j} N_{ij}^3(-),$$

$$B^- = \frac{\kappa_* |G_*|}{2} \left(e_{1-j} \hat{n}_{-i} N_{ij}^1(-) + e_{2-j} \hat{n}_{-i} N_{ij}^2(-) \right),$$

$$C^- = \frac{\kappa_* |G_*|}{2} \left(e_{2-j} \hat{n}_{-i} N_{ij}^1(-) - e_{1-j} \hat{n}_{-i} N_{ij}^2(-) \right).$$

If the Class II data are even under the matched opposite-valley transformation, then

$$\delta A^- = \delta A^+, \quad \delta B^- = \delta B^+, \quad \delta C^- = \delta C^+,$$

hence

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+.$$

Therefore

$$\lambda_{\parallel}^- = \lambda_{\parallel}^+, \quad \alpha^- = \alpha^+, \quad \beta^- = \beta^+,$$

and the negative-valley reduced Hamiltonian is

$$h_{-}^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2) = -h_{+}^{\text{lin}}(-p).$$

Hence the doubled Hamiltonian

$$H_D^{\text{lin}}(p) = \begin{pmatrix} h_{+}^{\text{lin}}(p) & 0 \\ 0 & h_{-}^{\text{lin}}(p) \end{pmatrix}$$

is of standard opposite-valley form and is unitarily equivalent to a massless Dirac Hamiltonian.

So at the shell-projected PDE level:

$$\text{matched opposite-valley evenness} \implies A_{\text{eff}}^- = A_{\text{eff}}^+, B_{\text{eff}}^- = B_{\text{eff}}^+, C_{\text{eff}}^- = C_{\text{eff}}^+ \implies h_{-}^{\text{lin}}(p) = -h_{+}^{\text{lin}}(-p).$$

. Minimal 2-state valley truncation from the Class II-enriched PDE

a. Honest scope. The full periodic Bloch-Hessian calculation proving the existence of an isolated 2-dimensional valley subspace is still open. What we do here is the exact finite calculation *once* the shell-projected 2-state valley carrier has been selected.

1. A. Minimal per-valley shell basis

Fix a selected shell vector $G_* \in S_{\text{BCC}}$, and write

$$k = G_* + p, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2, \quad \hat{n} = \frac{G_*}{|G_*|}. \quad (559)$$

Choose two orthonormal shell-envelope states spanning the valley carrier:

$$\Psi_{+,a}(x) = e^{iG_* \cdot x} u_a(x), \quad a = 1, 2, \quad (560)$$

where $u_a(x)$ span the internal periodic shell envelopes. This is exactly the minimal 2-state truncation at one valley.

Projecting the Bloch Hamiltonian onto this carrier gives a 2×2 Hermitian matrix

$$h_{+}^{\text{lin}}(p) = P_*(p_i K_i) P_* \Big|_{H_*}, \quad (561)$$

with

$$K_i = \partial_{k_i} L(k) \Big|_{k=G_*}, \quad H_* = \text{span}_{\mathbb{C}}\{\Psi_{+,1}, \Psi_{+,2}\}. \quad (562)$$

2. B. Pauli decomposition

Since $h_+^{\text{lin}}(p)$ is Hermitian on $H_* \cong \mathbb{C}^2$, it has the unique Pauli decomposition

$$h_+^{\text{lin}}(p) = a_0(p)\mathbf{1}_2 + a_\mu(p)\sigma^\mu, \quad \mu = 1, 2, 3. \quad (563)$$

The coefficients are extracted by

$$a_\mu(p) = \frac{1}{2}\text{Tr}_{H_*}(\sigma^\mu h_+^{\text{lin}}(p)), \quad a_0(p) = \frac{1}{2}\text{Tr}_{H_*}(h_+^{\text{lin}}(p)). \quad (564)$$

Equivalently,

$$a_\mu(p) = M_{\mu i} p_i, \quad M_{\mu i} := \frac{1}{2}\text{Tr}_{H_*}(\sigma^\mu P_* K_i P_*). \quad (565)$$

3. C. Class II-enriched first-order operator

The shell-relevant Class II-enriched term is

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a. \quad (566)$$

At the shell-projected level, its first-order contribution is completely encoded by the three contractions

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (567)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad (568)$$

$$\beta = \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2). \quad (569)$$

Therefore the minimal per-valley 2×2 Hamiltonian is

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2) + O(|p|^2). \quad (570)$$

4. D. Axial tensor family

Now impose the shell-adapted axial family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad (571)$$

$$N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad (572)$$

$$N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (573)$$

Substituting (571)–(573) into (567)–(569) yields

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C. \quad (574)$$

Hence

$$h_+^{\text{lin}}(p) = \kappa_* |G_*| \left[A p_{\parallel} \sigma_3 + B (p_1 \sigma_1 + p_2 \sigma_2) + C (-p_2 \sigma_1 + p_1 \sigma_2) \right] + O(|p|^2). \quad (575)$$

5. E. Weyl determinant of the minimal 2-state truncation

The corresponding Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad (576)$$

so

$$\det M = \lambda_{\parallel}(\alpha^2 + \beta^2). \quad (577)$$

Using (574),

$$\det M = (\kappa_* |G_*|)^3 A(B^2 + C^2). \quad (578)$$

Therefore the minimal 2-state valley truncation is Weyl iff

$$A \neq 0, \quad (B, C) \neq (0, 0). \quad (579)$$

6. F. Canonical Class II representative

For the canonical minimal Class II representative,

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad \mu = \frac{\lambda g_X}{M_X^2} m. \quad (580)$$

Hence the actual shell-projected 2×2 Hamiltonian becomes

$$h_{+,\text{eff}}^{\text{lin}}(p) = \kappa_* |G_*| \left[A_{\text{eff}} p_{\parallel} \sigma_3 + B_{\text{eff}} (p_1 \sigma_1 + p_2 \sigma_2) + C_{\text{eff}} (-p_2 \sigma_1 + p_1 \sigma_2) \right] + O(|p|^2), \quad (581)$$

with determinant

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2). \quad (582)$$

7. G. Doubling to the 4×4 Dirac block

At the opposite valley $-G_*$, assume the matched opposite-valley condition

$$h_-^{\text{lin}}(p) = -h_+^{\text{lin}}(-p). \quad (583)$$

Then the doubled four-state block

$$H_D^{\text{lin}}(p) = \begin{pmatrix} h_+^{\text{lin}}(p) & 0 \\ 0 & h_-^{\text{lin}}(p) \end{pmatrix} \quad (584)$$

is unitarily equivalent to a massless Dirac Hamiltonian.

Thus the exact role of the minimal 2-state truncation is:

$$\text{one valley} \Rightarrow 2 \times 2 \text{ Weyl block}, \quad \text{two opposite valleys} \Rightarrow 4 \times 4 \text{ Dirac block}.$$

8. H. Exact logical status

What is closed here is the explicit finite matrix calculation:

$$\text{Class II-enriched shell tensor } N_{ij}^a \Rightarrow \text{minimal } 2 \times 2 \text{ valley Hamiltonian} \Rightarrow \det M.$$

What remains open is not this calculation, but the prior spectral step: that the true microscopic periodic Bloch operator actually contains this isolated 2-state valley carrier.

. Summary

$$\Psi_{+,a}(x) = e^{iG_* \cdot x} u_a(x), \quad a = 1, 2, \quad H_* = \text{span}_{\mathbb{C}}\{\Psi_{+,1}, \Psi_{+,2}\}.$$

Projecting the Bloch operator to H_* gives the minimal per-valley 2×2 Hamiltonian

$$h_+^{\text{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}.$$

For the Class II shell-enriched tensor

$$\delta L_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i \partial_j) \otimes s^a,$$

the extracted coefficients are

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3,$$

$$\alpha = \frac{\kappa_* |G_*|}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2),$$

$$\beta = \frac{\kappa_* |G_*|}{2} (e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2).$$

Hence

$$h_+^{\text{lin}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2) + O(|p|^2).$$

For the axial family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

one gets

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C.$$

Thus

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det M = \lambda_{\parallel} (\alpha^2 + \beta^2) = (\kappa_* |G_*|)^3 A (B^2 + C^2).$$

For the canonical Class II representative,

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J,$$

so

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

Hence the minimal 2-state truncation is Weyl iff

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).$$

If the opposite valley satisfies

$$h_-^{\text{lin}}(p) = -h_+^{\text{lin}}(-p),$$

then the doubled 4×4 block is Dirac.

. **Minimal enriched 4×4 shell-pair matrix at the Bragg point**

a. Correction of the minimal truncation. The genuine minimal Bloch-pair truncation is centered not at $k = G_*$, but at the Brillouin-zone boundary point

$$k_* = \frac{G_*}{2}. \quad (585)$$

For the plain scalar periodic problem, the two-harmonic Bragg truncation gives

$$H_{\text{pair}}(p) = E_0(p)\mathbf{1} + v_{\parallel}p_{\parallel}\tau_3 + m\tau_1 + O(|p|^2), \quad k = k_* + p, \quad (586)$$

which is not yet Weyl because its linear traceless part is only one-dimensional.

Therefore the genuine minimal shell-Dirac candidate is the enriched shell space

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2. \quad (587)$$

1. *1. Shell basis and projected matrix*

Choose shell basis states

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2, \quad (588)$$

where $u_a(x)$ span the internal periodic shell envelopes. Projecting the Bloch Hamiltonian to this enriched shell space gives

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}, \quad (589)$$

where $E_{\pm}(p)$ and $B(p)$ are 2×2 internal matrices.

2. *2. Principal-order shell-Dirac conditions*

The shell-level Dirac theorem requires the following principal-order conditions:

$$B_0 := B(0) = 0, \quad (590)$$

$$E_+(p) - E_-(p) = 2v_{\parallel}p_{\parallel}\mathbf{1}_2 + O(|p|^2), \quad v_{\parallel} \neq 0, \quad (591)$$

$$B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2), \quad (592)$$

where W^1, W^2 are traceless Hermitian, nonzero, and anticommuting after internal basis reduction. Under these conditions,

$$H_{\text{proj}}(p) \sim v_{\parallel}p_{\parallel}\tau_3 \otimes \mathbf{1} + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2, \quad (593)$$

which is the massless Dirac principal operator on $\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$.

3. *3. Constant-envelope reduction*

For the simplest constant-envelope ansatz

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad \xi_a^\dagger \xi_b = \delta_{ab}, \quad (594)$$

the shell-level quantities reduce to primitive-cell Fourier coefficients:

$$B_0 = \hat{U}(G_*) \mathbf{1}_2, \quad (595)$$

$$W^\alpha = \hat{T}^\alpha(G_*) s_\alpha. \quad (596)$$

If the scalar periodic potential is generated quadratically from the first-shell BCC condensate,

$$U_{\text{per}}(x) \propto \phi_0(x)^2, \quad (597)$$

then the quadratic first-shell selection rule gives

$$\hat{U}(G_*) = 0, \quad (598)$$

hence

$$B_0 = 0. \quad (599)$$

4. 4. Minimal allowed enriched coupling

The most natural minimal enriched coefficient fields are linear derivative couplings

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x) \otimes s_\alpha, \quad \alpha = 1, 2. \quad (600)$$

Their first-shell Fourier coefficients are

$$\hat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha. \quad (601)$$

Therefore

$$W^\alpha \neq 0 \iff (C_\alpha \cdot G_*) A_{G_*} \neq 0. \quad (602)$$

5. 5. Explicit minimal 4×4 matrix

Choose the internal basis so that

$$W^1 = \nu_1 s_1, \quad W^2 = \nu_2 s_2, \quad (603)$$

with $\nu_1, \nu_2 \in \mathbb{R}$, $\nu_1 \nu_2 \neq 0$. Then the projected shell Hamiltonian takes the explicit form

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + p_1 \tau_1 \otimes W^1 + p_2 \tau_1 \otimes W^2 + O(|p|^2), \quad (604)$$

that is,

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2 + O(|p|^2). \quad (605)$$

In block-matrix form,

$$H_{\text{proj}}(p) = \begin{pmatrix} E_0(p) + v_{\parallel} p_{\parallel} & \nu_1 p_1 s_1 + \nu_2 p_2 s_2 \\ \nu_1 p_1 s_1 + \nu_2 p_2 s_2 & E_0(p) - v_{\parallel} p_{\parallel} \end{pmatrix} + O(|p|^2). \quad (606)$$

6. 6. Dirac form

Define

$$\Gamma_1 = \tau_1 \otimes s_1, \quad \Gamma_2 = \tau_1 \otimes s_2, \quad \Gamma_3 = \tau_3 \otimes \mathbf{1}_2. \quad (607)$$

Then

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} \mathbf{1}_4, \quad (608)$$

and the principal operator becomes

$$H_{\text{Dirac}}^{\text{pr}}(p) = \nu_1 p_1 \Gamma_1 + \nu_2 p_2 \Gamma_2 + v_{\parallel} p_{\parallel} \Gamma_3. \quad (609)$$

Thus the enriched shell-pair truncation is exactly Dirac at principal order.

7. 7. Honest status

What is explicitly computed here is the finite projected shell matrix once the enriched shell carrier

$$\mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2$$

is selected.

What remains open is the prior microscopic spectral step: to derive from the full periodic BCC Bloch Hessian that this enriched shell block is indeed spectrally isolated and that the coefficients $v_{\parallel}, \nu_1, \nu_2$ are nonzero in the actual TECT linearization.

. Summary

$$k_* = \frac{G_*}{2}, \quad \mathcal{H}_{\text{shell}} = \mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Choose basis states

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2.$$

Then the projected shell Hamiltonian is

$$H_{\text{proj}}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix},$$

with 2×2 internal blocks.

A shell-level Dirac operator emerges if

$$B_0 = B(0) = 0,$$

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2), \quad v_{\parallel} \neq 0,$$

$$B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2),$$

where W^1, W^2 are traceless Hermitian, nonzero, and anticommuting.

For the constant-envelope reduction,

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2, \quad W^\alpha = \widehat{T}^\alpha(G_*) s_\alpha.$$

If $U_{\text{per}}(x) \propto \phi_0(x)^2$, then

$$\widehat{U}(G_*) = 0 \quad \Rightarrow \quad B_0 = 0.$$

For the minimal derivative-enriched class

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x) \otimes s_\alpha,$$

one has

$$\widehat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha.$$

Hence

$$(C_1 \cdot G_*) A_{G_*} \neq 0, \quad (C_2 \cdot G_*) A_{G_*} \neq 0$$

imply two nonzero transverse matrices.

Choosing internal basis so that

$$W^1 = \nu_1 s_1, \quad W^2 = \nu_2 s_2,$$

the projected 4×4 shell Hamiltonian is

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \nu_1 p_1 \tau_1 \otimes s_1 + \nu_2 p_2 \tau_1 \otimes s_2 + O(|p|^2).$$

Defining

$$\Gamma_1 = \tau_1 \otimes s_1, \quad \Gamma_2 = \tau_1 \otimes s_2, \quad \Gamma_3 = \tau_3 \otimes \mathbf{1}_2,$$

one gets

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} \mathbf{1}_4,$$

so the principal part is exactly

$$H_{\text{Dirac}}^{\text{pr}}(p) = \nu_1 p_1 \Gamma_1 + \nu_2 p_2 \Gamma_2 + v_{\parallel} p_{\parallel} \Gamma_3.$$

Thus the genuine minimal finite-dimensional shell Dirac candidate is not the plain 2×2 scalar pair, but the enriched 4×4 shell-pair matrix

$$\boxed{\mathbb{C}_{\text{pair}}^2 \otimes \mathbb{C}_{\text{int}}^2}.$$

. Primitive-cell overlap integrals in the constant-envelope shell ansatz

a. Goal. We compute explicitly the shell-pair overlap integrals

$$B_0, \quad W^1, \quad W^2$$

for the constant-envelope enriched shell basis, and derive the exact Fourier formulas from the primitive-cell integrals.

1. A. Constant-envelope shell basis

Fix a selected reciprocal-shell vector $G_* \in S_{\text{BCC}}$, and set the Bragg quasimomentum

$$k_* = \frac{G_*}{2}. \quad (610)$$

Choose a constant orthonormal internal basis

$$\xi_1, \xi_2 \in \mathbb{C}_{\text{int}}^2, \quad \xi_a^\dagger \xi_b = \delta_{ab}, \quad (611)$$

and define the periodic envelopes

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad a = 1, 2. \quad (612)$$

The enriched shell-pair basis states are

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x). \quad (613)$$

2. B. Minimal enriched periodic operator

Consider the minimal enriched periodic operator

$$H = h_0(-i\nabla) \otimes \mathbf{1}_2 + U_{\text{per}}(x) \otimes \mathbf{1}_2 + \sum_{\alpha=1}^2 T^\alpha(x)(-i\partial_\alpha) \otimes s_\alpha, \quad (614)$$

where:

- $h_0(q)$ is the scalar bare kernel;
- $U_{\text{per}}(x)$ is a real periodic scalar coefficient;
- $T^\alpha(x)$ are real periodic enriched coefficient fields;
- s_α are Pauli matrices acting on the internal doublet.

3. C. Bragg mass B_0

By definition,

$$(B_0)_{ab} = \langle \Psi_{+,a}, H(k_*) \Psi_{-,b} \rangle. \quad (615)$$

The translation-invariant bare part $h_0(-i\nabla) \otimes \mathbf{1}_2$ does not contribute to pair mixing between Ψ_+ and Ψ_- , so the leading scalar contribution comes from U_{per} :

$$(B_0)_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \xi_a^\dagger e^{-iG_* \cdot x} U_{\text{per}}(x) \xi_b. \quad (616)$$

Since U_{per} acts trivially on the internal doublet,

$$(B_0)_{ab} = \left(\frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_{\text{per}}(x) \right) \delta_{ab}. \quad (617)$$

Define the primitive-cell Fourier coefficient

$$\hat{U}(G) := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG \cdot x} U_{\text{per}}(x). \quad (618)$$

Then

$$(B_0)_{ab} = \hat{U}(G_*) \delta_{ab}, \quad B_0 = \hat{U}(G_*) \mathbf{1}_2. \quad (619)$$

4. D. Transverse mixing matrices W^α

Now consider the enriched term

$$T^\alpha(x)(-i\partial_\alpha) \otimes s_\alpha.$$

At the shell-projected first-order level, the corresponding matrix elements are

$$(W^\alpha)_{ab} = \langle \Psi_{+,a}, T^\alpha(x)(-i\partial_\alpha) \otimes s_\alpha \Psi_{-,b} \rangle_{\text{cell}} \Big|_{\text{shell-projected first order}}. \quad (620)$$

In the constant-envelope ansatz, the shell-overlap reduction gives directly

$$(W^\alpha)_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} T^\alpha(x) \xi_a^\dagger s_\alpha \xi_b. \quad (621)$$

Since the internal matrix is constant over the cell,

$$(W^\alpha)_{ab} = \left(\frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} T^\alpha(x) \right) (s_\alpha)_{ab}. \quad (622)$$

Define

$$\hat{T}^\alpha(G) := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG \cdot x} T^\alpha(x). \quad (623)$$

Then

$$W^\alpha = \hat{T}^\alpha(G_*) s_\alpha. \quad (624)$$

5. E. Explicit derivative coupling computation

Now specialize to the minimal derivative-enriched class

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x), \quad \alpha = 1, 2, \quad (625)$$

where $C_{\alpha j}$ are constant coefficients and

$$\phi_0(x) = \sum_{Q \in S_{\text{BCC}}} A_Q e^{iQ \cdot x}, \quad A_{-Q} = A_Q. \quad (626)$$

Then

$$\partial_j \phi_0(x) = \sum_{Q \in S_{\text{BCC}}} iQ_j A_Q e^{iQ \cdot x}. \quad (627)$$

Substituting into (625),

$$T^\alpha(x) = \sum_{Q \in S_{\text{BCC}}} i(C_\alpha \cdot Q) A_Q e^{iQ \cdot x}. \quad (628)$$

Therefore its G_* -Fourier coefficient is

$$\hat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*}. \quad (629)$$

Hence the projected shell matrices are

$$W^\alpha = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha. \quad (630)$$

6. F. Exact nonvanishing criterion

Equation (630) immediately implies:

$$W^\alpha \neq 0 \iff (C_\alpha \cdot G_*) A_{G_*} \neq 0. \quad (631)$$

Thus a sufficient explicit shell-level criterion is

$$A_{G_*} \neq 0, \quad C_1 \cdot G_* \neq 0, \quad C_2 \cdot G_* \neq 0. \quad (632)$$

Under (632),

$$W^1 \neq 0, \quad W^2 \neq 0. \quad (633)$$

Since s_1 and s_2 anticommute,

$$\{W^1, W^2\} = 0 \quad (634)$$

up to the nonzero scalar coefficients.

7. G. About the Bragg-mass cancellation

For the scalar Bragg mass,

$$B_0 = \hat{U}(G_*) \mathbf{1}_2$$

is exact from (619). To go further and prove

$$\hat{U}(G_*) = 0, \quad (635)$$

one must specify the precise microscopic form of $U_{\text{per}}(x)$, not merely that it is “quadratic” in the first-shell condensate. At the present level, the exact direct calculation is

$$\hat{U}(G_*) = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_{\text{per}}(x). \quad (636)$$

If one inserts a specific quadratic Hessian form for U_{per} , then the first-shell quadratic selection rule established in the existing shell-Dirac analysis gives

$$\hat{U}(G_*) = 0.$$

So the new calculation here fully establishes the overlap formula itself; the vanishing of $\hat{U}(G_*)$ remains a further Fourier-selection computation tied to the detailed microscopic scalar potential.

8. H. Resulting projected 4×4 matrix

Combining the overlap formulas with the enriched shell-pair basis, the principal projected Hamiltonian is

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + p_1 \tau_1 \otimes W^1 + p_2 \tau_1 \otimes W^2 + O(|p|^2), \quad (637)$$

and for the derivative-enriched class this becomes

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + i(C_1 \cdot G_*) A_{G_*} p_1 \tau_1 \otimes s_1 + i(C_2 \cdot G_*) A_{G_*} p_2 \tau_1 \otimes s_2 + O(|p|^2). \quad (638)$$

After a harmless basis rephasing to make the coefficients real, this is the canonical enriched shell Dirac principal operator whenever (632) holds and $B_0 = 0$.

a. Exact logical status. What is fully computed here is the primitive-cell overlap reduction:

$$B_0 = \hat{U}(G_*) \mathbf{1}_2, \quad W^\alpha = \hat{T}^\alpha(G_*) s_\alpha, \quad \hat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*}.$$

What is not fully recomputed here is the separate first-shell quadratic cancellation proving $\hat{U}(G_*) = 0$ for the exact microscopic scalar Hessian.

. Summary

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad \Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad k_* = \frac{G_*}{2}.$$

For the scalar periodic potential,

$$(B_0)_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \xi_a^\dagger e^{-iG_* \cdot x} U_{\text{per}}(x) \xi_b = \hat{U}(G_*) \delta_{ab},$$

so

$$\boxed{B_0 = \hat{U}(G_*) \mathbf{1}_2.}$$

For the enriched coefficient field $T^\alpha(x)$,

$$(W^\alpha)_{ab} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} T^\alpha(x) \xi_a^\dagger s_\alpha \xi_b = \hat{T}^\alpha(G_*) (s_\alpha)_{ab},$$

hence

$$\boxed{W^\alpha = \hat{T}^\alpha(G_*) s_\alpha.}$$

For the derivative-enriched class

$$T^\alpha(x) = C_{\alpha j} \partial_j \phi_0(x), \quad \phi_0(x) = \sum_{Q \in S_{\text{BCC}}} A_Q e^{iQ \cdot x},$$

one finds

$$\partial_j \phi_0(x) = \sum_{Q \in S_{\text{BCC}}} iQ_j A_Q e^{iQ \cdot x},$$

so

$$\hat{T}^\alpha(G_*) = i(C_\alpha \cdot G_*) A_{G_*}.$$

Therefore

$$\boxed{W^\alpha = i(C_\alpha \cdot G_*) A_{G_*} s_\alpha.}$$

Hence the explicit shell-level nonvanishing criterion is

$$\boxed{A_{G_*} \neq 0, \quad C_1 \cdot G_* \neq 0, \quad C_2 \cdot G_* \neq 0.}$$

Under this condition,

$$W^1 \neq 0, \quad W^2 \neq 0,$$

and because s_1, s_2 anticommute, the transverse shell-Dirac matrices anticommute.

The resulting projected 4×4 shell Hamiltonian is

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + p_1 \tau_1 \otimes W^1 + p_2 \tau_1 \otimes W^2 + O(|p|^2),$$

i.e.

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + i(C_1 \cdot G_*) A_{G_*} p_1 \tau_1 \otimes s_1 + i(C_2 \cdot G_*) A_{G_*} p_2 \tau_1 \otimes s_2 + O(|p|^2).$$

So the primitive-cell overlap calculation is now explicit.

. Direct computation of $\hat{U}(G_*)$ for a quadratic scalar periodic potential

a. Goal. We compute explicitly the first-shell Fourier coefficient of a literal quadratic scalar periodic potential

$$U_{\text{per}}(x) = c \phi_0(x)^2$$

for the BCC first-shell condensate, and determine whether

$$\hat{U}(G_*) = 0$$

actually holds.

1. A. BCC first-shell condensate

Let the normalized BCC first shell be

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (639)$$

Write the equal-amplitude real condensate as

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}. \quad (640)$$

Hence the Fourier coefficient of ϕ_0 is

$$\widehat{\phi_0}(Q) = \begin{cases} A, & Q \in S_{\text{BCC}}, \\ 0, & Q \notin S_{\text{BCC}}. \end{cases} \quad (641)$$

2. B. Quadratic scalar periodic potential

Now take the literal quadratic scalar periodic potential

$$U_{\text{per}}(x) = c \phi_0(x)^2. \quad (642)$$

Then

$$\phi_0(x)^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 + Q_2) \cdot x}. \quad (643)$$

Therefore the Fourier coefficient at $G_* \in S_{\text{BCC}}$ is

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*), \quad N(G_*) := \# \{ (Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_* \}. \quad (644)$$

So the problem reduces to counting ordered first-shell pairs summing to a first-shell vector.

3. C. Explicit pair counting

Choose for concreteness

$$G_* = \frac{1}{\sqrt{2}}(1, 1, 0). \quad (645)$$

The first-shell vectors satisfying $Q_1 + Q_2 = G_*$ are:

$$\frac{1}{\sqrt{2}}(1, 0, 1) + \frac{1}{\sqrt{2}}(0, 1, -1) = \frac{1}{\sqrt{2}}(1, 1, 0), \quad (646)$$

$$\frac{1}{\sqrt{2}}(1, 0, -1) + \frac{1}{\sqrt{2}}(0, 1, 1) = \frac{1}{\sqrt{2}}(1, 1, 0). \quad (647)$$

Since ordered pairs are counted in (643), the reversed pairs also contribute:

$$(Q_1, Q_2), \quad (Q_2, Q_1). \quad (648)$$

Thus

$$N(G_*) = 4. \quad (649)$$

By cubic symmetry, the same count holds for every $G_* \in S_{\text{BCC}}$.

[First-shell square coefficient] For the equal-amplitude BCC first-shell condensate,

$$\widehat{\phi_0^2}(G_*) = 4A^2 \quad \forall G_* \in S_{\text{BCC}}. \quad (650)$$

Equation (649) proves the result for one representative G_* , and cubic symmetry propagates it to the full shell.

4. D. Consequence for the Bragg mass

From (642),

$$\widehat{U}(G_*) = c \widehat{\phi_0^2}(G_*) = 4cA^2. \quad (651)$$

Hence the projected constant Bragg mass is

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2 = 4cA^2 \mathbf{1}_2. \quad (652)$$

Therefore,

$$\boxed{U_{\text{per}}(x) = c \phi_0(x)^2 \implies \widehat{U}(G_*) \neq 0 \text{ generically.}} \quad (653)$$

5. E. Implication

Thus the statement

$$\widehat{U}(G_*) = 0$$

is *not* true for a literal pure square $U_{\text{per}} \propto \phi_0^2$. To make $B_0 = 0$, one needs one of the following:

1. a more complicated microscopic scalar Hessian in which distinct contributions cancel at first-shell momentum;
2. a non-scalar enriched mechanism so that the relevant shell Dirac structure comes from W^1, W^2 , not from vanishing scalar Bragg mass;
3. a modified choice of U_{per} with an explicit first-shell subtraction.

6. F. Example: scalar ϕ^4 Hessian

For a standard scalar quartic model

$$\mathcal{L} = \frac{1}{2} \phi \mathcal{K}(-i\nabla) \phi + \frac{g}{4} \phi^4, \quad (654)$$

linearization around ϕ_0 gives a scalar Hessian

$$H_{\text{lin}} = \mathcal{K}(-i\nabla) + 3g \phi_0(x)^2. \quad (655)$$

Then

$$\widehat{U}(G_*) = 3g \widehat{\phi_0^2}(G_*) = 12gA^2, \quad (656)$$

which is again nonzero generically.

So the direct computation confirms: a plain scalar quartic Hessian does *not* automatically eliminate the constant Bragg mass at the first shell.

7. G. Honest conclusion

What is proved here is stronger than a restatement:

$$\text{for the literal quadratic scalar example } U_{\text{per}} \propto \phi_0^2, \quad \widehat{U}(G_*) = 4cA^2 \neq 0.$$

Therefore any previous claim that a bare quadratic first-shell scalar potential automatically gives

$$\widehat{U}(G_*) = 0$$

must be interpreted as requiring an additional cancellation mechanism beyond the literal square.

. Summary

$$\phi_0(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}, \quad \widehat{\phi_0}(Q) = A \text{ for } Q \in S_{\text{BCC}}.$$

For the literal quadratic scalar periodic potential

$$U_{\text{per}}(x) = c \phi_0(x)^2,$$

one has

$$\phi_0(x)^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 + Q_2) \cdot x}.$$

Hence

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*), \quad N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}.$$

For every $G_* \in S_{\text{BCC}}$,

$$N(G_*) = 4.$$

Therefore

$$\widehat{\phi_0^2}(G_*) = 4A^2, \quad \widehat{U}(G_*) = 4cA^2.$$

So the constant Bragg mass is

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2 = 4cA^2 \mathbf{1}_2,$$

which is generically nonzero.

Thus:

$$\boxed{U_{\text{per}}(x) = c\phi_0(x)^2 \implies \widehat{U}(G_*) \neq 0.}$$

In particular, the statement

$$\widehat{U}(G_*) = 0$$

does not hold for the literal pure-square scalar example.

For a standard scalar ϕ^4 Hessian,

$$H_{\text{lin}} = \mathcal{K}(-i\nabla) + 3g\phi_0(x)^2,$$

one gets

$$\widehat{U}(G_*) = 12gA^2 \neq 0$$

generically, again confirming that the constant Bragg mass is not automatically removed in a plain scalar model.

. $B_0 \neq 0$ in the enriched 4×4 shell matrix: exact spectrum and obstruction theorem

a. Goal. We now accept the direct primitive-cell computation

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2, \quad \widehat{U}(G_*) = 4cA^2 \neq 0$$

for the literal quadratic scalar example $U_{\text{per}} = c\phi_0^2$, and determine the exact spectral effect of this term in the minimal enriched shell-pair Hamiltonian.

1. A. Minimal enriched shell Hamiltonian with nonzero scalar Bragg mass

The shell-level Dirac theorem requires $B_0 = 0$. Without imposing that condition, the minimal enriched 4×4 projected shell Hamiltonian has the form

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \tau_1 \otimes (B_0 \mathbf{1}_2 + \nu_1 p_1 s_1 + \nu_2 p_2 s_2) + O(|p|^2). \quad (657)$$

Here:

- τ_i act on the shell-pair sector,
- s_i act on the internal doublet,
- $B_0 = \widehat{U}(G_*)$ is the constant scalar Bragg mass,
- $\nu_1, \nu_2 \neq 0$ are the transverse enriched shell couplings.

Define

$$r_{\perp}(p) := \sqrt{\nu_1^2 p_1^2 + \nu_2^2 p_2^2}. \quad (658)$$

2. B. Exact diagonalization

Subtract the scalar part $E_0(p)\mathbf{1}_4$ and define

$$\tilde{H}(p) := H_{\text{proj}}(p) - E_0(p)\mathbf{1}_4 = v_{\parallel}p_{\parallel}\tau_3 \otimes \mathbf{1}_2 + \tau_1 \otimes (B_0\mathbf{1}_2 + \nu_1p_1s_1 + \nu_2p_2s_2). \quad (659)$$

Because τ_3 anticommutes with τ_1 ,

$$\tilde{H}(p)^2 = v_{\parallel}^2p_{\parallel}^2\mathbf{1}_4 + \mathbf{1}_2 \otimes (B_0\mathbf{1}_2 + \nu_1p_1s_1 + \nu_2p_2s_2)^2. \quad (660)$$

Now set

$$M_{\perp}(p) := B_0\mathbf{1}_2 + \nu_1p_1s_1 + \nu_2p_2s_2. \quad (661)$$

Since s_1, s_2 anticommute and square to $\mathbf{1}_2$, the internal 2×2 matrix

$$\nu_1p_1s_1 + \nu_2p_2s_2$$

has eigenvalues $\pm r_{\perp}(p)$. Therefore $M_{\perp}(p)$ has eigenvalues

$$\lambda_{\pm}(p) = B_0 \pm r_{\perp}(p). \quad (662)$$

Hence the eigenvalues of $\tilde{H}(p)^2$ are

$$v_{\parallel}^2p_{\parallel}^2 + (B_0 \pm r_{\perp}(p))^2. \quad (663)$$

Taking square roots, the full spectrum of $H_{\text{proj}}(p)$ is

$$\boxed{\text{spec}(H_{\text{proj}}(p)) = E_0(p) \pm \sqrt{v_{\parallel}^2p_{\parallel}^2 + (B_0 + r_{\perp}(p))^2}, \quad E_0(p) \pm \sqrt{v_{\parallel}^2p_{\parallel}^2 + (B_0 - r_{\perp}(p))^2}.} \quad (664)$$

3. C. Consequences at the expansion point

At $p = 0$, one has $r_{\perp}(0) = 0$, so

$$\text{spec}(H_{\text{proj}}(0)) = E_0(0) \pm |B_0|, \quad \text{each doubly degenerate.} \quad (665)$$

Thus the gap at the nominal shell expansion point is

$$\Delta_0 = 2|B_0|. \quad (666)$$

[Scalar Bragg mass obstruction theorem] If $B_0 \neq 0$, then the minimal enriched shell Hamiltonian (657) does *not* have a massless Dirac point at $p = 0$. In fact, the crossing is gapped by $2|B_0|$.

Immediate from (665) and (666).

4. D. Why a scalar energy shift cannot remove B_0

A scalar redefinition of the reference energy changes only the term

$$E_0(p)\mathbf{1}_4.$$

But the constant Bragg term is

$$B_0\tau_1 \otimes \mathbf{1}_2,$$

which is traceless in the shell-pair sector and therefore cannot be removed by adding or subtracting a multiple of the identity.

Hence:

$$\boxed{\text{a scalar energy subtraction cannot eliminate } B_0\tau_1 \otimes \mathbf{1}_2.} \quad (667)$$

5. E. Zero set of the principal symbol

The zero-energy condition for the traceless principal symbol $\tilde{H}(p)$ is

$$0 \in \text{spec}(\tilde{H}(p)).$$

By (663), this requires

$$p_{\parallel} = 0, \quad B_0 \pm r_{\perp}(p) = 0. \quad (668)$$

Equivalently,

$$p_{\parallel} = 0, \quad r_{\perp}(p) = |B_0|. \quad (669)$$

Thus the zero set is the ellipse

$$\nu_1^2 p_1^2 + \nu_2^2 p_2^2 = B_0^2, \quad p_{\parallel} = 0. \quad (670)$$

[No isolated Dirac point for $B_0 \neq 0$] If $B_0 \neq 0$, the zero set of the principal symbol is not an isolated point but an elliptic ring in the transverse plane. Hence the minimal enriched shell matrix does not produce a Dirac cone.

6. F. General classification of constant pair-mixing terms

We now go one step further. Replace the scalar constant Bragg mass $B_0 \mathbf{1}_2$ by the most general constant Hermitian internal matrix

$$M_0 = m_0 \mathbf{1}_2 + m_1 s_1 + m_2 s_2 + m_3 s_3. \quad (671)$$

Then the principal shell Hamiltonian becomes

$$\tilde{H}_{M_0}(p) = v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \tau_1 \otimes (M_0 + \nu_1 p_1 s_1 + \nu_2 p_2 s_2). \quad (672)$$

A zero of the principal symbol requires

$$p_{\parallel} = 0, \quad M_0 + \nu_1 p_1 s_1 + \nu_2 p_2 s_2 \text{ has a zero eigenvalue.} \quad (673)$$

Now observe:

- the identity component $m_0 \mathbf{1}_2$ cannot be canceled by momentum-dependent s_1, s_2 terms;
- the s_3 component $m_3 s_3$ cannot be canceled either, because the available momentum-dependent matrices lie only in the s_1, s_2 plane;
- the planar components $m_1 s_1 + m_2 s_2$ can be absorbed by shifting the transverse momentum origin:

$$p_1 \mapsto p_1 + \frac{m_1}{\nu_1}, \quad p_2 \mapsto p_2 + \frac{m_2}{\nu_2}. \quad (674)$$

Therefore:

[Classification of constant pair-mixing terms] For the minimal enriched shell Hamiltonian:

1. $m_1 s_1 + m_2 s_2$ is not a genuine mass; it merely shifts the transverse valley center.
2. $m_3 s_3$ is a genuine internal mass term.
3. $m_0 \mathbf{1}_2$ is a genuine scalar Bragg mass term.

Hence a massless Dirac point requires

$$m_0 = 0, \quad m_3 = 0, \quad (675)$$

while m_1, m_2 may be removed by redefining the transverse momentum origin.

7. *G. Application to the literal quadratic scalar example*

For the literal quadratic scalar example

$$U_{\text{per}}(x) = c \phi_0(x)^2, \quad (676)$$

the primitive-cell calculation gave

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2 = 4cA^2 \mathbf{1}_2. \quad (677)$$

So this corresponds precisely to the $m_0 \mathbf{1}_2$ case with

$$m_0 = 4cA^2, \quad m_1 = m_2 = m_3 = 0.$$

Therefore the constant term is a genuine scalar Bragg mass and cannot be removed by:

1. subtracting a scalar reference energy,
2. shifting the valley momentum origin,
3. or using only the transverse enriched linear couplings.

Hence:

the literal scalar quadratic example does not produce a massless shell Dirac point.

(678)

8. *H. Honest conclusion*

This computation sharpens the previous correction. Not only is it false that

$$U_{\text{per}} \propto \phi_0^2 \implies \widehat{U}(G_*) = 0,$$

but in fact the resulting nonzero $\widehat{U}(G_*)$ is a *genuine scalar Bragg mass obstruction* to shell Dirac emergence in the minimal enriched 4×4 truncation.

Therefore a consistent microscopic Dirac construction must do one of the following:

1. produce an exact cancellation so that the total scalar pair-mixing coefficient satisfies

$$B_0^{\text{tot}} = 0;$$

2. derive a different microscopic operator in which the constant pair-mixing block is not proportional to the identity and its removable part can be shifted away;
3. or realize a grading/odd-coupling mechanism in which the relevant projected principal operator is block-off-diagonal from the outset, bypassing the literal scalar Bragg mass channel.

. **Summary**

$$H_{\text{proj}}(p) = E_0(p) \mathbf{1}_4 + v_{\parallel} p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + \tau_1 \otimes (B_0 \mathbf{1}_2 + \nu_1 p_1 s_1 + \nu_2 p_2 s_2) + O(|p|^2).$$

Let

$$r_{\perp}(p) = \sqrt{\nu_1^2 p_1^2 + \nu_2^2 p_2^2}.$$

Then the exact spectrum is

$$\text{spec}(H_{\text{proj}}(p)) = E_0(p) \pm \sqrt{v_{\parallel}^2 p_{\parallel}^2 + (B_0 + r_{\perp}(p))^2}, \quad E_0(p) \pm \sqrt{v_{\parallel}^2 p_{\parallel}^2 + (B_0 - r_{\perp}(p))^2}.$$

At $p = 0$,

$$\text{spec}(H_{\text{proj}}(0)) = E_0(0) \pm |B_0|,$$

so the gap is

$$\Delta_0 = 2|B_0|.$$

Therefore:

$$\boxed{B_0 \neq 0 \implies \text{no massless Dirac point at } p = 0.}$$

A scalar energy subtraction cannot remove B_0 , because it changes only

$$E_0(p)\mathbf{1}_4,$$

whereas the constant Bragg term is

$$B_0 \tau_1 \otimes \mathbf{1}_2.$$

The zero set of the principal symbol is

$$p_{\parallel} = 0, \quad \nu_1^2 p_1^2 + \nu_2^2 p_2^2 = B_0^2,$$

which is an elliptic ring, not an isolated Dirac point.

More generally, for a constant pair-mixing block

$$M_0 = m_0 \mathbf{1}_2 + m_1 s_1 + m_2 s_2 + m_3 s_3,$$

one has:

$$m_1 s_1 + m_2 s_2 \quad \text{can be removed by shifting } (p_1, p_2),$$

$$m_3 s_3 \quad \text{is a genuine internal mass,}$$

$$m_0 \mathbf{1}_2 \quad \text{is a genuine scalar Bragg mass.}$$

Hence a massless Dirac point requires

$$\boxed{m_0 = 0, \quad m_3 = 0,}$$

while m_1, m_2 are removable by redefining the transverse momentum origin.

For the literal scalar quadratic example

$$U_{\text{per}}(x) = c \phi_0(x)^2, \quad B_0 = 4cA^2 \mathbf{1}_2,$$

so the constant term is precisely a genuine scalar Bragg mass obstruction.

. Actual microscopic scalar Hessian and the impossibility of scalar Bragg-mass cancellation

a. Goal. We now compute the *actual* microscopic scalar Hessian seen by the internal doublet sector, and determine whether the constant shell-pair Bragg mass

$$B_0^{\text{tot}}$$

can cancel inside the scalar Hessian itself.

1. *A. Exact microscopic linearization*

Consider the canonical complex condensate model

$$N(\Phi) = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u(\Phi^\dagger\Phi)\Phi. \quad (679)$$

Its exact first variation around a background Φ_0 is

$$\delta N_{\Phi_0}[\eta] = r\eta - \kappa\nabla^2\eta + \lambda\nabla^4\eta + u\left[(\Phi_0^\dagger\Phi_0)\eta + (\Phi_0^\dagger\eta)\Phi_0 + (\eta^\dagger\Phi_0)\Phi_0\right]. \quad (680)$$

Equivalently, in BdG/conservative form one may write

$$i\partial_t\eta = L_{\Phi_0}\eta + M_{\Phi_0}\bar{\eta}, \quad (681)$$

with

$$L_{\Phi_0}\eta = r\eta - \kappa\nabla^2\eta + \lambda\nabla^4\eta + u\left[(\Phi_0^\dagger\Phi_0)\eta + (\Phi_0^\dagger\eta)\Phi_0\right], \quad (682)$$

$$M_{\Phi_0}\bar{\eta} = u(\eta^\dagger\Phi_0)\Phi_0. \quad (683)$$

2. *B. Canonical symmetric BCC background*

Take the canonical locked BCC background

$$\Phi_0(x) = \phi_0(x)\zeta, \quad \zeta^\dagger\zeta = 1, \quad (684)$$

with equal-amplitude first-shell condensate

$$\phi_0(x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x} = 2A \sum_{a=1}^6 \cos(Q_a \cdot x). \quad (685)$$

Here

$$S_{\text{BCC}} = \frac{1}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (686)$$

3. *C. Restriction to the internal doublet sector*

Let

$$L = \mathbb{C}\zeta, \quad Q = (\mathbb{C}\zeta)^\perp, \quad \mathbb{C}^3 = L \oplus Q. \quad (687)$$

We now restrict the fluctuation η to the internal doublet sector:

$$\eta(x) \in \Gamma(Q), \quad \zeta^\dagger\eta(x) = 0. \quad (688)$$

Since $\Phi_0 = \phi_0\zeta$, we have

$$\Phi_0^\dagger\eta = \phi_0^*\zeta^\dagger\eta = 0, \quad (689)$$

and similarly

$$\eta^\dagger\Phi_0 = \phi_0\eta^\dagger\zeta = 0. \quad (690)$$

Therefore the two rank-one Hessian terms vanish identically on the Q -sector:

$$(\Phi_0^\dagger\eta)\Phi_0 = 0, \quad (\eta^\dagger\Phi_0)\Phi_0 = 0. \quad (691)$$

Hence the exact linearized operator on Q reduces to

$$H_Q = r - \kappa \nabla^2 + \lambda \nabla^4 + u |\phi_0(x)|^2. \quad (692)$$

Equivalently, the actual scalar periodic Hessian seen by the internal doublet sector is

$$U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2. \quad (693)$$

Moreover, the anomalous BdG block vanishes on Q :

$$M_{\Phi_0} \bar{\eta} = u (\eta^\dagger \Phi_0) \Phi_0 = 0. \quad (694)$$

So the Q -sector sees a purely scalar periodic Hessian.

4. *D. First-shell Fourier coefficient of $|\phi_0|^2$*

Because the canonical symmetric condensate is real,

$$|\phi_0(x)|^2 = \phi_0(x)^2. \quad (695)$$

Using (685),

$$\phi_0(x)^2 = A^2 \sum_{Q_1, Q_2 \in S_{\text{BCC}}} e^{i(Q_1 + Q_2) \cdot x}. \quad (696)$$

Therefore the first-shell Fourier coefficient at $G_* \in S_{\text{BCC}}$ is

$$\widehat{\phi_0^2}(G_*) = A^2 N(G_*), \quad N(G_*) := \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\}. \quad (697)$$

Choose the representative

$$G_* = \frac{1}{\sqrt{2}}(1, 1, 0). \quad (698)$$

Then the allowed first-shell decompositions are

$$\frac{1}{\sqrt{2}}(1, 0, 1) + \frac{1}{\sqrt{2}}(0, 1, -1) = \frac{1}{\sqrt{2}}(1, 1, 0), \quad (699)$$

$$\frac{1}{\sqrt{2}}(1, 0, -1) + \frac{1}{\sqrt{2}}(0, 1, 1) = \frac{1}{\sqrt{2}}(1, 1, 0). \quad (700)$$

Since ordered pairs are counted in (696), the reversed pairs also contribute. Hence

$$N(G_*) = 4. \quad (701)$$

By cubic symmetry, this holds for every $G_* \in S_{\text{BCC}}$. Therefore

$$\widehat{\phi_0^2}(G_*) = 4A^2. \quad (702)$$

Substituting into (693), we obtain

$$\widehat{U}_{\text{per}}^{\text{tot}}(G_*) = u \widehat{|\phi_0|^2}(G_*) = 4uA^2. \quad (703)$$

5. E. Primitive-cell shell overlap and Bragg mass

In the constant-envelope shell-pair reduction,

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2. \quad (704)$$

Therefore the *actual* constant pair-mixing Bragg mass generated by the microscopic scalar Hessian is

$$\boxed{B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2.} \quad (705)$$

In particular, if

$$uA^2 \neq 0, \quad (706)$$

then

$$B_0^{\text{tot}} \neq 0. \quad (707)$$

6. F. No scalar-Hessian cancellation

The key point is now immediate: inside the Q -sector scalar Hessian, the only periodic scalar contribution is

$$u|\phi_0|^2.$$

The two additional Hessian terms in (680) vanish identically on Q by (691). Therefore there are no remaining scalar terms available to cancel $\widehat{U}_{\text{per}}^{\text{tot}}(G_*)$.

Hence:

$$\boxed{\text{for the canonical symmetric BCC background, scalar-Hessian cancellation of } B_0^{\text{tot}} \text{ is impossible.}} \quad (708)$$

7. G. Consequence

Thus, within the actual microscopic scalar Hessian channel, the shell-pair constant mass does not cancel:

$$B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2 \neq 0. \quad (709)$$

So the plain scalar channel does not produce a massless shell Dirac point. Any consistent microscopic Dirac mechanism must therefore come from a different structure, e.g.

1. an exact cancellation involving additional non-scalar sectors not present in H_Q ,
2. a graded/projection mechanism in which the dangerous scalar pair-mixing block is absent from the relevant reduced operator,
3. or a genuinely different microscopic completion beyond the plain scalar Hessian channel.

a. Exact logical status. What is proved here is exact:

$$\text{actual scalar Hessian on } Q \implies U_{\text{per}}^{\text{tot}}(x) = u|\phi_0(x)|^2 \implies \widehat{U}_{\text{per}}^{\text{tot}}(G_*) = 4uA^2 \implies B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2 \neq 0.$$

So, for the canonical symmetric background, no cancellation is available inside the scalar Hessian itself.

. **Summary**

$$\delta N_{\Phi_0}[\eta] = r\eta - \kappa \nabla^2 \eta + \lambda \nabla^4 \eta + u \left[(\Phi_0^\dagger \Phi_0) \eta + (\Phi_0^\dagger \eta) \Phi_0 + (\eta^\dagger \Phi_0) \Phi_0 \right].$$

Let

$$\Phi_0(x) = \phi_0(x)\zeta, \quad \phi_0(x) = A \sum_{Q \in S_{\text{BCC}}} e^{iQ \cdot x}, \quad Q = (\mathbb{C}\zeta)^\perp.$$

For $\eta \in \Gamma(Q)$, one has

$$\Phi_0^\dagger \eta = 0, \quad \eta^\dagger \Phi_0 = 0,$$

so the rank-one Hessian terms vanish identically on Q .

Therefore the actual scalar periodic Hessian on the internal doublet sector is

$$H_Q = r - \kappa \nabla^2 + \lambda \nabla^4 + u |\phi_0(x)|^2, \quad U_{\text{per}}^{\text{tot}}(x) = u |\phi_0(x)|^2.$$

For the canonical symmetric BCC first-shell condensate,

$$\widehat{\phi}_0^2(G_*) = 4A^2,$$

hence

$$\widehat{U}_{\text{per}}^{\text{tot}}(G_*) = 4uA^2.$$

Using the shell-pair overlap formula

$$B_0 = \widehat{U}(G_*) \mathbf{1}_2,$$

we obtain

$$\boxed{B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2.}$$

Thus, if $uA^2 \neq 0$,

$$\boxed{B_0^{\text{tot}} \neq 0.}$$

Since the other Hessian contributions vanish exactly on Q , there is no available scalar-Hessian cancellation mechanism inside the canonical symmetric background.

Therefore:

$$\boxed{\text{plain scalar Hessian channel fails to eliminate the Bragg mass.}}$$

. **Graded projector-Hessian calculation: why the current refined model has $J = 0$**

a. Goal. We test whether the projector/ Q -sector can remove or grade away the scalar Bragg mass

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2$$

at quadratic order in the current refined microscopic model.

1. *A. Refined microscopic free energy*

The refined model is

$$F_{\text{ref}} = F_{\text{triplet}} + F_{\text{align}}, \tag{710}$$

with

$$F_{\text{triplet}} = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \tag{711}$$

and

$$F_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P) = \frac{\chi}{2} \int d^3x \rho^2 (D_i z)^\dagger (D_i z). \tag{712}$$

2. B. Locked constant-projector background

Take the polarization-locked background

$$\Psi_0(x) = \rho_0(x)z_0, \quad z_0^\dagger z_0 = 1, \quad z_0 = \text{const}, \quad P_0 = z_0 z_0^\dagger = \text{const}. \quad (713)$$

Hence

$$\partial_i P_0 = 0, \quad D_i z_0 = 0. \quad (714)$$

Now decompose fluctuations as

$$\Psi = (\rho_0 + \sigma)z_0 + \xi, \quad z_0^\dagger \xi = 0. \quad (715)$$

Equivalently, in projector language,

$$z = z_0 + \pi + O(\pi^2), \quad z_0^\dagger \pi = 0, \quad (716)$$

or

$$P = P_0 + \delta P + O((\delta P)^2), \quad \delta P = z_0 \pi^\dagger + \pi z_0^\dagger. \quad (717)$$

3. C. Quadratic expansion of the projector stiffness term

Since $\partial_i P_0 = 0$, one has

$$\partial_i P = \partial_i \delta P + O(\pi \partial \pi, \pi^2). \quad (718)$$

Therefore, to quadratic order,

$$F_{\text{align}}^{(2)} = \frac{\chi}{4} \int d^3 x \rho_0(x)^2 \text{Tr}(\partial_i \delta P \partial_i \delta P). \quad (719)$$

In the constant-amplitude approximation $\rho_0 \simeq \bar{\rho}$,

$$F_{\text{align}}^{(2)} \approx \frac{\chi \bar{\rho}^2}{2} \int d^3 x \partial_i \pi^\dagger \partial_i \pi. \quad (720)$$

This is crucial: $F_{\text{align}}^{(2)}$ contains only projector-tangent quadratic terms, with no piece linear in π , and no bilinear term involving σ or ξ from the shell sector.

4. D. Absence of shell-projector bilinear mixing

The shell-enriched tensor N_{ij}^a would have to come from a quadratic mixing source of schematic form

$$L_{\text{mix}}^{(2)} \sim J[\sigma, \psi] \cdot \xi \quad (721)$$

or, in projector variables,

$$L_{\text{mix}}^{(2)} \sim J[\sigma, \psi] \cdot \pi. \quad (722)$$

But from (720) we see that F_{align} begins at order π^2 , not at order π . Moreover, because P_0 is constant, there is no term of the schematic form

$$(\partial_i \rho_0) (\partial_j \psi) \pi^a$$

or

$$(\partial_i \phi_0) (-i \partial_j) \otimes s^a$$

generated at tree level from F_{align} .

Thus the quadratic source vanishes:

$$J = 0. \quad (723)$$

Equivalently, the quadratic Hessian is block diagonal:

$$\delta^2 F_{\text{ref}} = \delta^2 F_{\sigma} + \delta^2 F_{\xi}, \quad (724)$$

with no bilinear σ - ξ term.

[Block-diagonal projector Hessian theorem] For the current refined model $F_{\text{ref}} = F_{\text{triplet}} + F_{\text{align}}$, expanded around a locked constant-projector background $P_0 = \text{const}$, the quadratic Hessian is block diagonal between the shell sector and the tangent-projector sector:

$$\delta^2 F_{\text{ref}} = \delta^2 F_{\sigma} + \delta^2 F_{\xi}. \quad (725)$$

In particular,

$$J = 0. \quad (726)$$

The triplet part respects the locked decomposition $\Psi = (\rho_0 + \sigma)z_0 + \xi$ with z_0 constant, while the projector stiffness term begins at quadratic order in $\delta P \sim \pi$ because $\partial_i P_0 = 0$. Therefore no term linear in π and bilinear with the shell sector appears at quadratic order. This gives (725) and (726).

5. E. Consequence for induced shell-enriched tensors

The previously discussed mechanism for producing a correction δN_{ij}^a from the projector sector is Gaussian elimination of a heavy projector mode:

$$\delta L_{\text{eff}} = -\frac{1}{2} J G_{\pi} J. \quad (727)$$

But if $J = 0$, then

$$\delta L_{\text{eff}} = 0, \quad \delta N_{ij}^a = 0 \quad (728)$$

at quadratic order in the current refined model.

Thus the projector sector does not modify the leading witness triple:

$$\delta W_{\parallel} = \delta W_I = \delta W_J = 0 \quad (\text{at this order}). \quad (729)$$

6. F. Implication for the scalar Bragg mass

We previously proved that in the actual scalar Hessian channel on the internal doublet sector,

$$B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2 \neq 0. \quad (730)$$

The natural hope was that the projector/ Q -sector might generate a compensating graded or off-diagonal contribution. However, by (726)–(728), the current projector stiffness sector does not couple linearly to the shell block at quadratic order.

Therefore the projector sector does *not* cancel or grade away the scalar Bragg mass:

in the current refined model, B_0^{tot} survives unchanged at quadratic order.

(731)

7. G. What the projector sector actually does

Although it does not cure the scalar Bragg mass problem, the projector stiffness term has two direct effects:

1. it gives the tangent projector mode a positive inverse propagator of order

$$G_\pi^{-1}(q_*) \sim \chi \bar{\rho}^2 q_*^2, \quad (732)$$

so projector modes are pushed upward into the remote sector;

2. because the stiffness term is even under $G_* \leftrightarrow -G_*$, it tends to preserve opposite-valley matching rather than spoil it.

Thus its role is stabilizing, not mass-canceling.

8. H. Final conclusion

The graded projector-Hessian calculation gives a sharp negative result:

$$\text{current refined projector sector} \implies J = 0 \implies \delta N_{ij}^a = 0 \implies \text{no cancellation of } B_0^{\text{tot}}.$$

So if one wants a genuine microscopic Dirac mechanism, one must introduce an additional UV coupling whose quadratic expansion contains a nonzero shell-projector mixing source

$$L_{\text{mix}}^{(2)} \sim J[\sigma, \psi] \cdot \xi.$$

Only then can the projector sector induce a nontrivial δN_{ij}^a and potentially alter the constant shell-pair block.

a. Exact logical status. What is now closed is not merely that the current model fails heuristically, but the stronger statement:

for the current refined $U(3)$ -invariant triplet + projector-stiffness model, $J = 0$ at quadratic order.

(733)

Hence this model cannot remove the scalar Bragg mass through the projector sector.

. Summary

$$F_{\text{ref}} = F_{\text{triplet}} + F_{\text{align}}, \quad F_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P).$$

Expand around the locked constant-projector background

$$\Psi_0(x) = \rho_0(x) z_0, \quad z_0 = \text{const}, \quad P_0 = z_0 z_0^\dagger = \text{const}, \quad \partial_i P_0 = 0.$$

Write fluctuations as

$$\Psi = (\rho_0 + \sigma) z_0 + \xi, \quad z_0^\dagger \xi = 0,$$

or equivalently

$$P = P_0 + \delta P + O((\delta P)^2), \quad \delta P = z_0 \pi^\dagger + \pi z_0^\dagger.$$

Then

$$\partial_i P = \partial_i \delta P + O(\pi \partial \pi, \pi^2),$$

so to quadratic order

$$F_{\text{align}}^{(2)} = \frac{\chi}{4} \int d^3x \rho_0(x)^2 \text{Tr}(\partial_i \delta P \partial_i \delta P) \approx \frac{\chi \bar{\rho}^2}{2} \int d^3x \partial_i \pi^\dagger \partial_i \pi.$$

Therefore the projector stiffness sector begins at order π^2 , with no term linear in π , and hence no quadratic shell-projector mixing source

$$L_{\text{mix}}^{(2)} \sim J[\sigma, \psi] \cdot \pi.$$

Thus

$$J = 0,$$

and the quadratic Hessian is block diagonal:

$$\delta^2 F_{\text{ref}} = \delta^2 F_\sigma + \delta^2 F_\xi.$$

Consequently,

$$\delta L_{\text{eff}} = -\frac{1}{2} J G_\pi J = 0, \quad \delta N_{ij}^a = 0, \quad \delta W_{\parallel} = \delta W_I = \delta W_J = 0$$

at quadratic order in the current refined model.

Since the actual scalar Hessian channel already gave

$$B_0^{\text{tot}} = 4uA^2 \mathbf{1}_2 \neq 0,$$

the projector/ Q -sector does not cancel or grade away this scalar Bragg mass.

Hence:

$$\text{in the current refined } U(3)\text{-invariant triplet + projector-stiffness model, } B_0^{\text{tot}} \text{ survives unchanged at quadratic order.}$$

The projector sector improves remote-gap stability but does not solve the scalar Bragg-mass obstruction. A genuine induced correction requires an additional UV coupling producing

$$L_{\text{mix}}^{(2)} \sim J[\sigma, \psi] \cdot \xi$$

with $J \neq 0$.

. Class II mediator in projector language: explicit quadratic source with $J \neq 0$

a. Goal. We now revisit the quadratic shell-projector mixing problem in projector language and show that, although the current refined projector-stiffness model gives $J = 0$, the minimal Class II auxiliary mediator generates a genuine nonzero quadratic source

$$L_{\text{mix}}^{(2)} \sim J[\psi] \cdot \xi$$

and hence a nonzero induced shell-enriched tensor correction δN_{ij}^a .

1. A. What failed before

For the refined microscopic model

$$F_{\text{ref}} = F_{\text{triplet}} + F_{\text{align}}, \quad F_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P),$$

expanded around a locked constant-projector background

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger = \text{const}, \quad \partial_i P_0 = 0,$$

the quadratic projector piece begins at order $(\delta P)^2 \sim \xi^2$, so the shell-projector bilinear source vanishes:

$$J = 0.$$

Hence the current refined model alone cannot induce a nonzero δN_{ij}^a at quadratic order.

2. B. Minimal Class II UV extension

Introduce a heavy auxiliary internal vector mediator

$$X_i^a, \quad i = 1, 2, 3, \quad a = 1, 2, 3. \quad (734)$$

The minimal mediated quadratic extension is

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \Psi^\dagger T^a \Psi + \text{h.c.}) + \beta X_i^a \partial_i \delta P^a \psi. \quad (735)$$

This is exactly the lowest-order local UV term carrying both:

1. one spatial derivative index i ,
2. one internal shell-reduced index a .

So it is precisely of the type required by the shell-enriched target structure.

3. C. Expansion around the locked background

Expand around

$$\Psi_0 = \phi_0 z_0, \quad \delta P^a \sim \xi^a. \quad (736)$$

The shell current contains

$$\partial_i \Psi^\dagger T^a \Psi = (\partial_i \phi_0) z_0^\dagger T^a z_0 \phi_0 + (\partial_i \phi_0) C^a[\psi] + \cdots, \quad (737)$$

where the first term is a pure background contribution and the second term is the shell/test-mode contribution linear in the test fluctuation ψ .

Also,

$$\partial_i \delta P^a \sim \partial_i \xi^a. \quad (738)$$

Therefore the part of $L_{\text{UV}}^{(II)}$ relevant for quadratic shell-projector mixing is

$$L_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \phi_0) C^a[\psi] + \beta X_i^a \partial_i \xi^a. \quad (739)$$

4. D. Eliminating the heavy mediator

Since X_i^a is auxiliary, its Euler–Lagrange equation is algebraic:

$$\frac{\partial L_{\text{UV}}^{(2)}}{\partial X_i^a} = M_X^2 X_i^a + \alpha (\partial_i \phi_0) C^a[\psi] + \beta \partial_i \xi^a = 0. \quad (740)$$

Hence

$$X_i^a = -\frac{1}{M_X^2} \left[\alpha (\partial_i \phi_0) C^a[\psi] + \beta \partial_i \xi^a \right]. \quad (741)$$

Substituting back into (739) yields

$$L_{\text{eff}}^{(2)} = -\frac{1}{2M_X^2} \left[\alpha (\partial_i \phi_0) C^a[\psi] + \beta \partial_i \xi^a \right]^2. \quad (742)$$

The cross term is

$$L_{\text{mix}}^{(2)} = -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) C^a[\psi] \partial_i \xi^a. \quad (743)$$

Thus the quadratic source is

$$J_i^a[\psi] = -\frac{\alpha\beta}{M_X^2}(\partial_i\phi_0)C^a[\psi]. \quad (744)$$

In particular,

$$\boxed{J \neq 0} \quad (745)$$

whenever $C^a[\psi]$ is nontrivial.

5. E. Target shell-enriched structure

The source (743) is already of the schematic target form

$$L_{\text{mix}}^{(2)} \sim (\partial_i\phi_0)(\partial_j\psi)\xi^a, \quad (746)$$

after unpacking the derivative structure in $C^a[\psi]$ and integrating by parts when needed. Therefore integrating out ξ^a can now generate a genuine shell-enriched tensor correction

$$\delta N_{ij}^a \neq 0. \quad (747)$$

If the tangent projector mode has inverse propagator

$$G_\xi^{-1}(k) \sim \chi_{\text{eff}} k^2 + m_\xi^2, \quad (748)$$

then eliminating ξ yields schematically

$$\delta N_{ij}^a \sim \frac{\alpha\beta}{M_X^2} \frac{1}{\chi_{\text{eff}} q_*^2 + m_\xi^2} \times (\text{projected tensor coefficient}). \quad (749)$$

6. F. Equivalent shell-reduced Class II form

The same mechanism can be written more directly at shell-reduced quadratic order. Take

$$L_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (-i\partial_j) s^b \psi. \quad (750)$$

Then the algebraic equation is

$$M_X^2 X_i^a + \lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (-i\partial_j) s^b \psi = 0, \quad (751)$$

so

$$X_i^a = -\frac{1}{M_X^2} \left[\lambda m^a \partial_i \phi_0 + g_X C_{ij}^{ab} (-i\partial_j) s^b \psi \right]. \quad (752)$$

Substituting back gives the cross term

$$\delta L_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (-i\partial_j) s^b \psi. \quad (753)$$

Comparing with the target enriched operator

$$\delta L_{\text{enr}} = \delta N_{ij}^b (\partial_i \phi_0) (-i\partial_j) s^b \psi, \quad (754)$$

we read off

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (755)$$

7. G. Consequence for the current program

The current refined projector-stiffness model failed because

$$J = 0.$$

The Class II mediator succeeds because

$$J \neq 0$$

already at quadratic order around the same locked constant-projector background. Therefore:

Class II is the first mathematically consistent route from projector-sector physics to $\delta N_{ij}^a \neq 0$.

This is precisely the missing ingredient that the bare projector stiffness sector could not provide.

a. Exact logical status. What is now closed is:

$$\text{locked constant-projector background} + \text{Class II mediator} \implies L_{\text{mix}}^{(2)} \sim (\partial_i \phi_0) C^a[\psi] \partial_i \xi^a \implies J \neq 0 \implies \delta N_{ij}^a \neq 0.$$

What remains open is the full microscopic uniqueness question, not the existence of a nonzero quadratic source.

. Summary

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \Psi^\dagger T^a \Psi + \text{h.c.}) + \beta X_i^a \partial_i \delta P^a \psi.$$

Around the locked background

$$\Psi_0 = \phi_0 z_0, \quad \delta P^a \sim \xi^a,$$

the shell current contains

$$\partial_i \Psi^\dagger T^a \Psi = (\partial_i \phi_0) z_0^\dagger T^a z_0 \phi_0 + (\partial_i \phi_0) C^a[\psi] + \cdots,$$

and

$$\partial_i \delta P^a \sim \partial_i \xi^a.$$

Thus the quadratic shell-projector mixing sector is

$$L_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \phi_0) C^a[\psi] + \beta X_i^a \partial_i \xi^a.$$

Eliminating the auxiliary field X_i^a gives

$$X_i^a = -\frac{1}{M_X^2} \left[\alpha (\partial_i \phi_0) C^a[\psi] + \beta \partial_i \xi^a \right],$$

hence

$$L_{\text{mix}}^{(2)} = -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) C^a[\psi] \partial_i \xi^a.$$

Therefore the quadratic source is

$$J_i^a[\psi] = -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) C^a[\psi],$$

so

$$\boxed{J \neq 0.}$$

Equivalently, in shell-reduced form,

$$L_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (-i \partial_j) s^b \psi,$$

and integrating out X_i^a gives

$$\delta L_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (-i \partial_j) s^b \psi.$$

Hence the induced shell-enriched tensor is

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} .}$$

So, unlike the bare projector-stiffness model where $J = 0$, the Class II mediator yields a genuine quadratic shell-projector source and therefore a nonzero induced tensor correction

$$\delta N_{ij}^a \neq 0.$$

. Constant pair-mixing selection theorem for the Class II correction

a. Goal. We determine whether the nonzero Class II-induced shell tensor

$$\delta N_{ij}^a$$

can directly modify or cancel the scalar Bragg mass obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2$$

found in the actual scalar Hessian channel.

1. A. Scalar Bragg mass channel

For the enriched shell-pair basis

$$\Psi_{\pm,a}(x) = e^{\pm i k_* \cdot x} u_a(x), \quad k_* = \frac{G_*}{2},$$

the constant pair-mixing block is

$$(B_0)_{ab} = \langle \Psi_{+,a}, H(k_*) \Psi_{-,b} \rangle. \quad (756)$$

For a scalar periodic potential $U_{\text{per}}(x)$, the projected Bragg-mass formula is

$$(B_0)_{ab} = \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-i G_* \cdot x} U_{\text{per}}(x) u_b(x) \quad (757)$$

up to possible gradient-sector contributions. :contentReference[oaicite:3]index=3

Under the constant-envelope ansatz,

$$B_0 = \hat{U}(G_*) \mathbf{1}_2. \quad (758)$$

Thus the actual scalar obstruction found earlier is of the operator form

$$H_{\text{Bragg}}^{\text{scalar}} = B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2. \quad (759)$$

2. B. Class II shell-reduced operator

The shell-reduced Class II quadratic extension is

$$L_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (-i \partial_j) s^b \psi. \quad (760)$$

Eliminating the auxiliary field X_i^a gives the cross term

$$\delta L_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (-i \partial_j) s^b \psi, \quad (761)$$

hence

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (762)$$

This is a gradient-linear enriched correction, exactly of the shell-witness type. Therefore the induced shell operator has the form

$$\delta H_{\text{Class II}} = \delta N_{ij}^b (\partial_i \phi_0) (-i \partial_j) \otimes s^b. \quad (763)$$

3. C. Constant pair-mixing from a gradient-enriched operator

To test whether (763) can affect the scalar Bragg mass, compute its pair-block matrix element at $p = 0$:

$$(\delta B_0^{\text{Class II}})_{ab} = \langle \Psi_{+,a}, \delta H_{\text{Class II}}(k_*) \Psi_{-,b} \rangle. \quad (764)$$

Using the shell-pair basis and applying $(-i \partial_j)$ to $\Psi_{-,b} = e^{-ik_* \cdot x} u_b(x)$,

$$(-i \partial_j) \Psi_{-,b} = (-k_{*,j} - i \partial_j) e^{-ik_* \cdot x} u_b(x). \quad (765)$$

Thus, at principal order,

$$(\delta B_0^{\text{Class II}})_{ab} = - \int_{\text{cell}} d^3x u_a^\dagger(x) e^{-iG_* \cdot x} \left[\delta N_{ij}^b (\partial_i \phi_0) k_{*,j} \right] u_b(x) (s^b)_{ab} + \dots \quad (766)$$

where the omitted terms come from derivatives of the periodic envelopes $u_b(x)$.

Under the constant-envelope ansatz, those derivative-of-envelope terms vanish at principal order, so

$$\delta B_0^{\text{Class II}} = -\hat{\mathcal{T}}^b(G_*) s^b, \quad (767)$$

for some shell Fourier coefficients $\hat{\mathcal{T}}^b(G_*)$ determined by $\delta N_{ij}^b (\partial_i \phi_0) k_{*,j}$.

Therefore any constant pair-mixing produced by Class II lies in the traceless internal sector:

$$\boxed{\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\}.} \quad (768)$$

4. D. Identity-sector selection rule

The scalar Bragg obstruction lies in the identity channel:

$$B_0^{\text{tot}} \mathbf{1}_2.$$

The Class II correction lies in the traceless channel:

$$\delta B_0^{\text{Class II}} \in \mathfrak{su}(2).$$

These two sectors are orthogonal under the Hilbert–Schmidt pairing:

$$\text{Tr}(\mathbf{1}_2 s_a) = 0. \quad (769)$$

Hence:

[Identity-sector selection theorem] The Class II induced tensor δN_{ij}^a cannot directly cancel the scalar Bragg mass obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Any constant pair-mixing generated by the Class II derivative operator belongs to the traceless internal sector

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not to the identity sector

$$\tau_1 \otimes \mathbf{1}_2.$$

Equation (763) shows that the Class II correction always carries an explicit internal generator s^b . Therefore its projected pair block, constant or linear, remains in the traceless internal matrix algebra generated by the Pauli matrices. By contrast, the scalar Bragg mass is proportional to the internal identity. Orthogonality (769) forbids direct cancellation between the two channels.

5. E. Natural shell-Dirac parity and vanishing of the constant gradient block

The shell-level Dirac literature in the uploaded manuscript already states a sharper condition: the gradient-sector contribution to B_0 vanishes if the first-order mixing is odd and momentum-linear. Also, the transverse mixing matrices are principal-order derivatives,

$$W^\alpha = \hat{G}_\alpha(G_*) s^\alpha, \quad (770)$$

and the shell-Dirac criterion is formulated precisely in terms of vanishing B_0 together with nonzero W^1, W^2 .

Thus, under the natural shell-exchange/reflection parity used for shell Dirac,

$$\boxed{\delta B_0^{\text{Class II}} = 0} \quad (771)$$

at principal order.

So the stronger conclusion is:

- generically, Class II cannot cancel the scalar identity mass because it lives in the wrong internal channel;
- under the natural shell-Dirac odd-parity choice, Class II does not even generate a constant pair-mixing block at all.

6. F. Consequence for the role of Class II

The uploaded manuscript states explicitly that the canonical minimal Class II correction changes the first-order shell tensor coefficients A, B, C , and does not by construction insert an independent constant shell-level Dirac mass term.

Therefore the actual role of Class II is

$$(A, B, C) \mapsto (A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}), \quad (772)$$

not

$$B_0^{\text{tot}} \mapsto 0. \quad (773)$$

[Constant pair-mixing selection theorem for Class II] The canonical Class II mediator renormalizes the derivative shell witness sector (A, B, C) , but does not remove the scalar Bragg-mass obstruction $B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2$. Under the natural odd shell-parity realization, the Class II constant pair-mixing contribution vanishes altogether.

7. *G. Honest conclusion*

The current chain is now sharp:

$$\text{actual scalar Hessian} \implies B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2 \neq 0,$$

while

$$\text{Class II} \implies \delta N_{ij}^a \neq 0 \implies \text{renormalization of } (A, B, C),$$

but

$$\text{Class II} \not\Rightarrow \text{cancellation of } B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

So the scalar Bragg-mass obstruction and the Class II derivative witness mechanism live in different operator channels.

. Summary

$$B_0^{\text{tot}} = \widehat{U}_{\text{per}}^{\text{tot}}(G_*) \mathbf{1}_2$$

is the scalar Bragg-mass channel, hence in the enriched shell-pair block it appears as

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

By contrast, the shell-reduced Class II correction is

$$\delta L_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (-i \partial_j) s^b \psi, \quad \delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.$$

Therefore every Class II contribution carries an explicit internal Pauli generator s^b . Its projected constant pair-mixing block, if nonzero, must lie in

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

so in the shell-pair block it is of the form

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not

$$\tau_1 \otimes \mathbf{1}_2.$$

Since

$$\text{Tr}(\mathbf{1}_2 s_a) = 0,$$

the scalar identity channel and the traceless Pauli channel are orthogonal. Hence Class II cannot directly cancel

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Moreover, under the natural shell-Dirac odd/reflection parity, the gradient-sector constant pair-mixing vanishes at principal order:

$$\delta B_0^{\text{Class II}} = 0.$$

Therefore the actual role of Class II is

$$(A, B, C) \mapsto (A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}),$$

not

$$B_0^{\text{tot}} \mapsto 0.$$

So:

Class II renormalizes the derivative Weyl-witness sector, but does not remove the scalar Bragg-mass obstruction.

. Identity-channel cancellation theorem

a. Goal. We classify the only possible routes for removing the scalar Bragg-mass obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2$$

in the enriched shell-pair Hamiltonian.

1. A. Constant pair-mixing decomposition

Let the constant pair-mixing block on the internal doublet be the most general Hermitian matrix

$$M_0 = m_0 \mathbf{1}_2 + m_1 s_1 + m_2 s_2 + m_3 s_3. \quad (774)$$

Then the enriched shell-pair Hamiltonian has the principal constant part

$$H_0^{\text{pair}} = \tau_1 \otimes M_0. \quad (775)$$

The scalar Bragg-mass obstruction computed from the actual scalar Hessian is precisely the identity-channel case

$$M_0^{\text{scalar}} = B_0^{\text{tot}} \mathbf{1}_2, \quad B_0^{\text{tot}} \neq 0. \quad (776)$$

2. B. Identity/traceless decomposition is orthogonal

The Hermitian 2×2 algebra decomposes as

$$\text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \mathfrak{su}(2)_{\text{Herm}}, \quad (777)$$

with orthogonality under the Hilbert–Schmidt pairing

$$\langle A, B \rangle_{\text{HS}} := \text{Tr}(AB). \quad (778)$$

Because

$$\text{Tr}(\mathbf{1}_2 s_a) = 0, \quad a = 1, 2, 3, \quad (779)$$

the identity channel and the traceless channel are orthogonal.

Hence, if a UV correction produces only a traceless internal constant block

$$\delta M_0 \in \mathfrak{su}(2)_{\text{Herm}}, \quad (780)$$

then its identity coefficient vanishes:

$$\frac{1}{2} \text{Tr}(\delta M_0) = 0. \quad (781)$$

Therefore the corrected identity coefficient is unchanged:

$$m_0^{\text{eff}} = m_0. \quad (782)$$

[Orthogonality obstruction] Let

$$M_0 = m_0 \mathbf{1}_2 + \widetilde{M}_0, \quad \widetilde{M}_0 \in \mathfrak{su}(2)_{\text{Herm}}.$$

If a UV correction satisfies $\delta M_0 \in \mathfrak{su}(2)_{\text{Herm}}$, then it cannot cancel $m_0 \mathbf{1}_2$. Equivalently,

$$m_0^{\text{eff}} = m_0. \quad (783)$$

Take the identity projection by the normalized trace:

$$m_0 = \frac{1}{2} \text{Tr}(M_0).$$

If δM_0 is traceless, then

$$\frac{1}{2} \text{Tr}(\delta M_0) = 0.$$

Hence

$$m_0^{\text{eff}} = \frac{1}{2} \text{Tr}(M_0 + \delta M_0) = \frac{1}{2} \text{Tr}(M_0) = m_0.$$

3. C. Application to Class II

The shell-reduced Class II correction has the form

$$\delta L_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab} (\partial_i \phi_0) (-i \partial_j) s^b \psi, \quad (784)$$

hence

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (785)$$

Every Class II contribution carries an explicit shell-reduced internal generator s^b . Therefore any constant pair block produced by Class II, if present, lies in the traceless internal sector:

$$\delta M_0^{\text{Class II}} \in \mathfrak{su}(2)_{\text{Herm}}. \quad (786)$$

Moreover, the manuscript explicitly states that Class II is acceptable only if it does not create a dominant constant intervalley mass term, and that it merely renormalizes the derivative shell tensor coefficients. :contentReference[oaicite:6]index=6

Therefore:

[Class II cannot cancel the scalar Bragg mass] The Class II mediator cannot directly cancel the scalar Bragg-mass obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Its effect is restricted to the traceless internal channels and to the derivative witness sector (A, B, C) .

4. D. What kind of UV structure can cancel the identity channel?

To cancel the scalar Bragg mass, one needs a UV correction whose projected constant pair block has a nonzero identity component:

$$\delta M_0^{\text{id}} = \delta m_0 \mathbf{1}_2 + \delta \widetilde{M}_0, \quad \delta m_0 \neq 0. \quad (787)$$

Then cancellation requires

$$\delta m_0 = -B_0^{\text{tot}}. \quad (788)$$

This gives the first route:

[Identity-carrying mediator route] A direct cancellation of the scalar Bragg mass is possible only if the UV completion produces an identity-channel constant pair block

$$\delta m_0 \tau_1 \otimes \mathbf{1}_2$$

with

$$\delta m_0 = -B_0^{\text{tot}}.$$

Any UV completion whose projected constant pair block is purely traceless cannot achieve this.

Immediate from the decomposition

$$\text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \mathfrak{su}(2)_{\text{Herm}}$$

and the trace projection onto the identity channel.

a. Interpretation. This means the extra UV sector must carry not only the reduced-shell triplet label a , but also an identity-channel projection capable of surviving the shell-pair reduction as $\mathbf{1}_2$.

5. E. The second route: graded bypass rather than cancellation

The manuscripts already identify the genuinely different route: the plain scalar Bragg pair is not sufficient, and one needs either

1. an internal complex doublet,
2. a graded $2 + 2$ low-energy defect sector,
3. or a genuinely first-order projected operator whose off-diagonal 2×2 symbol is already full-rank.

:contentReference[oaicite:8]index=8

Also, the exact-grading and selection-rule criteria state that if the leading reduced operator is odd under an involution J , then the same-parity blocks vanish, and if the projected transport operator is first-order, the reduced principal operator is block-off-diagonal first-order.

This yields the second route:

[Graded bypass route] Suppose the physical low-energy carrier is not the plain scalar Bragg pair but a J -graded $2 + 2$ low-energy sector, and the leading reduced operator $D_1(p)$ is:

1. J -odd,
2. first-order in momentum.

Then the relevant reduced principal operator contains no constant term at all:

$$D_1(0) = 0. \quad (789)$$

In this case the scalar Bragg-mass channel is not cancelled; rather, it is absent from the physically relevant reduced block.

If D_1 is first-order in momentum, then

$$D_1(p) = p_i M_i + O(|p|^2),$$

hence $D_1(0) = 0$. If in addition D_1 is J -odd, then in the J -graded decomposition it is block-off-diagonal. Thus the reduced low-energy operator is a first-order chiral block with no constant scalar Bragg mass term.

a. Interpretation. This is not a cancellation mechanism inside the plain scalar shell-pair block. It is a *different low-energy reduction* in which the dangerous scalar pair channel is no longer the relevant infrared block.

6. F. Final classification

We can now state the full classification.

[Identity-channel cancellation theorem] Let the enriched shell-pair constant block be

$$H_0^{\text{pair}} = \tau_1 \otimes M_0, \quad M_0 = B_0^{\text{tot}} \mathbf{1}_2 + \widetilde{M}_0.$$

Then there are exactly two logically distinct ways to avoid the scalar Bragg-mass obstruction:

1. **Direct identity-channel cancellation:** introduce a UV completion whose projected constant pair block has a nonzero identity component $\delta m_0 \mathbf{1}_2$ with

$$\delta m_0 = -B_0^{\text{tot}}.$$

2. **Graded bypass:** replace the plain scalar Bragg-pair low-energy reduction by a J -graded first-order $2 + 2$ reduction in which the relevant principal operator is block-off-diagonal and first-order, so the constant scalar Bragg block never appears in the effective infrared operator.

Any UV completion whose projected constant pair block is purely traceless, including the canonical Class II mediator, cannot realize route (1).

Route (1) follows from the identity/traceless decomposition and the trace projection argument. Route (2) follows from the exact-grading / selection-rule theorems together with first-order reduction. No other possibility exists at the reduced operator level: one must either cancel the identity component directly, or work in a different reduced low-energy block where the constant scalar channel is absent.

7. *G. Honest conclusion*

The current state is therefore:

$$\text{actual scalar Hessian} \implies B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2 \neq 0,$$

$$\text{Class II} \implies \delta N_{ij}^a \neq 0 \implies (A, B, C) \text{ renormalization only,}$$

but

$$\text{Class II} \not\Rightarrow \delta m_0 \mathbf{1}_2.$$

So the next genuinely new microscopic step is not more Class II algebra. It is to choose between:

identity-carrying mediator	or	graded 2 + 2 low-energy bypass.
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(790)

. **Summary**

$$M_0 = m_0 \mathbf{1}_2 + m_1 s_1 + m_2 s_2 + m_3 s_3$$

is the most general constant internal pair-mixing block. The scalar Bragg-mass obstruction is

$$M_0^{\text{scalar}} = B_0^{\text{tot}} \mathbf{1}_2.$$

Since

$$\text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \mathfrak{su}(2)_{\text{Herm}}, \quad \text{Tr}(\mathbf{1}_2 s_a) = 0,$$

any traceless UV correction

$$\delta M_0 \in \mathfrak{su}(2)_{\text{Herm}}$$

cannot change the identity coefficient:

$$m_0^{\text{eff}} = m_0.$$

The canonical Class II mediator always carries an explicit reduced-shell Pauli generator s^a , so its projected correction is traceless in the internal sector. Therefore Class II cannot directly cancel

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Hence direct cancellation requires a UV completion with an identity-channel projected block

$$\delta m_0 \tau_1 \otimes \mathbf{1}_2, \quad \delta m_0 = -B_0^{\text{tot}}.$$

The only alternative is not cancellation but bypass: if the true low-energy carrier is a J -graded 2 + 2 sector and the leading reduced operator is J -odd and first-order, then

$$D_1(0) = 0$$

and the scalar Bragg-mass channel is absent from the relevant infrared block.

Therefore there are exactly two ways around the scalar Bragg-mass obstruction:

(1) identity-channel cancellation	or	(2) graded 2 + 2 bypass.
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Any purely traceless UV correction, including the canonical Class II mediator, realizes neither direct identity-channel cancellation nor removal of B_0^{tot} . It only renormalizes the derivative witness sector (A, B, C) .

. **Minimal graded $2+2$ low-energy carrier and the Dirac principal operator**

a. Goal. We now construct the minimal graded $2+2$ low-energy ansatz that bypasses the scalar Bragg-mass channel. The aim is to derive, at the reduced infrared level,

$$D_{\text{eff}}(p) = D_1(p) + O(|p|^2), \quad D_1(0) = 0,$$

with

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu.$$

1. *A. Graded infrared sector*

Let $L(p)$ denote the linearized microscopic or effective operator near the infrared point $p = 0$. Assume there exists an isolated four-dimensional infrared sector

$$K \subset H, \quad \dim_{\mathbb{C}} K = 4. \quad (791)$$

Assume further that K carries an involutive grading operator

$$J : K \rightarrow K, \quad J^\dagger = J, \quad J^2 = 1. \quad (792)$$

Then

$$K = K_+ \oplus K_-, \quad K_\pm = \ker(J \mp 1), \quad \dim_{\mathbb{C}} K_\pm = 2. \quad (793)$$

Let Π_K be the spectral projector onto K , and define the reduced operator

$$D_{\text{eff}}(p) := \Pi_K L(p) \Pi_K. \quad (794)$$

Expanding near $p = 0$,

$$D_{\text{eff}}(p) = D_1(p) + O(|p|^2). \quad (795)$$

2. *B. Exact grading criterion*

Assume the leading reduced operator is odd under the grading:

$$J D_1(p) J^{-1} = -D_1(p). \quad (796)$$

Let $\Pi_\pm$ denote the projectors onto $K_\pm$. In the graded basis $K = K_+ \oplus K_-$, write

$$D_1(p) = \begin{pmatrix} D_{++}(p) & D_{+-}(p) \\ D_{-+}(p) & D_{--}(p) \end{pmatrix}. \quad (797)$$

Then (796) implies

$$D_{++}(p) = 0, \quad D_{--}(p) = 0, \quad (798)$$

hence

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad (799)$$

for some 2×2 mixed symbol $A_1(p)$.

In particular, because D_1 is first-order,

$$D_1(0) = 0. \quad (800)$$

3. C. Minimal projector/defect ansatz

Now realize this grading concretely in the defect/worldvolume setting. Let ξ denote the transverse defect coordinate and y^μ the slow worldvolume coordinates. Assume there exist four low-energy modes

$$u_a^+(\xi), \quad a = 1, 2, \quad u_b^-(\xi), \quad b = 1, 2, \quad (801)$$

such that

$$Ju_a^\pm = \pm u_a^\pm. \quad (802)$$

Use the envelope ansatz

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a^+(y) u_a^+(\xi) + \sum_{b=1}^2 \psi_b^-(y) u_b^-(\xi) + \cdots. \quad (803)$$

Assume the leading transport operator has the first-order form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (804)$$

and is odd under the grading,

$$JL_1J^{-1} = -L_1. \quad (805)$$

A sufficient concrete realization is a reflection-odd transverse kernel

$$V^\mu(-\xi) = -V^\mu(\xi). \quad (806)$$

Then the projected matrix elements in the low-energy sector satisfy

$$\langle u_a^\pm, L_1 u_b^\pm \rangle_\xi = 0, \quad (807)$$

while the mixed ones define

$$(M^\mu)_{ab} := \langle u_a^+, V^\mu u_b^- \rangle_\xi. \quad (808)$$

Hence

$$A_1(p) = p_\mu M^\mu, \quad (809)$$

and therefore

$$D_1(p) = \begin{pmatrix} 0 & p_\mu M^\mu \\ p_\mu (M^\mu)^\dagger & 0 \end{pmatrix}. \quad (810)$$

4. D. Nondegeneracy criterion

The reduced principal operator is Dirac-type iff the mixed symbol is nondegenerate:

$$\det A_1(p) \neq 0 \quad \text{for generic } p \neq 0. \quad (811)$$

Equivalently,

$$\det(p_\mu M^\mu) \neq 0 \quad \text{for generic } p \neq 0. \quad (812)$$

5. E. Explicit full-rank realization

Choose the mixed symbol matrices to be a Pauli triplet

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (813)$$

Then

$$A_1(p) = p_\mu M^\mu = I_g (p_1 \sigma_1 + p_2 \sigma_2 + p_3 \sigma_3). \quad (814)$$

Therefore

$$\det A_1(p) = -I_g^2 (p_1^2 + p_2^2 + p_3^2), \quad (815)$$

which is nonzero for generic $p \neq 0$.

Thus the reduced principal operator becomes

$$D_1(p) = I_g \begin{pmatrix} 0 & p_i \sigma_i \\ p_i \sigma_i & 0 \end{pmatrix}. \quad (816)$$

6. F. Dirac form

Define the 4×4 matrices

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}. \quad (817)$$

Then

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4. \quad (818)$$

Hence

$$D_1(p) = I_g p_i \alpha_i, \quad (819)$$

which is exactly the massless Dirac principal operator.

In position space, after inverse Fourier transform $p_i \mapsto -i\partial_i$,

$$i\partial_t \Psi = -iI_g \alpha_i \partial_i \Psi + O(\partial^2) \Psi. \quad (820)$$

7. G. Why this bypasses the scalar Bragg-mass obstruction

The scalar Bragg-mass obstruction arose in the plain shell-pair carrier as a constant identity-channel block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

But in the graded $2+2$ reduction, the relevant principal infrared operator is *defined* by the first-order projected sector $D_1(p)$, and the grading condition forces

$$D_1(0) = 0.$$

Thus the scalar constant pair-mixing block is not cancelled inside this reduced operator; rather, it is absent from the graded first-order infrared principal block.

So this route is a *bypass*, not a cancellation.

8. *H. Honest logical status*

What is proved here is the exact reduced-operator theorem:

$$\text{isolated graded } 2 + 2 \text{ sector} + \text{first-order } J\text{-odd transport} \implies D_1(p) = \begin{pmatrix} 0 & p_\mu M^\mu \\ p_\mu (M^\mu)^\dagger & 0 \end{pmatrix}, \quad D_1(0) = 0.$$

And if the mixed symbol is full rank, then

$$D_1(p) \sim p_i \alpha_i$$

is Dirac.

What remains open is the microscopic realization problem: that the full TECT defect/projector dynamics actually produces this isolated graded $2 + 2$ sector and the required nondegenerate mixed first-order symbol.

. Summary

$$K \subset H, \quad \dim_{\mathbb{C}} K = 4, \quad K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

with an involution

$$J^\dagger = J, \quad J^2 = 1.$$

Let

$$D_{\text{eff}}(p) = \Pi_K L(p) \Pi_K = D_1(p) + O(|p|^2).$$

If the leading reduced operator is J -odd,

$$J D_1(p) J^{-1} = -D_1(p),$$

then in the graded basis

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad D_1(0) = 0.$$

For a defect/worldvolume ansatz

$$\Phi(y, \xi) = \sum_{a=1}^2 \psi_a^+(y) u_a^+(\xi) + \sum_{b=1}^2 \psi_b^-(y) u_b^-(\xi) + \cdots, \quad J u_a^\pm = \pm u_a^\pm,$$

with leading transport

$$L_1 = -i V^\mu(\xi) \partial_{y^\mu}, \quad J L_1 J^{-1} = -L_1,$$

the same-parity projected matrix elements vanish:

$$\langle u_a^\pm, L_1 u_b^\pm \rangle_\xi = 0.$$

The mixed symbol is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle_\xi,$$

so

$$A_1(p) = p_\mu M^\mu, \quad D_1(p) = \begin{pmatrix} 0 & p_\mu M^\mu \\ p_\mu (M^\mu)^\dagger & 0 \end{pmatrix}.$$

If one chooses an explicit full-rank realization

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0,$$

then

$$A_1(p) = I_g p_i \sigma_i, \quad \det A_1(p) = -I_g^2 |p|^2 \neq 0 \quad (p \neq 0),$$

and

$$D_1(p) = I_g \begin{pmatrix} 0 & p_i \sigma_i \\ p_i \sigma_i & 0 \end{pmatrix} = I_g p_i \alpha_i, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4.$$

Therefore the leading infrared operator is exactly massless Dirac:

$$\boxed{D_1(0) = 0, \quad D_1(p) \sim p_i \alpha_i.}$$

This avoids the scalar Bragg-mass obstruction not by cancellation, but by working in a different graded first-order infrared block.

. Actual TECT defect-sector realization of the graded 2 + 2 bypass

a. Goal. We now embed the abstract graded 2 + 2 low-energy Dirac theorem into the actual CP^2 projector-defect fluctuation problem of TECT. The purpose is to identify the low modes $u_a^\pm(\xi)$, the odd transport kernel $V^\mu(\xi)$, and the projected mixed symbol

$$A_1(p) = p_\mu M^\mu$$

directly in projector-defect variables.

1. A. Tangent projector fluctuation as a rank-two Q_B -bundle field

Let the static defect background be

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B(\xi)^\dagger z_B(\xi) = 1, \quad (821)$$

with transverse coordinate ξ and worldvolume coordinates y^μ .

Because $P^2 = P$, linearized projector fluctuations satisfy

$$P_B \delta P + \delta P P_B = \delta P. \quad (822)$$

Using a local normalized representative z_B , a convenient tangent parametrization is

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0. \quad (823)$$

Hence the physical fluctuation field is a rank-two complex field

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp. \quad (824)$$

Thus the defect fluctuation problem is not a scalar one: it is already a rank-two bundle problem on Q_B .

2. B. Quadratic Hessian as a transverse matrix Schrödinger operator

The quadratic fluctuation action has the generic form

$$\delta^2 S = \int d^{d+1}x \eta^\dagger \left[-\rho D_\mu^{(B)} D^{(B)\mu} + M_B(\xi) \right] \eta, \quad (825)$$

where $D_\mu^{(B)}$ is the induced connection on Q_B and $M_B(\xi)$ is a Hermitian 2×2 endomorphism built from the defect background and its derivatives.

Because the background depends only on ξ , the operator splits into transverse and worldvolume pieces. In a local orthonormal Q_B -frame

$$e(\xi) = (e_1(\xi), e_2(\xi)), \quad (826)$$

writing

$$\eta(\xi, y) = e_a(\xi) \chi^a(\xi, y), \quad (827)$$

the static transverse Hessian becomes a 2×2 matrix Schrödinger operator

$$H_B^{\text{stat}} = -\partial_\xi^2 + W_B(\xi) \quad (\text{up to frame connection terms}). \quad (828)$$

This is the actual TECT projector-Hessian realization of the abstract low-mode operator class.

3. C. Reflection symmetry and parity channels

Assume the defect background is reflection-symmetric:

$$P_B(-\xi) = P_B(\xi). \quad (829)$$

Then the static Hessian commutes with reflection

$$(J\chi)(\xi) = \chi(-\xi). \quad (830)$$

Therefore its low transverse spectrum decomposes into even and odd parity sectors.

Assume that in each internal channel there exists one localized even low mode and one localized odd low mode:

$$f_+(\xi), \quad f_-(\xi), \quad (831)$$

with

$$f_+(-\xi) = +f_+(\xi), \quad f_-(-\xi) = -f_-(\xi). \quad (832)$$

Define the four low modes

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (833)$$

They span the low-energy sector

$$K = K_+ \oplus K_-, \quad K_\pm = \text{span}_{\mathbb{C}}\{u_1^\pm, u_2^\pm\}, \quad \dim_{\mathbb{C}} K_\pm = 2. \quad (834)$$

Thus the actual projector Hessian naturally produces the required graded $2 + 2$ carrier.

4. D. Odd first-order worldvolume transport kernel

Restoring slow worldvolume dependence y^μ , the full linearized operator acquires a first-order transport term

$$L_1 = -iV^\mu(\xi) \partial_{y^\mu}. \quad (835)$$

The local tensors from which $V^\mu(\xi)$ is built are linear in the transverse defect gradient, e.g.

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi), \quad (836)$$

or equivalent frame-covariant combinations.

Since $P_B(\xi)$ is even, all such first derivatives are odd:

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi), \quad \theta'(-\xi) = -\theta'(\xi). \quad (837)$$

Hence the transport kernel is reflection-odd:

$$V^\mu(-\xi) = -V^\mu(\xi). \quad (838)$$

This is exactly the structural J -oddness required by the graded reduction theorem.

5. *E. Projected overlap and the mixed symbol*

Project L_1 onto the low-energy sector K . The same-parity matrix elements vanish:

$$\langle u_a^\pm, L_1 u_b^\pm \rangle_\xi = 0. \quad (839)$$

Indeed, $f_\pm^2$ is even while V^μ is odd, so the integrand is odd.

The mixed block is

$$(M^\mu)_{ab} := \langle u_a^+, V^\mu u_b^- \rangle_\xi = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (840)$$

Because $f_+ f_-$ is odd and V^μ is odd, the integrand is even, so this overlap is generically nonzero.

Therefore the leading reduced operator on $K = K_+ \oplus K_-$ is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu. \quad (841)$$

In particular,

$$D_1(0) = 0. \quad (842)$$

Thus the dangerous constant scalar Bragg block is absent from the relevant first-order graded defect operator.

6. *F. Minimal explicit full-rank realization inside the actual operator class*

The previous step is structural and exact. To get an explicit Dirac symbol, choose a local Q_B -frame in which the projected mixed matrices form a Pauli triplet:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (843)$$

Then

$$A_1(p) = I_g p_i \sigma_i, \quad (844)$$

and

$$D_1(p) = I_g \begin{pmatrix} 0 & p_i \sigma_i \\ p_i \sigma_i & 0 \end{pmatrix}. \quad (845)$$

This is the massless Dirac principal operator.

Equivalently, with

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4, \quad (846)$$

one has

$$D_1(p) = I_g p_i \alpha_i. \quad (847)$$

7. *G. What is actual and what is still open*

What is rigorously matched to the actual TECT defect projector Hessian is:

1. the rank-two Q_B -bundle fluctuation field,
2. the matrix Schrödinger transverse Hessian,
3. the even/odd parity channel decomposition,
4. the odd first-order transport kernel,

5. the off-diagonal projected overlap formula.

Therefore the graded 2 + 2 Dirac bypass is not ad hoc; it belongs to the actual local $\mathbb{CP}^2$ projector-Hessian operator class.

What is still open for the full microscopic TECT dynamics is stronger:

1. proving that the actual defect background $P_B(\xi)$ indeed has the required isolated even/odd low modes,
2. proving that the projected mixed symbol M^μ is full rank for the actual $V^\mu(\xi)$,
3. deriving the explicit Pauli-triplet form (843) directly from the full microscopic transport kernel rather than from an explicit representative in the same operator class.

So the strongest honest statement is:

$$\text{actual projector Hessian} \implies \text{graded } 2 + 2 \text{ operator class,}$$

and

$$\text{full-rank Pauli-triplet realization} \implies \text{explicit Dirac principal symbol.}$$

. Summary

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \quad z_B^\dagger \eta = 0, \quad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp.$$

Thus the actual $\mathbb{CP}^2$ defect fluctuation problem is a rank – two bundle problem on Q_B . Its quadratic Hessian is a transverse matrix Schrödinger operator on Q_B .

For a reflection-symmetric defect background

$$P_B(-\xi) = P_B(\xi),$$

the static transverse spectrum decomposes into even/odd low channels

$$f_+(\xi), \quad f_-(\xi), \quad f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi).$$

Using a local Q_B -frame $e_a(\xi)$, define

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2.$$

These span a graded low-energy sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_\pm = 2.$$

The leading worldvolume transport operator has the form

$$L_1 = -i V^\mu(\xi) \partial_{y^\mu},$$

and because it is built linearly from $\partial_\xi P_B, \partial_\xi z_B, \theta'(\xi)$, it is reflection-odd:

$$V^\mu(-\xi) = -V^\mu(\xi).$$

Hence same-parity overlaps vanish,

$$\langle u_a^\pm, L_1 u_b^\pm \rangle_\xi = 0,$$

while the mixed projected symbol is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi).$$

Therefore the reduced principal operator on K is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad D_1(0) = 0.$$

So the graded $2 + 2$ bypass is structurally realized by the actual $\text{CP}^2 \text{projector} - \text{defectHessian}$.
If, in addition, one chooses an explicit full-rank representative in this operator class

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0,$$

then

$$A_1(p) = I_g p_i \sigma_i,$$

and

$$D_1(p) = I_g \begin{pmatrix} 0 & p_i \sigma_i \\ p_i \sigma_i & 0 \end{pmatrix} = I_g p_i \alpha_i, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4.$$

Thus the actual TECT defect-sector realization reaches:

$$\boxed{\text{projector Hessian} \Rightarrow \text{graded } 2 + 2 \text{ carrier} \Rightarrow \text{off-diagonal first-order symbol}}$$

and, with full-rank mixed overlap,

$$\boxed{D_1(p) \sim p_i \alpha_i \quad (\text{Dirac principal operator}).}$$

. Explicit evaluation of the graded mixed overlap integral in the CP^2 defect sector

a. Goal. We now evaluate explicitly the graded mixed overlap integral

$$(M^\mu)_{ab} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi)$$

for a concrete reflection-symmetric CP^2 defect representative in the actual TECT projector-Hessian operator class.

1. A. Explicit reflection-symmetric defect background

Take the explicit CP^2 -type defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad \theta(\xi) = \theta_0 \operatorname{sech}(\kappa \xi). \quad (848)$$

Then

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger \quad (849)$$

is reflection-even:

$$P_B(-\xi) = P_B(\xi). \quad (850)$$

An explicit even orthonormal frame of

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp \quad (851)$$

is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (852)$$

Because $\theta(\xi)$ is even, both $e_1(\xi)$ and $e_2(\xi)$ are even under $\xi \mapsto -\xi$.

2. B. Explicit even/odd low modes

Take the standard normalized localized even/odd pair

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad (853)$$

$$f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi). \quad (854)$$

Then

$$f_+(-\xi) = f_+(\xi), \quad f_-(-\xi) = -f_-(\xi). \quad (855)$$

Define the graded low modes

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2. \quad (856)$$

3. C. Odd transport kernel

From the linearized defect operator class, the first-order transport term has the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (857)$$

with

$$V^\mu(-\xi) = -V^\mu(\xi), \quad (858)$$

because V^μ is built linearly from $\partial_\xi P_B$, $\partial_\xi z_B$, or $\theta'(\xi)$, and

$$\theta'(-\xi) = -\theta'(\xi).$$

To obtain an explicit solvable representative in the same operator class, choose a factorized odd kernel

$$V^i(\xi) = g(\xi)\Sigma^i, \quad g(\xi) = 3g_0 \tanh(\kappa\xi), \quad (859)$$

where the Σ^i are constant Hermitian endomorphisms acting on the Q_B -frame. Choose them as

$$\Sigma^i = v_i \sigma_i, \quad v_i \in \mathbb{R}. \quad (860)$$

In the frame (e_1, e_2) , this means

$$e_a^\dagger(\xi)\Sigma^i e_b(\xi) = v_i(\sigma_i)_{ab}. \quad (861)$$

4. D. Same-parity overlaps vanish

The same-parity projected overlap is

$$\langle u_a^\pm, V^i u_b^\pm \rangle = \int d\xi f_\pm(\xi)^2 e_a^\dagger(\xi) V^i(\xi) e_b(\xi). \quad (862)$$

Since $f_\pm^2$ is even, $e_a^\dagger \Sigma^i e_b$ is even, and $g(\xi)$ is odd, the integrand is odd. Hence

$$\langle u_a^\pm, V^i u_b^\pm \rangle = 0. \quad (863)$$

5. E. Mixed overlap integral

Now compute the mixed block:

$$(M^i)_{ab} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^i(\xi) e_b(\xi). \quad (864)$$

Using (859) and (861),

$$(M^i)_{ab} = v_i(\sigma_i)_{ab} \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) g(\xi). \quad (865)$$

Thus

$$M^i = I_g v_i \sigma_i, \quad (866)$$

where

$$I_g := \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) g(\xi). \quad (867)$$

Substitute (853), (854), and $g(\xi) = 3g_0 \tanh(\kappa\xi)$:

$$f_+(\xi) f_-(\xi) = \sqrt{\frac{\kappa}{2}} \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}^2(\kappa\xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa\xi) \operatorname{sech}^2(\kappa\xi). \quad (868)$$

Therefore

$$I_g = \frac{3g_0\sqrt{3}\kappa}{2} \int_{-\infty}^{\infty} d\xi \tanh^2(\kappa\xi) \operatorname{sech}^2(\kappa\xi). \quad (869)$$

Set

$$t = \tanh(\kappa\xi), \quad dt = \kappa \operatorname{sech}^2(\kappa\xi) d\xi. \quad (870)$$

Then

$$I_g = \frac{3g_0\sqrt{3}}{2} \int_{-1}^1 t^2 dt = \frac{3g_0\sqrt{3}}{2} \cdot \frac{2}{3} = g_0\sqrt{3}. \quad (871)$$

Hence

$$\boxed{M^i = g_0\sqrt{3} v_i \sigma_i.} \quad (872)$$

6. F. Reduced symbol and nondegeneracy

The graded mixed symbol is

$$A_1(p) = p_i M^i = g_0\sqrt{3} (v_1 p_1 \sigma_1 + v_2 p_2 \sigma_2 + v_3 p_3 \sigma_3). \quad (873)$$

Therefore

$$\det A_1(p) = -3g_0^2 (v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2). \quad (874)$$

In particular, if

$$v_1 v_2 v_3 \neq 0, \quad (875)$$

then

$$\det A_1(p) \neq 0 \quad \text{for generic } p \neq 0. \quad (876)$$

Thus the reduced principal operator on the graded $2 + 2$ carrier is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad (877)$$

and since $A_1(p)$ is Hermitian here,

$$D_1(p) = g_0\sqrt{3} \begin{pmatrix} 0 & v_i p_i \sigma_i \\ v_i p_i \sigma_i & 0 \end{pmatrix}. \quad (878)$$

7. *G. Dirac form*

Define

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4. \quad (879)$$

Then

$$D_1(p) = g_0 \sqrt{3} v_i p_i \alpha_i. \quad (880)$$

This is exactly the massless Dirac principal operator on the graded $2 + 2$ low-energy sector.

8. *H. Honest status*

What is explicit here is a concrete representative inside the actual TECT projector-defect operator class:

1. explicit reflection-symmetric CP^2 defect profile $z_B(\xi)$,
2. explicit even Q_B -frame $e_a(\xi)$,
3. explicit even/odd low modes $f_{\pm}(\xi)$,
4. explicit odd transport kernel $V^i(\xi) = 3g_0 \tanh(\kappa\xi) \Sigma^i$,
5. explicit overlap integral $I_g = g_0 \sqrt{3}$,
6. explicit full-rank mixed symbol $A_1(p)$.

What remains open is stronger: deriving this very same $V^\mu(\xi)$ and the same full-rank Pauli-triplet form directly from the complete microscopic TECT transport kernel without choosing an explicit representative in the same operator class.

. Summary

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi),$$

with even Q_B -frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Take the explicit even/odd low modes

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi),$$

and define

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi), \quad a = 1, 2.$$

Choose the odd transport representative

$$V^i(\xi) = g(\xi) \Sigma^i, \quad g(\xi) = 3g_0 \tanh(\kappa\xi), \quad \Sigma^i = v_i \sigma_i.$$

Then same-parity overlaps vanish,

$$\langle u_a^\pm, V^i u_b^\pm \rangle = 0,$$

while the mixed overlap is

$$(M^i)_{ab} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^i(\xi) e_b(\xi) = I_g v_i(\sigma_i)_{ab},$$

with

$$I_g = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) g(\xi).$$

Using the explicit profiles,

$$I_g = \frac{3g_0\sqrt{3}\kappa}{2} \int_{-\infty}^{\infty} d\xi \tanh^2(\kappa\xi) \text{sech}^2(\kappa\xi).$$

With $t = \tanh(\kappa\xi)$, $dt = \kappa \text{sech}^2(\kappa\xi) d\xi$,

$$I_g = \frac{3g_0\sqrt{3}}{2} \int_{-1}^1 t^2 dt = g_0\sqrt{3}.$$

Therefore

$$\boxed{M^i = g_0\sqrt{3} v_i \sigma_i.}$$

Hence

$$A_1(p) = p_i M^i = g_0\sqrt{3} (v_1 p_1 \sigma_1 + v_2 p_2 \sigma_2 + v_3 p_3 \sigma_3),$$

so

$$\det A_1(p) = -3g_0^2 (v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2).$$

If $v_1 v_2 v_3 \neq 0$, then $\det A_1(p) \neq 0$ for generic $p \neq 0$.

Thus the graded $2+2$ reduced principal operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix} = g_0\sqrt{3} v_i p_i \alpha_i, \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij} \mathbf{1}_4.$$

So, for this explicit representative inside the actual projector-defect operator class, the graded $2+2$ bypass yields an explicit massless Dirac principal symbol.

• **Actual local transport kernel profile: the leading odd scalar is $\theta'(\xi)$**

a. Goal. We now determine the leading scalar profile $g(\xi)$ in the actual local transport kernel

$$L_1 = -iV^\mu(\xi) \partial_{y^\mu}$$

for the CP^2 projector defect, and evaluate the resulting mixed overlap integral explicitly.

1. *A. Explicit local defect background and frame*

Take the explicit reflection-symmetric defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad \theta(\xi) = \theta_0 \text{sech}(\kappa\xi). \quad (881)$$

An orthonormal local frame of

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (882)$$

Differentiate:

$$\partial_\xi z_B(\xi) = \theta'(\xi) \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix} = \theta'(\xi) e_1(\xi). \quad (883)$$

Also,

$$\partial_\xi e_1(\xi) = -\theta'(\xi) z_B(\xi), \quad \partial_\xi e_2(\xi) = 0. \quad (884)$$

Therefore the frame connection along ξ vanishes:

$$A_\xi = -ie^\dagger \partial_\xi e = 0. \quad (885)$$

2. B. Local gradient structure of the transport kernel

The actual local defect-Hessian analysis states that the leading transport kernel is built linearly from the transverse defect gradient, i.e. from quantities such as

$$\partial_\xi P_B(\xi), \quad \partial_\xi z_B(\xi), \quad \theta'(\xi),$$

and is therefore reflection-odd. Using (883), every such local first-order building block is proportional to $\theta'(\xi)$ times an ξ -even matrix determined by $\theta(\xi)$.

Indeed,

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad (886)$$

so

$$\partial_\xi P_B = (\partial_\xi z_B) z_B^\dagger + z_B (\partial_\xi z_B)^\dagger = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (887)$$

Thus any local transport endomorphism built linearly from the background gradient has the schematic form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (888)$$

where $\mathcal{M}^\mu(\theta)$ is an even 2×2 Hermitian matrix in the Q_B -frame.

Expanding in the Pauli basis,

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i. \quad (889)$$

Since $\theta(\xi)$ is even, all coefficients $a_0^\mu(\theta(\xi)), a_i^\mu(\theta(\xi))$ are even functions of ξ . Hence the leading odd scalar profile is exactly

$$g_{\text{lead}}(\xi) = \theta'(\xi). \quad (890)$$

3. C. Explicit low modes

Use the explicit even/odd low modes

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad (891)$$

$$f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi), \quad (892)$$

and define

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi), \quad a = 1, 2. \quad (893)$$

The mixed projected symbol is

$$(M^\mu)_{ab} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (894)$$

At leading order in the local gradient expansion, using (888),

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)). \quad (895)$$

If one freezes the even matrix coefficient at leading order to a constant traceless Hermitian matrix

$$\mathcal{M}^\mu(\theta) \approx \Sigma^\mu = v_\mu \sigma_\mu, \quad (896)$$

then

$$M^\mu = I_\theta \Sigma^\mu, \quad I_\theta := \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) \theta'(\xi). \quad (897)$$

4. D. Exact evaluation of I_θ

First compute

$$\theta'(\xi) = \theta_0 \frac{d}{d\xi} \operatorname{sech}(\kappa\xi) = -\theta_0 \kappa \operatorname{sech}(\kappa\xi) \tanh(\kappa\xi). \quad (898)$$

Next,

$$f_+(\xi) f_-(\xi) = \sqrt{\frac{\kappa}{2}} \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}^2(\kappa\xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa\xi) \operatorname{sech}^2(\kappa\xi). \quad (899)$$

Therefore

$$\begin{aligned} I_\theta &= \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \\ &= -\frac{\sqrt{3}\theta_0\kappa^2}{2} \int_{-\infty}^{\infty} d\xi \tanh^2(\kappa\xi) \operatorname{sech}^3(\kappa\xi). \end{aligned} \quad (900)$$

Now set

$$t = \tanh(\kappa\xi), \quad dt = \kappa \operatorname{sech}^2(\kappa\xi) d\xi, \quad \operatorname{sech}(\kappa\xi) = \sqrt{1-t^2}. \quad (901)$$

Then

$$I_\theta = -\frac{\sqrt{3}\theta_0\kappa}{2} \int_{-1}^1 t^2 \sqrt{1-t^2} dt. \quad (902)$$

Using

$$\int_{-1}^1 t^2 \sqrt{1-t^2} dt = \frac{\pi}{8}, \quad (903)$$

we obtain

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (904)$$

Hence

$$I_\theta \neq 0 \quad \text{whenever } \theta_0 \kappa \neq 0. \quad (905)$$

5. E. Consequence for the mixed symbol

Substituting (904) into (897),

$$M^\mu = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Sigma^\mu. \quad (906)$$

If the three frozen matrices Σ^μ form a full-rank Pauli triple, e.g.

$$\Sigma^1 = v_1 \sigma_1, \quad \Sigma^2 = v_2 \sigma_2, \quad \Sigma^3 = v_3 \sigma_3, \quad v_1 v_2 v_3 \neq 0, \quad (907)$$

then

$$A_1(p) = p_\mu M^\mu = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (v_1 p_1 \sigma_1 + v_2 p_2 \sigma_2 + v_3 p_3 \sigma_3), \quad (908)$$

and therefore

$$\det A_1(p) = -\frac{3\pi^2}{256} \theta_0^2 \kappa^2 (v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2). \quad (909)$$

Thus the mixed symbol is nonzero and full rank for generic $p \neq 0$.

6. F. Honest conclusion

What is now directly computed from the actual local CP^2 defect geometry is the leading odd scalar profile:

$$g(\xi) \propto \theta'(\xi).$$

For the explicit profile $\theta(\xi) = \theta_0 \text{sech}(\kappa\xi)$, the mixed overlap integral is

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0.$$

Therefore the actual local defect-gradient transport mechanism produces a nonzero graded mixed symbol already at the level of the explicit local projector representative.

What remains open is stronger: deriving the full matrix coefficient $\mathcal{M}^\mu(\theta(\xi))$ from the complete microscopic TECT linearization, rather than freezing it to a constant representative Σ^μ .

. Summary

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(\xi) = \theta_0 \text{sech}(\kappa\xi), \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Therefore every leading local transport kernel built linearly from the defect gradient has the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

with $\mathcal{M}^\mu$ even in ξ .

So the actual leading odd scalar profile is

$$g_{\text{lead}}(\xi) = \theta'(\xi).$$

Using

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \text{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \text{sech}(\kappa\xi),$$

the mixed overlap integral

$$I_\theta = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) \theta'(\xi)$$

is exactly

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Hence $I_\theta \neq 0$ whenever $\theta_0 \kappa \neq 0$.

If one freezes the even matrix coefficient to a Pauli-triplet representative

$$\mathcal{M}^\mu(\theta) \approx \Sigma^\mu = v_\mu \sigma_\mu,$$

then

$$M^\mu = I_\theta \Sigma^\mu,$$

so

$$A_1(p) = p_\mu M^\mu = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (v_1 p_1 \sigma_1 + v_2 p_2 \sigma_2 + v_3 p_3 \sigma_3),$$

and

$$\det A_1(p) = -\frac{3\pi^2}{256} \theta_0^2 \kappa^2 (v_1^2 p_1^2 + v_2^2 p_2^2 + v_3^2 p_3^2).$$

So, in the actual local CP^2 projector-defect geometry, the natural transport scalar profile is $\theta'(\xi)$, and its graded overlap is explicitly nonzero.

. Unfreezing $\mathcal{M}^\mu(\theta)$: identity part, traceless part, and the full-rank criterion

a. Goal. We now unfreeze the matrix coefficient in the actual local transport kernel

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

and determine:

1. whether the identity component of $\mathcal{M}^\mu(\theta)$ must vanish,
2. whether the traceless Pauli components can span three independent directions.

1. A. General Hermitian decomposition on the rank-two Q_B bundle

In the local orthonormal Q_B -frame $(e_1(\xi), e_2(\xi))$, every Hermitian endomorphism of the rank-two fiber $Q_B(\xi)$ decomposes uniquely as

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i, \quad i = 1, 2, 3, \quad (910)$$

with real coefficient functions $a_0^\mu(\theta), a_i^\mu(\theta)$.

Thus the actual local transport kernel has the general form

$$V^\mu(\xi) = \theta'(\xi) \left(a_0^\mu(\theta(\xi)) \mathbf{1}_2 + a_i^\mu(\theta(\xi)) \sigma_i \right). \quad (911)$$

2. B. Parity of the coefficient functions

The defect profile satisfies

$$\theta(-\xi) = \theta(\xi), \quad \theta'(-\xi) = -\theta'(\xi).$$

Since $a_0^\mu(\theta), a_i^\mu(\theta)$ are functions of the even variable $\theta(\xi)$, they are even in ξ :

$$a_0^\mu(\theta(-\xi)) = a_0^\mu(\theta(\xi)), \quad a_i^\mu(\theta(-\xi)) = a_i^\mu(\theta(\xi)). \quad (912)$$

Therefore the transport kernel is odd:

$$V^\mu(-\xi) = -V^\mu(\xi), \quad (913)$$

exactly as required by the local matching theorem for the projector Hessian operator class. :contentReference[oaicite:3]index=3

Importantly, parity alone does *not* force

$$a_0^\mu(\theta) \equiv 0.$$

So the identity component is not forbidden by reflection symmetry.

3. C. Projected mixed symbol with the unfrozen coefficient

The mixed overlap matrix is

$$(M^\mu)_{ab} = \int_{-\infty}^{\infty} d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (914)$$

Using (911),

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) \left(a_0^\mu(\theta(\xi)) \mathbf{1}_2 + a_i^\mu(\theta(\xi)) \sigma_i \right). \quad (915)$$

Define the coefficient integrals

$$C_0^\mu := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) a_0^\mu(\theta(\xi)), \quad (916)$$

$$C_i^\mu := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) a_i^\mu(\theta(\xi)). \quad (917)$$

Then

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i. \quad (918)$$

Because f_+ is even, f_- is odd, and θ' is odd, the product

$$f_+ f_- \theta'$$

is even. Hence both C_0^μ and C_i^μ are generically allowed by parity. There is no parity reason forcing $C_0^\mu = 0$.

4. D. Why the identity part is not a mass obstruction

The reduced first-order mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i. \quad (919)$$

Thus the full graded principal operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}. \quad (920)$$

Since $A_1(p)$ is linear in momentum,

$$A_1(0) = 0, \quad D_1(0) = 0. \quad (921)$$

Therefore the identity component $C_0^\mu \mathbf{1}_2$ contributes only a *first-order* term

$$p_\mu C_0^\mu \mathbf{1}_2,$$

not a constant mass term. This is completely different from the scalar Bragg-mass obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2,$$

which is momentum-independent.

Hence:

the identity part $C_0^\mu \mathbf{1}_2$ is not a mass obstruction in the graded $2 + 2$ bypass.

(922)

5. E. Canonical reduction and irrelevance of the identity part

The graded infrared theorem already established in the manuscript states that any nondegenerate mixed symbol

$$A_1(p) = p_\mu M^\mu$$

with M^μ arbitrary 2×2 complex matrices can, after unitary basis changes in $K_\pm$ and an invertible linear transformation of momentum coordinates, be brought to the canonical form

$$A_1(p) = p_i \sigma_i. \quad (923)$$

This means that the identity component is not load-bearing at the principal level; it can be absorbed into the canonical reduction once the full symbol is nondegenerate.

So the correct statement is:

C_0^μ need not vanish, but it is principal-level irrelevant once $A_1(p)$ is nondegenerate.

(924)

6. F. The actual load-bearing data: the traceless coefficient vectors

Write

$$b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3. \quad (925)$$

Then

$$M^\mu = C_0^\mu \mathbf{1}_2 + b_i^\mu \sigma_i. \quad (926)$$

The manuscript states explicitly that the mixed symbol is nondegenerate iff the vectors $b^\mu \in \mathbb{R}^3$ span $\mathbb{R}^3$. Equivalently,

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3 \iff A_1(p) \text{ is full rank for generic } p \neq 0. \quad (927)$$

Thus the actual load-bearing condition is not $C_0^\mu = 0$, but

$\det[b_i^\mu] \neq 0.$

(928)

7. G. Structural realization in the actual local projector-Hessian class

The honest local matching theorem states that the actual CP^2 projector Hessian around a reflection-symmetric defect background belongs to the same operator class as the minimal graded defect operator: rank-two matrix Schrödinger static part, even/odd parity channels, and odd first-order transport kernel. :contentReference[oaicite:6]index=6

Moreover, the same operator class admits an explicit full-rank realization

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0, \quad (929)$$

which is exactly the case

$$C_0^\mu = 0, \quad b^\mu = I_g \hat{e}_\mu.$$

This proves that there is *no structural obstruction* inside the actual local operator class to having three independent traceless directions. :contentReference[oaicite:7]index=7

Therefore:

$$\boxed{\text{in the actual local projector-Hessian operator class, } b^\mu \text{ can span } \mathbb{R}^3.} \quad (930)$$

8. H. What is still open

What is not yet proved microscopically is stronger: that the *actual full TECT transport kernel* produces coefficient functions $a_i^\mu(\theta(\xi))$ whose projected integrals b^μ satisfy

$$\det[b_i^\mu] \neq 0$$

for the physically realized defect background.

So the honest status is:

$$\text{actual local operator class} \implies \text{no obstruction to full rank,}$$

but

$$\text{full microscopic TECT dynamics} \implies ?$$

a. *Exact logical status.* What is now closed is:

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu = a_0^\mu \mathbf{1}_2 + a_i^\mu \sigma_i,$$

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i,$$

$$C_0^\mu \text{ is allowed but not a mass term,} \quad \text{span}\{b^\mu\} = \mathbb{R}^3 \text{ is the real nondegeneracy criterion.}$$

And the actual local projector-Hessian operator class already contains explicit full-rank representatives.

. Summary

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i.$$

Because $\theta(\xi)$ is even and $\theta'(\xi)$ is odd, parity does not force $a_0^\mu(\theta)$ to vanish. The projected mixed matrices are

$$M^\mu = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)) = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i,$$

with

$$C_0^\mu = \int d\xi f_+ f_- \theta' a_0^\mu(\theta), \quad C_i^\mu = \int d\xi f_+ f_- \theta' a_i^\mu(\theta).$$

Thus the mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i.$$

Since this is first-order in p ,

$$A_1(0) = 0, \quad D_1(0) = 0.$$

Therefore the identity part $C_0^\mu \mathbf{1}_2$ is not a constant mass term. It is principal-level irrelevant once the full mixed symbol is nondegenerate.

The actual load-bearing data are the traceless coefficient vectors

$$b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3.$$

The exact nondegeneracy criterion is

$$\boxed{\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.}$$

Equivalently,

$$\det[b_i^\mu] \neq 0.$$

The actual local CP^2 projector-Hessian operator class already contains an explicit full-rank realization

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0,$$

so there is no structural obstruction to obtaining three independent traceless directions.

Hence:

$$\boxed{a_0^\mu \text{ need not vanish, but it is not the load-bearing issue.}}$$

$$\boxed{\text{The real Dirac criterion is whether the traceless vectors } b^\mu \text{ span } \mathbb{R}^3.}$$

. Three independent traceless transport channels from the actual local Q_B -geometry

a. Goal. We now derive, directly from the actual local CP^2 projector-defect geometry, why there are naturally three independent traceless transport channels available in the graded $2+2$ reduction.

1. A. Local rank-two bundle and its endomorphism algebra

For the reflection-symmetric defect background

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (931)$$

the physical tangent fluctuation lives in the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp. \quad (932)$$

An explicit orthonormal local frame is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (933)$$

In this frame the transverse Q_B -connection vanishes:

$$A_\xi = -ie^\dagger \partial_\xi e = 0. \quad (934)$$

Therefore the local fiber geometry is completely standard:

$$\text{Herm}(Q_B) \cong \text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \mathfrak{su}(2)_{\text{Herm}}. \quad (935)$$

A canonical traceless Hermitian basis is given by

$$\Sigma^{(1)} := e_1 e_2^\dagger + e_2 e_1^\dagger, \quad (936)$$

$$\Sigma^{(2)} := -ie_1 e_2^\dagger + ie_2 e_1^\dagger, \quad (937)$$

$$\Sigma^{(3)} := e_1 e_1^\dagger - e_2 e_2^\dagger. \quad (938)$$

In the local frame (e_1, e_2) , these are exactly the Pauli matrices:

$$\Sigma^{(1)} \leftrightarrow \sigma_1, \quad \Sigma^{(2)} \leftrightarrow \sigma_2, \quad \Sigma^{(3)} \leftrightarrow \sigma_3. \quad (939)$$

2. B. Actual local transport kernel

The local projector-Hessian analysis gives the first-order transport operator

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (940)$$

with

$$V^\mu(\xi) \sim (\partial_\xi P_B(\xi)) T^\mu \quad (941)$$

for constant matrices T^μ determined by the elastic/worldvolume couplings. :contentReference[oaicite:3]index=3 Because $P_B(-\xi) = P_B(\xi)$, one has

$$\partial_\xi P_B(-\xi) = -\partial_\xi P_B(\xi), \quad (942)$$

hence

$$V^\mu(-\xi) = -V^\mu(\xi). \quad (943)$$

Using

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger), \quad (944)$$

every leading local transport kernel built linearly from the background gradient has the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (945)$$

where $\mathcal{M}^\mu(\theta)$ is an even Hermitian endomorphism of Q_B .

Expanding in the canonical basis $\{\mathbf{1}_2, \Sigma^{(1)}, \Sigma^{(2)}, \Sigma^{(3)}\}$,

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + h_1^\mu(\theta) \Sigma^{(1)} + h_2^\mu(\theta) \Sigma^{(2)} + h_3^\mu(\theta) \Sigma^{(3)}. \quad (946)$$

So the traceless local transport channels are exactly

$$\Sigma^{(1)}, \Sigma^{(2)}, \Sigma^{(3)}.$$

3. C. Projected mixed symbol

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2. \quad (947)$$

Then the mixed projected matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (948)$$

Substituting (945) and (946),

$$M^\mu = \int d\xi f_+f_- \theta'(\xi) \left[a_0^\mu(\theta)\mathbf{1}_2 + h_i^\mu(\theta)\Sigma^{(i)} \right]. \quad (949)$$

Define

$$C_0^\mu := \int d\xi f_+f_- \theta'(\xi) a_0^\mu(\theta(\xi)), \quad (950)$$

$$C_i^\mu := \int d\xi f_+f_- \theta'(\xi) h_i^\mu(\theta(\xi)). \quad (951)$$

Then

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \Sigma^{(i)}. \quad (952)$$

The load-bearing traceless coefficient vectors are therefore

$$b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3. \quad (953)$$

By the exact mixed-symbol criterion of the graded effective model, the reduced symbol is nondegenerate iff

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3. \quad (954)$$

4. D. Why three independent channels are structurally natural

There is no local geometric obstruction to having three independent traceless channels because:

1. the fiber $Q_B(\xi)$ is complex rank 2, so $\mathfrak{su}(2)$ is the full traceless Hermitian algebra;
2. the chosen frame is smooth and $A_\xi = 0$, so no covariant-derivative obstruction mixes away the three basis elements;
3. the actual local transport kernel already takes values in $\text{End}(Q_B)$, so all three traceless basis directions are admissible.

Hence the strongest structural statement is

the actual local projector-defect operator class naturally contains three independent traceless transport channels.
(955)

5. E. Generic spanning argument

Suppose the coefficient matrix

$$H(\theta) := (h_i^\mu(\theta))_{\mu,i} \quad (956)$$

has full rank at some range of θ values visited by the defect profile. Then the integrated coefficients

$$C_i^\mu = \int d\xi w(\xi) h_i^\mu(\theta(\xi)), \quad w(\xi) := f_+(\xi)f_-(\xi)\theta'(\xi),$$

are generically full rank unless a nontrivial cancellation occurs. Since $w(\xi)$ is even and not identically zero, such cancellation requires nongeneric fine tuning of the functions $h_i^\mu(\theta)$.

Thus, in the absence of special symmetry or tuning,

$$\det[C_i^\mu] \neq 0 \tag{957}$$

is the generic expectation.

a. Important honesty. This is a genericity statement, not yet a theorem for the full microscopic TECT transport kernel.

6. F. Explicit realizable point inside the same operator class

The same actual local operator class already admits an explicit full-rank realization:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0, \tag{958}$$

which corresponds to

$$C_0^\mu = 0, \quad b^1 = I_g(1, 0, 0), \quad b^2 = I_g(0, 1, 0), \quad b^3 = I_g(0, 0, 1). \tag{959}$$

Therefore

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3. \tag{960}$$

So, while the full microscopic coefficient functions $h_i^\mu(\theta)$ are still open, the actual local projector-Hessian operator class does contain explicit full-rank points.

7. G. Final conclusion

What is now closed is:

$$Q_B \text{ rank-2 geometry} \implies \text{Herm}(Q_B) = \mathbb{R}\mathbf{1}_2 \oplus \text{span}\{\Sigma^{(1)}, \Sigma^{(2)}, \Sigma^{(3)}\},$$

$$V^\mu(\xi) = \theta'(\xi) \left(a_0^\mu \mathbf{1}_2 + h_i^\mu \Sigma^{(i)} \right),$$

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \Sigma^{(i)}, \quad b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu).$$

The actual nondegeneracy criterion is

$$\boxed{\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.}$$

And the local operator class already contains explicit points where this is realized.

What remains open is the final microscopic computation of the actual coefficient functions $h_i^\mu(\theta)$ from the full TECT transport kernel.

. **Summary**

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

is a complex rank-two bundle, so in a local orthonormal frame (e_1, e_2) ,

$$\text{Herm}(Q_B) \cong \mathbb{R} \mathbf{1}_2 \oplus \text{span}\{\Sigma^{(1)}, \Sigma^{(2)}, \Sigma^{(3)}\},$$

with canonical traceless basis

$$\Sigma^{(1)} = e_1 e_2^\dagger + e_2 e_1^\dagger, \quad \Sigma^{(2)} = -i e_1 e_2^\dagger + i e_2 e_1^\dagger, \quad \Sigma^{(3)} = e_1 e_1^\dagger - e_2 e_2^\dagger.$$

In the local frame these are exactly

$$\Sigma^{(1)} \leftrightarrow \sigma_1, \quad \Sigma^{(2)} \leftrightarrow \sigma_2, \quad \Sigma^{(3)} \leftrightarrow \sigma_3.$$

The actual local transport kernel has the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

and therefore can be expanded as

$$V^\mu(\xi) = \theta'(\xi) \left(a_0^\mu(\theta) \mathbf{1}_2 + h_i^\mu(\theta) \Sigma^{(i)} \right).$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$ gives

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) \left(a_0^\mu(\theta) \mathbf{1}_2 + h_i^\mu(\theta) \Sigma^{(i)} \right) = C_0^\mu \mathbf{1}_2 + C_i^\mu \Sigma^{(i)}.$$

The load-bearing traceless coefficient vectors are

$$b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3.$$

The exact nondegeneracy criterion is

$$\boxed{\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.}$$

Therefore the actual local Q_B -geometry naturally provides exactly three independent traceless transport channels. Moreover, the same local operator class already contains an explicit full-rank point

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0,$$

so there is no structural obstruction to full rank inside the actual projector-defect operator class.

. **Minimal actual parametrization of the microscopic transport tensor T^μ in the local projector-defect sector**

a. Goal. We now write the strongest actual local form currently available for the transport kernel

$$V^\mu(\xi) \sim (\partial_\xi P_B) T^\mu$$

inside the CP^2 projector-defect sector, and identify the true load-bearing coefficients for the graded $2 + 2$ mixed symbol.

1. A. Actual local projector geometry

Let the reflection-symmetric defect background be

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (961)$$

with

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger. \quad (962)$$

The tangent fluctuation lives in the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp. \quad (963)$$

An explicit orthonormal frame is

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (964)$$

In this frame the transverse connection vanishes:

$$A_\xi = -ie^\dagger \partial_\xi e = 0. \quad (965)$$

2. B. Actual local gradient source

Differentiating the background gives

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi), \quad (966)$$

and therefore

$$\partial_\xi P_B(\xi) = (\partial_\xi z_B) z_B^\dagger + z_B (\partial_\xi z_B)^\dagger = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (967)$$

Hence every leading local transport kernel built linearly from the background gradient is necessarily of the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (968)$$

where $\mathcal{M}^\mu(\theta)$ is a Hermitian endomorphism of the rank-two fiber Q_B .

3. C. Endomorphism algebra of the rank-two fiber

Since $Q_B(\xi)$ is complex rank 2, in the local frame (e_1, e_2) one has

$$\text{Herm}(Q_B) \cong \text{Herm}(2) = \mathbb{R} \mathbf{1}_2 \oplus \mathfrak{su}(2)_{\text{Herm}}. \quad (969)$$

Thus the most general local Hermitian transport tensor is

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i, \quad i = 1, 2, 3, \quad (970)$$

with real coefficient functions $a_0^\mu(\theta), a_i^\mu(\theta)$.

Equivalently,

$$V^\mu(\xi) = \theta'(\xi) \left(a_0^\mu(\theta(\xi)) \mathbf{1}_2 + a_i^\mu(\theta(\xi)) \sigma_i \right). \quad (971)$$

4. D. Projected mixed symbol

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2. \quad (972)$$

Then the projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (973)$$

Substituting (971),

$$M^\mu = \int d\xi f_+f_- \theta'(\xi) \left(a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i \right). \quad (974)$$

Define

$$C_0^\mu := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi) a_0^\mu(\theta(\xi)), \quad (975)$$

$$C_i^\mu := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi) a_i^\mu(\theta(\xi)). \quad (976)$$

Then

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i. \quad (977)$$

5. E. Load-bearing data

The mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i. \quad (978)$$

Because it is first-order in momentum,

$$A_1(0) = 0, \quad D_1(0) = 0. \quad (979)$$

Therefore the identity coefficient C_0^μ is not a constant mass term. The actual nondegeneracy data are the traceless coefficient vectors

$$b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3. \quad (980)$$

The exact criterion is

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3. \quad (981)$$

6. F. Natural three-channel basis

A canonical traceless Hermitian basis on Q_B is

$$\Sigma^{(1)} := e_1 e_2^\dagger + e_2 e_1^\dagger, \quad (982)$$

$$\Sigma^{(2)} := -ie_1 e_2^\dagger + ie_2 e_1^\dagger, \quad (983)$$

$$\Sigma^{(3)} := e_1 e_1^\dagger - e_2 e_2^\dagger. \quad (984)$$

In the local frame these are exactly $\sigma_1, \sigma_2, \sigma_3$. Therefore the actual local projector geometry naturally supplies exactly three traceless transport channels.

7. *G. Honest status*

What is genuinely derived from the actual local projector-defect geometry is:

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad \mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i. \quad (985)$$

What is not yet derived from the final microscopic TECT PDE is the explicit formula for the coefficient functions

$$a_i^\mu(\theta).$$

Thus the remaining microscopic gap is coefficient extraction, not operator-class identification.

. **Summary**

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Hence every leading local transport kernel built linearly from the defect gradient has the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Because $Q_B(\xi)$ is a complex rank-two bundle,

$$\text{Herm}(Q_B) \cong \mathbb{R} \mathbf{1}_2 \oplus \text{span}\{\sigma_1, \sigma_2, \sigma_3\}.$$

Therefore the most general local Hermitian transport tensor is

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i.$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$ gives

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) (a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i) = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i.$$

The mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i,$$

so

$$A_1(0) = 0.$$

Thus C_0^μ is not a constant mass term.

The actual load-bearing data are the traceless coefficient vectors

$$b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3,$$

and the exact nondegeneracy criterion is

$$\boxed{\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.}$$

A canonical traceless basis is

$$\Sigma^{(1)} = e_1 e_2^\dagger + e_2 e_1^\dagger, \quad \Sigma^{(2)} = -i e_1 e_2^\dagger + i e_2 e_1^\dagger, \quad \Sigma^{(3)} = e_1 e_1^\dagger - e_2 e_2^\dagger,$$

which are exactly $\sigma_1, \sigma_2, \sigma_3$ in the local frame.

So the actual local projector geometry naturally provides three independent traceless transport channels. What remains open is the explicit microscopic extraction of the coefficient functions $a_i^\mu(\theta)$ from the full TECT linearized projector equation.

. **The strongest honest local form of the microscopic transport tensor T^μ**

a. Goal. We now derive the strongest local form currently justified for the transport kernel in the projector-defect sector, without overclaiming a full microscopic derivation from the final TECT PDE.

1. *A. Local projector-defect setting*

Let the defect background depend only on the transverse coordinate ξ :

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \quad z_B(\xi)^\dagger z_B(\xi) = 1. \quad (986)$$

The physical fluctuation field is a section of the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp. \quad (987)$$

Choose a local orthonormal frame

$$e(\xi) = (e_1(\xi), e_2(\xi)) \quad (988)$$

for $Q_B(\xi)$, and write

$$\eta(\xi, y) = e_a(\xi)\chi^a(\xi, y). \quad (989)$$

For the explicit defect representative already used,

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (990)$$

with

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad (991)$$

one has

$$A_\xi := -ie^\dagger \partial_\xi e = 0. \quad (992)$$

2. *B. General local form of the quadratic operator*

The local quadratic fluctuation operator on Q_B has the generic form

$$L = L_0 + L_1 + \cdots, \quad (993)$$

where

- L_0 is the static transverse operator (matrix Schrödinger type),
- L_1 is the first-order worldvolume transport operator,
- the omitted terms are higher order in the slow worldvolume gradients.

The graded reduction theorem only needs the first-order projected part of L_1 , not the full microscopic closed form of L . This distinction is essential.

3. C. Origin of the leading transport kernel

At the local projector-defect level, the first-order worldvolume transport must be linear in:

1. one worldvolume derivative ∂_{y^μ} ,
2. one background gradient tensor in the transverse direction.

The only actual local odd building blocks available at lowest order are objects linear in

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi). \quad (994)$$

Indeed, for the explicit profile,

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad (995)$$

and

$$\partial_\xi P_B = (\partial_\xi z_B) z_B^\dagger + z_B (\partial_\xi z_B)^\dagger = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (996)$$

Hence every leading local transport endomorphism is necessarily proportional to $\theta'(\xi)$.

Therefore the strongest justified local form is

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)). \quad (997)$$

4. D. Endomorphism-valued transport tensor

Because $Q_B(\xi)$ is complex rank 2, every Hermitian endomorphism of the fiber decomposes as

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i, \quad i = 1, 2, 3, \quad (998)$$

with real coefficient functions $a_0^\mu(\theta), a_i^\mu(\theta)$.

Thus the actual local transport kernel has the general form

$$V^\mu(\xi) = \theta'(\xi) \left[a_0^\mu(\theta(\xi)) \mathbf{1}_2 + a_i^\mu(\theta(\xi)) \sigma_i \right]. \quad (999)$$

This is the strongest honest local parametrization of the microscopic transport tensor presently available.

5. E. Projection to the graded 2 + 2 carrier

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (1000)$$

Then the mixed symbol is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (1001)$$

Substituting (999),

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) \left[a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i \right]. \quad (1002)$$

Define

$$C_0^\mu := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) a_0^\mu(\theta(\xi)), \quad (1003)$$

$$C_i^\mu := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) a_i^\mu(\theta(\xi)). \quad (1004)$$

Then

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i. \quad (1005)$$

6. *F. Why the identity part is not the load-bearing issue*

The reduced mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i. \quad (1006)$$

Because this is first-order in momentum,

$$A_1(0) = 0. \quad (1007)$$

So the identity contribution $C_0^\mu \mathbf{1}_2$ is not a constant mass term. It belongs to the same first-order graded mixed symbol as the traceless part.

Hence the genuine nondegeneracy data are the traceless vectors

$$b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3. \quad (1008)$$

The exact criterion is

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3. \quad (1009)$$

7. *G. What is actually proved and what remains open*

What is actually proved at this point is:

1. the actual local defect geometry forces the transport kernel to be of the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi));$$

2. the transport tensor $\mathcal{M}^\mu(\theta)$ must be a Q_B -endomorphism-valued Hermitian matrix, hence admits the Pauli decomposition

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i;$$

3. the reduced Dirac criterion depends only on the projected traceless coefficient vectors b^μ .

What remains open is the full microscopic coefficient extraction problem:

derive the actual functions $a_i^\mu(\theta)$ from the final TECT linearized projector equation.

(1010)

So the remaining frontier is no longer operator-class identification, parity, or graded reduction. It is the actual coefficient computation.

. **Summary**

$$\partial_\xi z_B = \theta'(\xi) e_1, \quad \partial_\xi P_B = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger).$$

Therefore every leading local transport kernel built linearly from the defect gradient has the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)).$$

Because $Q_B(\xi)$ is complex rank 2,

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i.$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$ gives

$$M^\mu = \int d\xi f_+ f_- \theta'(\xi) \left[a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i \right] = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i.$$

The mixed symbol is

$$A_1(p) = p_\mu M^\mu = p_\mu C_0^\mu \mathbf{1}_2 + p_\mu C_i^\mu \sigma_i, \quad A_1(0) = 0.$$

Thus C_0^μ is not a constant mass term. The actual load-bearing data are the traceless coefficient vectors

$$b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3,$$

and the exact nondegeneracy criterion is

$$\boxed{\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.}$$

So the strongest honest local result is:

$$\boxed{\text{the actual local projector geometry fixes the transport kernel to be } Q_B\text{-endomorphism valued,}}$$

$$\boxed{\text{and the remaining open problem is the explicit microscopic computation of } a_i^\mu(\theta).}$$

. A minimal microscopic ansatz for the projector transport equation

a. Goal. We now formulate a minimal microscopic ansatz for the full projector transport equation that is consistent with all previously established local structural results and reduces the remaining open problem to a finite coefficient test.

1. A. Structural input already fixed

The actual local CP^2 projector-defect analysis establishes the following operator class:

1. the physical fluctuation field is a section

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp, \quad (1011)$$

2. the static transverse Hessian is a matrix Schrödinger operator on the rank-two bundle Q_B ,
3. the leading worldvolume transport is first-order and odd under reflection,

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1012)$$

4. for the explicit defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (1013)$$

one has

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger), \quad (1014)$$

so every leading local transport kernel linear in the defect gradient must factor through $\theta'(\xi)$.

Therefore the strongest local form already justified is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (1015)$$

with $\mathcal{M}^\mu(\theta) \in \text{Herm}(Q_B)$.

2. B. Minimal microscopic projector transport equation

The minimal microscopic transport ansatz consistent with the above structure is

$$i\partial_t \eta = \left[-\rho \partial_\xi^2 + U_B(\xi) - i\theta'(\xi) \mathcal{M}^\mu(\theta(\xi)) \partial_{y^\mu} \right] \eta + O(\partial_y^2) \eta, \quad (1016)$$

where $U_B(\xi)$ is the static Hermitian 2×2 transverse potential already fixed by the local projector Hessian.

This is not yet a derivation from the final full TECT PDE; it is the minimal microscopic ansatz compatible with all local structural theorems already proved.

3. C. Pauli decomposition of the transport tensor

In a local orthonormal Q_B -frame,

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i, \quad i = 1, 2, 3. \quad (1017)$$

Hence

$$V^\mu(\xi) = \theta'(\xi) \left[a_0^\mu(\theta(\xi)) \mathbf{1}_2 + a_i^\mu(\theta(\xi)) \sigma_i \right]. \quad (1018)$$

4. D. Minimal even- θ expansion

Because the physical projector P_B is invariant under the local sign reversal $\theta \mapsto -\theta$ up to equivalent background relabeling, the minimal scalar coefficient functions are naturally taken to be even in θ :

$$a_0^\mu(\theta) = \Lambda_0^\mu + O(\theta^2), \quad (1019)$$

$$a_i^\mu(\theta) = \Lambda_i^\mu + O(\theta^2). \quad (1020)$$

Thus, to leading order,

$$\mathcal{M}^\mu(\theta) = \Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i + O(\theta^2), \quad (1021)$$

and

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta^2 \theta'). \quad (1022)$$

5. E. Projection to the graded $2 + 2$ carrier

Let the even/odd low modes be

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (1023)$$

Then the mixed overlap matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (1024)$$

Substituting (1022),

$$M^\mu = I_\theta \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta_0^3 \kappa), \quad (1025)$$

where

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi). \quad (1026)$$

From the previous explicit computation,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1027)$$

Hence $I_\theta \neq 0$ whenever $\theta_0 \kappa \neq 0$.

6. F. Reduced mixed symbol

The projected first-order mixed symbol is therefore

$$A_1(p) = p_\mu M^\mu = I_\theta p_\mu \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta_0^3 \kappa |p|). \quad (1028)$$

The load-bearing traceless coefficient vectors are

$$b^\mu = I_\theta (\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3 \kappa). \quad (1029)$$

Thus, to leading order,

$$\det[b_i^\mu] = I_\theta^3 \det[\Lambda_i^\mu] + O(\theta_0^5 \kappa^3). \quad (1030)$$

[Finite-dimensional reduction of the coefficient problem] Assume the minimal transport ansatz (1016) and the even- θ expansion (1020). If

$$\det[\Lambda_i^\mu] \neq 0, \quad (1031)$$

then for sufficiently small but nonzero defect amplitude θ_0 ,

$$\det[b_i^\mu] \neq 0, \quad (1032)$$

and hence the graded reduced mixed symbol is nondegenerate.

From (1029) and (1027), the leading term in b^μ is

$$I_\theta (\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu),$$

with $I_\theta \neq 0$ for $\theta_0 \kappa \neq 0$. Therefore

$$\det[b_i^\mu] = I_\theta^3 \det[\Lambda_i^\mu] + \text{higher-order corrections.}$$

If $\det[\Lambda_i^\mu] \neq 0$, then continuity implies the determinant remains nonzero for sufficiently small nonzero θ_0 .

7. G. Canonical full-rank choice

A canonical full-rank choice is

$$\Lambda_i^\mu = \lambda \delta_i^\mu, \quad \lambda \neq 0, \quad (1033)$$

so

$$\mathcal{M}^\mu(\theta) = \lambda \sigma_\mu + O(\theta^2), \quad (1034)$$

and

$$A_1(p) = \lambda I_\theta p_i \sigma_i + O(\theta_0^3 \kappa |p|). \quad (1035)$$

Hence the reduced principal operator is Dirac to leading order.

8. H. Honest status

What is now closed is not the exact coefficient extraction from the full TECT PDE, but a sharper reduction:

$$\text{full microscopic coefficient problem} \implies \text{determine the finite matrix } \Lambda_i^\mu.$$

The remaining frontier is therefore: derive the actual Λ_i^μ from the final microscopic projector transport equation of TECT.

. **Summary**

$$i\partial_t\eta = \left[-\rho\partial_\xi^2 + U_B(\xi) - i\theta'(\xi)\mathcal{M}^\mu(\theta(\xi))\partial_{y^\mu} \right]\eta + O(\partial_y^2)\eta$$

is the minimal microscopic projector transport ansatz compatible with the actual local CP^2 defect geometry.
Write

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i.$$

Using the minimal even- θ expansion,

$$a_0^\mu(\theta) = \Lambda_0^\mu + O(\theta^2), \quad a_i^\mu(\theta) = \Lambda_i^\mu + O(\theta^2),$$

one gets

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta^2 \theta').$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$,

$$M^\mu = I_\theta \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta_0^3 \kappa), \quad I_\theta = \int d\xi f_+ f_- \theta' = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Hence the reduced mixed symbol is

$$A_1(p) = p_\mu M^\mu = I_\theta p_\mu \left[\Lambda_0^\mu \mathbf{1}_2 + \Lambda_i^\mu \sigma_i \right] + O(\theta_0^3 \kappa |p|).$$

The load-bearing traceless vectors are

$$b^\mu = I_\theta (\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3 \kappa).$$

Therefore

$$\det[b_i^\mu] = I_\theta^3 \det[\Lambda_i^\mu] + O(\theta_0^5 \kappa^3).$$

So if

$$\boxed{\det[\Lambda_i^\mu] \neq 0},$$

then for sufficiently small nonzero θ_0 ,

$$\boxed{\det[b_i^\mu] \neq 0},$$

and the graded reduced symbol is nondegenerate.

Thus the full microscopic coefficient problem has been reduced to the finite matrix problem of determining

$$\Lambda_i^\mu.$$

. **Local symmetry classification of Λ_i^μ**

a. Goal. We classify which residual local symmetries force rank deficiency of the leading transport matrix

$$\Lambda_i^\mu,$$

and why full rank is the generic case once those extra symmetries are absent.

1. A. Reduced coefficient problem

From the minimal microscopic transport ansatz already derived,

$$i\partial_t\eta = \left[-\rho\partial_\xi^2 + U_B(\xi) - i\theta'(\xi)\mathcal{M}^\mu(\theta(\xi))\partial_{y^\mu} \right]\eta + O(\partial_y^2)\eta, \quad (1036)$$

with

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i, \quad (1037)$$

the minimal even- θ expansion gives

$$a_i^\mu(\theta) = \Lambda_i^\mu + O(\theta^2). \quad (1038)$$

Projecting onto the graded low modes,

$$b^\mu = I_\theta(\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3\kappa), \quad (1039)$$

with

$$I_\theta \neq 0 \quad (\theta_0\kappa \neq 0). \quad (1040)$$

Therefore the Dirac nondegeneracy problem reduces to the finite matrix condition

$$\det[\Lambda_i^\mu] \neq 0. \quad (1041)$$

2. B. Representation-theoretic meaning of Λ_i^μ

The index $i = 1, 2, 3$ labels the three traceless Hermitian generators of $\text{End}(Q_B)$,

$$\sigma_1, \sigma_2, \sigma_3,$$

while $\mu = 1, 2, 3$ labels the three worldvolume derivative directions ∂_{y^μ} . Hence

$$\Lambda : \mathbb{R}_{\text{wv}}^3 \rightarrow \mathbb{R}_{\text{tr}}^3, \quad (\Lambda p)_i = \Lambda_i^\mu p_\mu, \quad (1042)$$

is the linear map from worldvolume momentum space to the traceless transport space.

The mixed symbol is nondegenerate iff this map has full rank:

$$\text{rank } \Lambda = 3. \quad (1043)$$

3. C. Symmetry constraints that force rank deficiency

A residual symmetry G acting simultaneously on the worldvolume directions and on the Q_B -fiber imposes the intertwining condition

$$R_{\text{tr}}(g)\Lambda = \Lambda R_{\text{wv}}(g), \quad g \in G, \quad (1044)$$

where R_{wv} is the representation on momentum directions and R_{tr} is the representation on the traceless Hermitian fiber sector.

Rank deficiency occurs when this intertwining condition leaves only a proper invariant subspace.

a. Case 1: residual axial $U(1)$ in the fiber. Suppose the local defect/transport structure preserves a distinguished internal axis, so that only σ_3 is invariant and (σ_1, σ_2) rotate as a doublet. If the worldvolume representation has no matching doublet, then only

$$\Lambda_3^\mu \quad (1045)$$

can survive. Hence

$$\text{rank } \Lambda \leq 1. \quad (1046)$$

b. Case 2: residual axial symmetry in physical space. Suppose a worldvolume axis is distinguished and the symmetry preserves only the longitudinal direction $\mu = 3$, while $(\mu = 1, \mu = 2)$ form a transverse doublet. If the transport tensor couples only this distinguished direction to one fixed traceless channel, then

$$\Lambda_i{}^\mu \sim v_i n^\mu \quad (1047)$$

is rank 1.

c. Case 3: planar symmetry. Suppose the symmetry preserves a two-dimensional plane and forbids coupling to the third direction. Then

$$\Lambda_i{}^\mu = \begin{pmatrix} * & * & 0 \\ * & * & 0 \\ * & * & 0 \end{pmatrix}, \quad (1048)$$

so

$$\text{rank } \Lambda \leq 2. \quad (1049)$$

d. Case 4: mirror/reflection selection. If a reflection symmetry kills one odd transport direction $\mu = \mu_0$, then the corresponding column of Λ vanishes:

$$\Lambda_i{}^{\mu_0} = 0. \quad (1050)$$

Again

$$\text{rank } \Lambda \leq 2. \quad (1051)$$

Thus:

[Symmetry-forced rank deficiency] If a residual symmetry G constrains the intertwiner space

$$\text{Hom}_G(\mathbb{R}_{\text{wv}}^3, \mathbb{R}_{\text{tr}}^3)$$

to lie in a proper algebraic subset of $\text{Mat}_{3 \times 3}(\mathbb{R})$, then $\det \Lambda = 0$ is forced. In particular, any symmetry that preserves only one traceless channel or only one worldvolume direction forces $\text{rank } \Lambda \leq 1$, and any symmetry that removes one worldvolume direction forces $\text{rank } \Lambda \leq 2$.

4. D. Why full rank is generic once those symmetries are absent

Now assume there is no extra residual symmetry beyond:

1. reflection parity already used to construct the $2 + 2$ grading,
2. Hermiticity of the transport tensor,
3. locality and first-order worldvolume transport.

Then Λ is simply a real 3×3 matrix with no further algebraic restriction. The set of singular matrices

$$\mathcal{S} := \{\Lambda \in \text{Mat}_{3 \times 3}(\mathbb{R}) : \det \Lambda = 0\} \quad (1052)$$

is a proper algebraic hypersurface. Hence its complement

$$\text{GL}(3, \mathbb{R}) = \{\Lambda : \det \Lambda \neq 0\} \quad (1053)$$

is open and dense.

Therefore:

[Generic full-rank criterion] If no additional residual symmetry constrains $\Lambda_i{}^\mu$ beyond the basic graded transport structure, then

$$\det \Lambda \neq 0$$

is the generic case. Equivalently, rank deficiency is nongeneric and requires either symmetry or fine tuning.

5. *E. Consistency with the explicit full-rank representative*

The uploaded manuscript already gives an explicit full-rank point in the same operator class:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (1054)$$

This corresponds exactly to

$$\Lambda_i^\mu = \lambda \delta_i^\mu \quad (1055)$$

for some $\lambda \neq 0$, up to normalization. Thus the actual local operator class contains at least one point with

$$\det \Lambda \neq 0. \quad (1056)$$

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This proves there is no structural obstruction inside the operator class.

6. *F. Honest final status*

What is now closed is:

1. the coefficient problem reduces to a finite real 3×3 matrix Λ_i^μ ;
2. symmetry-forced rank deficiency can be classified explicitly;
3. absent such extra symmetry, full rank is generic;
4. the actual local operator class already contains explicit full-rank representatives.

What remains open is the strongest microscopic statement: that the *actual full TECT linearized projector transport equation* yields a specific Λ_i^μ lying in the full-rank region.

. **Summary**

$$a_i^\mu(\theta) = \Lambda_i^\mu + O(\theta^2), \quad b^\mu = I_\theta(\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3 \kappa),$$

so the mixed symbol is nondegenerate iff

$$\det[\Lambda_i^\mu] \neq 0.$$

Here Λ is a linear map

$$\Lambda : \mathbb{R}_{\text{wv}}^3 \rightarrow \mathbb{R}_{\text{tr}}^3,$$

from worldvolume momentum directions to the three traceless Hermitian transport channels $\sigma_1, \sigma_2, \sigma_3$.

Rank deficiency is forced only when extra residual symmetry constrains the intertwiner space

$$\text{Hom}_G(\mathbb{R}_{\text{wv}}^3, \mathbb{R}_{\text{tr}}^3)$$

to a proper subset. In particular:

$$\text{one distinguished traceless axis} \Rightarrow \text{rank } \Lambda \leq 1,$$

$$\text{one forbidden worldvolume direction} \Rightarrow \text{rank } \Lambda \leq 2,$$

$$\text{planar or axial symmetry} \Rightarrow \det \Lambda = 0.$$

If no such additional residual symmetry is imposed, then Λ is just a real 3×3 matrix, and

$$\det \Lambda \neq 0$$

is open and dense in $\text{Mat}_{3 \times 3}(\mathbb{R})$. So full rank is generic, and rank deficiency is nongeneric.

Moreover, the actual local operator class already contains an explicit full-rank point

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3,$$

which corresponds to

$$\Lambda_i^\mu \propto \delta_i^\mu.$$

Therefore:

full rank of Λ_i^μ is not obstructed structurally;

it fails only if a special residual symmetry or fine tuning forces $\det \Lambda = 0$.

. **Why both branches must be analyzed: symmetry-forced rank deficiency versus generic full rank**

a. Claim. At the present stage, both possibilities for the transport matrix

$$\Lambda_i^\mu$$

must be analyzed:

1. residual-symmetry-forced rank deficiency,
2. generic full-rank behavior in the absence of extra symmetry.

1. *A. Where the remaining problem has been reduced*

The strongest honest reduction already established in the manuscript is the following:

graded low-energy sector + first-order odd reduced symbol + nondegenerate mixed 2×2 symbol $\implies$ Dirac operator. (1057)

More precisely, the reduced operator has the form

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad (1058)$$

and Dirac emergence depends on the nondegeneracy of $A_1(p)$ for generic $p \neq 0$. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

In the local projector-defect reduction, this was further reduced to the finite coefficient problem

$$M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i, \quad b^\mu := (C_1^\mu, C_2^\mu, C_3^\mu) \in \mathbb{R}^3, \quad (1059)$$

with exact nondegeneracy criterion

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3. \quad (1060)$$

Under the minimal transport ansatz, one has

$$b^\mu = I_\theta (\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3 \kappa), \quad (1061)$$

so the remaining finite-dimensional problem is

$$\det[\Lambda_i^\mu] \neq 0 \quad ? \quad (1062)$$

2. *B. Why symmetry-forced rank deficiency must be tested*

A residual local symmetry G acting on worldvolume directions and on the traceless transport channels imposes the intertwining condition

$$R_{\text{tr}}(g) \Lambda = \Lambda R_{\text{wv}}(g), \quad g \in G. \quad (1063)$$

This can force Λ to lie in a proper linear or algebraic subspace of $\text{Mat}_{3 \times 3}(\mathbb{R})$.

Typical possibilities are:

1. **Single distinguished traceless axis:** only one Pauli channel survives, so

$$\text{rank } \Lambda \leq 1.$$

2. **One forbidden worldvolume direction:** one column vanishes, so

$$\text{rank } \Lambda \leq 2.$$

3. **Axial or planar symmetry:** the map Λ is forced to preserve a proper subspace, hence

$$\det \Lambda = 0.$$

Therefore symmetry-forced rank deficiency is a genuine possibility, not a negligible corner case.

3. *C. Why generic full rank must also be tested*

On the other hand, if no additional residual symmetry survives beyond the basic parity/grading structure already used in the construction, then Λ is simply a real 3×3 matrix. The set of singular matrices

$$\mathcal{S} = \{\Lambda \in \text{Mat}_{3 \times 3}(\mathbb{R}) : \det \Lambda = 0\} \quad (1064)$$

is a proper algebraic hypersurface, while

$$\text{GL}(3, \mathbb{R}) = \{\Lambda : \det \Lambda \neq 0\} \quad (1065)$$

is open and dense.

Hence, absent extra symmetry or fine tuning,

$$\det \Lambda \neq 0 \quad (1066)$$

is the generic case.

Moreover, the manuscript already exhibits an explicit full-rank point in the same operator class:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0, \quad (1067)$$

which corresponds to

$$\Lambda_i{}^\mu \propto \delta_i^\mu. \quad (1068)$$

So there is no structural obstruction to full rank inside the actual local operator class. :contentReference[oacite:6]index=6

4. *D. Why both branches are logically necessary*

The open microscopic task is to determine which case is realized by the actual TECT defect/projector transport kernel. That requires answering two logically distinct questions:

$$\text{Q1: Does the actual microscopic background preserve an extra residual symmetry forcing } \det \Lambda = 0? \quad (1069)$$

$$\text{Q2: If not, does the actual microscopic coefficient lie in the generic full-rank region } \det \Lambda \neq 0? \quad (1070)$$

Without answering Q1, an appeal to genericity is not rigorous. Without answering Q2, ruling out special symmetry does not itself prove nondegeneracy.

Therefore both branches must be checked.

5. E. Operational consequence

The remaining microscopic problem is thus not merely:

compute Λ ,

but rather the two-step program

$$\boxed{\text{(i) classify the residual symmetry of the actual defect/projector background;}} \quad (1071)$$

$$\boxed{\text{(ii) within the allowed intertwiner space, determine whether the actual } \Lambda \text{ is full rank.}} \quad (1072)$$

This is exactly consistent with the manuscript's current strongest honest status: the theorem-level architecture is closed, and what remains is a finite list of explicit microscopic spectral / coefficient checks. :contentReference[oaicite:7]index=7

. Summary

$$A_1(p) = p_\mu M^\mu, \quad M^\mu = C_0^\mu \mathbf{1}_2 + C_i^\mu \sigma_i, \quad b^\mu = (C_1^\mu, C_2^\mu, C_3^\mu).$$

The exact nondegeneracy criterion is

$$\text{span}\{b^1, b^2, b^3\} = \mathbb{R}^3.$$

Under the minimal transport reduction,

$$b^\mu = I_\theta(\Lambda_1^\mu, \Lambda_2^\mu, \Lambda_3^\mu) + O(\theta_0^3 \kappa),$$

so the remaining finite-dimensional test is

$$\det[\Lambda_i^\mu] \neq 0.$$

There are two logically distinct possibilities:

(A) residual symmetry forces $\det \Lambda = 0$,

(B) no such symmetry remains, so $\det \Lambda \neq 0$ is generic.

Residual symmetry can force rank deficiency by preserving only:

one traceless channel $\Rightarrow \text{rank } \Lambda \leq 1$,

one or two worldvolume directions $\Rightarrow \text{rank } \Lambda \leq 2$,

axial/planar invariant subspaces $\Rightarrow \det \Lambda = 0$.

But if no additional residual symmetry survives, then $\Lambda \in \text{Mat}_{3 \times 3}(\mathbb{R})$ is generic, and

$$\det \Lambda \neq 0$$

is open and dense.

Therefore both questions must be checked:

$$\boxed{\text{(1) does the actual microscopic background leave an extra residual symmetry?}}$$

$$\boxed{\text{(2) if not, is the actual } \Lambda \text{ in the full-rank region?}}$$

So yes:

$$\boxed{\text{both branches must be analyzed.}}$$

. **Residual symmetry classification of the actual defect/projector background**

a. Goal. We classify the residual symmetry of the explicit local CP^2 defect background and determine when it forces rank deficiency of

$$\Lambda_i^\mu,$$

and when it still allows a full-rank Dirac symbol.

1. *A. Explicit local defect background and the distinguished internal axis*

Consider the explicit reflection-symmetric local defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (1073)$$

with the local Q_B -frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (1074)$$

Since

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi), \quad (1075)$$

the background gradient singles out the e_1 -direction inside the rank-two fiber Q_B . Thus the full $U(2)$ fiber symmetry of Q_B is generically broken by the defect gradient.

a. Residual fiber symmetry. The subgroup of $U(2)$ preserving both the decomposition

$$Q_B = \mathbb{C}e_1 \oplus \mathbb{C}e_2$$

and the distinguished real direction e_1 is the phase rotation of e_2 :

$$U(1)_{\text{res}} : \quad e_1 \mapsto e_1, \quad e_2 \mapsto e^{i\varphi} e_2. \quad (1076)$$

In the (e_1, e_2) frame, this acts by

$$R_{\text{fib}}(\varphi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}. \quad (1077)$$

2. *B. Action on the traceless Hermitian channels*

The traceless Hermitian basis of $\text{End}(Q_B)$ is

$$\Sigma^{(1)} = \sigma_1, \quad \Sigma^{(2)} = \sigma_2, \quad \Sigma^{(3)} = \sigma_3. \quad (1078)$$

Under $U(1)_{\text{res}}$,

$$R_{\text{fib}}(\varphi) \sigma_3 R_{\text{fib}}(\varphi)^{-1} = \sigma_3, \quad (1079)$$

while

$$R_{\text{fib}}(\varphi) \sigma_1 R_{\text{fib}}(\varphi)^{-1} = (\cos \varphi) \sigma_1 + (\sin \varphi) \sigma_2, \quad (1080)$$

$$R_{\text{fib}}(\varphi) \sigma_2 R_{\text{fib}}(\varphi)^{-1} = -(\sin \varphi) \sigma_1 + (\cos \varphi) \sigma_2. \quad (1081)$$

Thus the traceless transport space decomposes as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}, \quad (1082)$$

i.e. one invariant singlet plus one real two-dimensional rotation doublet.

3. C. Two logically distinct symmetry branches

Let

$$\Lambda : \mathbb{R}_{\text{wv}}^3 \rightarrow \mathbb{R}_{\text{tr}}^3 \quad (1083)$$

be the transport matrix in the leading reduced symbol. The allowed Λ must satisfy the intertwining condition

$$R_{\text{tr}}(g) \Lambda = \Lambda R_{\text{wv}}(g), \quad g \in G_{\text{res}}. \quad (1084)$$

Now there are two different possibilities depending on the actual action R_{wv} on worldvolume directions.

a. Branch S1: internal $U(1)_{\text{res}}$ survives, but there is no matching doublet on worldvolume momentum. Suppose the worldvolume representation is trivial or contains only singlets under the same symmetry. Then the doublet

$$\text{span}\{\sigma_1, \sigma_2\}$$

has no matching source, and only the invariant channel σ_3 is allowed. Hence

$$\Lambda_i{}^\mu = v^\mu \delta_{i3}, \quad (1085)$$

so

$$\text{rank } \Lambda \leq 1. \quad (1086)$$

This branch forces rank deficiency.

b. Branch S2: matched doublet symmetry. Suppose instead the worldvolume directions decompose under the residual symmetry as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\}, \quad (1087)$$

with $p_{\parallel}$ a singlet and (p_1, p_2) a doublet. Then the intertwiner condition allows:

1. a singlet-to-singlet map

$$p_{\parallel} \mapsto \sigma_3,$$

2. a doublet-to-doublet map

$$(p_1, p_2) \mapsto \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2).$$

Therefore the most general symmetry-allowed mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2). \quad (1088)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}. \quad (1089)$$

Hence

$$\det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (1090)$$

Thus this branch is full rank iff

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1091)$$

4. D. Interpretation

So the same residual $U(1)$ can lead to two different outcomes:

1. if it acts only on the fiber and has no matching worldvolume doublet, it forces rank deficiency;
2. if the worldvolume derivative sector carries the matching singlet $\oplus$ doublet decomposition, it still allows full-rank Dirac form.

Therefore one must not say merely “a residual $U(1)$ survives”. One must specify the pair of representations

$$(R_{\text{wv}}, R_{\text{tr}}).$$

5. E. Which branch is more natural for the actual defect geometry?

For the explicit defect background, the most natural worldvolume decomposition is precisely

$$\text{one distinguished direction } p_{\parallel} \quad + \quad \text{a transverse two-plane } (p_1, p_2),$$

because the defect gradient selects one local axis while the worldvolume tangential derivatives naturally split into longitudinal and transverse parts relative to that axis.

On the transport side, the actual local Q_B -endomorphism algebra already splits as

$$\mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

Hence the matched-doublet branch (1088) is the natural symmetry-compatible realization.

This is exactly consistent with the explicit full-rank realization already exhibited in the same operator class,

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad (1092)$$

which shows there is no structural obstruction to full rank. :contentReference[oaicite:3]index=3

6. F. Honest conclusion

What is now actually classified is:

Residual symmetry does *not* automatically force rank deficiency.

Instead:

rank deficiency occurs only when the residual fiber symmetry has no matching worldvolume doublet.

 (1093)

if a matched singlet $\oplus$ doublet structure exists, then the symmetry-allowed form is $A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2)$

 (1094)

which is full rank iff

$$\lambda_{\parallel}(\alpha^2 + \beta^2) \neq 0.$$

So the next microscopic question is no longer

“Is there residual symmetry?”

but rather

“Which representation pair $(R_{\text{wv}}, R_{\text{tr}})$ is realized by the actual TECT transport kernel?”

. **Summary**

$$Q_B = \mathbb{C}e_1 \oplus \mathbb{C}e_2, \quad \partial_\xi z_B = \theta' e_1,$$

so the defect gradient singles out e_1 in the rank-two fiber. The residual fiber symmetry is therefore

$$U(1)_{\text{res}} : \quad e_1 \mapsto e_1, \quad e_2 \mapsto e^{i\varphi} e_2.$$

On the traceless Hermitian transport channels,

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\},$$

with σ_3 invariant and (σ_1, σ_2) rotating as a doublet.

This leads to two distinct branches:

$$(S1) \text{ no matching worldvolume doublet} \implies \Lambda_i{}^\mu = v^\mu \delta_{i3}, \quad \text{rank } \Lambda \leq 1.$$

(S2) matched singlet $\oplus$ doublet on worldvolume

$$\implies A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

equivalently

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Hence residual symmetry by itself does not force rank deficiency. It forces rank deficiency only if the residual fiber symmetry has no matching worldvolume doublet.

If the matched singlet $\oplus$ doublet structure is realized, then the symmetry-allowed mixed symbol is already of Weyl/Dirac type, and it is full rank iff

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).}$$

So the real microscopic question is:

$$\boxed{\text{which representation pair } (R_{\text{wv}}, R_{\text{tr}}) \text{ is realized by the actual TECT transport kernel?}}$$

. **Why the matched singlet $\oplus$ doublet branch S2 is the natural local realization**

a. Goal. We determine which of the two residual-symmetry branches is more natural for the actual local TECT projector-defect background:

S1: residual fiber symmetry without a matching worldvolume doublet,

S2: matched singlet $\oplus$ doublet decomposition on both sides.

1. *A. Local defect geometry already singles out one distinguished axis*

For the explicit local defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad (1095)$$

one has

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi), \quad (1096)$$

so the defect gradient selects the e_1 -direction in the rank-two fiber Q_B . Thus the local fiber symmetry is reduced from the full $U(2)$ to a residual axial subgroup preserving the decomposition

$$Q_B = \mathbb{C}e_1 \oplus \mathbb{C}e_2.$$

In the local frame, the traceless Hermitian fiber sector decomposes as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R}\sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}, \quad (1097)$$

where σ_3 is the distinguished singlet and (σ_1, σ_2) form a doublet under the residual $U(1)$ rotation. This is exactly the standard axial decomposition of the traceless transport channels.

2. *B. The worldvolume side naturally has the same singlet $\oplus$ doublet decomposition*

The first-order transport term in the microscopic TECT linearization is

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (1098)$$

with worldvolume coordinates $y^\mu = (t, x^1, x^2)$ in the local defect reduction.

Once the defect gradient selects a local distinguished axis, the worldvolume momentum directions naturally split into:

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R}p_{\parallel} \oplus E_{\perp}, \quad E_{\perp} = \text{span}\{p_1, p_2\}. \quad (1099)$$

Thus the worldvolume side also carries a singlet $\oplus$ doublet decomposition:

$$p_{\parallel} \quad (\text{singlet}), \quad (p_1, p_2) \quad (\text{doublet}).$$

So the local defect geometry itself already supplies the matching representation pattern required for S2.

3. *C. Symmetry-compatible intertwiner space*

Let

$$\Lambda : \mathbb{R}_{\text{wv}}^3 \rightarrow \mathbb{R}_{\text{tr}}^3$$

be the leading transport map. Under the matched residual axial symmetry, both domain and codomain decompose as

$$\mathbb{R} \oplus \mathbb{R}^2.$$

Therefore the intertwiner space contains:

1. a singlet-to-singlet map

$$p_{\parallel} \longmapsto \sigma_3,$$

2. the full equivariant algebra on the transverse doublet plane

$$E_{\perp} \rightarrow \text{span}\{\sigma_1, \sigma_2\},$$

which is spanned by the identity and the planar complex structure J .

This is exactly the minimal axial tensor basis already derived in the microscopic Weyl witness analysis:

$$P_{\parallel} = \hat{n}\hat{n}, \quad P_{\perp}, \quad J_{ij} = \epsilon_{ijk}\hat{n}_k.$$

Restricted to the transverse plane, $\{P_{\perp}, J\}$ spans the full algebra of equivariant endomorphisms.

Hence the most general symmetry-compatible reduced mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (1100)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad (1101)$$

with determinant

$$\det \Lambda = \lambda_{\parallel}(\alpha^2 + \beta^2). \quad (1102)$$

Thus the S2 branch is not merely possible; it is exactly the natural symmetry-compatible branch once the defect geometry splits both sides into singlet $\oplus$ doublet sectors.

4. D. Why S1 is less natural locally

S1 would require that the residual fiber symmetry survives, but the worldvolume derivative sector carries no matching doublet. In that case only the invariant singlet channel survives:

$$A_1(p) \sim p_{\parallel} \sigma_3, \quad (1103)$$

so

$$\text{rank } \Lambda \leq 1.$$

However, in the actual local defect reduction, the worldvolume derivative sector automatically contains the transverse two-plane $E_{\perp}$, and the local transport operator $L_1 = -iV^{\mu}(\xi)\partial_{y^{\mu}}$ acts on all worldvolume directions. Therefore S1 is not the generic local outcome; it would require an additional symmetry or truncation that suppresses the transverse doublet coupling.

So:

S1 is possible only after an additional restriction; it is not the natural branch of the local defect geometry.

(1104)

5. E. Consistency with the explicit full-rank realization

The manuscript already constructs an explicit full-rank point in the same operator class:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (1105)$$

This is exactly the isotropic special case of the S2 branch:

$$\lambda_{\parallel} = I_g, \quad \alpha = I_g, \quad \beta = 0. \quad (1106)$$

Hence the local operator class already contains an explicit S2 full-rank realization.

6. *F. Honest conclusion*

The strongest honest local conclusion is:

The actual defect/worldvolume decomposition naturally realizes the matched singlet $\oplus$ doublet branch S2. (1107)

At the same time:

This still does not prove that the final full microscopic TECT transport kernel must land in the full-rank region. (1108)

What it proves is that, at the local defect/projector level, S2 is the geometrically natural branch, while S1 requires extra symmetry or truncation beyond the local defect geometry itself.

. Summary

$$\partial_\xi z_B = \theta' e_1$$

singles out one distinguished fiber direction e_1 , so the local traceless transport sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

At the same time, the local defect/worldvolume momentum directions split naturally as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\}.$$

Therefore the actual local defect geometry naturally supplies a matched singlet $\oplus$ doublet decomposition on both sides.

The corresponding symmetry-compatible mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

equivalently

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Hence the matched branch S2 is full rank iff

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

By contrast, the rank-deficient branch S1 would require that the residual fiber symmetry has no matching world-volume doublet, so only

$$A_1(p) \sim p_{\parallel} \sigma_3$$

survives. This is not the natural outcome of the local defect/worldvolume decomposition; it would require an additional symmetry or truncation.

Therefore:

S2 is the geometrically natural local branch, while S1 is an additional restricted branch.

. Actual residual symmetry diagnosis: the local defect geometry allows S2 and does not force S1

a. Goal. We determine, using only the actually established local CP^2 defect/projector geometry, whether the residual symmetry forces the rank-deficient branch S1, or whether it allows the matched singlet $\oplus$ doublet branch S2.

1. *A. Actual local operator class already fixed*

From the local projector-Hessian analysis around a reflection-symmetric defect background, the following are already established:

1. the fluctuation field is a complex doublet on the rank-two bundle

$$Q_B = (\mathbb{C}z_B)^\perp, \quad (1109)$$

2. the static Hessian is a transverse matrix Schrödinger operator on Q_B ,
3. even/odd low channels f_+, f_- arise under reflection symmetry,
4. the leading worldvolume transport kernel is odd,

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1110)$$

5. same-parity overlaps vanish, while the mixed overlap block is generically nonzero:

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (1111)$$

This is the exact local matching theorem for the $\mathbb{CP}^2$ projector Hessian operator class.

2. *B. Residual fiber symmetry of the explicit defect background*

Take the explicit local representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi). \quad (1112)$$

Then

$$\partial_\xi z_B(\xi) = \theta'(\xi)e_1(\xi), \quad (1113)$$

so the defect gradient singles out one preferred direction e_1 in the rank-two fiber Q_B .

Thus the full $U(2)$ fiber symmetry is reduced to a residual axial subgroup preserving the decomposition

$$Q_B = \mathbb{C}e_1 \oplus \mathbb{C}e_2.$$

On the traceless Hermitian sector this gives the splitting

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R}\sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}, \quad (1114)$$

where σ_3 is a singlet and (σ_1, σ_2) transform as a doublet under the residual $U(1)$.

3. *C. Worldvolume side also has a natural singlet+doublet split*

The same local defect geometry also singles out a distinguished worldvolume direction associated with the defect-gradient axis. Hence the worldvolume momentum space naturally decomposes as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R}p_\parallel \oplus \text{span}\{p_1, p_2\}. \quad (1115)$$

So the worldvolume representation has the same singlet+doublet pattern as the traceless fiber transport sector:

$$p_\parallel \leftrightarrow \sigma_3, \quad (p_1, p_2) \leftrightarrow (\sigma_1, \sigma_2).$$

This means the local geometry itself provides the matched representation pattern needed for S2.

4. *D. The symmetry-allowed mixed symbol is exactly S2*

When the singlet $\oplus$ doublet decompositions match on both sides, the intertwiner space allows:

1. a singlet-to-singlet map

$$p_{\parallel} \mapsto \sigma_3,$$

2. the full two-parameter equivariant map from the transverse worldvolume doublet to the transverse Pauli doublet.

Therefore the most general symmetry-allowed mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (1116)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad (1117)$$

with determinant

$$\det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (1118)$$

Thus S2 is not only possible; it is exactly the symmetry-compatible branch selected by the natural local singlet $\oplus$ doublet decomposition.

5. *E. Why S1 is not forced by the actual local geometry*

S1 would mean that only the invariant singlet survives:

$$A_1(p) \sim p_{\parallel} \sigma_3, \quad (1119)$$

so

$$\text{rank } \Lambda \leq 1. \quad (1120)$$

But this requires the transverse worldvolume doublet (p_1, p_2) to fail to couple to the transverse Pauli doublet (σ_1, σ_2) . The actual local defect geometry by itself does not enforce such a decoupling. It already contains:

1. a transverse two-plane in worldvolume momentum space,
2. a transverse two-plane in the traceless fiber transport sector,
3. a mixed overlap formula with generically nonzero off-diagonal block.

Therefore S1 is not the default local outcome. To realize S1 one needs an additional restriction beyond the basic defect geometry, such as:

1. an extra mirror or axial symmetry killing one or both transverse channels,
2. an additional truncation that removes the worldvolume doublet,
3. or special fine tuning of the transport coefficients.

6. *F. Consistency with the manuscript's symmetry theorem*

The manuscript's reduced symmetry theorem already gives the precise logic:

1. symmetry forbids inequivalent mixed linear terms,
2. symmetry allows the longitudinal scalar term $v_{\parallel} p_{\parallel} \sigma_3$,
3. symmetry allows the transverse doublet term $p_a T_{ab} \sigma_b$ whenever the two doublets are representation-equivalent,
4. symmetry alone does not force $v_{\parallel} \neq 0$ or $\det T \neq 0$.

So symmetry is a reduction theorem, not a nondegeneracy theorem. This is exactly what the local defect analysis is showing here as well.

7. *G. Explicit full-rank point in the same actual operator class*

The same local operator class already contains an explicit full-rank point:

$$M^1 = I_g \sigma_1, \quad M^2 = I_g \sigma_2, \quad M^3 = I_g \sigma_3, \quad I_g \neq 0. \quad (1121)$$

This is simply the isotropic special point of the S2 family. Therefore there is no structural obstruction inside the actual local defect/operator class to full rank.

8. *H. Honest conclusion*

The strongest honest local conclusion is:

The actual local defect/projector geometry is naturally compatible with S2.

(1122)

At the same time:

S1 is not forced by local geometry; it requires an additional symmetry, truncation, or fine tuning.

(1123)

And finally:

This still does not prove that the full microscopic TECT transport kernel lands in the full-rank region.

(1124)

So the remaining microscopic task is to determine the actual coefficients $(\lambda_{\parallel}, \alpha, \beta)$ of the S2 branch in the full theory.

. **Summary**

$$\partial_{\xi} z_B = \theta' e_1$$

singles out one preferred fiber direction, so the traceless transport sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

At the same time, the actual defect/worldvolume momentum directions split naturally as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\}.$$

Thus the local defect geometry naturally provides a matched singlet $\oplus$ doublet decomposition on both sides. The corresponding symmetry-compatible mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

equivalently

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Hence the local symmetry-compatible branch is S2.

By contrast, S1 would keep only

$$A_1(p) \sim p_{\parallel} \sigma_3,$$

which requires additional symmetry, truncation, or fine tuning to remove the transverse doublet coupling.

Therefore:

S2 is the natural local branch, while S1 is an additional restricted branch.

The remaining microscopic question is not whether S2 is allowed, but whether the actual full TECT transport kernel realizes

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

. **Actual local formulas for the S2 two-witness condition**

a. Goal. We now write the actual local formulas for the S2-branch witnesses

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

directly in terms of the projected local transport kernel of the CP^2 projector defect.

1. *A. Projected mixed transport matrices*

For a reflection-symmetric defect background, the leading transport term is

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (1125)$$

with odd kernel $V^\mu(-\xi) = -V^\mu(\xi)$. Projecting onto the graded low modes

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2, \quad (1126)$$

gives

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (1127)$$

The strongest honest local transport form already obtained is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (1128)$$

with

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i. \quad (1129)$$

Hence

$$M^\mu = \int d\xi f_+f_- \theta'(\xi) \left[a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i \right]. \quad (1130)$$

Write

$$M^\mu = A^\mu\mathbf{1}_2 + B_i^\mu\sigma_i, \quad (1131)$$

where

$$A^\mu = \int d\xi f_+f_- \theta'(\xi) a_0^\mu(\theta(\xi)), \quad (1132)$$

$$B_i^\mu = \int d\xi f_+f_- \theta'(\xi) a_i^\mu(\theta(\xi)). \quad (1133)$$

2. *B. Singlet $\oplus$ doublet decomposition of the S2 branch*

In the S2 branch, the worldvolume momentum directions split as

$$p = (p_1, p_2, p_{\parallel}) \in 2_{\perp} \oplus 1_{\parallel}, \quad (1134)$$

while the traceless transport sector splits as

$$(\sigma_1, \sigma_2) \oplus \sigma_3. \quad (1135)$$

The most general symmetry-compatible S2 mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1136)$$

equivalently

$$A_1(p) = p_\mu B_i^\mu \sigma_i \quad \text{with} \quad \Lambda = \begin{pmatrix} B_1^1 & B_1^2 & B_1^\parallel \\ B_2^1 & B_2^2 & B_2^\parallel \\ B_3^1 & B_3^2 & B_3^\parallel \end{pmatrix} = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix}. \quad (1137)$$

Thus the S2 coefficients are read off from the projected traceless matrices as

$$\lambda_\parallel = B_3^\parallel, \quad (1138)$$

$$\alpha = \frac{1}{2}(B_1^1 + B_2^2), \quad (1139)$$

$$\beta = \frac{1}{2}(B_2^1 - B_1^2). \quad (1140)$$

3. C. Actual local overlap formulas for the two witnesses

It is convenient to rewrite the traceless coefficient functions in the S2-adapted basis:

$$a_i^\mu(\theta) \sigma_i = h_\parallel(\theta) \delta_\parallel^\mu \sigma_3 + h_{11}(\theta) \delta_1^\mu \sigma_1 + h_{21}(\theta) \delta_1^\mu \sigma_2 + h_{12}(\theta) \delta_2^\mu \sigma_1 + h_{22}(\theta) \delta_2^\mu \sigma_2 + \dots, \quad (1141)$$

where the omitted terms denote symmetry-forbidden or higher-order pieces if the S2 reduction is imposed. Then the longitudinal witness is

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_\parallel(\theta(\xi)). \quad (1142)$$

For the transverse doublet, the 2×2 reduced intertwiner matrix is

$$T = \begin{pmatrix} B_1^1 & B_1^2 \\ B_2^1 & B_2^2 \end{pmatrix} = \int d\xi f_+ f_- \theta'(\xi) \begin{pmatrix} h_{11}(\theta) & h_{12}(\theta) \\ h_{21}(\theta) & h_{22}(\theta) \end{pmatrix}. \quad (1143)$$

Equivalently, the two standard S2 coefficients are

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (1144)$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (1145)$$

The nondegeneracy condition is

$$\det \Lambda = \lambda_\parallel (\alpha^2 + \beta^2) \neq 0. \quad (1146)$$

4. D. What symmetry does and does not imply

Symmetry does two things:

1. it forbids mismatched channels, e.g. longitudinal momentum coupling to transverse Pauli directions and vice versa;
2. it reduces the possible first-order symbol to the S2 template (1136).

But symmetry does *not* prove the actual nonvanishing of

$$\lambda_\parallel \quad \text{or} \quad (\alpha, \beta).$$

It only shows that these coefficients are allowed, not ruled out, by the selection rules.

This is the same logic already established in the shell-level Weyl witness reduction: symmetry determines the allowed invariant tensor space, but the actual coefficients must still be checked microscopically.

5. E. Leading small- θ reduction

If one uses the minimal even- θ expansion

$$h_{\parallel}(\theta) = \Lambda_{\parallel} + O(\theta^2), \quad h_{ab}(\theta) = \Lambda_{ab} + O(\theta^2), \quad (1147)$$

then, since

$$I_{\theta} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \neq 0, \quad (1148)$$

one gets

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel} + O(\theta_0^3 \kappa), \quad (1149)$$

$$\alpha = \frac{I_{\theta}}{2} (\Lambda_{11} + \Lambda_{22}) + O(\theta_0^3 \kappa), \quad (1150)$$

$$\beta = \frac{I_{\theta}}{2} (\Lambda_{21} - \Lambda_{12}) + O(\theta_0^3 \kappa). \quad (1151)$$

Thus the S2 two-witness problem reduces to a finite coefficient test even in the actual local transport formulation.

6. F. Honest conclusion

The strongest honest actual-local result is:

$$\boxed{\lambda_{\parallel}, \alpha, \beta \text{ admit explicit local overlap formulas in terms of } f_+ f_- \theta'(\xi) \times h(\theta(\xi)).} \quad (1152)$$

What remains open is not the formula, but the microscopic coefficient extraction: the actual functions

$$h_{\parallel}(\theta), \quad h_{11}(\theta), \quad h_{12}(\theta), \quad h_{21}(\theta), \quad h_{22}(\theta)$$

have not yet been derived from the final full TECT linearized projector equation.

. Summary

$$(M^{\mu})_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^{\dagger}(\xi) V^{\mu}(\xi) e_b(\xi), \quad V^{\mu}(\xi) = \theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)).$$

Writing

$$\mathcal{M}^{\mu}(\theta) = a_0^{\mu}(\theta) \mathbf{1}_2 + a_i^{\mu}(\theta) \sigma_i,$$

one gets

$$M^{\mu} = A^{\mu} \mathbf{1}_2 + B_i^{\mu} \sigma_i, \quad B_i^{\mu} = \int d\xi f_+ f_- \theta'(\xi) a_i^{\mu}(\theta(\xi)).$$

In the S2 singlet $\oplus$ doublet decomposition, the symmetry-compatible mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

equivalently

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Therefore the actual local overlap formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))),$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))).$$

Thus the S2 two-witness condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Symmetry determines that these are the only relevant first-order coefficients, but symmetry alone does not prove their nonvanishing. The remaining microscopic task is to derive the coefficient functions

$$h_{\parallel}(\theta), \quad h_{11}(\theta), \quad h_{12}(\theta), \quad h_{21}(\theta), \quad h_{22}(\theta)$$

from the final full TECT linearized projector transport equation.

• Strongest honest statement at the current stage

a. Purpose. We summarize, without overclaiming, what has been rigorously closed, what has been structurally reduced, and what remains genuinely open in the current TECT Dirac-emergence program.

1. A. What is already closed

a. (1) Plain scalar shell-pair route is insufficient. The scalar shell-pair channel produces a constant intervalley Bragg block of the form

$$H_{\text{Bragg}}^{\text{scalar}} = B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2, \quad (1153)$$

with $B_0^{\text{tot}} \neq 0$ generically in the actual scalar Hessian channel. This is a genuine constant pair-mixing obstruction and cannot be removed by a scalar energy shift.

b. (2) The refined projector-stiffness model alone does not solve the problem. For the current refined $U(3)$ -invariant triplet plus projector-stiffness model expanded around a locked constant-projector background, the quadratic shell-projector source vanishes:

$$J = 0. \quad (1154)$$

Hence the refined projector stiffness sector alone does not induce a nontrivial shell-enriched tensor at quadratic order.

c. (3) Class II produces $J \neq 0$ and a genuine derivative witness correction. Once the auxiliary Class II mediator is included, one obtains a nonzero quadratic shell-projector mixing source and therefore

$$\delta N_{ij}^a \neq 0. \quad (1155)$$

This renormalizes the derivative witness sector (A, B, C) , but it does not directly cancel the scalar Bragg mass because the Class II correction lives in the traceless internal channel, not in the identity channel.

d. (4) The graded $2 + 2$ bypass is the natural remaining route. The exact reduced Dirac-emergence theorem is already established: if the infrared sector contains an isolated four-dimensional graded subspace

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2, \quad (1156)$$

and the leading projected symbol is grading-odd and first-order with nondegenerate mixed block, then the reduced principal operator is Dirac type. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

e. (5) The actual local CP^2 projector-defect geometry belongs to this operator class. For a reflection-symmetric defect background, the projector Hessian reduces to a rank-two bundle problem on

$$Q_B = (\mathbb{C}z_B)^\perp, \quad (1157)$$

with parity decomposition into even/odd low modes and a first-order odd transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi). \quad (1158)$$

The projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (1159)$$

This is the exact local matching statement of the projector-defect operator class. :contentReference[oaicite:9]index=9
:contentReference[oaicite:10]index=10

f. (6) The leading odd scalar profile is $\theta'(\xi)$. For the explicit local defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi), \quad (1160)$$

one has

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger), \quad (1161)$$

so every leading local transport kernel linear in the defect gradient is of the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)). \quad (1162)$$

Moreover, with the explicit low modes used in the improved effective realization,

$$I_\theta := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi) \neq 0 \quad (1163)$$

for nontrivial defect amplitude.

2. B. The natural local branch is S2

The local defect geometry singles out one distinguished axis through $\partial_\xi z_B$, so the traceless transport sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R}\sigma_3 \oplus \operatorname{span}\{\sigma_1, \sigma_2\}. \quad (1164)$$

At the same time the worldvolume derivative directions split naturally as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R}p_\parallel \oplus \operatorname{span}\{p_1, p_2\}. \quad (1165)$$

Hence the matched singlet $\oplus$ doublet branch S2 is the natural local symmetry-compatible branch. Its most general first-order traceless symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1166)$$

equivalently

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix}, \quad \det \Lambda = \lambda_\parallel(\alpha^2 + \beta^2). \quad (1167)$$

This is exactly consistent with the shell-level symmetry classification theorem. :contentReference[oaicite:11]index=11

3. C. Actual local overlap formulas for the two witnesses

In the actual local projector reduction, the two S2 witnesses admit explicit overlap formulas:

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (1168)$$

$$\alpha = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (1169)$$

$$\beta = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (1170)$$

Thus the reduced nondegeneracy condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1171)$$

4. D. What is not yet closed

At the present stage, the following remains genuinely open:

1. the explicit full microscopic TECT condensate/projector PDE in final closed form suitable for direct spectral analysis;
2. the direct derivation of the microscopic transport kernel coefficients

$$h_{\parallel}(\theta), h_{11}(\theta), h_{12}(\theta), h_{21}(\theta), h_{22}(\theta);$$

3. the explicit verification that the actual microscopic TECT projected overlaps satisfy

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

rather than merely allowing them by symmetry.

This is exactly the manuscript's strongest honest status: the reduced Dirac-emergence theorem is rigorous, an improved effective CP²-defect realization is explicitly constructed, but the full microscopic matching to the complete TECT operator remains open. :contentReference[oaicite:13]index=13

5. E. Strongest honest conclusion

The strongest honest statement currently supported is therefore:

$$\boxed{\text{Dirac emergence is rigorously proved as a reduced spectral theorem,}} \quad (1172)$$

$$\boxed{\text{its hypotheses are explicitly realizable in an improved effective CP}^2\text{-defect model,}} \quad (1173)$$

$$\boxed{\text{and the actual local projector-defect geometry is naturally compatible with the S2 branch,}} \quad (1174)$$

but

$$\boxed{\text{the full microscopic coefficient extraction and verification of the S2 two-witness condition in complete TECT remains open.}} \quad (1175)$$

. **Summary**

Closed:

1. Plain scalar shell-pair dynamics is insufficient because it produces a genuine constant Bragg block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2 \neq 0.$$

2. The refined projector-stiffness model alone gives

$$J = 0,$$

so it does not induce a shell-enriched tensor at quadratic order.

3. Class II yields

$$\delta N_{ij}^a \neq 0,$$

hence renormalizes the derivative witness sector, but it does not cancel the scalar Bragg mass.

4. The graded 2+2 reduced theorem is rigorous: if the projected low-energy operator is grading-odd and first-order with nondegenerate mixed symbol, then the principal operator is Dirac type.

5. The actual local CP^2 projector-defect geometry belongs to this operator class:

$$Q_B = (\mathbb{C}z_B)^\perp, \quad L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

6. The natural local symmetry-compatible branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\det \Lambda = \lambda_{\parallel}(\alpha^2 + \beta^2).$$

7. The actual local overlap formulas are

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))),$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))).$$

Still open:

1. the final full microscopic TECT projector/condensate PDE in a form suitable for direct spectral analysis;
2. the explicit microscopic extraction of

$$h_{\parallel}(\theta), h_{11}(\theta), h_{12}(\theta), h_{21}(\theta), h_{22}(\theta);$$

3. the direct microscopic proof that

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Therefore the strongest honest current conclusion is

Dirac emergence is rigorously proved as a reduced theorem and explicitly realizable in an improved effective CP^2 -defect model,

but

the full microscopic matching to complete TECT, including the actual S2 coefficient extraction, remains open.

I. TECT DIRAC-EMERGENCE MASTER CONTEXT (CURRENT STRONGEST HONEST STATEMENT)

A. Project goal

The current goal is to determine whether TECT admits a *microscopically justified* emergent Dirac operator in the defect/projector sector.

The program has now split into two logically distinct layers:

1. a *rigorous reduced spectral theorem* for Dirac emergence;
2. a *microscopic matching problem* for the actual TECT projector/defect transport coefficients.

B. I. What is already closed

1. I.1 Plain scalar shell-pair route is insufficient

The scalar shell-pair channel produces a genuine constant intervalley Bragg block

$$H_{\text{Bragg}}^{\text{scalar}} = B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2, \quad B_0^{\text{tot}} \neq 0 \text{ generically.} \quad (1176)$$

This cannot be removed by a scalar energy subtraction, so the plain scalar shell-pair route does *not* yield a massless Dirac point.

2. I.2 The refined projector-stiffness model alone does not induce the needed shell mixing

For the current refined $U(3)$ -invariant triplet plus projector-stiffness model around a locked constant-projector background, the quadratic shell-projector mixing source vanishes:

$$J = 0. \quad (1177)$$

Hence the projector-stiffness sector alone does not induce a nontrivial shell-enriched tensor at quadratic order.

3. I.3 Class II yields a genuine derivative witness correction but does not remove the scalar Bragg mass

With the auxiliary Class II mediator included, one obtains

$$\delta N_{ij}^a \neq 0, \quad (1178)$$

so the derivative witness sector (A, B, C) is renormalized.

However, the Class II correction lives in the traceless internal sector and therefore does not directly cancel the identity-channel scalar Bragg obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

4. I.4 The natural remaining route is the graded $2+2$ bypass

There is a rigorous reduced theorem:

If the low-energy sector contains an isolated graded four-dimensional space

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2, \quad (1179)$$

and the leading projected symbol is grading-odd and first-order, with nondegenerate mixed block, then the reduced principal operator is Dirac type.

Equivalently, the reduced operator has the form

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad (1180)$$

and if $A_1(p)$ is nondegenerate for generic $p \neq 0$, then $D_1(p)$ is a Dirac principal operator.

C. II. Actual local $\mathbb{CP}^2$ projector-defect realization

1. II.1 Rank-two bundle structure

For a local projector defect background

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \quad z_B(\xi)^\dagger z_B(\xi) = 1, \quad (1181)$$

the physical tangent fluctuation lives in the rank-two complex bundle

$$Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp. \quad (1182)$$

Thus the defect fluctuation problem is not scalar; it is already a rank-two bundle problem.

2. II.2 Reflection-symmetric explicit representative

A useful explicit local representative is

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad \theta(\xi) = \theta_0 \operatorname{sech}(\kappa \xi). \quad (1183)$$

Then

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi), \quad (1184)$$

and

$$\partial_\xi P_B(\xi) = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (1185)$$

Hence every leading local transport kernel linear in the defect gradient factors through $\theta'(\xi)$.

3. II.3 Leading transport kernel

The actual local first-order transport operator is of the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1186)$$

and the strongest honest local form is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (1187)$$

with

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i. \quad (1188)$$

4. II.4 Even/odd low modes and mixed overlap

Using a local Q_B -frame $e_a(\xi)$, define

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2, \quad (1189)$$

with f_+ even and f_- odd.

Then the projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (1190)$$

Same-parity overlaps vanish, while the mixed block is generically nonzero.

5. *II.5 Explicit nonzero overlap of the leading odd scalar profile*

The leading odd scalar profile is

$$g_{\text{lead}}(\xi) = \theta'(\xi).$$

For the explicit low modes used in the effective realization,

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (1191)$$

so

$$I_\theta \neq 0 \quad (\theta_0 \kappa \neq 0). \quad (1192)$$

Thus the actual local geometry already gives a nontrivial graded overlap scale.

D. III. Residual symmetry and the natural S2 branch

1. *III.1 Fiber-side singlet $\oplus$ doublet decomposition*

Because $\partial_\xi z_B = \theta' e_1$ singles out e_1 , the traceless Hermitian transport sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}. \quad (1193)$$

2. *III.2 Worldvolume-side singlet $\oplus$ doublet decomposition*

The local defect/worldvolume geometry also naturally splits momentum directions into

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_{\parallel} \oplus \text{span}\{p_1, p_2\}. \quad (1194)$$

3. *III.3 Natural S2 mixed symbol*

Therefore the natural local symmetry-compatible branch is the matched singlet $\oplus$ doublet branch S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (1195)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (1196)$$

Hence the S2 two-witness condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1197)$$

4. *III.4 Why S1 is not the natural local branch*

The rank-deficient branch S1 would keep only

$$A_1(p) \sim p_{\parallel} \sigma_3.$$

But the local defect geometry already contains both the transverse worldvolume doublet (p_1, p_2) and the transverse traceless transport doublet (σ_1, σ_2) . Therefore S1 is not forced by the local geometry itself; it would require an additional symmetry, truncation, or fine tuning.

E. IV. Actual local formulas for the S2 witnesses

The local overlap formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (1198)$$

$$\alpha = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (1199)$$

$$\beta = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (1200)$$

Thus the two-witness problem has been reduced to a coefficient-extraction problem for the local transport tensor entries

$$h_{\parallel}(\theta), h_{11}(\theta), h_{12}(\theta), h_{21}(\theta), h_{22}(\theta).$$

F. V. What remains genuinely open

The remaining frontier is now sharply isolated:

1. the final full microscopic TECT condensate/projector PDE in a form suitable for direct spectral analysis;
2. the direct extraction of the microscopic coefficient functions

$$h_{\parallel}(\theta), h_{11}(\theta), h_{12}(\theta), h_{21}(\theta), h_{22}(\theta);$$

3. the direct microscopic proof that

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

in complete TECT rather than only in the explicit improved effective realization.

G. VI. Strongest honest conclusion

The strongest currently justified statement is:

$$\boxed{\text{Dirac emergence is rigorously proved as a reduced spectral theorem,}} \quad (1201)$$

$$\boxed{\text{its hypotheses are explicitly realizable in an improved effective CP}^2\text{-defect model,}} \quad (1202)$$

$$\boxed{\text{and the actual local projector-defect geometry is naturally compatible with the S2 branch,}} \quad (1203)$$

but

$$\boxed{\text{the full microscopic matching to complete TECT, including the actual S2 coefficient extraction, remains open.}} \quad (1204)$$

H. Summary

Closed:

1. The plain scalar shell-pair route is insufficient because it yields a genuine constant Bragg block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2 \neq 0.$$

2. The refined projector-stiffness model alone gives

$$J = 0,$$

so it does not induce the needed shell-enriched tensor.

3. Class II gives

$$\delta N_{ij}^a \neq 0,$$

hence renormalizes the derivative witness sector, but it does not cancel the scalar Bragg mass.

4. The graded $2 + 2$ reduced theorem is rigorous: if the leading reduced operator is grading-odd, first-order, and has nondegenerate mixed symbol, then the principal operator is Dirac type.

5. The actual local CP^2 projector-defect geometry belongs to this operator class:

$$Q_B = (\mathbb{C}z_B)^\perp, \quad L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

6. The natural local symmetry-compatible branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with nondegeneracy condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

7. The actual local witness formulas are

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))),$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))).$$

Still open:

1. the final full microscopic TECT projector/condensate PDE suitable for direct spectral analysis;
2. the explicit microscopic extraction of

$$h_{\parallel}(\theta), h_{11}(\theta), h_{12}(\theta), h_{21}(\theta), h_{22}(\theta);$$

3. the direct microscopic proof that

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

in complete TECT.

Therefore:

Dirac emergence is rigorously proved as a reduced theorem and explicitly realizable in an improved effective CP^2 -defect model,

but

the full microscopic S2 coefficient extraction in complete TECT remains open.

I. Concrete microscopic ansatz for the projector transport equation

a. Goal. We formulate a concrete microscopic ansatz for the transport sector of the projector-defect fluctuation equation, consistent with all previously established local structural results, and reduce the remaining coefficient problem to explicit overlap integrals.

1. A. Structural input already fixed

The local CP^2 projector-defect analysis has already established the following:

1. the physical fluctuation field is a section

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp, \quad (1205)$$

2. the static quadratic Hessian is a transverse matrix Schrödinger operator on Q_B ,
3. upon restoring slow worldvolume dependence, the leading transport term has the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1206)$$

with odd kernel built linearly from the transverse defect gradient,

4. for the explicit defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi), \quad (1207)$$

one has

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger), \quad (1208)$$

so the leading local transport kernel must factor through $\theta'(\xi)$.

Therefore the strongest honest local transport form is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)). \quad (1209)$$

2. B. Concrete microscopic ansatz

We now take the minimal first-order microscopic transport ansatz

$$i\partial_t \eta = \left[-\rho D_\xi^2 + U_B(\xi) - i\theta'(\xi) \mathcal{M}^\mu(\theta(\xi)) \partial_{y^\mu} \right] \eta + O(\partial_y^2) \eta. \quad (1210)$$

This is not yet derived from the final complete TECT PDE; it is the minimal microscopic ansatz consistent with all established local projector-defect structure.

In a local Q_B -frame, decompose

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta) \mathbf{1}_2 + a_i^\mu(\theta) \sigma_i. \quad (1211)$$

3. C. S2-adapted decomposition

For the natural local S2 branch, decompose the worldvolume directions into

$$p = (p_1, p_2, p_\parallel),$$

and write the traceless part in the S2 basis:

$$a_i^\mu(\theta) \sigma_i = h_\parallel(\theta) \delta_\parallel^\mu \sigma_3 + h_{11}(\theta) \delta_1^\mu \sigma_1 + h_{21}(\theta) \delta_1^\mu \sigma_2 + h_{12}(\theta) \delta_2^\mu \sigma_1 + h_{22}(\theta) \delta_2^\mu \sigma_2 + \cdots. \quad (1212)$$

Then the local S2 witness formulas are

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (1213)$$

$$\alpha = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (1214)$$

$$\beta = \frac{1}{2} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (1215)$$

This is the actual local form of the S2 two-witness condition.

4. D. Minimal even- θ expansion

Because the local transport data are built from the projector background and the physical projector class does not distinguish θ from $-\theta$ at the level of the local scalar coefficient functions, the minimal consistent ansatz is an even expansion in θ :

$$h_{\parallel}(\theta) = \Lambda_{\parallel} + \Xi_{\parallel} \theta^2 + O(\theta^4), \quad (1216)$$

$$h_{ab}(\theta) = \Lambda_{ab} + \Xi_{ab} \theta^2 + O(\theta^4), \quad a, b = 1, 2. \quad (1217)$$

Similarly,

$$a_0^{\mu}(\theta) = \Lambda_0^{\mu} + \Xi_0^{\mu} \theta^2 + O(\theta^4). \quad (1218)$$

5. E. Reduction to explicit overlap integrals

Insert (1216)–(1217) into (1213)–(1215). Define

$$I_{\theta} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi), \quad (1219)$$

$$I_{\theta^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \theta(\xi)^2. \quad (1220)$$

Then

$$\lambda_{\parallel} = \Lambda_{\parallel} I_{\theta} + \Xi_{\parallel} I_{\theta^3} + O(\theta_0^5 \kappa), \quad (1221)$$

$$\alpha = \frac{1}{2} (\Lambda_{11} + \Lambda_{22}) I_{\theta} + \frac{1}{2} (\Xi_{11} + \Xi_{22}) I_{\theta^3} + O(\theta_0^5 \kappa), \quad (1222)$$

$$\beta = \frac{1}{2} (\Lambda_{21} - \Lambda_{12}) I_{\theta} + \frac{1}{2} (\Xi_{21} - \Xi_{12}) I_{\theta^3} + O(\theta_0^5 \kappa). \quad (1223)$$

6. F. Explicit evaluation of I_{θ} and I_{θ^3}

Using the explicit profiles

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa \xi), \quad (1224)$$

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi), \quad (1225)$$

one already has

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1226)$$

Now compute I_{θ^3} . Since

$$\theta'(\xi) = -\theta_0 \kappa \operatorname{sech}(\kappa\xi) \tanh(\kappa\xi), \quad (1227)$$

and

$$\theta(\xi)^2 = \theta_0^2 \operatorname{sech}^2(\kappa\xi), \quad (1228)$$

while

$$f_+(\xi)f_-(\xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa\xi) \operatorname{sech}^2(\kappa\xi), \quad (1229)$$

we get

$$\begin{aligned} I_{\theta^3} &= \int d\xi f_+ f_- \theta'(\xi) \theta(\xi)^2 \\ &= -\frac{\sqrt{3}\theta_0^3 \kappa^2}{2} \int_{-\infty}^{\infty} d\xi \tanh^2(\kappa\xi) \operatorname{sech}^5(\kappa\xi). \end{aligned} \quad (1230)$$

Set

$$t = \tanh(\kappa\xi), \quad dt = \kappa \operatorname{sech}^2(\kappa\xi) d\xi, \quad \operatorname{sech}(\kappa\xi) = \sqrt{1-t^2}.$$

Then

$$I_{\theta^3} = -\frac{\sqrt{3}\theta_0^3 \kappa}{2} \int_{-1}^1 t^2 (1-t^2)^{3/2} dt. \quad (1231)$$

Using

$$\int_{-1}^1 t^2 (1-t^2)^{3/2} dt = \frac{\pi}{16}, \quad (1232)$$

we obtain

$$\boxed{I_{\theta^3} = -\frac{\sqrt{3}\pi}{32} \theta_0^3 \kappa.} \quad (1233)$$

7. G. Concrete witness expansion

Substituting (1226) and (1233) into (1221)–(1223),

$$\boxed{\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{\parallel} - \frac{\sqrt{3}\pi}{32} \theta_0^3 \kappa \Xi_{\parallel} + O(\theta_0^5 \kappa),} \quad (1234)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (\Lambda_{11} + \Lambda_{22}) - \frac{\sqrt{3}\pi}{64} \theta_0^3 \kappa (\Xi_{11} + \Xi_{22}) + O(\theta_0^5 \kappa),} \quad (1235)$$

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (\Lambda_{21} - \Lambda_{12}) - \frac{\sqrt{3}\pi}{64} \theta_0^3 \kappa (\Xi_{21} - \Xi_{12}) + O(\theta_0^5 \kappa).} \quad (1236)$$

8. H. Interpretation

The remaining full microscopic coefficient problem has now been reduced to determining the finite coefficient data

$$\Lambda_{\parallel}, \Lambda_{ab}, \Xi_{\parallel}, \Xi_{ab}.$$

At leading order in small defect amplitude θ_0 , the two-witness condition becomes simply

$$\Lambda_{\parallel} \neq 0, \quad (\Lambda_{11} + \Lambda_{22}, \Lambda_{21} - \Lambda_{12}) \neq (0, 0), \quad (1237)$$

since $I_{\theta} \neq 0$.

a. Honest status. What has been achieved here is a concrete microscopic transport ansatz plus an explicit reduction of the S2 witness coefficients to finite overlap integrals with fully computed θ -profile factors. What remains open is the derivation of the actual microscopic numbers $\Lambda_{\parallel}, \Lambda_{ab}, \Xi_{\parallel}, \Xi_{ab}$ from the final complete TECT projector transport PDE.

J. Summary

$$i\partial_t \eta = \left[-\rho D_{\xi}^2 + U_B(\xi) - i\theta'(\xi) \mathcal{M}^{\mu}(\theta(\xi)) \partial_{y^{\mu}} \right] \eta + O(\partial_y^2) \eta$$

is the minimal concrete microscopic ansatz consistent with the actual local projector-defect operator class.

Using the S2 decomposition,

$$\lambda_{\parallel} = \int d\xi f_+ f_- \theta'(\xi) h_{\parallel}(\theta(\xi)),$$

$$\alpha = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))),$$

$$\beta = \frac{1}{2} \int d\xi f_+ f_- \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))).$$

With the minimal even- θ expansion

$$h_{\parallel}(\theta) = \Lambda_{\parallel} + \Xi_{\parallel} \theta^2 + O(\theta^4), \quad h_{ab}(\theta) = \Lambda_{ab} + \Xi_{ab} \theta^2 + O(\theta^4),$$

the witness coefficients reduce to

$$\lambda_{\parallel} = \Lambda_{\parallel} I_{\theta} + \Xi_{\parallel} I_{\theta^3} + O(\theta_0^5 \kappa),$$

$$\alpha = \frac{1}{2} (\Lambda_{11} + \Lambda_{22}) I_{\theta} + \frac{1}{2} (\Xi_{11} + \Xi_{22}) I_{\theta^3} + O(\theta_0^5 \kappa),$$

$$\beta = \frac{1}{2} (\Lambda_{21} - \Lambda_{12}) I_{\theta} + \frac{1}{2} (\Xi_{21} - \Xi_{12}) I_{\theta^3} + O(\theta_0^5 \kappa).$$

For the explicit defect and low-mode profiles,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta^3} = -\frac{\sqrt{3}\pi}{32} \theta_0^3 \kappa.$$

Hence

$$\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{\parallel} - \frac{\sqrt{3}\pi}{32} \theta_0^3 \kappa \Xi_{\parallel} + O(\theta_0^5 \kappa),$$

$$\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (\Lambda_{11} + \Lambda_{22}) - \frac{\sqrt{3}\pi}{64} \theta_0^3 \kappa (\Xi_{11} + \Xi_{22}) + O(\theta_0^5 \kappa),$$

$$\beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (\Lambda_{21} - \Lambda_{12}) - \frac{\sqrt{3}\pi}{64} \theta_0^3 \kappa (\Xi_{21} - \Xi_{12}) + O(\theta_0^5 \kappa).$$

Therefore, at leading order, the full microscopic S2 witness problem has been reduced to the finite coefficient test

$$\boxed{\Lambda_{\parallel} \neq 0, \quad (\Lambda_{11} + \Lambda_{22}, \Lambda_{21} - \Lambda_{12}) \neq (0, 0).}$$

K. A minimal S2-compatible concrete ansatz for the microscopic projector transport operator

a. Goal. We propose a concrete microscopic transport operator compatible with all previously established local projector-defect structure and the S2 symmetry reduction, and extract the three Weyl/Dirac witness coefficients explicitly.

1. A. Structural constraints already fixed

The actual local CP^2 projector-defect analysis gives:

1. the transport term is first-order and odd:

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1238)$$

2. the defect gradient factorizes through

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger), \quad (1239)$$

hence the transport kernel must take the form

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (1240)$$

3. the local fiber $Q_B(\xi)$ is complex rank 2, so the traceless Hermitian transport channels form a three-dimensional space.

The shell/local symmetry reduction further shows that the natural symmetry-compatible Weyl form is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1241)$$

with nondegeneracy condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1242)$$

2. B. Canonical local fiber basis

Let the local orthonormal Q_B -frame be (e_1, e_2) . Define the canonical traceless Hermitian basis

$$\Sigma_1 := e_1 e_2^\dagger + e_2 e_1^\dagger, \quad (1243)$$

$$\Sigma_2 := -ie_1 e_2^\dagger + ie_2 e_1^\dagger, \quad (1244)$$

$$\Sigma_3 := e_1 e_1^\dagger - e_2 e_2^\dagger. \quad (1245)$$

In the local frame, these are exactly

$$\Sigma_i \leftrightarrow \sigma_i, \quad i = 1, 2, 3. \quad (1246)$$

3. C. Canonical worldvolume splitting

Let the worldvolume momentum directions split as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} n^\mu \oplus E_\perp, \quad (1247)$$

where n^μ is the distinguished longitudinal direction and $E_\perp$ is the transverse two-plane.

Choose an orthonormal transverse frame E^μ_a ($a = 1, 2$) on $E_\perp$, and define the transverse complex-structure rotated frame

$$J^\mu_a := \epsilon_{ab} E^\mu_b. \quad (1248)$$

Then the three canonical symmetry-compatible channels are:

1. longitudinal singlet:

$$n^\mu \Sigma_3,$$

2. transverse even doublet:

$$E^\mu{}_a \Sigma_a, \quad a = 1, 2,$$

3. transverse odd doublet:

$$J^\mu{}_a \Sigma_a, \quad a = 1, 2.$$

4. *D. Minimal concrete transport ansatz*

We now propose the minimal S2-compatible microscopic transport kernel:

$$\boxed{V^\mu(\xi) = \theta'(\xi) \left[\Lambda_\parallel n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right]}. \quad (1249)$$

Here:

- $\Lambda_\parallel \in \mathbb{R}$ is the longitudinal singlet coupling,
- $\Lambda_E \in \mathbb{R}$ is the transverse even coupling,
- $\Lambda_O \in \mathbb{R}$ is the transverse odd coupling.

This is the minimal concrete ansatz because:

1. it is linear in the actual local odd scalar profile $\theta'(\xi)$,
2. it uses the actual local rank-two fiber endomorphism basis,
3. it realizes exactly the S2 singlet $\oplus$ doublet structure and nothing more.

5. *E. Projection to the graded 2 + 2 carrier*

Let

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (1250)$$

Then the mixed projected matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (1251)$$

Substituting (1249),

$$M^\mu = I_\theta \left[\Lambda_\parallel n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right], \quad (1252)$$

where

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi). \quad (1253)$$

For the explicit defect and low-mode profiles already used,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1254)$$

6. *F. Reading off the S2 witnesses*

Choose worldvolume coordinates adapted to the splitting:

$$p = (p_1, p_2, p_{\parallel}).$$

Then the reduced mixed symbol is

$$\begin{aligned} A_1(p) &= p_{\mu} M^{\mu} \\ &= I_{\theta} \left[\Lambda_{\parallel} p_{\parallel} \sigma_3 + \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2) \right]. \end{aligned} \quad (1255)$$

Therefore the actual S2 witnesses are

$$\boxed{\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (1256)$$

Since $I_{\theta} \neq 0$ for nontrivial defect amplitude, the nondegeneracy condition is equivalent to

$$\boxed{\Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (1257)$$

7. *G. Determinant and Weyl/Dirac condition*

The coefficient matrix is

$$\Lambda = \begin{pmatrix} \Lambda_E & -\Lambda_O & 0 \\ \Lambda_O & \Lambda_E & 0 \\ 0 & 0 & \Lambda_{\parallel} \end{pmatrix}, \quad (1258)$$

hence

$$\det \Lambda = \Lambda_{\parallel} (\Lambda_E^2 + \Lambda_O^2). \quad (1259)$$

So the reduced symbol is nondegenerate iff

$$\Lambda_{\parallel} (\Lambda_E^2 + \Lambda_O^2) \neq 0. \quad (1260)$$

If the opposite valley or doubled graded partner satisfies the standard pairing relation, this yields the Dirac principal operator after doubling.

8. *H. Honest status*

What is proved here is not the final microscopic TECT PDE, but something sharper than an abstract theorem:

1. a concrete S2-compatible microscopic transport ansatz has been written,
2. its projected witnesses are explicitly computable,
3. the Dirac nondegeneracy problem is reduced to the finite coefficient test

$$\Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

What remains open is the derivation of the actual values of

$$\Lambda_{\parallel}, \Lambda_E, \Lambda_O$$

from the complete microscopic TECT projector transport equation.

L. Summary

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi))$$

must be odd and Q_B -endomorphism valued.

Using the canonical local Q_B -fiber basis

$$\Sigma_1, \Sigma_2, \Sigma_3 \cong \sigma_1, \sigma_2, \sigma_3,$$

and the canonical worldvolume singlet $\oplus$ doublet splitting

$$n^\mu, \quad E^\mu{}_a, \quad J^\mu{}_a = \epsilon_{ab} E^\mu{}_b,$$

the minimal S2-compatible concrete transport ansatz is

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_\parallel n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right].$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$ gives

$$M^\mu = I_\theta \left[\Lambda_\parallel n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right], \quad I_\theta = \int d\xi f_+ f_- \theta'(\xi).$$

Hence

$$A_1(p) = I_\theta \left[\Lambda_\parallel p_\parallel \sigma_3 + \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2) \right].$$

Therefore the actual S2 witnesses are

$$\lambda_\parallel = I_\theta \Lambda_\parallel, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Since

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0$$

for nontrivial defect amplitude, the nondegeneracy condition is equivalently

$$\Lambda_\parallel \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

The coefficient matrix is

$$\Lambda = \begin{pmatrix} \Lambda_E & -\Lambda_O & 0 \\ \Lambda_O & \Lambda_E & 0 \\ 0 & 0 & \Lambda_\parallel \end{pmatrix}, \quad \det \Lambda = \Lambda_\parallel (\Lambda_E^2 + \Lambda_O^2).$$

Thus the full microscopic S2 witness problem has been reduced to a three-parameter finite coefficient test.

M. A minimal microscopic transport operator generating the S2 witnesses

a. Goal. We propose a concrete microscopic transport operator consistent with the actual local projector-defect structure and the S2 symmetry reduction, and extract the three witness coefficients

$$\Lambda_\parallel, \quad \Lambda_E, \quad \Lambda_O$$

explicitly.

1. *A. Structural constraints from the actual local defect geometry*

The local CP^2 projector-defect Hessian already fixes the following:

1. the low-energy transport operator has the first-order form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad (1261)$$

2. for a reflection-symmetric defect background, the transport kernel is odd,

$$V^\mu(-\xi) = -V^\mu(\xi), \quad (1262)$$

3. the leading local odd building blocks are linear in

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

hence the strongest honest local form is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)), \quad (1263)$$

4. the local fiber $Q_B(\xi)$ is complex rank 2, so its traceless Hermitian endomorphism algebra is three-dimensional.

These are exactly the structural ingredients proved in the local projector-Hessian matching theorem.

2. *B. Canonical local basis on the fiber and worldvolume*

Let (e_1, e_2) be the local orthonormal frame of Q_B , and define the canonical traceless Hermitian basis

$$\Sigma_1 := e_1 e_2^\dagger + e_2 e_1^\dagger, \quad (1264)$$

$$\Sigma_2 := -ie_1 e_2^\dagger + ie_2 e_1^\dagger, \quad (1265)$$

$$\Sigma_3 := e_1 e_1^\dagger - e_2 e_2^\dagger. \quad (1266)$$

In the local frame,

$$\Sigma_i \leftrightarrow \sigma_i. \quad (1267)$$

On the worldvolume side, use the natural S2-adapted splitting

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} n^\mu \oplus E_\perp, \quad (1268)$$

where n^μ is the distinguished longitudinal direction and $E_\perp$ is the transverse two-plane. Choose an orthonormal transverse frame E^μ_a ($a = 1, 2$) and define the rotated frame

$$J^\mu_a := \epsilon_{ab} E^\mu_b. \quad (1269)$$

3. *C. Minimal S2-compatible transport ansatz*

The minimal S2-compatible transport kernel is then

$$\boxed{V^\mu(\xi) = \theta'(\xi) \left[\Lambda_\parallel n^\mu \Sigma_3 + \Lambda_E E^\mu_a \Sigma_a + \Lambda_O J^\mu_a \Sigma_a \right], \quad a = 1, 2.} \quad (1270)$$

This ansatz is minimal because it keeps exactly the three symmetry-allowed channels:

1. a longitudinal singlet $n^\mu \Sigma_3$,
2. a transverse even doublet $E^\mu_a \Sigma_a$,
3. a transverse odd doublet $J^\mu_a \Sigma_a$.

No additional first-order channel is required by the already established local geometry and S2 symmetry reduction.

4. D. Projection onto the graded low-energy carrier

Let the graded low modes be

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2. \quad (1271)$$

The projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (1272)$$

Substituting (1270),

$$M^\mu = I_\theta \left[\Lambda_\parallel n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right], \quad (1273)$$

where

$$I_\theta := \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi). \quad (1274)$$

For the explicit defect profile and low modes already used in the improved effective realization,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1275)$$

5. E. Reduced mixed symbol and direct identification of the witnesses

Using worldvolume momenta

$$p = (p_1, p_2, p_\parallel),$$

the reduced mixed symbol becomes

$$\begin{aligned} A_1(p) &= p_\mu M^\mu \\ &= I_\theta \left[\Lambda_\parallel p_\parallel \sigma_3 + \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2) \right]. \end{aligned} \quad (1276)$$

Comparing with the canonical S2 template

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad (1277)$$

we read off

$$\boxed{\lambda_\parallel = I_\theta \Lambda_\parallel, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (1278)$$

Since $I_\theta \neq 0$ for nontrivial defect amplitude, the S2 two-witness condition is equivalent to

$$\boxed{\Lambda_\parallel \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (1279)$$

6. F. Determinant and nondegeneracy

The effective transport matrix in the S2 basis is

$$\Lambda = \begin{pmatrix} \Lambda_E & -\Lambda_O & 0 \\ \Lambda_O & \Lambda_E & 0 \\ 0 & 0 & \Lambda_\parallel \end{pmatrix}, \quad (1280)$$

with determinant

$$\det \Lambda = \Lambda_\parallel (\Lambda_E^2 + \Lambda_O^2). \quad (1281)$$

Thus the reduced symbol is nondegenerate exactly when

$$\Lambda_\parallel (\Lambda_E^2 + \Lambda_O^2) \neq 0. \quad (1282)$$

7. *G. Interpretation and honest status*

This construction is not yet a full microscopic derivation from the final TECT PDE. Rather, it is the minimal concrete ansatz consistent with all currently established local structural facts:

1. oddness from the reflection-symmetric defect gradient,
2. rank-two local projector fiber Q_B ,
3. canonical three-dimensional traceless Hermitian fiber algebra,
4. natural S2 singlet $\oplus$ doublet symmetry reduction.

Therefore the remaining microscopic problem is no longer one of operator-class identification. It is the finite coefficient problem:

$$\boxed{\text{determine the actual values of } \Lambda_{\parallel}, \Lambda_E, \Lambda_O \text{ from the full TECT projector transport dynamics.}} \quad (1283)$$

N. Summary

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi))$$

must be odd and Q_B -endomorphism valued.

Using the canonical local fiber basis

$$\Sigma_1, \Sigma_2, \Sigma_3 \cong \sigma_1, \sigma_2, \sigma_3,$$

and the canonical worldvolume singlet $\oplus$ doublet splitting

$$n^\mu, \quad E^\mu{}_a, \quad J^\mu{}_a = \epsilon_{ab} E^\mu{}_b,$$

the minimal concrete S2-compatible transport kernel is

$$\boxed{V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right].}$$

Projecting onto the graded low modes $u_a^\pm = f_\pm e_a$ gives

$$M^\mu = I_\theta \left[\Lambda_{\parallel} n^\mu \sigma_3 + \Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right], \quad I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Hence

$$A_1(p) = I_\theta \left[\Lambda_{\parallel} p_{\parallel} \sigma_3 + \Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2) \right].$$

Therefore the actual S2 witnesses are

$$\boxed{\lambda_{\parallel} = I_\theta \Lambda_{\parallel}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.}$$

Since $I_\theta \neq 0$, nondegeneracy is equivalent to

$$\boxed{\Lambda_{\parallel} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).}$$

The coefficient matrix is

$$\Lambda = \begin{pmatrix} \Lambda_E & -\Lambda_O & 0 \\ \Lambda_O & \Lambda_E & 0 \\ 0 & 0 & \Lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \Lambda_{\parallel} (\Lambda_E^2 + \Lambda_O^2).$$

So the full microscopic S2 witness problem is reduced to a three-parameter finite coefficient test.

O. A more concrete microscopic transport ansatz using the primitive projector channel $T(P) = 3P - \mathbf{1}_3$

a. Goal. We refine the previous minimal S2-compatible transport ansatz by using the most canonical internal tensor induced by the CP^2 defect projector,

$$T(P) = 3P - \mathbf{1}_3,$$

and determine how much of the desired S2 structure it can actually generate on the rank-two bundle Q_B .

1. A. Microscopic origin of the transport term

In the local projector-defect linearization, the first-order transport contribution arises because derivatives act on the background projector. The microscopic linearized transport term is schematically

$$T(\delta P) = -C_{ij} \left[(\partial_i P_B) \partial_j \delta P + (\partial_j P_B) \partial_i \delta P \right]. \quad (1284)$$

Thus the transport kernel is built from two ingredients:

1. the background projector gradient $\partial_\xi P_B$,
2. an internal tensor channel acting on projector fluctuations.

:contentReference[oaicite:2]index=2

2. B. The primitive internal channel

A canonically distinguished internal Hermitian traceless tensor induced by a CP^2 projector is

$$T(P) = 3P - \mathbf{1}_3. \quad (1285)$$

Since P is a rank-one Hermitian projector, $T(P)$ has eigenvalues

$$2, -1, -1,$$

is Hermitian, traceless, and pointwise invertible, and distinguishes the projector line $\text{im } P$ from its orthogonal complement $\ker P$. :contentReference[oaicite:3]index=3

Therefore the most canonical concrete transport operator one can try is a defect-gradient coupling dressed by the primitive channel.

3. C. Explicit defect background and restriction to Q_B

Take the explicit local defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger. \quad (1286)$$

Let

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp \quad (1287)$$

with local orthonormal frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (1288)$$

These satisfy

$$P_B e_1 = 0, \quad P_B e_2 = 0, \quad (1289)$$

because $e_1, e_2 \in Q_B$.

Therefore, restricted to the Q_B fiber,

$$T(P_B)|_{Q_B} = (3P_B - \mathbf{1}_3)|_{Q_B} = -\mathbf{1}_{Q_B}. \quad (1290)$$

4. *D. Consequence: the primitive channel alone gives only the identity on Q_B*

Equation (1290) is decisive. It implies that the primitive channel $T(P_B)$ does *not* generate a traceless Pauli-triplet structure on the rank-two fiber Q_B . Instead, on Q_B it collapses to a scalar multiple of the identity.

Thus any transport ansatz of the form

$$V^\mu(\xi) = \theta'(\xi) \Lambda^\mu T(P_B)|_{Q_B} \quad (1291)$$

becomes

$$V^\mu(\xi) = -\theta'(\xi) \Lambda^\mu \mathbf{1}_2. \quad (1292)$$

After projection onto the graded $2 + 2$ carrier,

$$M^\mu = -I_\theta \Lambda^\mu \mathbf{1}_2, \quad (1293)$$

with

$$I_\theta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \neq 0. \quad (1294)$$

Hence the mixed symbol is

$$A_1(p) = -I_\theta p_\mu \Lambda^\mu \mathbf{1}_2. \quad (1295)$$

This is first-order and not a constant mass term, but it carries no traceless Pauli structure.

5. *E. Why this is insufficient for the full S2 branch*

The natural S2 branch requires the mixed symbol

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1296)$$

which needs one singlet traceless channel σ_3 and a transverse traceless doublet (σ_1, σ_2) .

But the primitive projector channel restricted to Q_B yields only

$$\mathbf{1}_2,$$

not $\sigma_1, \sigma_2, \sigma_3$.

Therefore:

$T(P) = 3P - \mathbf{1}_3$ alone cannot generate the full S2 Pauli-triplet transport structure on Q_B .

(1297)

In particular, for the minimal S2 parameterization

$$V^\mu(\xi) = \theta'(\xi) \left[\Lambda_{\parallel} n^\mu \Sigma_3 + \Lambda_E E^\mu{}_a \Sigma_a + \Lambda_O J^\mu{}_a \Sigma_a \right],$$

the primitive channel alone effectively gives

$$\Lambda_E = \Lambda_O = 0 \quad \text{generically}, \quad (1298)$$

because it contains no off-diagonal or traceless splitting inside Q_B .

6. *F. Minimal refinement needed*

To obtain the full S2 structure, one needs a more refined internal channel than $T(P)$ alone. The minimal possibilities are:

1. **frame-sensitive commutator channel:** an operator built from P_B and $\partial_\xi P_B$, such as

$$[P_B, \partial_\xi P_B],$$

which acts nontrivially on Q_B ;

2. **connection/gradient channel:** an endomorphism built from the induced Q_B -connection or its frame-covariant derivative, which can distinguish e_1 and e_2 ;
3. **direct shell-projector derivative coupling:** a UV term of the type already classified in the shell-enriched analysis,

$$L_{\text{mix}}^{(2)} \sim \Lambda_{ij}^a (\partial_i \phi_0) (D_j \psi) \xi^a,$$

whose internal label a directly spans the reduced Pauli sector.

:contentReference[oaicite:4]index=4

7. *G. Honest conclusion*

The primitive channel $T(P) = 3P - \mathbf{1}_3$ is canonical and extremely useful for topological and mirror-sector questions, but it is too coarse to generate the full S2 transport triplet on the rank-two tangent bundle Q_B .

So the honest conclusion is:

$$\boxed{\text{primitive channel alone} \Rightarrow \text{identity-valued first-order transport only;}} \quad (1299)$$

$$\boxed{\text{full S2 Weyl/Dirac transport requires an additional gradient-sensitive traceless } Q_B\text{-endomorphism channel.}} \quad (1300)$$

P. Summary

$$T(P) = 3P - \mathbf{1}_3$$

is the canonical primitive Hermitian traceless tensor induced by a rank-one CP^2 projector.

For the explicit defect background

$$P_B = z_B z_B^\dagger, \quad Q_B = (\mathbb{C} z_B)^\perp,$$

and the local Q_B -frame e_1, e_2 , one has

$$P_B e_1 = 0, \quad P_B e_2 = 0.$$

Therefore

$$T(P_B)|_{Q_B} = (3P_B - \mathbf{1}_3)|_{Q_B} = -\mathbf{1}_{Q_B}.$$

Hence a transport kernel built only from the primitive channel takes the form

$$V^\mu(\xi) = \theta'(\xi) \Lambda^\mu T(P_B)|_{Q_B} = -\theta'(\xi) \Lambda^\mu \mathbf{1}_2.$$

After projection,

$$M^\mu = -I_\theta \Lambda^\mu \mathbf{1}_2, \quad A_1(p) = -I_\theta p_\mu \Lambda^\mu \mathbf{1}_2.$$

This is first-order but carries no traceless Pauli structure. Therefore the primitive channel alone cannot generate the full S2 symbol

$$A_1(p) = \lambda_{||} p_{||} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

So:

$$T(P) = 3P - \mathbf{1}_3 \text{ alone is too coarse for the full S2 Weyl/Dirac transport triplet on } Q_B.$$

A full S2 realization requires an additional gradient-sensitive traceless Q_B -endomorphism channel, such as one built from

$$[P_B, \partial_\xi P_B],$$

the induced frame/connection structure, or a direct shell-projector derivative coupling carrying the reduced Pauli label.

Q. The commutator channel $\mathcal{T}_\xi = i[P_B, \partial_\xi P_B]$ and its induced action on the tangent bundle

a. Goal. We now analyze the first genuinely gradient-sensitive internal channel

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]$$

for the explicit CP^2 defect background, and determine what structure it induces on the rank-two tangent bundle Q_B .

1. A. Why this is the natural next channel

The primitive Hermitian traceless tensor

$$T(P) = 3P - \mathbf{1}_3$$

is canonical but too coarse on Q_B , since

$$T(P_B)|_{Q_B} = -\mathbf{1}_{Q_B}.$$

So it cannot produce the full traceless transport triplet on the tangent bundle. :contentReference[oaicite:0]index=0

On the other hand, the microscopic defect connection in the Callias construction is explicitly

$$A_i = [P, \partial_i P],$$

and for the affine Higgs field $\Phi = \mu(3P - \mathbf{1}_3)$ one has

$$D_i \Phi = 0.$$

Thus the commutator channel $[P, \partial_i P]$ is not ad hoc; it is already a canonically distinguished gradient-sensitive object in the microscopic projector geometry. :contentReference[oaicite:1]index=1

At the same time, the local projector-Hessian transport kernel is known to be built linearly from $\partial_\xi P_B$ and related gradient tensors. Hence

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B]$$

is the next minimal channel to test.

2. B. Explicit defect background and frame

Take the explicit reflection-symmetric defect representative

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger. \quad (1301)$$

Let the orthonormal local frame of

$$Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp$$

be

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (1302)$$

Then

$$\partial_\xi z_B(\xi) = \theta'(\xi) e_1(\xi), \quad (1303)$$

and therefore

$$\partial_\xi P_B = (\partial_\xi z_B) z_B^\dagger + z_B (\partial_\xi z_B)^\dagger = \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger). \quad (1304)$$

3. C. Exact computation of the commutator channel

Now compute

$$[P_B, \partial_\xi P_B] = z_B z_B^\dagger \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger) - \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger) z_B z_B^\dagger. \quad (1305)$$

Using

$$z_B^\dagger e_1 = 0, \quad e_1^\dagger z_B = 0, \quad z_B^\dagger z_B = 1,$$

we get

$$P_B \partial_\xi P_B = \theta'(\xi) z_B e_1^\dagger, \quad (\partial_\xi P_B) P_B = \theta'(\xi) e_1 z_B^\dagger. \quad (1306)$$

Hence

$$[P_B, \partial_\xi P_B] = \theta'(\xi) (z_B e_1^\dagger - e_1 z_B^\dagger). \quad (1307)$$

Therefore

$$\boxed{\mathcal{T}_\xi = i[P_B, \partial_\xi P_B] = i\theta'(\xi) (z_B e_1^\dagger - e_1 z_B^\dagger).} \quad (1308)$$

4. D. Geometric meaning

The operator $\mathcal{T}_\xi$ exchanges the projector line $\mathbb{C} z_B$ and the distinguished tangent direction $\mathbb{C} e_1$. Indeed,

$$\mathcal{T}_\xi z_B = i\theta'(\xi) (z_B e_1^\dagger z_B - e_1 z_B^\dagger z_B) = -i\theta'(\xi) e_1, \quad (1309)$$

while

$$\mathcal{T}_\xi e_1 = i\theta'(\xi) (z_B e_1^\dagger e_1 - e_1 z_B^\dagger e_1) = +i\theta'(\xi) z_B. \quad (1310)$$

Also,

$$\mathcal{T}_\xi e_2 = 0. \quad (1311)$$

Thus $\mathcal{T}_\xi$ is not an endomorphism of Q_B ; it is an off-block generator between $\mathbb{C} z_B$ and $\mathbb{C} e_1$, annihilating e_2 .

5. *E. Direct restriction to Q_B*

Let $\Pi_Q = \mathbf{1}_3 - P_B$ be the projector onto Q_B . Then

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0. \quad (1312)$$

Indeed, for any $v \in Q_B$, $\mathcal{T}_\xi v \in \mathbb{C}z_B$, so projecting back to Q_B gives zero.

Therefore the commutator channel itself is not yet a Q_B -endomorphism transport tensor.

6. *F. Induced quadratic channel on Q_B*

However, $\mathcal{T}_\xi^2$ is block-preserving. Using (1309)–(1311),

$$\mathcal{T}_\xi^2 z_B = \theta'(\xi)^2 z_B, \quad (1313)$$

$$\mathcal{T}_\xi^2 e_1 = \theta'(\xi)^2 e_1, \quad (1314)$$

$$\mathcal{T}_\xi^2 e_2 = 0. \quad (1315)$$

Hence, restricted to Q_B ,

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \theta'(\xi)^2 \Pi_{e_1}, \quad (1316)$$

where

$$\Pi_{e_1} = e_1 e_1^\dagger.$$

In the local Q_B -frame (e_1, e_2) ,

$$\Pi_{e_1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}(\mathbf{1}_2 + \sigma_3). \quad (1317)$$

Therefore

$$\boxed{\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}(\mathbf{1}_2 + \sigma_3).} \quad (1318)$$

7. *G. Consequence for transport structure*

This shows that the commutator channel is much stronger than the primitive channel $3P - \mathbf{1}_3$.

- The primitive channel restricted to Q_B gives only

$$-\mathbf{1}_2.$$

- The induced quadratic commutator channel gives

$$\frac{1}{2}(\mathbf{1}_2 + \sigma_3),$$

so it contains a genuine traceless splitting σ_3 on Q_B .

Hence the commutator channel naturally generates the longitudinal singlet part of the S2 structure.

8. *H. But not yet the full S2 triplet*

Equation (1318) still singles out only the e_1/e_2 splitting. So it naturally produces the σ_3 channel, but not the transverse doublet (σ_1, σ_2) .

Therefore:

$$\boxed{\mathcal{T}_\xi = i[P_B, \partial_\xi P_B] \text{ is sufficient to generate a gradient-sensitive } \sigma_3\text{-type channel,}} \quad (1319)$$

but

$$\boxed{\text{it is not by itself sufficient to produce the full } (\sigma_1, \sigma_2, \sigma_3) \text{ S2 triplet.}} \quad (1320)$$

To obtain the transverse doublet, one still needs an additional frame-sensitive or worldvolume-plane-sensitive channel, e.g.

1. a connection/frame-twist channel,
2. a worldvolume transverse complex-structure channel,
3. or a direct shell-projector derivative coupling carrying the reduced Pauli doublet label.

9. *I. Honest conclusion*

The commutator channel is the first genuinely successful refinement beyond the primitive $3P - \mathbf{1}_3$ channel:

$$3P - \mathbf{1}_3 \Rightarrow \text{identity only on } Q_B,$$

whereas

$$i[P_B, \partial_\xi P_B] \Rightarrow \text{after induced projection, } \mathbf{1}_2 + \sigma_3.$$

So the gradient-sensitive commutator channel captures the distinguished tangent splitting of the defect geometry and provides a natural microscopic source for the longitudinal witness $\lambda_\parallel$.

But a full Weyl/Dirac S2 symbol still requires one more ingredient for the transverse doublet.

R. Summary

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]$$

is the natural first gradient-sensitive internal channel beyond the primitive tensor $3P - \mathbf{1}_3$.

For the explicit $\mathbb{CP}^2$ defect background,

$$\partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger),$$

hence

$$\boxed{\mathcal{T}_\xi = i\theta'(\xi)(z_B e_1^\dagger - e_1 z_B^\dagger).}$$

Its action is

$$\mathcal{T}_\xi z_B = -i\theta'(\xi)e_1, \quad \mathcal{T}_\xi e_1 = +i\theta'(\xi)z_B, \quad \mathcal{T}_\xi e_2 = 0.$$

So $\mathcal{T}_\xi$ is an off-block operator between the projector line $\mathbb{C}z_B$ and the distinguished tangent direction $\mathbb{C}e_1$.

Direct restriction to Q_B gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.$$

But the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \theta'(\xi)^2 \Pi_{e_1} = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3).$$

Therefore the commutator channel, unlike the primitive channel, generates a genuine traceless splitting on Q_B , namely a natural σ_3 -type channel.

So:

$$i[P_B, \partial_\xi P_B] \text{ is a genuine gradient-sensitive microscopic source for the longitudinal singlet witness.}$$

However, it still does not generate the full transverse doublet (σ_1, σ_2) by itself. Hence:

the commutator channel improves the primitive channel, but a full S2 Weyl/Dirac triplet still needs one additional transverse channel.

S. Minimal composite transport operator generating the full S2 triplet

a. Goal. We construct the minimal composite transport operator that generates the full S2 triplet

$$(\sigma_1, \sigma_2, \sigma_3)$$

on the graded low-energy sector, combining the commutator-induced singlet channel with the transverse doublet source.

1. A. What the previous channels achieved

a. Primitive projector channel. The canonical primitive internal tensor

$$T(P) = 3P - \mathbf{1}_3 \tag{1321}$$

is Hermitian, traceless, and pointwise invertible on $\mathbb{C}^3$. However, on the tangent bundle

$$Q_B = (\mathbb{C}z_B)^\perp,$$

it reduces to

$$T(P_B)|_{Q_B} = -\mathbf{1}_{Q_B}, \tag{1322}$$

so it produces only an identity-valued first-order transport and no Pauli triplet.

b. Commutator channel. The canonical gradient-sensitive channel is

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]. \tag{1323}$$

This is natural because the Callias connection is exactly

$$A_i = [P, \partial_i P].$$

For the explicit defect background,

$$\mathcal{T}_\xi = i\theta'(\xi)(z_B e_1^\dagger - e_1 z_B^\dagger), \tag{1324}$$

so $\mathcal{T}_\xi$ is an off-block operator exchanging $\mathbb{C}z_B$ and $\mathbb{C}e_1$, while annihilating e_2 .

Direct restriction gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0, \tag{1325}$$

but the induced quadratic channel gives

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \theta'(\xi)^2 \Pi_{e_1} = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3). \tag{1326}$$

Thus the commutator channel naturally produces a σ_3 -type splitting, but not the transverse Pauli doublet (σ_1, σ_2) .

2. B. Need for a transverse doublet source

The symmetry-compatible S2 reduced symbol has the exact form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1327)$$

with nondegeneracy criterion

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1328)$$

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Hence a full microscopic transport operator must supply:

1. a longitudinal singlet channel σ_3 ,
2. a transverse even channel $(p_1 \sigma_1 + p_2 \sigma_2)$,
3. a transverse odd channel $(-p_2 \sigma_1 + p_1 \sigma_2)$.

The commutator-induced channel provides the first one naturally. The remaining two require an explicit transverse doublet source.

3. C. Canonical transverse doublet source

Let n^μ denote the distinguished longitudinal worldvolume direction, and let E^μ_a ($a = 1, 2$) be an orthonormal frame of the transverse two-plane. Define the rotated transverse frame

$$J^\mu_a := \epsilon_{ab} E^\mu_b. \quad (1329)$$

On the fiber, use the canonical traceless basis

$$\Sigma_1, \Sigma_2, \Sigma_3 \cong \sigma_1, \sigma_2, \sigma_3. \quad (1330)$$

The minimal transverse doublet source is then

$$V^\mu_{\perp}(\xi) = \theta'(\xi) \left[\Lambda_E E^\mu_a \Sigma_a + \Lambda_O J^\mu_a \Sigma_a \right], \quad a = 1, 2. \quad (1331)$$

This is the minimal S2-compatible first-order transverse doublet source.

4. D. Why the singlet must carry an odd prefactor

A naive singlet term from $\Pi_Q \mathcal{T}_\xi^2 \Pi_Q$ would be

$$V^\mu_{\parallel, \text{naive}}(\xi) = \lambda_c \theta'(\xi)^2 n^\mu \Sigma_3. \quad (1332)$$

But after projection onto the graded low modes, the overlap integral would be proportional to

$$\int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^2. \quad (1333)$$

Now:

$$f_+ \text{ is even, } \quad f_- \text{ is odd, } \quad \theta'^2 \text{ is even,}$$

so the integrand is odd. Therefore

$$\int d\xi f_+ f_- \theta'^2 = 0. \quad (1334)$$

Hence the naive quadratic commutator singlet does *not* contribute to the first-order mixed symbol.

Therefore a viable longitudinal singlet source must include one additional odd factor. The simplest choice is

$$V_{\parallel}^{\mu}(\xi) = \lambda_c \theta'(\xi) \Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q n^{\mu}. \quad (1335)$$

Using (1326),

$$V_{\parallel}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} (\mathbf{1}_2 + \sigma_3). \quad (1336)$$

Its traceless part is

$$V_{\parallel, \text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (1337)$$

Now $f_+ f_- \theta'^3$ is even, so the overlap need not vanish.

5. E. Minimal composite S2 transport operator

Combining the longitudinal commutator-induced singlet with the transverse doublet source, the minimal composite transport operator is

$$V_{\text{comp}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3 + \theta'(\xi) \left[\Lambda_E E^{\mu}_a \sigma_a + \Lambda_O J^{\mu}_a \sigma_a \right], \quad a = 1, 2. \quad (1338)$$

This is the minimal first-order transport operator that uses:

1. the actual commutator geometry for the longitudinal singlet,
2. the canonical transverse doublet structure for the Pauli doublet.

6. F. Projection to the graded low-energy carrier

Project onto the graded low modes

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (1339)$$

The mixed matrices are

$$(M^{\mu})_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^{\dagger}(\xi) V_{\text{comp}}^{\mu}(\xi) e_b(\xi). \quad (1340)$$

Define the overlap integrals

$$I_{\theta} := \int d\xi f_+ f_- \theta'(\xi), \quad (1341)$$

$$I_{\theta^3} := \int d\xi f_+ f_- \theta'(\xi)^3. \quad (1342)$$

Then

$$M^{\mu} = \frac{\lambda_c}{2} I_{\theta^3} n^{\mu} \sigma_3 + I_{\theta} \left[\Lambda_E E^{\mu}_a \sigma_a + \Lambda_O J^{\mu}_a \sigma_a \right]. \quad (1343)$$

Therefore the reduced mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2), \quad (1344)$$

with

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (1345)$$

7. *G. Explicit evaluation of I_{θ^3}*

Using the explicit profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}}(\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi)(\kappa\xi),$$

and

$$\theta'(\xi) = -\theta_0 \kappa (\kappa\xi) \tanh(\kappa\xi),$$

we compute

$$\begin{aligned} I_{\theta^3} &= \int d\xi f_+ f_- \theta'^3 \\ &= -\frac{\sqrt{3} \theta_0^3 \kappa^4}{2} \int_{-\infty}^{\infty} d\xi \tanh^4(\kappa\xi)^5 (\kappa\xi). \end{aligned} \quad (1346)$$

Set

$$t = \tanh(\kappa\xi), \quad dt = \kappa^2 (\kappa\xi) d\xi, \quad (\kappa\xi) = \sqrt{1-t^2}.$$

Then

$$I_{\theta^3} = -\frac{\sqrt{3} \theta_0^3 \kappa^3}{2} \int_{-1}^1 t^4 (1-t^2)^{3/2} dt. \quad (1347)$$

The integral is strictly positive, so

$$I_{\theta^3} \neq 0 \quad (\theta_0 \kappa \neq 0). \quad (1348)$$

Thus the longitudinal witness produced by the commutator-induced channel is generically nonzero whenever $\lambda_c \neq 0$.

8. *H. Final nondegeneracy condition*

Since

$$I_\theta \neq 0, \quad I_{\theta^3} \neq 0$$

for nontrivial defect amplitude, the S2 two-witness condition becomes

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0)}. \quad (1349)$$

So the full S2 triplet is generated by:

1. the commutator-induced channel for the longitudinal singlet,
2. the transverse doublet source for the Pauli doublet.

a. Honest status. This is still a concrete microscopic ansatz, not the final derived TECT PDE. But it is stronger than the earlier abstract reduction because it identifies a specific minimal composite operator that produces all three S2 witness coefficients explicitly.

T. Summary

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B]$$

is the first genuinely gradient-sensitive internal channel beyond the primitive tensor $3P - \mathbf{1}_3$.

Direct restriction gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3).$$

A naive singlet source proportional to $\theta'^2 \sigma_3$ fails at first order because

$$\int d\xi f_+ f_- \theta'^2 = 0$$

by parity.

Therefore the minimal viable longitudinal singlet source is

$$\frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3.$$

Combining this with the transverse doublet source, the minimal composite S2 transport operator is

$$V_{\text{comp}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right].$$

Projecting onto the graded low modes gives

$$M^\mu = \frac{\lambda_c}{2} I_{\theta'^3} n^\mu \sigma_3 + I_\theta \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right],$$

hence

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

Since

$$I_\theta \neq 0, \quad I_{\theta'^3} \neq 0$$

for nontrivial defect amplitude, the final S2 nondegeneracy condition reduces to

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Thus the minimal full S2 triplet is generated by:

longitudinal singlet from the commutator-induced channel,

transverse doublet from the worldvolume-plane traceless source.

U. A concrete minimal microscopic action term generating the three S2 witnesses

a. Goal. We now choose one explicit microscopic transport term, compatible with the actual local projector-defect geometry and the S2 reduction, and compute the three witness coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

as explicit constants.

1. A. Structural inputs already fixed

The local CP^2 projector-defect analysis establishes:

1. the leading transport is first-order and odd,

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi), \quad (1350)$$

with the transport kernel built linearly from the background projector gradient, :contentReference[oaicite:3]index=3

2. the canonical gradient-sensitive internal channel is

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad (1351)$$

which is natural because the projector connection itself is

$$A_i = [P, \partial_i P], \quad (1352)$$

in the microscopic Callias/projector geometry, :contentReference[oaicite:4]index=4

3. the reduced symmetry-compatible S2 symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (1353)$$

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2. B. Canonical local basis

Let (e_1, e_2) be the local orthonormal frame of

$$Q_B = (\mathbb{C}z_B)^\perp.$$

Define the canonical traceless Hermitian basis

$$\Sigma_1 := e_1 e_2^\dagger + e_2 e_1^\dagger, \quad (1354)$$

$$\Sigma_2 := -ie_1 e_2^\dagger + ie_2 e_1^\dagger, \quad (1355)$$

$$\Sigma_3 := e_1 e_1^\dagger - e_2 e_2^\dagger. \quad (1356)$$

In the local frame these are exactly $\sigma_1, \sigma_2, \sigma_3$.

On the worldvolume side, choose the natural S2-adapted basis:

$$n^\mu \quad (\text{longitudinal singlet}), \quad E^\mu{}_a \quad (a = 1, 2) \quad (\text{transverse frame}), \quad J^\mu{}_a := \epsilon_{ab} E^\mu{}_b. \quad (1357)$$

3. C. Commutator-induced singlet

From the explicit defect computation,

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3). \quad (1358)$$

Its traceless part is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^2}{2} \sigma_3. \quad (1359)$$

Therefore

$$\theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^3}{2} \sigma_3 \quad (1360)$$

is the minimal odd longitudinal singlet source.

4. D. Minimal concrete transport action

We now propose the concrete microscopic transport term

$$\boxed{\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta.} \quad (1361)$$

Using (1360), this becomes

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[\frac{g_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + g_E \theta'(\xi) E^\mu{}_a \sigma_a + g_O \theta'(\xi) J^\mu{}_a \sigma_a \right] \partial_{y^\mu} \eta. \quad (1362)$$

This is the minimal concrete S2-compatible transport action because:

1. the longitudinal singlet comes from the actual commutator geometry,
2. the transverse doublet comes from the minimal worldvolume-plane traceless source,
3. no additional first-order channel is included.

5. E. Projection onto the graded low modes

Let

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2. \quad (1363)$$

The projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (1364)$$

Define

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi), \quad (1365)$$

$$I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (1366)$$

Then

$$M^\mu = \frac{g_c}{2} I_{\theta'^3} n^\mu \sigma_3 + I_\theta \left[g_E E^\mu{}_a \sigma_a + g_O J^\mu{}_a \sigma_a \right]. \quad (1367)$$

Hence the reduced mixed symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1368)$$

with

$$\boxed{\lambda_\parallel = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta.} \quad (1369)$$

6. F. Explicit evaluation of the overlap integrals

For the explicit defect profile

$$\theta(\xi) = \theta_0(\kappa\xi), \quad (1370)$$

and the explicit low modes

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} (\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi)(\kappa\xi), \quad (1371)$$

one already has

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1372)$$

Now compute $I_{\theta'^3}$. Using

$$\theta'(\xi) = -\theta_0 \kappa (\kappa\xi) \tanh(\kappa\xi), \quad (1373)$$

and

$$f_+(\xi)f_-(\xi) = \frac{\sqrt{3}\kappa}{2} \tanh(\kappa\xi)^2(\kappa\xi), \quad (1374)$$

we obtain

$$\begin{aligned} I_{\theta'^3} &= \int d\xi f_+(\xi)f_-(\xi)\theta'(\xi)^3 \\ &= -\frac{\sqrt{3}\theta_0^3\kappa^4}{2} \int_{-\infty}^{\infty} d\xi \tanh^4(\kappa\xi)^5(\kappa\xi). \end{aligned} \quad (1375)$$

Set

$$t = \tanh(\kappa\xi), \quad dt = \kappa^2(\kappa\xi) d\xi, \quad (\kappa\xi) = \sqrt{1-t^2}.$$

Then

$$I_{\theta'^3} = -\frac{\sqrt{3}\theta_0^3\kappa^3}{2} \int_{-1}^1 t^4(1-t^2)^{3/2} dt. \quad (1376)$$

Using

$$\int_{-1}^1 t^4(1-t^2)^{3/2} dt = \frac{3\pi}{128}, \quad (1377)$$

we obtain

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \quad (1378)$$

7. *G. Explicit witness coefficients*

Substituting (1372) and (1378) into (1369),

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,} \quad (1379)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.} \quad (1380)$$

Thus the S2 nondegeneracy condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

reduces exactly to

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).} \quad (1381)$$

8. H. Honest conclusion

This is still a concrete microscopic ansatz, not the final derived complete TECT PDE. But it is now a very sharp one: a single explicit transport action term has been chosen, and the three S2 witnesses have been computed all the way down to explicit numerical profile factors.

Therefore the remaining open microscopic problem is no longer structural. It is simply whether complete TECT dynamics actually generates nonzero

$$g_c, \quad g_E, \quad g_O.$$

V. Summary

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta$$

is a concrete minimal microscopic transport action compatible with the local projector-defect geometry and the S2 symmetry reduction.

Using

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^2}{2} \sigma_3,$$

this becomes

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[\frac{g_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + g_E \theta'(\xi) E^\mu{}_a \sigma_a + g_O \theta'(\xi) J^\mu{}_a \sigma_a \right] \partial_{y^\mu} \eta.$$

After projection onto the graded low modes,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta,$$

with

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Therefore

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Hence the full S2 nondegeneracy condition reduces to the simple finite test

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

So the remaining microscopic question is just whether complete TECT dynamics actually generates nonzero g_c, g_E, g_O .

II. TECT DIRAC-EMERGENCE MASTER CONTEXT (LATEST SNAPSHOT WITH CONCRETE TRANSPORT ANSATZ)

A. 0. Working rules for the project

- Conversation language: Korean.
- Mathematical derivations: English L^AT_EX.
- Focus: rigorous derivation, honest separation of what is proved / reduced / still open.

B. 1. Global research target

The current target is:

derive an emergent massless Dirac operator from the actual TECT defect/projector sector.

The program now has two layers:

1. a *reduced spectral theorem layer*, already essentially closed;
2. a *microscopic coefficient extraction layer*, still open.

C. 2. What is already closed

1. 2.1 Plain scalar shell-pair route fails

The scalar shell-pair channel yields a genuine constant Bragg block

$$H_{\text{Bragg}}^{\text{scalar}} = B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2, \quad B_0^{\text{tot}} \neq 0 \text{ generically.} \quad (1382)$$

This is a true constant pair-mixing obstruction and cannot be removed by a simple scalar shift. Hence the plain scalar shell-pair route does *not* yield a massless Dirac point.

2. 2.2 Refined projector-stiffness model alone fails

For the current refined $U(3)$ -invariant triplet plus projector-stiffness model around a locked constant-projector background,

$$J = 0. \quad (1383)$$

Therefore the projector-stiffness sector alone does not induce the needed shell-projector mixing at quadratic order.

3. 2.3 Class II gives derivative witnesses but not identity-channel cancellation

With the auxiliary Class II mediator included, one obtains a genuine shell-enriched correction

$$\delta N_{ij}^a \neq 0. \quad (1384)$$

Thus the derivative witness sector is renormalized. However, this correction lives in the traceless internal channel and therefore does not directly cancel the identity-channel scalar Bragg obstruction

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

4. 2.4 The natural remaining route is the graded $2+2$ bypass

The reduced theorem is:

if there exists an isolated four-dimensional infrared sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2, \quad (1385)$$

and the projected leading symbol is grading-odd, first-order, and nondegenerate, then the reduced principal operator is Dirac.

Equivalently,

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_\mu M^\mu, \quad (1386)$$

and nondegeneracy of $A_1(p)$ for generic $p \neq 0$ implies Dirac emergence.

D. 3. Actual local $\mathbb{CP}^2$ projector-defect geometry

1. 3.1 Rank-two tangent bundle

For a defect background

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1, \quad (1387)$$

the physical tangent fluctuation lives in

$$Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp. \quad (1388)$$

Thus the defect fluctuation problem is already a rank-two bundle problem, not a scalar one.

2. 3.2 Explicit reflection-symmetric representative

A useful explicit local representative is

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(\xi) = \theta_0(\kappa\xi), \quad \theta(-\xi) = \theta(\xi). \quad (1389)$$

Choose the local Q_B -frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (1390)$$

Then

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger). \quad (1391)$$

Hence every leading local transport kernel linear in the defect gradient must factor through $\theta'(\xi)$.

3. 3.3 Local transport kernel

The actual local first-order transport operator has the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi). \quad (1392)$$

The strongest honest local form is therefore

$$V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)), \quad (1393)$$

with

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i. \quad (1394)$$

4. 3.4 Even/odd low modes and projected overlap

Let

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \quad u_a^-(\xi) = f_-(\xi)e_a(\xi), \quad a = 1, 2, \quad (1395)$$

with f_+ even and f_- odd. Then the projected mixed matrices are

$$(M^\mu)_{ab} = \int d\xi f_+(\xi)f_-(\xi) e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi). \quad (1396)$$

Same-parity overlaps vanish, while the mixed even/odd overlap survives.

5. 3.5 Explicit nonzero overlap scale

For the explicit profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} (\kappa\xi), \quad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi)(\kappa\xi), \quad (1397)$$

one finds

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (1398)$$

Hence

$$I_\theta \neq 0 \quad (\theta_0 \kappa \neq 0). \quad (1399)$$

E. 4. Residual symmetry and the natural local branch

1. 4.1 Traceless transport splitting

Because $\partial_\xi z_B = \theta' e_1$ singles out e_1 , the traceless Hermitian transport sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R} \sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}. \quad (1400)$$

2. 4.2 Worldvolume splitting

The local defect/worldvolume momentum directions split naturally as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R} p_\parallel \oplus \text{span}\{p_1, p_2\}. \quad (1401)$$

3. 4.3 Natural S2 branch

Therefore the natural local symmetry-compatible branch is S2:

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (1402)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix}, \quad \det \Lambda = \lambda_\parallel (\alpha^2 + \beta^2). \quad (1403)$$

Hence the S2 two-witness condition is

$$\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (1404)$$

S1 would require keeping only $p_\parallel \sigma_3$, which is not forced by the local geometry and instead needs an extra symmetry, truncation, or fine tuning.

F. 5. Primitive and commutator channels

1. 5.1 Primitive channel is too coarse

The canonical primitive traceless tensor is

$$T(P) = 3P - \mathbf{1}_3. \quad (1405)$$

But on the tangent bundle Q_B ,

$$T(P_B)|_{Q_B} = -\mathbf{1}_{Q_B}. \quad (1406)$$

So the primitive channel gives only identity-valued transport on Q_B , not a full Pauli triplet.

2. 5.2 Commutator channel is the first genuine gradient-sensitive refinement

Define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]. \quad (1407)$$

Using the explicit background,

$$\mathcal{T}_\xi = i\theta'(\xi)(z_B e_1^\dagger - e_1 z_B^\dagger). \quad (1408)$$

This exchanges the projector line $\mathbb{C}z_B$ and the distinguished tangent line $\mathbb{C}e_1$, and annihilates e_2 .

Direct restriction gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0, \quad (1409)$$

but the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3). \quad (1410)$$

Thus the commutator channel naturally produces a σ_3 -type splitting, but not the full $(\sigma_1, \sigma_2, \sigma_3)$ triplet.

G. 6. Minimal composite transport operator producing the full S2 triplet

1. 6.1 The concrete transport action

The minimal composite microscopic transport action is

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta. \quad (1411)$$

Using

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^2}{2} \sigma_3, \quad (1412)$$

this becomes

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[\frac{g_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + g_E \theta'(\xi) E^\mu{}_a \sigma_a + g_O \theta'(\xi) J^\mu{}_a \sigma_a \right] \partial_{y^\mu} \eta. \quad (1413)$$

2. 6.2 Projection and witness coefficients

Projecting onto the graded low modes gives

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta. \quad (1414)$$

Here

$$I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (1415)$$

For the explicit defect profile one finds

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (1416)$$

Therefore the explicit witnesses are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (1417)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (1418)$$

Hence the full S2 nondegeneracy condition is exactly

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0)}. \quad (1419)$$

H. 7. What is actually proved vs. what remains open

1. 7.1 Proved / reduced

The following are now sharply established:

1. plain scalar Bragg route is insufficient;
2. refined projector-stiffness model alone is insufficient;
3. Class II gives derivative witnesses but not identity-channel cancellation;
4. graded $2 + 2$ reduced Dirac theorem is rigorous;
5. the actual local CP^2 projector-defect geometry belongs to this operator class;
6. the natural local symmetry-compatible branch is S2;
7. a concrete microscopic transport action producing the three S2 witnesses has been written;
8. the final local witness problem is reduced to three constants g_c, g_E, g_O .

2. 7.2 Still open

The genuine remaining frontier is:

1. derive the final full microscopic TECT projector/condensate PDE in a form suitable for direct spectral analysis;
2. derive the above transport term from complete TECT rather than postulate it as the minimal concrete ansatz;
3. compute the actual microscopic values of

$$g_c, \quad g_E, \quad g_O;$$

4. verify that in complete TECT one indeed has

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

I. 8. Strongest honest conclusion

The strongest justified current statement is:

$$\boxed{\text{Dirac emergence is rigorously proved as a reduced spectral theorem,}} \quad (1420)$$

$$\boxed{\text{the actual local } \text{CP}^2 \text{ projector-defect geometry is naturally compatible with the S2 branch,}} \quad (1421)$$

$$\boxed{\text{and the full local witness problem is reduced to the finite coefficient test } g_c \neq 0, (g_E, g_O) \neq (0, 0),} \quad (1422)$$

but

$$\boxed{\text{the actual values of } g_c, g_E, g_O \text{ from complete microscopic TECT remain open.}} \quad (1423)$$

J. Summary

Closed:

1. The plain scalar shell-pair route is insufficient because it yields a genuine constant Bragg block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2 \neq 0.$$

2. The refined projector-stiffness model alone gives

$$J = 0,$$

so it does not generate the needed shell/projector mixing.

3. Class II yields

$$\delta N_{ij}^a \neq 0,$$

hence renormalizes the derivative witness sector, but it does not cancel the scalar Bragg mass.

4. The graded $2 + 2$ reduced theorem is rigorous: if the projected infrared sector is grading-odd, first-order, and has nondegenerate mixed symbol, then the principal operator is Dirac.

5. The actual local CP^2 projector-defect geometry belongs to this operator class:

$$Q_B = (\mathbb{C}z_B)^\perp, \quad L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).$$

6. The natural local branch is S2:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

7. The primitive projector channel is too coarse on Q_B , while the commutator channel

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B]$$

naturally provides the longitudinal σ_3 -type splitting.

8. A concrete minimal microscopic transport action is

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta.$$

9. Its projected witnesses are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Therefore the full local S2 nondegeneracy condition is exactly

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).}$$

Still open:

1. the final full microscopic TECT projector/condensate PDE suitable for direct spectral analysis;
2. the derivation of the concrete transport term above from complete TECT rather than choosing it as the minimal local ansatz;
3. the actual microscopic values of

$$g_c, \quad g_E, \quad g_O;$$

4. the direct proof that complete TECT indeed gives

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Thus the strongest honest current conclusion is

$$\boxed{\text{Dirac emergence is rigorously proved at the reduced spectral level,}}$$

$$\boxed{\text{the local } \text{CP}^2 \text{ defect geometry supports the required S2 structure,}}$$

but

$$\boxed{\text{the complete microscopic closure of TECT is now reduced to the finite coefficient extraction } (g_c, g_E, g_O).}$$

K. Finite witness closure for the concrete S2 transport ansatz

a. Goal. We now show that the remaining microscopic coefficient problem for

$$g_c, \quad g_E, \quad g_O$$

is not an infinite-dimensional PDE problem anymore, but a finite witness-construction problem on the symmetry-allowed microscopic tensor space.

1. A. Concrete S2 transport ansatz and the reduced witnesses

From the previously constructed concrete microscopic transport action,

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu_a \Sigma_a + g_O \theta'(\xi) J^\mu_a \Sigma_a \right] \partial_{y^\mu} \eta, \quad (1424)$$

one obtains the reduced mixed symbol

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1425)$$

with explicit coefficients

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (1426)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (1427)$$

Hence the S2 nondegeneracy condition

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

is equivalent to

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (1428)$$

2. B. Finite-dimensional tensor-space viewpoint

Let V_{mic} denote the finite-dimensional real vector space of all symmetry-allowed microscopic transport tensors that survive the linearization of the actual TECT condensate around the chosen defect background. This is the exact structural viewpoint already adopted in the shell-level Weyl/Dirac analysis: the remaining microscopic data live in a finite-dimensional allowed tensor space, not in an unconstrained infinite-dimensional function space. :contentReference[oaicite:3]index=3 :contentReference[oaicite:4]index=4

Inside V_{mic} , define the three witness functionals corresponding to the concrete S2 ansatz:

$$W_{\parallel}(N) := g_c(N), \quad (1429)$$

$$W_E(N) := g_E(N), \quad W_O(N) := g_O(N). \quad (1430)$$

Equivalently, after projection to the reduced symbol,

$$\lambda_{\parallel}(N) = c_{\parallel} W_{\parallel}(N), \quad \alpha(N) = c_{\perp} W_E(N), \quad \beta(N) = c_{\perp} W_O(N), \quad (1431)$$

where

$$c_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \theta_0^3 \kappa^3, \quad c_{\perp} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

For nontrivial defect amplitude, both constants are nonzero.

Thus the reduced Dirac witness problem is finite-dimensional from the start.

3. C. Longitudinal and transverse witness functionals

Define the longitudinal witness functional

$$\mathfrak{W}_{\parallel}(N) := \lambda_{\parallel}(N). \quad (1432)$$

This is linear in the microscopic tensor N , because the projected mixed overlap is linear in the transport kernel V^{μ} , and V^{μ} is linear in the microscopic transport tensor by construction. This is precisely the same linear-functional structure emphasized in the finite-dimensional shell genericity theorem. :contentReference[oaicite:5]index=5

For the transverse sector, define the 2×2 reduced transverse matrix

$$T_{\perp}(N) := \begin{pmatrix} \alpha(N) & -\beta(N) \\ \beta(N) & \alpha(N) \end{pmatrix}. \quad (1433)$$

Then

$$\det T_{\perp}(N) = \alpha(N)^2 + \beta(N)^2. \quad (1434)$$

This is a polynomial functional on V_{mic} .

Therefore the full S2 witness condition is

$$\mathfrak{W}_{\parallel}(N) \neq 0, \quad \det T_{\perp}(N) \neq 0. \quad (1435)$$

This is exactly parallel to the shell-level Weyl criterion

$$v_{\parallel} \neq 0, \quad \det T \neq 0,$$

which the manuscript already proved is the correct finite witness reduction. :contentReference[oaicite:6]index=6

4. D. Exact dichotomy: symmetry-forbidden vs generic nonvanishing

The finite-dimensional genericity theorem says:

1. if a linear witness functional vanishes identically on the allowed space, then the channel is symmetry-forbidden;
2. if it is nonzero on even a single admissible tensor, then it is nonzero on an open dense full-measure subset.

:contentReference[oaicite:7]index=7

Applying this to the present concrete S2 witnesses gives the exact dichotomy:

a. Longitudinal channel. Either

$$\mathfrak{W}_{\parallel}|_{V_{\text{mic}}} \equiv 0, \quad (1436)$$

in which case the commutator-induced singlet channel is symmetry-forbidden in the actual microscopic TECT linearization,

or else there exists a single admissible seed tensor $N_{\text{seed}}^{(\parallel)} \in V_{\text{mic}}$ such that

$$\mathfrak{W}_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0, \quad (1437)$$

and then

$$\mathfrak{W}_{\parallel}(N) \neq 0 \quad \text{for an open dense full-measure subset of } V_{\text{mic}}. \quad (1438)$$

b. Transverse sector. Either

$$\det T_{\perp}(N) \equiv 0 \quad \forall N \in V_{\text{mic}}, \quad (1439)$$

in which case the transverse doublet channel is symmetry-forbidden or projected out exactly,

or else there exists a single admissible seed tensor $N_{\text{seed}}^{(\perp)} \in V_{\text{mic}}$ such that

$$\det T_{\perp}(N_{\text{seed}}^{(\perp)}) \neq 0, \quad (1440)$$

and then

$$\det T_{\perp}(N) \neq 0 \quad \text{for an open dense full-measure subset of } V_{\text{mic}}. \quad (1441)$$

This is the direct S2 analogue of the shell-level witness theorem

$$W_{\parallel} \neq 0, \quad (W_I, W_J) \neq (0, 0),$$

or equivalently

$$v_{\parallel} \neq 0, \quad \det T \neq 0,$$

already proved in the uploaded manuscripts.

5. E. Finite witness closure theorem for the concrete transport ansatz

We can now state the concrete closure result.

[Finite witness closure for the concrete S2 transport ansatz] Let V_{mic} be the finite-dimensional space of symmetry-allowed microscopic transport tensors in the actual TECT linearization around the chosen CP^2 defect background. Let

$$\mathfrak{W}_{\parallel}(N) = \lambda_{\parallel}(N)$$

be the longitudinal witness functional induced by the commutator-based singlet channel, and let

$$T_{\perp}(N) = \begin{pmatrix} \alpha(N) & -\beta(N) \\ \beta(N) & \alpha(N) \end{pmatrix}$$

be the transverse reduced matrix induced by the doublet channels.

Assume there exist:

1. one admissible longitudinal seed tensor $N_{\text{seed}}^{(\parallel)} \in V_{\text{mic}}$ with

$$\mathfrak{W}_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0;$$

2. one admissible transverse seed tensor $N_{\text{seed}}^{(\perp)} \in V_{\text{mic}}$ with

$$\det T_{\perp}(N_{\text{seed}}^{(\perp)}) \neq 0.$$

Then, for an open dense full-measure subset of V_{mic} ,

$$g_c(N) \neq 0, \quad (g_E(N), g_O(N)) \neq (0, 0), \quad (1442)$$

equivalently,

$$\lambda_{\parallel}(N) \neq 0, \quad (\alpha(N), \beta(N)) \neq (0, 0), \quad (1443)$$

and the reduced S2 symbol is nondegenerate.

The longitudinal witness is linear on the finite-dimensional space V_{mic} , so if it is nonzero on one seed tensor, it is not identically zero; hence its zero set is a proper hyperplane and its complement is open dense full measure. :contentReference[oaicite:9]index=9

The transverse determinant is a polynomial functional on V_{mic} . If it is nonzero at one point, it is not identically zero; therefore its zero set is a proper algebraic subset with empty interior and measure zero. This is the same finite-dimensional logic used in the shell-level Weyl determinant theorem. :contentReference[oaicite:10]index=10

Intersecting the two open dense full-measure good sets proves the claim.

6. F. What remains genuinely open

The remaining microscopic task is therefore no longer:

“solve the full PDE and compute everything exactly”.

Instead, it is the finite explicit witness problem:

construct one admissible longitudinal seed tensor and one admissible transverse doublet seed tensor in the actual TECT microspace

(1444)

Once this is done, the concrete transport ansatz closes generically at the witness level.

a. Honest status. What is still not proved is that the complete microscopic TECT dynamics actually generates such seeds. What *is* now proved is that *if* those two seeds exist, then the nonvanishing of g_c and (g_E, g_O) is generic and the concrete S2 transport ansatz is enough to produce the required local Weyl/Dirac witnesses.

L. Summary

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Thus the concrete S2 witness condition is

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Let V_{mic} be the finite-dimensional space of symmetry-allowed microscopic transport tensors in the actual TECT linearization. Define the longitudinal witness

$$\mathfrak{W}_{\parallel}(N) := \lambda_{\parallel}(N),$$

and the transverse reduced matrix

$$T_{\perp}(N) = \begin{pmatrix} \alpha(N) & -\beta(N) \\ \beta(N) & \alpha(N) \end{pmatrix}, \quad \det T_{\perp}(N) = \alpha(N)^2 + \beta(N)^2.$$

Then the remaining microscopic problem is exactly the finite witness problem:

$$\text{find one } N_{\text{seed}}^{(\parallel)} \in V_{\text{mic}} \text{ with } \mathfrak{W}_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0,$$

$$\text{find one } N_{\text{seed}}^{(\perp)} \in V_{\text{mic}} \text{ with } \det T_{\perp}(N_{\text{seed}}^{(\perp)}) \neq 0.$$

If such seeds exist, then by finite-dimensional genericity:

$$g_c(N) \neq 0, \quad (g_E(N), g_O(N)) \neq (0, 0)$$

hold on an open dense full-measure subset of V_{mic} .

Therefore the complete microscopic closure of the concrete S2 transport ansatz is reduced to:

one longitudinal seed witness + one transverse doublet seed witness.

Nothing more is needed at the theorem level.

M. Explicit seed construction inside the concrete S2 transport ansatz space

a. Goal. We now explicitly construct one longitudinal seed tensor and one transverse seed tensor inside the concrete microscopic transport ansatz space, and show that the corresponding witnesses are manifestly nonzero.

1. A. Concrete transport ansatz recalled

We work with the concrete microscopic transport action

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta, \quad (1445)$$

where

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad \Pi_Q = \mathbf{1}_3 - P_B.$$

From the previous computation,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^2}{2} \sigma_3, \quad (1446)$$

so the transport kernel is

$$V^\mu(\xi) = \frac{g_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + g_E \theta'(\xi) E^\mu{}_a \sigma_a + g_O \theta'(\xi) J^\mu{}_a \sigma_a, \quad a = 1, 2. \quad (1447)$$

Projecting onto the graded low modes gives the reduced witnesses

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_\theta, \quad \beta = g_O I_\theta, \quad (1448)$$

with

$$I_\theta = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi), \quad I_{\theta'^3} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (1449)$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0, \quad (1450)$$

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0 \quad (1451)$$

for nontrivial defect amplitude $\theta_0 \kappa \neq 0$.

2. B. Longitudinal seed tensor

Define the explicit longitudinal seed transport kernel

$$\boxed{V_{(\parallel)}^\mu(\xi) := \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3.} \quad (1452)$$

Equivalently, this is the concrete ansatz (1447) with

$$g_c = 1, \quad g_E = 0, \quad g_O = 0.$$

Therefore the projected witnesses are

$$\lambda_{\parallel}(N_{\text{seed}}^{(\parallel)}) = \frac{1}{2} I_{\theta'^3}, \quad (1453)$$

$$\alpha(N_{\text{seed}}^{(\parallel)}) = 0, \quad \beta(N_{\text{seed}}^{(\parallel)}) = 0. \quad (1454)$$

Since $I_{\theta'^3} \neq 0$,

$$\boxed{\lambda_{\parallel}(N_{\text{seed}}^{(\parallel)}) \neq 0.} \quad (1455)$$

Thus an explicit longitudinal seed witness exists.

3. C. Transverse seed tensor

Define the explicit transverse-even seed transport kernel

$$\boxed{V_{(\perp, E)}^\mu(\xi) := \theta'(\xi) E^\mu{}_a \sigma_a.} \quad (1456)$$

Equivalently, this is (1447) with

$$g_c = 0, \quad g_E = 1, \quad g_O = 0.$$

Then the projected witnesses are

$$\lambda_{\parallel}(N_{\text{seed}}^{(\perp, E)}) = 0, \quad (1457)$$

$$\alpha(N_{\text{seed}}^{(\perp, E)}) = I_\theta, \quad \beta(N_{\text{seed}}^{(\perp, E)}) = 0. \quad (1458)$$

Hence the reduced transverse matrix is

$$T_{\perp}(N_{\text{seed}}^{(\perp, E)}) = \begin{pmatrix} I_\theta & 0 \\ 0 & I_\theta \end{pmatrix}, \quad (1459)$$

so

$$\det T_{\perp}(N_{\text{seed}}^{(\perp, E)}) = I_\theta^2 > 0. \quad (1460)$$

Thus an explicit transverse seed witness exists.

4. D. Optional transverse-odd seed

Similarly, define

$$\boxed{V_{(\perp, O)}^\mu(\xi) := \theta'(\xi) J^\mu{}_a \sigma_a.} \quad (1461)$$

This corresponds to

$$g_c = 0, \quad g_E = 0, \quad g_O = 1,$$

and gives

$$\lambda_{\parallel}(N_{\text{seed}}^{(\perp, O)}) = 0, \quad \alpha(N_{\text{seed}}^{(\perp, O)}) = 0, \quad \beta(N_{\text{seed}}^{(\perp, O)}) = I_\theta \neq 0. \quad (1462)$$

So the transverse witness may be realized in either the even or odd doublet channel.

5. E. Closure inside the concrete ansatz subspace

Now define the concrete seed subspace

$$V_{\text{seed}} := \text{span}_{\mathbb{R}} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\}. \quad (1463)$$

An arbitrary element of this subspace has the form

$$N = c_{\parallel} N_{\text{seed}}^{(\parallel)} + c_E N_{\text{seed}}^{(\perp, E)} + c_O N_{\text{seed}}^{(\perp, O)}. \quad (1464)$$

By linearity of the projected transport kernel,

$$\lambda_{\parallel}(N) = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad (1465)$$

$$\alpha(N) = c_E I_\theta, \quad \beta(N) = c_O I_\theta. \quad (1466)$$

Therefore

$$\boxed{\lambda_{\parallel}(N) \neq 0 \iff c_{\parallel} \neq 0,} \quad (1467)$$

and

$$\boxed{(\alpha(N), \beta(N)) \neq (0, 0) \iff (c_E, c_O) \neq (0, 0).} \quad (1468)$$

Hence, inside the concrete ansatz subspace V_{seed} , the S2 nondegeneracy condition is *exactly*

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).} \quad (1469)$$

6. F. Exact closure statement

This yields the following exact closure result.

[Exact closure inside the concrete seed subspace] Inside the concrete transport ansatz subspace

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\},$$

the reduced S2 symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1470)$$

with

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_\theta, \quad \beta = c_O I_\theta. \quad (1471)$$

Since $I_\theta \neq 0$ and $I_{\theta'^3} \neq 0$ for nontrivial defect amplitude, the S2 symbol is nondegenerate if and only if

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0). \quad (1472)$$

Immediate from (1465)–(1466).

7. G. Honest conclusion

This is stronger than the earlier generic witness theorem in one important sense:

we no longer merely know that seed witnesses would imply generic closure; we now have explicit seed tensors that realize the longitudinal and transverse witnesses *inside the chosen concrete ansatz space itself*.

What remains open is more specific:

$$\boxed{\text{Does the complete microscopic TECT transport space actually contain } V_{\text{seed}} \text{ (or at least its two seed directions)?}} \quad (1473)$$

Thus the remaining frontier is no longer witness existence inside the concrete ansatz. It is the embedding of this explicit seed subspace into the complete microscopic TECT transport space.

N. Summary

$$\text{Explicit longitudinal seed:} \quad V_{(\parallel)}^\mu(\xi) = \theta'(\xi)(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3.$$

It gives

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \neq 0, \quad \alpha = \beta = 0.$$

Explicit transverse seed: $V_{(\perp,E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a.$

It gives

$$\lambda_{\parallel} = 0, \quad \alpha = I_{\theta} \neq 0, \quad \beta = 0,$$

hence

$$\det T_{\perp} = I_{\theta}^2 > 0.$$

Optionally,

$$V_{(\perp,O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a$$

gives

$$\lambda_{\parallel} = 0, \quad \alpha = 0, \quad \beta = I_{\theta} \neq 0.$$

Define

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp,E)}, N_{\text{seed}}^{(\perp,O)}\}.$$

For

$$N = c_{\parallel} N_{\text{seed}}^{(\parallel)} + c_E N_{\text{seed}}^{(\perp,E)} + c_O N_{\text{seed}}^{(\perp,O)},$$

the reduced S2 witnesses are exactly

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

Since

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0,$$

the S2 symbol is nondegenerate if and only if

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).}$$

So inside the concrete ansatz space itself, the finite witness problem is already exactly closed.

The only remaining open question is whether complete microscopic TECT actually contains this seed subspace (or at least the two seed directions) in its true transport tensor space.

O. Embedding of the explicit seed subspace into the allowed microscopic transport space

a. Goal. We show that the explicit seed subspace

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp,E)}, N_{\text{seed}}^{(\perp,O)}\}$$

is not an arbitrary effective construction, but can be embedded into the class of symmetry-allowed microscopic TECT transport couplings already identified in the uploaded derivations.

1. A. Allowed microscopic carrier structure

The microscopic classification theorem already established that any UV source able to induce a first-order enriched tensor must carry simultaneously:

1. one spatial derivative index i or j ,
2. one internal reduced-shell/tangent index a .

More precisely, the generic target quadratic mixing form is

$$L_{\text{mix}}^{(2)} = \Lambda_{ij}^a (\partial_i \phi_0) (D_j \psi) \xi^a + \text{h.c.} \quad (1474)$$

and this already implies that the UV extension must introduce an (i, a) -carrier. Pure scalar or pure projector corrections are insufficient. :contentReference[oaicite:3]index=3

At lowest derivative order, the minimal viable local classes are:

1. direct shell–projector derivative couplings (Class I),
2. heavy auxiliary internal vector mediators X_i^a (Class II),
3. connection-like mediators A_i^a (Class III).

Thus the allowed microscopic transport space already contains exactly the type of tensor carrier our explicit seeds require.

2. B. Shell-level minimal witness basis

The shell-level Weyl reduction already identified the minimal symmetry-compatible microscopic tensor family:

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad (1475)$$

$$N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad (1476)$$

$$N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (1477)$$

for which the reduced coefficients satisfy

$$\lambda_{\parallel} \propto A, \quad \alpha \propto B, \quad \beta \propto C. \quad (1478)$$

Equivalently, the local Weyl/Dirac problem is exactly the finite coefficient test

$$A \neq 0, \quad (B, C) \neq (0, 0). \quad (1479)$$

:contentReference[oaicite:5]index=5

This provides the precise microscopic template with which the explicit defect/projector seeds should be compared.

3. C. Defect/projector version of the same three channels

In the local CP^2 defect geometry, the natural S2 splitting is

$$\text{longitudinal singlet} \leftrightarrow \sigma_3, \quad \text{transverse doublet} \leftrightarrow (\sigma_1, \sigma_2).$$

The explicit seed transport kernels we constructed are:

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3, \quad (1480)$$

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a, \quad (1481)$$

$$V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a. \quad (1482)$$

These are the exact defect/projector analogues of the shell-level tensors (1475)–(1477):

$$A \leftrightarrow N_{\text{seed}}^{(\parallel)}, \quad B \leftrightarrow N_{\text{seed}}^{(\perp, E)}, \quad C \leftrightarrow N_{\text{seed}}^{(\perp, O)}. \quad (1483)$$

So the seed subspace V_{seed} is exactly the defect/projector realization of the already-proved three-parameter microscopic witness basis.

4. D. Explicit embedding through Class I

Class I already allows direct local derivative couplings of the form

$$L_{\text{UV}}^{(I)} = \Lambda_{ij}^a (\partial_i \Psi^\dagger) T_a \Psi (T_a \partial_j P) + \text{h.c.} \quad (1484)$$

:contentReference[oaicite:6]index=6

After expansion around the locked background and reduction to the tangent channel ξ^a , this produces the target structure

$$(\partial_i \phi_0)(D_j \psi) \xi^a.$$

Now choose the coefficient tensors Λ_{ij}^a in the three basis forms

$$\Lambda_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad (1485)$$

$$\Lambda_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad (1486)$$

$$\Lambda_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (1487)$$

These are symmetry-allowed by the microscopic shell-level analysis and already generate the three reduced coefficients $(\lambda_\parallel, \alpha, \beta)$.

Therefore Class I already contains a microscopic three-dimensional subspace whose reduced defect/projector image is isomorphic to V_{seed} .

5. E. Explicit embedding through Class II

Class II introduces a heavy auxiliary internal vector mediator X_i^a :

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \Psi^\dagger T_a \Psi + \text{h.c.}) + \beta X_i^a \partial_i \delta P^a \psi. \quad (1488)$$

Eliminating X_i^a generates precisely the target first-order enriched structure

$$(\partial_i \phi_0)(\partial_j \psi) \xi^a,$$

hence a nonzero δN_{ij}^a . :contentReference[oaicite:8]index=8

Now choose the shell/projector current carried by X_i^a to project onto the same three basis tensors (1485)–(1487). Then the reduced low-energy transport coefficients again span the same three-dimensional space (A, B, C) , hence the same seed subspace.

Therefore Class II also admits an embedding of V_{seed} .

6. F. Embedding theorem

We may now state the exact result.

[Embedding of the explicit seed subspace] Assume the true microscopic TECT transport sector contains at least one of the minimal viable UV extension classes identified in the classification theorem, namely Class I or Class II.

Then there exists a three-dimensional microscopic tensor subspace

$$V_{\text{mic}}^{(3)} \subset V_{\text{mic}}$$

whose reduced defect/projector image is isomorphic to

$$V_{\text{seed}} = \text{span}\left\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\right\}.$$

Equivalently, the longitudinal seed witness and the transverse doublet seed witnesses can be realized by admissible microscopic transport tensors.

Class I and Class II both supply the required (i, a) -carrier structure (1474). The shell-level witness analysis already proved that the three tensors (1485)–(1487) form the minimal symmetry-compatible microscopic basis controlling the reduced coefficients $(\lambda_{\parallel}, \alpha, \beta)$.

The explicit defect/projector seeds (1480)–(1482) are exactly the local projector realization of those three channels under the dictionary (1483). Hence the reduced image of the three-dimensional microscopic basis is isomorphic to V_{seed} .

7. G. Honest final status

What is now genuinely closed:

1. the finite witness problem is explicitly solved inside the concrete ansatz space;
2. the seed subspace V_{seed} is compatible with the already-classified allowed microscopic UV carrier structures;
3. if complete TECT realizes Class I or Class II, then the explicit seed directions are admissible microscopic transport directions.

What is still open:

$$\boxed{\text{which minimal UV class (if any) is actually realized by the true full microscopic TECT dynamics.}} \quad (1489)$$

So the remaining frontier is no longer witness construction or local operator structure. It is a model-selection / microscopic-realization question.

P. Summary

$$L_{\text{mix}}^{(2)} = \Lambda_{ij}^a (\partial_i \phi_0) (D_j \psi) \xi^a + \text{h.c.}$$

is the generic target microscopic transport structure, and any viable UV extension must carry both a spatial index i/j and an internal index a .

The already-classified minimal viable classes are:

Class I: direct shell–projector derivative coupling,

Class II: heavy auxiliary internal vector mediator X_i^a ,

Class III: connection-like mediator.

At the shell-level, the minimal symmetry-compatible microscopic tensor basis is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j,$$

$$N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j},$$

$$N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

and these reduce exactly to

$$\lambda_{\parallel} \propto A, \quad \alpha \propto B, \quad \beta \propto C.$$

The explicit defect/projector seed kernels are

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,$$

$$V_{(\perp, E)}^{\mu}(\xi) = \theta'(\xi) E^{\mu}{}_a \sigma_a, \quad V_{(\perp, O)}^{\mu}(\xi) = \theta'(\xi) J^{\mu}{}_a \sigma_a.$$

These are exactly the defect/projector realization of the three shell-level channels A, B, C . Therefore the explicit seed subspace

$$V_{\text{seed}} = \text{span}\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\}$$

can be embedded into any microscopic TECT transport space that realizes at least one of the minimal viable UV classes (especially Class I or Class II).

Hence:

the explicit seed witnesses are not merely formal; they are compatible with the already-allowed microscopic carrier structure.

The only remaining open question is:

which minimal UV class is actually realized by the true full microscopic TECT dynamics.

Q. Why Class II is the natural microscopic choice for TECT at the current stage

a. Goal. We now justify why the Class II UV completion — the heavy auxiliary internal vector mediator X_i^a — is the most natural microscopic carrier for the remaining TECT Dirac-emergence problem, and reinterpret the previously constructed concrete witness coefficients in that language.

1. A. The remaining problem requires an (i, a) carrier

The final local witness problem was reduced to three coefficients

$$g_c, \quad g_E, \quad g_O,$$

appearing in the concrete transport action

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^{\dagger} \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} + g_E \theta'(\xi) E^{\mu}{}_a \Sigma_a + g_O \theta'(\xi) J^{\mu}{}_a \Sigma_a \right] \partial_{y^{\mu}} \eta. \quad (1490)$$

This action requires a microscopic source carrying simultaneously:

1. a spatial / derivative index i or j ,
2. an internal reduced-shell / tangent index a .

The classification theorem already established that any viable UV completion must provide exactly such an (i, a) -carrier, and that pure scalar or pure projector corrections are insufficient. :contentReference[oaicite:2]index=2

2. B. Why Class II is the most natural realization of the required carrier

The minimal viable classes already identified are:

1. Class I: direct shell–projector derivative coupling,
2. Class II: heavy auxiliary internal vector mediator X_i^a ,
3. Class III: connection-like mediator.

Among them, Class II is the most natural present choice because the mediator

$$X_i^a \quad (1491)$$

already carries the exact index structure needed for the enriched first-order tensor: one spatial index i and one internal index a .

The Class II UV action has the schematic form

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a(\Psi) + \beta X_i^a K_i^a(P, \delta P), \quad (1492)$$

where $J_i^a(\Psi)$ is the shell/current source and $K_i^a(P, \delta P)$ is the projector/tangent source. Eliminating X_i^a generates an effective interaction of the target type

$$(\partial_i \phi_0)(D_j \psi) \xi^a,$$

and the document already verified that this leads to a nonzero shell-enriched tensor

$$\delta N_{ij}^a \neq 0. \quad (1493)$$

:contentReference[oaicite:4]index=4

Thus Class II is not merely allowed; it is precisely the minimal UV carrier structure matching the final reduced witness problem.

3. C. Comparison with the other classes

a. Pure scalar / pure projector corrections. These fail because they do not carry the required (i, a) structure. They cannot produce the enriched mixed tensor needed for the first-order reduced symbol. :contentReference[oaicite:5]index=5

b. Primitive projector channel $3P - \mathbf{1}_3$. This is a useful internal tensor, but on Q_B it reduces to the identity, so by itself it is too coarse to generate the full S2 Pauli triplet.

c. Class I. Class I also works in principle:

$$L_{\text{mix}}^{(2)} = \Lambda_{ij}^a (\partial_i \phi_0)(D_j \psi) \xi^a + \text{h.c.}$$

But it inserts the needed carrier tensor Λ_{ij}^a directly by hand. It is therefore less explanatory microscopically than Class II, where the same tensor is generated dynamically by integrating out X_i^a . :contentReference[oaicite:6]index=6

d. Class III. Class III is plausible but currently less explicit in the present derivation chain. At this stage it is weaker as a concrete closure candidate than Class II.

Therefore the natural priority ordering is:

$$\text{Class II} > \text{Class I} > \text{Class III},$$

for the current TECT closure problem.

4. D. Reinterpretation of the seed subspace in Class II language

Previously we constructed the explicit seed subspace

$$V_{\text{seed}} = \text{span}\left\{N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}\right\}. \quad (1494)$$

The shell-level minimal witness basis is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (1495)$$

and it reduces exactly to

$$\lambda_{\parallel} \propto A, \quad \alpha \propto B, \quad \beta \propto C.$$

:contentReference[oaicite:7]index=7

In the Class II interpretation, these three channels correspond naturally to the three internal mediator directions:

$$X_i^3 \leftrightarrow \text{longitudinal singlet seed}, \quad (1496)$$

$$X_i^1 \leftrightarrow \text{transverse-even seed}, \quad X_i^2 \leftrightarrow \text{transverse-odd seed}. \quad (1497)$$

Hence the explicit defect/projector seed basis is exactly the local realization of the Class II carrier decomposition.

5. E. Interpretation of g_c, g_E, g_O

Under this reading, the three witness coefficients in the concrete transport action

$$g_c, \quad g_E, \quad g_O$$

are no longer abstract placeholders. They are the reduced effective couplings induced by the three Class II mediator channels after integrating out X_i^a and projecting to the graded defect sector.

Thus:

$$\boxed{g_c \sim \text{reduced effective coupling of the } X_i^3 \text{ channel},} \quad (1498)$$

$$\boxed{g_E \sim \text{reduced effective coupling of the } X_i^1 \text{ channel}, \quad g_O \sim \text{reduced effective coupling of the } X_i^2 \text{ channel}.} \quad (1499)$$

The previously derived witness formulas then become

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (1500)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (1501)$$

Therefore the S2 nondegeneracy condition is exactly

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (1502)$$

6. F. Strongest honest conclusion

At the current stage, the strongest honest microscopic interpretation is:

Class II is the most natural UV realization of the carrier structure needed by the TECT Dirac-emergence problem. (1503)

The explicit seed subspace V_{seed} is naturally interpreted as the reduced image of the three mediator channels X_i^a , $a = 1, 2, 3$. (1504)

The remaining open question is no longer whether suitable seeds can exist, but whether the true complete TECT action actual (1505)

R. Summary

Why choose Class II?

The remaining TECT Dirac-emergence problem needs a microscopic carrier with both:

a spatial index i/j , an internal index a .

The already-classified minimal viable UV classes are:

Class I: direct shell–projector derivative coupling,

Class II: heavy auxiliary internal vector mediator X_i^a ,

Class III: connection-like mediator.

Among these, Class II is the most natural present choice because the mediator

$$X_i^a$$

already carries exactly the needed (i, a) -structure, and integrating it out is known to generate

$$\delta N_{ij}^a \neq 0.$$

The explicit defect/projector seed subspace

$$V_{\text{seed}} = \text{span} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\}$$

is naturally interpreted as the reduced image of the three Class II mediator channels:

$$X_i^3 \leftrightarrow \text{longitudinal singlet seed},$$

$$X_i^1 \leftrightarrow \text{transverse-even seed}, \quad X_i^2 \leftrightarrow \text{transverse-odd seed}.$$

Accordingly, the witness coefficients are reinterpreted as reduced Class II couplings:

$$g_c \sim X_i^3\text{-channel}, \quad g_E \sim X_i^1\text{-channel}, \quad g_O \sim X_i^2\text{-channel}.$$

The explicit reduced witnesses remain

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Therefore the full local S2 nondegeneracy condition is

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).}$$

So the strongest honest microscopic conclusion is:

$$\boxed{\text{Class II is the leading closure candidate for complete TECT.}}$$

The only remaining open issue is whether the true full TECT action actually generates nonzero reduced Class II couplings g_c, g_E, g_O .

III. DIRAC EMERGENCE VIA THE CLASS II MICROSCOPIC CARRIER

A. Logical status

In this section we separate three layers of the argument:

1. a rigorously proved reduced spectral theorem;
2. a local geometric compatibility statement for the CP^2 projector-defect sector;
3. a concrete Class II microscopic ansatz yielding explicit low-energy witnesses.

Only the first item is claimed as a theorem in the strict reduced-operator sense. The second is a local structural proposition. The third is a concrete microscopic ansatz proposition, not yet a derivation from the final complete TECT PDE.

B. The reduced graded Dirac theorem

[Reduced graded 2 + 2 Dirac emergence] Let $K \subset H$ be an isolated four-dimensional infrared sector with a grading

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2.$$

Assume the projected leading operator takes the form

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^{\dagger} & 0 \end{pmatrix}, \quad A_1(p) = p_{\mu} M^{\mu},$$

and assume $A_1(p)$ is nondegenerate for generic $p \neq 0$. Then the reduced principal symbol is Dirac type.

The grading forces the diagonal blocks to vanish. The first-order mixed symbol defines a Clifford-type linearization once it is nondegenerate. After a linear change of momenta and a unitary change of graded basis, one obtains the canonical Dirac form.

C. Local $\mathbb{CP}^2$ projector-defect geometry

Let the explicit local defect background be

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad \theta(-\xi) = \theta(\xi), \quad \theta(\xi) = \theta_0(\kappa\xi),$$

with projector

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger.$$

Its tangent bundle is

$$Q_B(\xi) := (\mathbb{C}z_B(\xi))^\perp.$$

Choose the local orthonormal frame

$$e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then

$$\partial_\xi z_B = \theta'(\xi)e_1, \quad \partial_\xi P_B = \theta'(\xi)(e_1 z_B^\dagger + z_B e_1^\dagger).$$

[Local odd transport structure] In the local projector-defect sector, the leading first-order transport operator has the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \quad V^\mu(-\xi) = -V^\mu(\xi),$$

and the strongest honest local form is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

where $\mathcal{M}^\mu(\theta)$ is a Hermitian Q_B -endomorphism.

Reflection symmetry makes P_B even and $\partial_\xi P_B$ odd. The leading transport term is linear in the background gradient and first-order in worldvolume derivatives. Therefore it factors through $\theta'(\xi)$, and its fiber action is an endomorphism of Q_B .

In a local Q_B -frame, write

$$\mathcal{M}^\mu(\theta) = a_0^\mu(\theta)\mathbf{1}_2 + a_i^\mu(\theta)\sigma_i.$$

D. Natural S2 local branch

The defect gradient singles out one preferred tangent direction e_1 , so the traceless fiber sector splits as

$$\mathbb{R}_{\text{tr}}^3 = \mathbb{R}\sigma_3 \oplus \text{span}\{\sigma_1, \sigma_2\}.$$

The worldvolume momentum directions split naturally as

$$\mathbb{R}_{\text{wv}}^3 = \mathbb{R}p_\parallel \oplus \text{span}\{p_1, p_2\}.$$

[Natural local S2 branch] The natural local symmetry-compatible reduced symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Its nondegeneracy condition is

$$\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

This is the most general intertwiner between the singlet $\oplus$ doublet decomposition of worldvolume momenta and the singlet $\oplus$ doublet decomposition of the traceless fiber sector.

E. Primitive and commutator channels

The canonical primitive internal tensor

$$T(P) = 3P - \mathbf{1}_3$$

is too coarse on Q_B , since

$$T(P_B)|_{Q_B} = -\mathbf{1}_{Q_B}.$$

So it does not generate a full Pauli triplet on Q_B .

Next define the commutator channel

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

Direct computation gives

$$\mathcal{T}_\xi = i\theta'(\xi)(z_B e_1^\dagger - e_1 z_B^\dagger),$$

so $\mathcal{T}_\xi$ exchanges $\mathbb{C}z_B$ and $\mathbb{C}e_1$, while annihilating e_2 . Hence

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}(\mathbf{1}_2 + \sigma_3).$$

Thus the commutator channel naturally provides the longitudinal σ_3 -type splitting, but not the full transverse doublet.

F. Class II microscopic ansatz

We now choose the Class II heavy auxiliary internal vector mediator X_i^a as the minimal microscopic carrier.

[Class II microscopic carrier ansatz] Assume the microscopic UV sector contains a heavy internal vector mediator X_i^a carrying one spatial index i and one internal index a . Assume its low-energy reduction induces the effective transport action

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu{}_a \Sigma_a + g_O \theta'(\xi) J^\mu{}_a \Sigma_a \right] \partial_{y^\mu} \eta.$$

Then the reduced S2 witnesses are explicitly computable.

Using

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{\theta'(\xi)^2}{2} \sigma_3,$$

this becomes

$$\mathcal{L}_{\text{trans}}^{\min} = -i\eta^\dagger \left[\frac{g_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + g_E \theta'(\xi) E^\mu{}_a \sigma_a + g_O \theta'(\xi) J^\mu{}_a \sigma_a \right] \partial_{y^\mu} \eta.$$

Project onto the graded low modes

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi).$$

Define

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi), \quad I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.$$

For the explicit defect profile,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

[Explicit witness coefficients] Under the Class II microscopic ansatz,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

Equivalently,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Hence the reduced S2 nondegeneracy condition is exactly

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Immediate from the projected overlap formula and the explicit evaluations of I_{θ} and $I_{\theta'^3}$.

G. Finite witness closure

[Finite witness closure for Class II] Inside the concrete Class II transport ansatz space, define the explicit seed directions:

$$N_{\text{seed}}^{(\parallel)} : \quad V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,$$

$$N_{\text{seed}}^{(\perp, E)} : \quad V_{(\perp, E)}^{\mu}(\xi) = \theta'(\xi) E^{\mu}_a \sigma_a,$$

$$N_{\text{seed}}^{(\perp, O)} : \quad V_{(\perp, O)}^{\mu}(\xi) = \theta'(\xi) J^{\mu}_a \sigma_a.$$

For

$$N = c_{\parallel} N_{\text{seed}}^{(\parallel)} + c_E N_{\text{seed}}^{(\perp, E)} + c_O N_{\text{seed}}^{(\perp, O)},$$

the reduced witnesses are exactly

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

Therefore the reduced S2 symbol is nondegenerate if and only if

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).$$

Linearity of the projected transport kernel in the seed coefficients.

H. Honest conclusion

At the current stage, the logically strongest honest statement is:

1. the reduced graded 2 + 2 Dirac theorem is rigorous;
2. the actual local CP^2 projector-defect geometry is compatible with the natural S2 branch;
3. Class II provides the most natural microscopic carrier for the remaining (i, a) -indexed transport problem;
4. under the concrete Class II ansatz, the entire local Dirac witness problem is reduced to the finite coefficient test

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0);$$

5. the only remaining open question is whether complete TECT actually generates nonzero reduced Class II couplings g_c, g_E, g_O .

I. Summary

Theorem layer:

If an isolated graded $2+2$ infrared sector exists and its projected first-order mixed symbol is nondegenerate, then the reduced principal operator is Dirac type.

Local geometry layer:

For the explicit CP^2 defect background,

$$P_B = z_B z_B^\dagger, \quad \partial_\xi z_B = \theta' e_1,$$

the leading transport kernel is

$$V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta(\xi)),$$

and the natural local symmetry-compatible reduced symbol is the S2 form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Microscopic ansatz layer:

Choosing Class II with heavy internal vector mediator X_i^a , the concrete effective transport action is

$$\mathcal{L}_{\text{trans}}^{\min} = -i \eta^\dagger \left[g_c \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu + g_E \theta'(\xi) E^\mu_a \Sigma_a + g_O \theta'(\xi) J^\mu_a \Sigma_a \right] \partial_{y^\mu} \eta.$$

Its reduced witnesses are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa.$$

Therefore the full local S2 nondegeneracy condition is exactly

$$\boxed{g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).}$$

Inside the concrete seed subspace, the witness problem is already exactly closed.

Hence the only remaining open issue is:

$$\boxed{\text{does complete microscopic TECT actually generate nonzero reduced Class II couplings } g_c, g_E, g_O ?}$$

J. Generic nonvanishing of the Class II witness triple

a. Goal. We now upgrade the concrete Class II transport ansatz to a finite-dimensional witness theorem showing that the nonvanishing of

$$g_c, \quad g_E, \quad g_O$$

is generic once the corresponding Class II channels are not identically forbidden on the allowed microscopic tensor space.

1. A. Class II preserves the shell-relevant axial tensor structure

The Class II UV mediator X_i^a generates the induced enriched tensor

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}. \quad (1506)$$

This is already derived explicitly in the Class II construction. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

Projecting onto the shell-adapted invariant basis,

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (1507)$$

the induced correction has the same form, with coefficient shifts

$$\delta A = -\frac{\lambda g_X}{M_X^2} m^a \hat{n}_i \hat{n}_j C_{ij}^{a3}, \quad (1508)$$

$$\delta B = -\frac{\lambda g_X}{2M_X^2} m^a (e_{1j} \hat{n}_i C_{ij}^{a1} + e_{2j} \hat{n}_i C_{ij}^{a2}), \quad (1509)$$

$$\delta C = -\frac{\lambda g_X}{2M_X^2} m^a (e_{2j} \hat{n}_i C_{ij}^{a1} - e_{1j} \hat{n}_i C_{ij}^{a2}). \quad (1510)$$

So Class II does not change the structure of the reduced Weyl/Dirac theorem; it only renormalizes the three axial coefficients. :contentReference[oaicite:7]index=7

2. B. Dictionary between the shell coefficients and the local defect witnesses

In the local defect/projector S2 reduction, we already identified the reduced symbol

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2), \quad (1511)$$

with the concrete Class II transport action giving

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (1512)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (1513)$$

The shell-level structure and the defect-level structure are therefore identified by the dictionary

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O. \quad (1514)$$

Thus the Class II shell witness triple $(A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}})$ is exactly the same finite witness triple as (g_c, g_E, g_O) in the local defect transport formulation.

3. C. Finite-dimensional genericity theorem

Let V_{II} denote the finite-dimensional real vector space of all symmetry-allowed Class II microscopic tensors surviving the linearization around the chosen background.

The shell-level genericity theorem already states:

1. the longitudinal coefficient $v_{\parallel}$ is a linear functional on V ;

2. the transverse intertwiner $T(N)$ depends linearly on N ;
3. therefore $\det T(N)$ is a polynomial on V ;
4. if neither is identically zero, then their simultaneous nonvanishing holds on an open dense full-measure subset.

:contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

Applying this directly to the Class II tensor space V_{II} , we obtain:

$$g_c : N \mapsto g_c(N) \quad \text{is linear on } V_{II}, \quad (1515)$$

and

$$\det T_{\perp}(N) = g_E(N)^2 + g_O(N)^2 \quad \text{is polynomial on } V_{II}. \quad (1516)$$

Therefore the exact dichotomy is:

- either $g_c \equiv 0$ on V_{II} , in which case the longitudinal Class II witness is symmetry-forbidden;
- or $g_c \not\equiv 0$, in which case $g_c(N) \neq 0$ on an open dense full-measure subset of V_{II} ;
- either $g_E^2 + g_O^2 \equiv 0$ on V_{II} , in which case the transverse Class II doublet is symmetry-forbidden;
- or $g_E^2 + g_O^2 \not\equiv 0$, in which case $(g_E(N), g_O(N)) \neq (0, 0)$ on an open dense full-measure subset of V_{II} .

4. D. Minimal Class II choice supplies explicit seeds

The minimal Class II choice already written in the manuscript is

$$m^a(P_0) = m \delta^{a3}, \quad (1517)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (1518)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (1519)$$

Under this choice, the manuscript states that opposite-valley pairing is preserved, $V(0) = 0$, $\Delta > 0$, and mass protection is satisfied. :contentReference[oaicite:10]index=10

Now compare (1518)–(1519) with (1508)–(1510). One reads immediately:

$$\delta A \propto c_{\parallel}, \quad \delta B \propto c_I, \quad \delta C \propto c_J. \quad (1520)$$

Hence, through the dictionary (1514),

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J. \quad (1521)$$

So the minimal Class II choice itself already gives explicit seed directions:

$$c_{\parallel} \neq 0, \quad c_I = c_J = 0 \quad \implies \quad g_c \neq 0, \quad (1522)$$

$$c_{\parallel} = 0, \quad c_I \neq 0, \quad c_J = 0 \quad \implies \quad g_E \neq 0, \quad (1523)$$

$$c_{\parallel} = 0, \quad c_I = 0, \quad c_J \neq 0 \quad \implies \quad g_O \neq 0. \quad (1524)$$

Thus the witness problem is not merely abstractly reducible: the minimal Class II parameterization already contains explicit longitudinal and transverse seed witnesses.

5. E. Class II witness theorem

We can now state the actual closure theorem for the chosen microscopic carrier.

[Class II witness theorem for the local S2 transport ansatz] Let V_{II} be the finite-dimensional space of symmetry-allowed Class II microscopic tensors around the chosen TECT defect background. Assume the reduced local transport coefficients (g_c, g_E, g_O) are read from the induced Class II tensor via the shell/defect dictionary above.

If the minimal Class II parameter space contains:

1. one admissible longitudinal seed with $c_{||} \neq 0$;
2. one admissible transverse seed with $(c_I, c_J) \neq (0, 0)$;

then:

$$g_c(N) \neq 0 \quad (1525)$$

and

$$(g_E(N), g_O(N)) \neq (0, 0) \quad (1526)$$

hold simultaneously on an open dense full-measure subset of V_{II} .

Equivalently, the reduced S2 symbol is nondegenerate generically in the allowed Class II tensor space.

The longitudinal statement follows because g_c is a nonzero linear functional once one seed with $g_c \neq 0$ exists. Its zero set is then a proper hyperplane. :contentReference[oaicite:11]index=11

The transverse statement follows because $g_E^2 + g_O^2$ is a nonzero polynomial once one seed with $(g_E, g_O) \neq (0, 0)$ exists. Its zero set is then a proper algebraic subset. :contentReference[oaicite:12]index=12

Intersecting the two good sets proves the claim.

6. F. Strong honest conclusion

At this point, the status is:

$$\boxed{\text{inside the Class II truncation, the finite witness problem is closed.}} \quad (1527)$$

What remains open is not witness existence, nor local geometric compatibility, nor finite-dimensional genericity. The only remaining microscopic question is:

$$\boxed{\text{does the true complete TECT action actually realize a nontrivial Class II sector in the physically relevant defect background?}} \quad (1528)$$

K. Summary

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}$$

is the Class II induced enriched tensor.

After shell-adapted projection,

$$\delta A \propto m^a \hat{n}_i \hat{n}_j C_{ij}^{a3}, \quad \delta B \propto m^a (e_{1j} \hat{n}_i C_{ij}^{a1} + e_{2j} \hat{n}_i C_{ij}^{a2}), \quad \delta C \propto m^a (e_{2j} \hat{n}_i C_{ij}^{a1} - e_{1j} \hat{n}_i C_{ij}^{a2}),$$

so Class II preserves the exact axial witness structure and only renormalizes the three coefficients. :contentReference[oaicite:13]index=13

Using the local defect/projector dictionary,

$$A_{\text{eff}} \leftrightarrow g_c, \quad B_{\text{eff}} \leftrightarrow g_E, \quad C_{\text{eff}} \leftrightarrow g_O.$$

The minimal Class II choice

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}$$

already supplies explicit seed directions, and is stated to be compatible with opposite-valley pairing, $V(0) = 0$, $\Delta > 0$, and mass protection. :contentReference[oaicite:14]index=14

Hence

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Since g_c is linear on the allowed Class II tensor space and $g_E^2 + g_O^2$ is polynomial, finite-dimensional genericity implies:

$$c_{\parallel} \neq 0 \implies g_c \neq 0 \text{ generically,}$$

$$(c_I, c_J) \neq (0, 0) \implies (g_E, g_O) \neq (0, 0) \text{ generically.}$$

Therefore:

$$\text{inside the Class II truncation, the local S2 witness problem is generically closed.}$$

The only remaining open issue is whether the true complete TECT action actually realizes a nontrivial Class II sector in the relevant defect background.